QUANTITATIVE METHODS LEVEL 1

Trustville

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Lời giới thiệu

Tài liệu này được biên soạn và tổng hợp chủ yếu từ nguồn học liệu Curriculum level 1. Bên cạnh những trích dẫn, công thức và biểu đồ, đôi khi bạn sẽ gặp một số đoạn văn tiếng Việt in nghiêng và màu xanh là phần diễn giải bổ sung của Giảng viên biên soạn, hy vọng giúp giải thích rõ ràng hơn.
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Mặc dù tài liệu biên soạn được chuẩn bị một cách kỹ lưỡng và cẩn trọng, nhưng cũng không thể tránh khỏi sai sót và lỗi diễn đạt. Vậy rất mong những ý kiến phản hồi từ các bạn khi sử dụng tài liệu, để giúp hoàn thiện nội dung hơn.
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Chúc các bạn có những giờ học bổ ích và hoàn thành mục tiêu!

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Module 1

Returns of Financial Assets and Instruments

LEARNING OUTCOMES	
a	describe, compare, and interpret returns
b	describe, compare, and interpret required rates of return, risk-free rates, risk premia, and inflation

Financial Returns

Total Return
Arithmetic and Geometric Holding Period Return
Compounding: Daily, Monthly, Quarterly, and Annual Returns
Continuous Compounding

Common Return Measures

The Required Rate of Return
Excess Returns
Real Returns
Gross and Net Return
Pre-Tax and After-Tax Nominal Return
Leveraged Returns

1.1 Financial Returns

a | describe, compare, and interpret returns

Lãi suất (lợi suất) là giá của Vốn - hàng hóa đặc biệt trên thị trường tài chính. Hãy bắt đầu tìm hiểu cách đo lường mức giá này.

1.1.1 Total Return

$$\text{Total Return} = \text{Capital Appreciation Return} + \text{Capital Distribution Return}$$

$$\begin{aligned} r &= r_{\text{price}} + r_{\text{distribution}} \\ &= \frac{P_1 - P_0}{P_0} + \frac{\text{Inc}}{P_0} \end{aligned} \tag{1.1}$$

For capital distributions on equities, we focus on the dividend yield; for bonds, we look at current yield.

- Some expressions of Return:

Expected returns (ex ante returns): are the anticipated returns on an investment over a specified period.

Actual returns (ex post returns): are the returns on an investment after the investment period has ended.

Unrealized returns: changes in value represent while an investor holds an investment.

Realized returns: when the investment is ultimately sold, gains and losses would be realized at the current market value.

1.1.2 Arithmetic and Geometric Holding Period Return

To ensure accurate comparisons across different investments and time periods, it's crucial that these returns are calculated using a uniform time period.

- **arithmetic mean return**:

$$\bar{r}_A = \frac{r_1 + r_2 + \cdots + r_T}{T} = \frac{1}{T}\sum_{t=1}^{T} r_t \tag{1.2}$$

- **geometric mean return**:

$$\bar{r}_G = \sqrt[T]{(1+r_1)\times(1+r_2)\times\cdots\times(1+r_T)} - 1 = \left[\prod_{t=1}^{T}(1+r_t)\right]^{1/T} - 1 \tag{1.3}$$

Geometric mean is a more accurate reflection of investment performance over time. Arithmetic average return calculations assume that the amount invested at the beginning of the period does not change, but the base amount changes due to capital appreciation or depreciation. The arithmetic-geometric mean inequality highlights the impact of volatility on investment returns.

Exam Tip

Remember: - **Arithmetic Mean** is the best estimator of return for a single future period (ex-ante). - **Geometric Mean** represents the actual historical compound rate of return over multiple periods (ex-post). - Unless all period returns are identical, **Arithmetic Mean is always strictly greater than Geometric Mean** due to volatility ($\bar{r}_A > \bar{r}_G$).

1.1.3 Compounding: Daily, Monthly, Quarterly, and Annual Returns

With c is the number of compounding periods in a year,

$$r_{\text{annualized}} = (1 + r_{\text{period}})^c - 1 \tag{1.4}$$

One major limitation of annualizing returns is the implicit assumption that earned returns can be repeated.

Example: Compounding and Annualizing Returns

Consider a bank deposit with a 3-month periodic interest rate of 1.50%.

- **Compounding periods per year:** $c = 4$.
- **Annualized return calculation:**

$$r_{\text{annualized}} = (1 + 0.015)^4 - 1 \approx 6.136\%$$

Thực tế: Lãi suất 6.136% lớn hơn mức 6.0% nhân đơn (1.5% × 4), phản ánh lãi kép khi tiền lãi kỳ trước được tái đầu tư sinh lãi kỳ sau.

1.1.4 Continuous Compounding

The more frequently we compound returns, the higher the annual return will be; however, there is a limit. As shown in Exhibit 1.1:

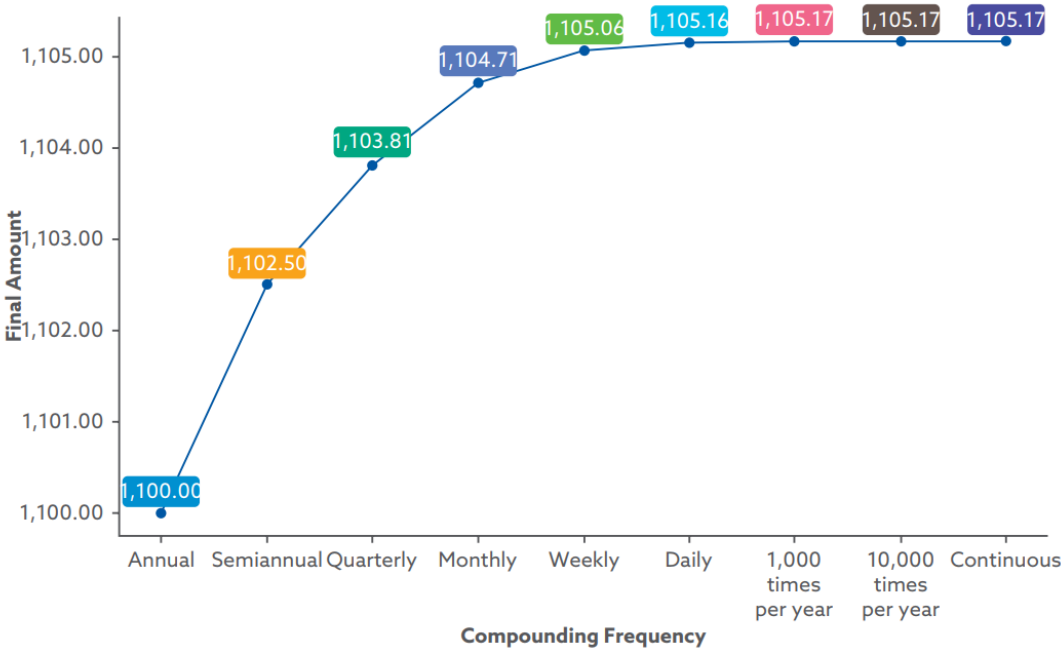


Figure 1.1: The Impact of Compounding

- **continuously compounded return**:

$$\tilde{r}_{t,t+1} = \ln\left(\frac{P_{t+1}}{P_t}\right) = \ln(1 + r_{t,t+1}) \tag{1.5}$$

The additive property of continuously compounded returns provides calculation advantages:

$$\begin{aligned}
&\frac{P_T}{P_0} = \frac{P_T}{P_{T-1}} \cdot \frac{P_{T-1}}{P_{T-2}} \cdots \frac{P_2}{P_1} \cdot \frac{P_1}{P_0} \\
\Rightarrow &\ln\left(\frac{P_T}{P_0}\right) = \ln\left(\frac{P_T}{P_{T-1}}\right) + \ln\left(\frac{P_{T-1}}{P_{T-2}}\right) + \cdots + \ln\left(\frac{P_2}{P_1}\right) + \ln\left(\frac{P_1}{P_0}\right) \\
\Rightarrow &\tilde{r}_{0,T} = \tilde{r}_{T-1,T} + \tilde{r}_{T-2,T-1} + \cdots + \tilde{r}_{1,2} + \tilde{r}_{0,1} = \sum_{t=1}^{T} \tilde{r}_{t-1,t}
\end{aligned}$$

Tại sao suất sinh lời ghép lãi liên tục (log return) được ưa chuộng hơn suất sinh lời rời rạc? Nhờ tính cộng gộp thời gian (time-additivity): tổng log return các ngày bằng log return cả năm, giúp đơn giản hóa mô hình toán học.

1.2 Common Return Measures

b | describe, compare, and interpret required rates of return, risk-free rates, risk premia, and inflation

1.2.1 The Required Rate of Return

- **opportunity cost**: that incurs when delaying consumption, or the value that investors forgo by choosing a course of action.
- **required rate of return**: the minimum rate of return an investor must receive to accept the investment.

Components of Required Rate of Return:

- **risk-free rate**: is the return on an investment without any default or reinvestment risk.

  It refers to returns on short-term government bonds, such as the 30-day US Treasury bill. The bonds have a short maturity, which reduces their sensitivity to interest rates changes.

  real risk-free rate reflects the time preferences of individuals for current versus future real consumption.

$$r_{\text{real_rf}} = \frac{1 + r_{\text{rf}}}{1 + \text{infl}} - 1 \quad \text{or quick approximation: } r_{\text{real_rf}} \approx r_{\text{rf}} - \text{infl} \tag{1.6}$$

- Total required returns for any asset can then be approximated as:

$$r = r_{\text{rf}} + \text{infl} + \sum_{j=1}^{n}(f_j \times \text{rp}_j) \tag{1.7}$$

n is the number of risk factors

f_j is the exposure of the security to the jth risk factor

rp_j is the specific risk premium associated with the jth risk factor

$\sum_{j=1}^{n}(f_j \times \text{rp}_j)$ is the sum of all risk premia.

As shown in Exhibit 1.2:

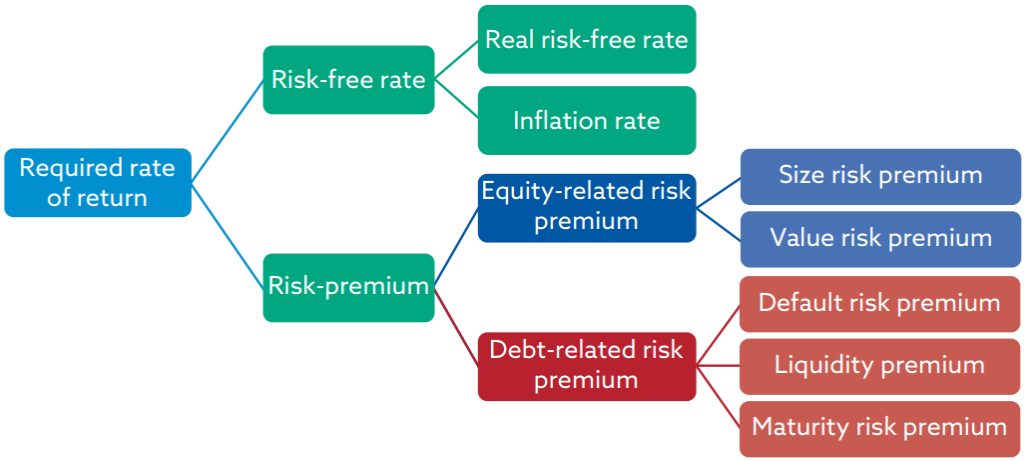


Figure 1.2: Components of Required Rate of Return and Examples of Different Risk Premia

Examples of risk premia for bonds:

- **Default risk premium**: the additional return for buying a riskier corporate bond instead of a less-risky government bond.
- **Liquidity premium**: the additional return for holding a bond that may be harder to sell quickly.
- **Maturity risk premium**: the additional return for holding more interest rate sensitive, longer maturity bonds.

Examples of risk premia for stocks:

- **Size risk premium**: the return spread between small and large market-capitalization stocks.
- **Value risk premium**: the return spread between value stocks (high book-to-market) and growth stocks (low book-to-market).

1.2.2 Excess Returns

Excess returns: the difference between an asset's return and the return on some reference rate or benchmark.

$$\dot{r}_i = \frac{1 + r_i}{1 + r_B} - 1 \quad \text{or quick approximation: } \dot{r}_i \approx r_i - r_B \tag{1.8}$$

1.2.3 Real Returns

Real returns are inflation-adjusted returns.

$$r_{\text{real}} = \frac{1 + r_i}{1 + \text{infl}} - 1 \quad \text{or quick approximation: } r_{\text{real}} \approx r_i - \text{infl} \tag{1.9}$$

1.2.4 Gross and Net Return

Gross return = Total return - Transaction costs

An appropriate measure for evaluating and comparing the investment skill of managers.

Excludes indirect expenses related to the management and administration of an investment.

Net return = Gross return - Management fees - Administrative expenses

The return the investor actually earns.

Exhibit 1.3 depicts the relationship and shows how costs erode investment returns:

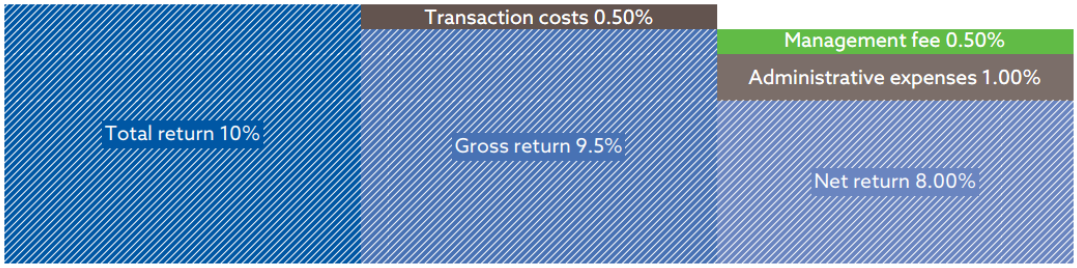


Figure 1.3: The Relationship between Total Return, Gross Return, and Net Return

1.2.5 Pre-Tax and After-Tax Nominal Return

After-tax nominal return:

$$r_{\text{net-tax}} = r_{\text{price}} \times (1 - \text{tax}_{\text{capital gains}}) + r_{\text{capital}} \times (1 - \text{tax}_{\text{capital distribution}}) \tag{1.10}$$

Capital gains tax is a tax on a positive price return. Two forms of capital gains tax rate:

Short-term capital gains are levied on capital gains on short-term holdings.

Long-term capital gains are levied on longer-term holdings and receive preferential tax treatment in a number of countries.

Forms of capital distribution tax rate:

Interest income is taxed as ordinary income in most countries.

Dividend income may be taxed as ordinary income, at a lower rate, or be exempt from taxes depending on the jurisdiction.

Exam Tip

In examinations, note the different tax rates applied to price returns (capital gains tax) versus income distributions (dividend or interest income tax). Always apply them to their respective components of the return formula.

1.2.6 Leveraged Returns

Using leverage to increase the total amount of investible assets is a common feature of investments, whether through borrowing or derivatives. Leverage increases investible capital, magnifying both returns and risks.

$$R_L = R_A + \frac{D}{E}(R_A - R_D) \tag{1.11}$$

R_L is the leveraged return (Return on Equity)

R_A is the return on total assets (unleveraged return)

R_D is the cost of debt (borrowing interest rate)

D/E is the debt-to-equity ratio.

Đòn bẩy là dao hai lưỡi. Khi suất sinh lời tài sản (R_A) lớn hơn chi phí đi vay (R_D), đòn bẩy phóng đại suất sinh lời vốn tự có (R_L). Nhưng nếu $R_A < R_D$, tỷ suất sinh lời của chúng ta sẽ sụt giảm rất nhanh!

The potential financial outcomes for owning a stock outright versus owning a call option on the stock (another form of leverage) are illustrated in Exhibit 1.4:

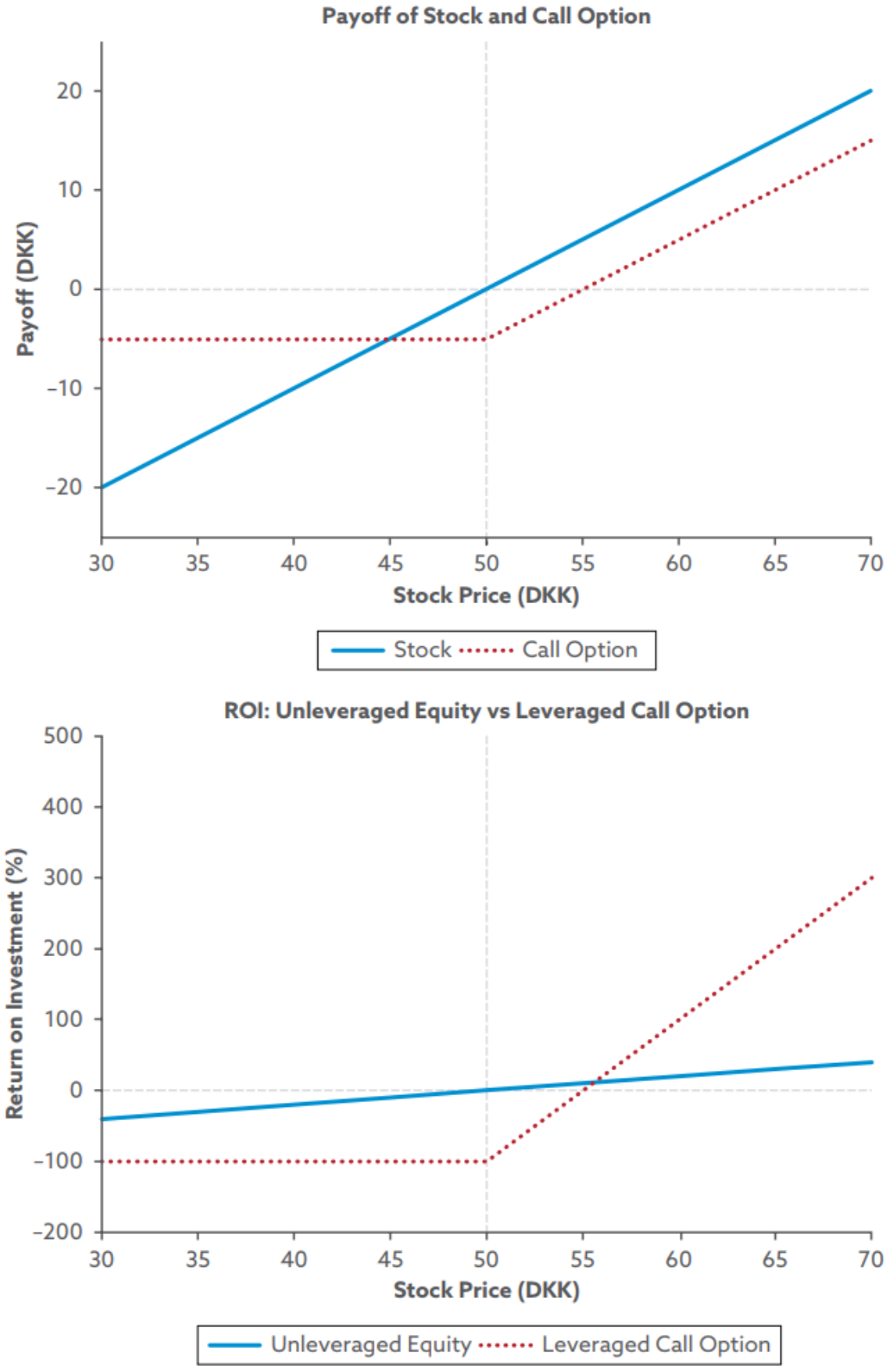


Figure 1.4: The Payoff Relationship between Stock Price, Unleveraged Return, and Leveraged Return

Module 2

Types of Returns for Financial Assets, Instruments, and Indicators

LEARNING OUTCOMES	
a	calculate, compare, and interpret different types of returns for financial assets, instruments, and indicators

Types of Returns

Types of Returns: Financial Assets

Types of Returns: Financial Instruments

Types of Returns: Financial Indicators

Long-Term Financial Asset Returns

Historical Long-Term Asset Returns

Long-Term Asset Return Components

Risks and Returns

Systematic and Unsystematic Risk

The Risk–Return Trade-Off

Portfolio Return Calculation

2.1 Types of Returns

a	calculate, compare, and interpret different types of returns for financial assets, instruments, and indicators

Total Return gồm Price Return (lãi/lỗ vốn) và Capital Distribution Return (cổ tức, tiền lãi). Cấu phần này thay đổi tùy thuộc vào việc chúng ta đánh giá tài sản, công cụ hay chỉ báo tài chính.

2.1.1 Matrix Mapping of Return Types

To provide a scientific and intuitive overview, Table 2.1 maps the characteristics of price return and capital distribution return across all assets, instruments, and indicators.

Table 2.1: Matrix Mapping of Return Types across Assets, Instruments, and Indicators

Category	Price Return	Capital Distribution Return
Financial Assets		
- Cash	None (zero price change).	None (unless held in interest-bearing accounts).
- Equities	Highly volatile and uncertain (Capital Appreciation).	Variable and discretionary (Dividends).
- Bonds (Debt)	Inversely related to interest rates; higher volatility for long-term bonds.	Fixed and contractual (Coupon payments).
- Hybrids	Moderate appreciation potential.	Fixed or structured (Preferred dividends).
Financial Instruments		
- Derivatives	Determined by price movements of underlying assets.	None (does not distribute cash flows).
Financial Indicators		
- Interest Rates	Inverse relation to debt instrument prices.	None (acts as valuation driver).
- FX Rates	Currency appreciation/depreciation (r_{FX}).	None (acts as return multiplier: $1 + r_{\text{FX}}$).
- Market Indexes	Reflects aggregate price movements.	Reflects weighted cash distributions of constituents.

2.1.2 Types of Returns: Financial Assets

The main financial assets are cash, equity, and debt, as well as hybrids that combine their characteristics.

Khi chi trả cổ tức X, lý thuyết giá giảm ngay đúng bằng X, khiến phần sinh lời phân phối bù trừ Price Return trong ngắn hạn. Chúng ta đã bao giờ quan sát hiện tượng này tại sàn chứng khoán Việt Nam vào ngày giao dịch không hưởng quyền chưa?

Let's detail the characteristics across major financial assets:

- **Cash:** does not generate any return unless invested. In the short term, excess cash is typically placed in interest-bearing bank accounts or short-term debt instruments.
- **Equities:** generate returns from capital appreciation and discretionary dividends (variable capital distributions). Rapidly growing companies often choose not to distribute dividends and instead reinvest earnings to fuel growth.
- **Bonds (Debt):** represent contractual lending agreements that pay fixed interest coupons (fixed capital distributions). Shorter-term bonds are dominated by capital distributions, whereas longer-term bonds exhibit substantial price volatility due to market interest rate movements.
- **Hybrids:** preferred stock and convertible bonds blend fixed-income features of debt with the growth potential of equity.

2.1.3 Types of Returns: Financial Instruments

Financial instruments are standardized legal agreements for financial assets (such as equity and bonds), as well as derivatives and securitized debt. The standardization facilitates public trading.

- **Cash Instruments:** represent direct ownership claims on financial assets (e.g., actual shares of stock or actual corporate bonds).
- **Derivatives:** financial instruments whose value is derived from the performance of another underlying financial asset or indicator (e.g., stock price, interest rate, index, exchange rate). A call option is a typical example.

2.1.4 Types of Returns: Financial Indicators

Financial indicators do not directly generate cash flows, but are critical to financial market operations.

- **Interest Rates:** traded through risk-free government bonds. Their level varies inversely with bond prices.
- **The Risk-Free Rate:** represented by short-term, risk-free government debt (e.g., US Treasury bills).

Exam Tip

Short-term discount instruments (like T-bills) do not pay regular coupons ($\text{Inc}_t = 0$). Instead, they sell at a discount to face value, and the return is earned purely from the price appreciation when the bond matures at par value.

Let's review the annualized risk-free rate calculation using a Treasury Bill:

$$r_{\text{period}} = \frac{P_1 - P_0 + \text{Inc}_t}{P_0} = \frac{\text{Face Value} - P_0}{P_0} \tag{2.1}$$

To annualize this period return for a T-bill with 90 days to maturity using 365 days in a year:

$$r_{\text{annual}} = (1 + r_{\text{period}})^{\frac{365}{90}} - 1 \tag{2.2}$$

Example: T-Bill Period and Annualized Returns

Consider a 90-day US Treasury Bill with a face value of USD 1,000 purchased at USD 990.

- **Period return calculation:**

$$r_{\text{period}} = \frac{1000 - 990}{990} \approx 1.01\%$$

- **Annualized return calculation (using 365 days):**

$$r_{\text{annual}} = (1 + 0.0101)^{\frac{365}{90}} - 1 \approx 4.16\%$$

Thực tế: So sánh lãi suất năm thực tế (4.16% ghép lãi) và lãi suất đơn ($1.01\% \times \frac{365}{90} \approx 4.10\%$). Trong phòng thi, chúng ta hãy luôn ưu tiên phương pháp ghép lãi lũy thừa.

- **Foreign Exchange (FX) Rates:** represent the value of one currency expressed in terms of another currency.

When investing internationally, the home-currency return (r_{home}) depends on both the investment return in foreign currency (r_{foreign}) and the currency appreciation/depreciation rate (r_{FX}):

$$r_{\text{home}} = (1 + r_{\text{foreign}})(1 + r_{\text{FX}}) - 1 \approx r_{\text{foreign}} + r_{\text{FX}} \tag{2.3}$$

Where:

$$r_{\text{FX}} = \frac{\text{Ending FX Rate} - \text{Beginning FX Rate}}{\text{Beginning FX Rate}} \tag{2.4}$$

Example: Foreign Exchange FX Adjustment on Returns

A US dollar-based investor purchases a bond denominated in Euros (EUR).

- **Foreign asset return:** $r_{\text{foreign}} = -2.91\%$.
- **FX rate change (EUR to USD):** $r_{\text{FX}} = -6.90\%$ (the Euro depreciated from 1.2216 to 1.1373 USD/EUR).
- **Home-currency return calculation:**

$$r_{\text{home}} = (1 - 0.0291)(1 - 0.0690) - 1 = -9.61\%$$

Thực tế: Tác động kép từ trái phiếu giảm giá và EUR mất giá so với USD, khuếch đại khoản lỗ bằng USD từ -2.91% thành -9.61%.

- **Market Indexes:** reflect the aggregate value of a basket of assets (e.g., S&P 500, OMX Copenhagen 25 Index) to indicate market trends.

To evaluate and compare performance, we look at the arithmetic and geometric averages of index returns:

$$\bar{r}_{\text{arithmetic}} = \frac{1}{T}\sum_{t=1}^{T} r_t \tag{2.5}$$

$$\bar{r}_{\text{geometric}} = \left[\prod_{t=1}^{T}(1+r_t)\right]^{\frac{1}{T}} - 1 \tag{2.6}$$

Let's review the historical performance numbers of the OMX Copenhagen 25 Index versus Ørsted A/S equity between 2017 and 2023:

Table 2.2: OMX Copenhagen 25 Index vs. Ørsted A/S Equity Total Returns (2017–2023)

Year	OMX Copenhagen 25 Return	Ørsted A/S Equity Return
2017	15.09%	28.81%
2018	-11.19%	31.30%
2019	28.71%	60.37%
2020	35.60%	82.00%
2021	18.60%	-31.91%
2022	-11.54%	-22.92%
2023	9.63%	-38.57%
Arithmetic Mean	**12.13%**	**15.58%**
Geometric Mean	**10.81%**	**6.86%**

Chúng ta thấy gì thú vị? Arithmetic Mean của Ørsted (15.58%) cao hơn OMX (12.13%), nhưng Geometric Mean lại thấp hơn nhiều (6.86% so với 10.81%), chứng minh tác động của biến động (volatility drag) lên lãi kép khi tăng giảm sâu.

2.2 Long-Term Financial Asset Returns

Chúng ta hãy cùng rà soát số liệu thực tế 45 năm (1979–2023) tại thị trường Mỹ để so sánh lợi nhuận và cơ cấu sinh lời của các loại tài sản thay đổi thế nào qua thời gian.

2.2.1 Historical Long-Term Asset Returns

Based on historical data for selected major US asset classes between 1979 and 2023 (large-cap stocks, small-cap stocks, corporate bonds, government bonds, and Treasury bills), several trends emerge:

- **Equities Outperform Bonds:** driven primarily by capital appreciation, equities deliver higher long-term average returns than fixed-income indexes.
- **Small-Cap Premium:** small-capitalization stocks have periodically outperformed large-capitalization stocks (size premium), though this premium varies over time and can occasionally be negative.
- **Corporate vs. Government Debt:** corporate bonds offer higher coupon yields on paper, but government bonds can occasionally match or exceed corporate bond performance over specific horizons due to the impact of default losses.

Let's review the relative outperformance calculations using geometric average annual returns from the 1979–2023 historical record:

- Large-cap equity (12.26%) versus corporate debt (7.43%):
- Small-cap equity (11.23%) versus large-cap equity (12.26%):
- US Treasury bonds (7.62%) versus corporate bonds (7.43%):

2.2.2 Long-Term Asset Return Components

The relative contribution of capital appreciation and capital distributions has evolved significantly:

- **Equities:** dividend yields have trended downward over time, particularly following a 1982 regulatory change that established a safe harbor for share buybacks. Companies increasingly repurchase shares to return cash to shareholders, which reduces share count, increases stock value, and provides tax advantages (capital gains tax is generally lower than dividend tax).
- **Government Bonds:** income return (coupon payments) remains the dominant driver of total returns, historically accounting for approximately 85% of total government bond returns. Capital appreciation plays a minor role, and bond prices suffer capital losses during periods of rising interest rates (such as the early 2020s).

Mua lại cổ phiếu (Share Buybacks) được tính vào Price Return thay vì Capital Distribution. Điều này giải thích lợi nhuận tăng giá chiếm tỷ trọng ngày càng lớn ở các nước phát triển. Chúng ta có nghĩ đây là xu thế tất yếu nhằm tối ưu hóa thuế cho cổ đông?

2.3 Risks and Returns

Chúng ta cùng rà soát định nghĩa, cách tính suất sinh lời danh mục, phân loại rủi ro (hệ thống, phi hệ thống) và nguyên lý đánh đổi rủi ro - lợi nhuận (Risk-Return Trade-Off).

Risk represents the uncertainty of investment returns and can be categorized into macro and micro risks:

- **Systematic Risk (Macro Risks):** broad, economy-wide risks that affect all financial assets (e.g., inflation risk, market interest rate risk, and market risk). Systematic risk cannot be diversified away.
- **Unsystematic Risk (Micro Risks):** risks specific to a particular issuer or sector (e.g., credit/default risk, business risk, financial leverage, sector/industry risk). Unsystematic risk can be mitigated through diversification.

2.3.1 The Risk–Return Trade-Off

Investors demand a higher expected return (risk premium) to compensate for taking on higher volatility and uncertainty. Exhibit 2.1 shows that historically, higher average annual returns are generally associated with a wider range of annual returns (higher volatility).

2.3.2 Portfolio Return Calculation

The expected return of a portfolio is the weighted average of the returns of the individual assets in the portfolio:

$$r_p = \sum_{i=1}^{N} w_i r_i = w_1 r_1 + w_2 r_2 + \cdots + w_N r_N \tag{2.7}$$

Where:

w_i is the weight of asset i in the portfolio (where $\sum_{i=1}^{N} w_i = 1$).

r_i is the rate of return of asset i.

Let's review the asset allocations and long-term returns for three different investors (Investor A: Aggressive, Investor B: Balanced, and Investor C: Conservative):

Table 2.3: Asset Class Returns and Allocation Summary

Asset Class	Total Return	Capital Appreciation	Capital Distribution
Large-cap Equities	12.00%	8.50%	3.50%
Long-term Gov Bonds	5.75%	0.75%	5.00%
30-Day T-Bills (Cash)	3.00%	0.00%	3.00%

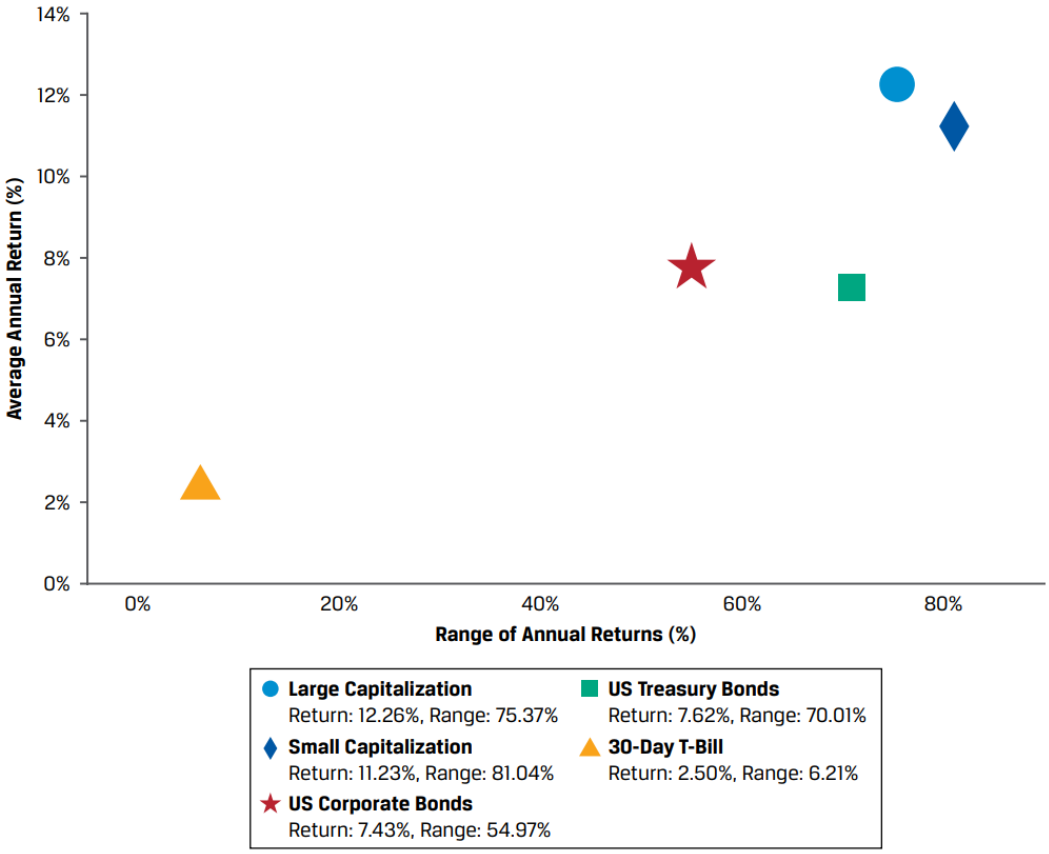


Figure 2.1: Historical Geometric Average Return for Different Assets (1979–2023)

Table 2.4: Investor Asset Allocations and Portfolio Returns

Investor	Equities Weight	Bonds Weight	T-Bills Weight	Total Return	Capital Apprec.	Capital Distrib.
A	80%	10%	10%	10.48%	6.88%	3.60%
B	60%	35%	5%	9.36%	5.36%	4.00%
C	50%	45%	5%	8.74%	4.59%	4.15%

Example: Detailed Portfolio Return Calculations (Investor A)

Using the asset allocations and returns from Table 2.3 and Table 2.4:

- **Asset allocation for Investor A:** 80% Equities, 10% Long-term Gov Bonds, 10% T-Bills (Cash).
- **Total portfolio return calculation:**

$$r_p = 80\%(12.00\%) + 10\%(5.75\%) + 10\%(3.00\%) = 10.48\%$$

- **Capital appreciation component:**

$$r_{\text{price}} = 80\%(8.50\%) + 10\%(0.75\%) + 10\%(0.00\%) = 6.88\%$$

- **Capital distribution component:**

$$r_{\text{dist}} = 80\%(3.50\%) + 10\%(5.00\%) + 10\%(3.00\%) = 3.60\%$$

Thực tế: Investor A tối đa hóa tăng trưởng giá (6.88%), còn Investor C ưu tiên dòng thu nhập đều đặn (4.15%). Chúng ta hãy thử tự giải cho Investor B và C để thực hành!

Module 3

Benchmarking Returns

LEARNING OUTCOMES	
a	calculate and compare money-weighted and time-weighted rates of return
b	describe the choices and the implications of the different weighting methods used in index construction and management, and calculate, interpret, and explain the value and the returns of an index

Money-Weighted and Time-Weighted Rates of Return

Index Definition and Calculation

Price Return and Total Return Indexes

Index Weightings

Index Management

Comparing Index Performance

3.1 Money-Weighted and Time-Weighted Rates of Return

a	calculate and compare money-weighted and time-weighted rates of return

Chúng ta cần phân biệt Suất sinh lời theo thời gian (TWR) và theo dòng tiền (MWR) để đánh giá chính xác hiệu quả quản lý quỹ lẫn đầu tư thực tế cá nhân.

3.1.1 Time-Weighted Rate of Return (TWR)

The time-weighted rate of return measures the compounded rate of growth of an investment portfolio over a specified horizon. To eliminate the distorting effects of external cash flows (which the investment manager typically does not control), the total investment horizon is broken down into subperiods at each cash flow event. The returns for each subperiod are calculated and then geometrically linked:

$$r_{\text{TW}} = \prod_{t=1}^{T}(1 + r_t) - 1 \tag{3.1}$$

Where:

r_{TW} is the time-weighted rate of return.

T is the total number of subperiods.

r_t is the holding period return for subperiod t, calculated between two successive cash flow events.

Because TWR measures performance between periods and filters out investor-driven contributions or withdrawals, it is the standard metric used to evaluate the investment skill of public market fund managers.

3.1.2 Money-Weighted Rate of Return (MWR)

The money-weighted rate of return (or dollar-weighted rate of return) measures the internal rate of return (IRR) of the investment. It accounts for both the size and timing of all cash flows. The MWR is the discount rate that sets the net present value of all cash flows equal to zero:

$$0 = \sum_{t=0}^{T} \frac{\mathrm{CF}_t}{(1 + r_{\mathrm{MW}})^t} \tag{3.2}$$

Where:

r_{MW} is the money-weighted rate of return.

CF_t is the net cash flow at time t.

Quy ước dấu dòng tiền: - Tiền đầu tư ban đầu (CF_0) và góp thêm mang dấu âm (−). - Cổ tức, rút vốn hoặc giá trị danh mục cuối kỳ mang dấu dương (+). - Để tính MWR (giải phương trình bậc cao), chúng ta cần dùng máy tính tài chính hoặc Excel.

3.1.3 Case Study: The Albright Fund

Case Study: The Albright Fund

The Albright Fund has a beginning asset value of EUR 100 million on 1 January. We evaluate its performance over a year with three 4-month subperiods:

- **1 January ($t = 0$):** Initial portfolio value is EUR 100 million.
- **30 April ($t = 1$):** Fund value before contribution is EUR 115 million. An additional EUR 20 million contribution is received (total value EUR 135 million). Received EUR 2 million in dividends (reinvested) and realized EUR 10 million in capital gains (reinvested).
- **31 August ($t = 2$):** Received EUR 3 million in dividends (reinvested). Total fund value is EUR 130 million.
- **31 December ($t = 3$):** Received EUR 3 million in dividends (not reinvested). Ending fund value is EUR 140 million (including the dividend).

Dòng tiền Albright Fund: - $t = 0$: $CF_0 = -100$ triệu EUR. - $t = 1$: $CF_1 = -30$ triệu EUR (gồm -20 triệu góp thêm, -10 triệu tái đầu tư lãi vốn; cổ tức 2 triệu tái đầu tư ngay nên triệt tiêu). - $t = 2$: $CF_2 = 0$ (tái đầu tư cổ tức). - $t = 3$: $CF_3 = +140$ triệu EUR (gồm giá trị cuối 137 triệu và 3 triệu cổ tức rút ra).

Exam Tip

Texas Instruments BA II Plus Keystrokes for MWR:

1. Press [**CF**] → [**2nd**] → [**CLR CF**] to clear worksheet memory.
2. Enter $\mathrm{CF}_0 = -100$: Press **100** → [+/-] → [**ENTER**].
3. Press [↓] to display **C01**.
4. Enter $\mathrm{CF}_1 = -30$: Press **30** → [+/-] → [**ENTER**].
5. Press [↓] → [↓] to display **C02** (F01 defaults to 1).
6. Enter $\mathrm{CF}_2 = 0$: Press **0** → [**ENTER**].
7. Press [↓] → [↓] to display **C03** (F02 defaults to 1).
8. Enter $\mathrm{CF}_3 = 140$: Press **140** → [**ENTER**].
9. Press [**IRR**] then [**CPT**].

Periodic 4-month IRR is $r_{\mathrm{MW,\ period}} = 2.71\%$. Annualized MWR:

$$r_{\mathrm{MW,\ annual}} = (1 + 0.027126)^3 - 1 = 8.35\%$$

Time-Weighted Rate of Return (TWR) Calculation:

1. **Subperiod 1 (1 Jan to 30 Apr):** $r_1 = \frac{115-100}{100} = 15.00\%$
2. **Subperiod 2 (1 May to 31 Aug):** $r_2 = \frac{130-135}{135} = -3.70\%$
3. **Subperiod 3 (1 Sep to 31 Dec):** $r_3 = \frac{140-130}{130} = 7.69\%$

Geometrically linking the returns:

$$r_{\text{TW}} = (1 + 0.1500)(1 - 0.0370)(1 + 0.0769) - 1 = 19.26\%$$

So sánh: $r_{TW} = 19.26\%$, $r_{MW} = 8.35\%$. MWR thấp vì quỹ thêm EUR 20 triệu ngay trước kỳ âm (−3.70%). Vốn lớn chịu lỗ làm giảm mạnh MWR, trong khi TWR phản ánh khách quan hiệu quả đầu tư.

3.1.4 Summary of Differences

Table 3.1 summarizes the key comparisons between the two return methodologies.

Table 3.1: Comparison of Money-Weighted and Time-Weighted Returns

Dimension	Money-Weighted Return (MWR)	Time-Weighted Return (TWR)
Core Objective	Measures the actual growth rate of the investor's funds.	Measures the compound growth rate of the portfolio assets.
Cash Flow Sensitivity	Highly sensitive to the size and timing of external cash flows.	Completely unaffected by external cash flows.
Calculation Base	Internal Rate of Return (IRR) equation.	Geometric linkage of subperiod returns.
Performance Bias	Can misrepresent manager skill if they cannot control cash flow timings.	Provides an objective measure of the manager's investment selection skill.
Primary Suitability	Recommended for private equity (where managers call capital) and individual investors.	Mandated by GIPS for public market mutual funds and active asset managers.

3.2 Index Definition and Calculation

b | describe the choices and the implications of the different weighting methods used in index construction and management, and calculate, interpret, and explain the value and the returns of an index

Market Index làm Benchmark và cơ sở cho ETF. Khi thiết kế chỉ số, chúng ta lưu ý: (1) Cách tính (Price Return/Total Return) và (2) Phương pháp phân bổ tỷ trọng (Weighting Scheme).

3.2.1 Price Return vs. Total Return Indexes

- **Price Return Index (r_{PR}):** reflects only the prices of the constituent securities and measures capital appreciation:

$$r_{\text{PR}} = \frac{P_{t+1}}{P_t} - 1 \tag{3.3}$$

- **Total Return Index (r_{TR}):** reflects both the prices and the reinvestment of all capital distributions (dividends/interest):

$$r_{\text{TR}} = \frac{P_{t+1} + \text{Div}_{t+1}}{P_t} - 1 \tag{3.4}$$

Over time, the total return index will significantly outperform the price return index due to the compounding effect of reinvested cash distributions.

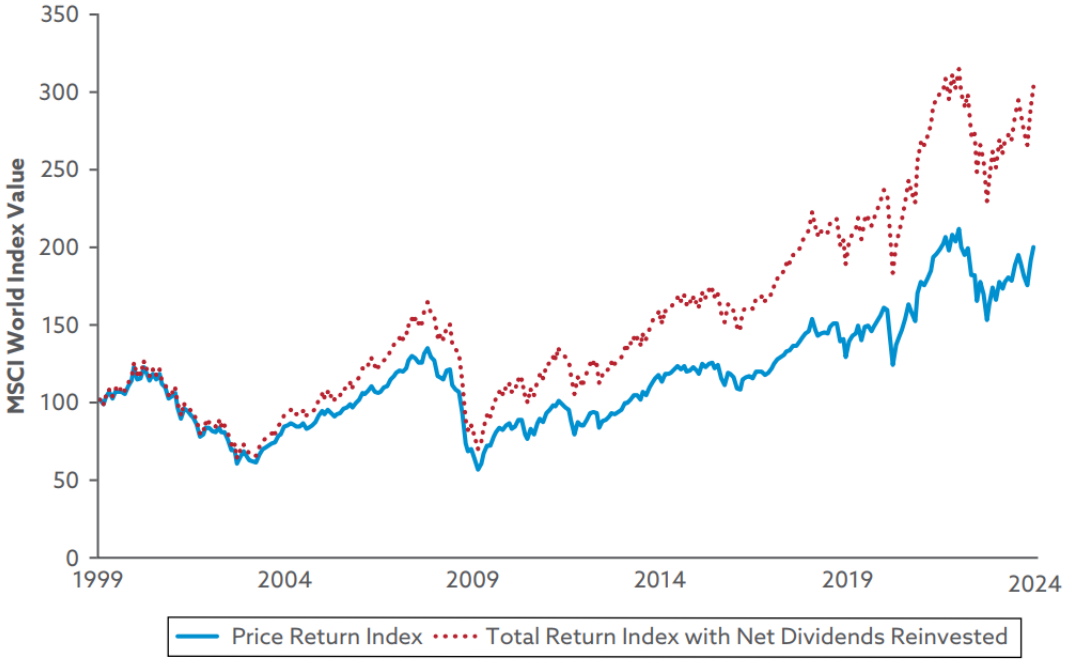


Figure 3.1: MSCI World Index Price Return vs. Total Return (1999–2023)

3.2.2 Index Weighting Methodologies

The general formula for calculating an index value at time t is:

$$\text{Index Value}_t = \sum_{i=1}^{N} w_{i,t} \times P_{i,t} \tag{3.5}$$

Where:

$w_{i,t}$ is the weight of constituent i at time t.

$P_{i,t}$ is the price of constituent i at time t.

Let's review the five primary weighting methodologies:

1. **Market Capitalization Weighting:** Weights are proportional to the total market capitalization ($Q_i P_i$):

$$w_i^M = \frac{Q_i P_i}{\sum_{j=1}^{N} Q_j P_j} \tag{3.6}$$

The index value is often calculated using a divisor (D_t) to normalize the value:

$$V_t = \frac{\sum_{i=1}^{N} Q_{i,t} P_{i,t}}{D_t} \tag{3.7}$$

2. **Float-Adjusted Market Cap Weighting:** Accounts only for shares available to the public ($f_i Q_i$), excluding strategic holdings:

$$w_i^{M,\text{Float}} = \frac{f_i Q_i P_i}{\sum_{j=1}^{N} f_j Q_j P_j} \tag{3.8}$$

3. **Fundamental Weighting:** Weights are based on economic or accounting indicators (e.g., book value, revenues, earnings, or dividends) instead of market prices:

$$w_i^F = \frac{F_i}{\sum_{j=1}^{N} F_j} \tag{3.9}$$

4. **Price Weighting:** Weights are proportional to the security price per share:

$$w_i^P = \frac{P_i}{\sum_{j=1}^{N} P_j} \tag{3.10}$$

The index value is calculated by dividing the sum of constituent prices by a divisor (D_t):

$$V_{\text{PRI},t} = \frac{\sum_{i=1}^{N} P_{i,t}}{D_t} \tag{3.11}$$

5. **Equal Weighting:** Each security is assigned an equal share at the inception/rebalancing date:

$$w_i^E = \frac{1}{N} \tag{3.12}$$

The index value is:

$$V_{\text{EQI},t} = \frac{\sum_{i=1}^{N} n_i P_{i,t}}{D_t} \tag{3.13}$$

3.2.3 Summary of Weighting Methodologies

Table 3.2 compares the characteristics of these methodologies.

Table 3.2: Comparison of Index-Weighting Methodologies

Index Type	Weighting Basis	Examples	Pros	Cons
Market Cap	Proportional to market capitalization.	S&P 500, MSCI World, S&P BSE SENSEX	Reflects relative sizes; low turnover; automatic rebalancing.	Concentrated in top firms; overweights overpriced stocks.
Equal Weighted	Identical weight ($1/N$).	S&P 500 Equal Weight, MSCI USA Equal Weight	Reduces concentration risk; enhances exposure to smaller firms.	Requires frequent rebalancing; high turnover and transaction costs.
Fundamental	Economic metrics (sales, dividends, book value).	FTSE RAFI US 1000	Focuses on firm size over market sentiment; value bias.	Sector bias; excludes firms with zero earnings or dividends.
Price Weighted	Share price (P_i).	Dow Jones Industrial Average (DJIA), Nikkei 225	Simple to calculate and interpret.	High price stocks dominate index; stock splits distort weights.

Tự điều chỉnh (Self-Rebalancing) của chỉ số Vốn hóa: - Ưu điểm: Tỷ trọng tự động điều chỉnh theo vốn hóa, luân chuyển danh mục rất thấp. - Nhược điểm: Bị ảnh hưởng bởi đà tăng (momentum bias), ép mua nhiều cổ phiếu định giá quá cao.

3.2.4 Index Management: Rebalancing and Reconstitution

- **Periodic Rebalancing:** adjusts the weights of the constituents back to their target levels. This is critical for equal-weighted and fundamentally weighted indexes because price variations cause weights to drift. Equal-weighted indexes require counter-trend trading (selling winners and buying losers), which increases transaction costs and portfolio turnover.
- **Reconstitution:** the periodic process of adding or removing constituent securities from the index based on eligibility rules. Widely followed indexes often apply a *banding strategy* (e.g., Russell 2000's 5% band) to reduce unnecessary index turnover at the breakpoints.

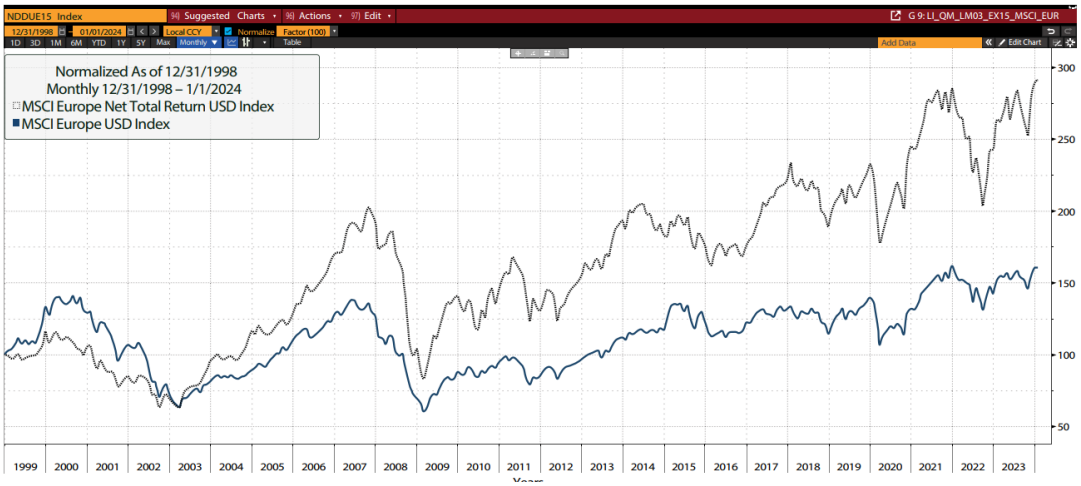


Figure 3.2: MSCI Europe Equal-Weighted vs. MSCI Europe Market Cap-Weighted (1999–2023)

3.2.5 Divisor Adjustment for Stock Splits

In a price-weighted index, when a constituent stock splits, its price per share drops mechanically without any change in its actual economic value or the firm's market capitalization. To prevent the index value from dropping artificially, the divisor (D) must be adjusted downward:

$$D_{\text{new}} = \frac{\sum P_{\text{new}}}{V_{\text{PRI, before}}} \tag{3.14}$$

Example: Divisor Adjustment for Stock Splits

Consider a price-weighted index with 5 stocks: A (50), B (40), C (30), D (25), E (5). Divisor $D_{\text{old}} = 1.5$.

- **Before Split:** The index value is:

$$V_{\text{PRI}} = \frac{50 + 40 + 30 + 25 + 5}{1.5} = 100$$

- **After 2-for-1 Split of A:** Stock A's price drops from 50 to 25. The new price sum is $25 + 40 + 30 + 25 + 5 = 125$.
- **Divisor Adjustment:** To keep the index value unchanged at 100:

$$D_{\text{new}} = \frac{125}{100} = 1.25$$

Khi cổ phiếu A chia tách giá giảm, Divisor phải giảm từ 1.5 xuống 1.25 để giữ chỉ số ổn định, gián tiếp làm tăng tỷ trọng ảnh hưởng của 4 cổ phiếu còn lại.

Module 4

The Time Value of Money in Finance

LEARNING OUTCOMES	
a	calculate and interpret the present value of fixed-income and equity instruments based on expected future cash flows
b	calculate and interpret the implied return of fixed-income instruments and required return and implied growth of equity instruments given their present value and cash flows
c	explain the cash flow additivity principle and its importance for the condition of no arbitrage, and explain its use in calculating implied forward interest rates, forward exchange rates, and option values

Time Value of Money in Fixed Income and Equity

Fixed-Income Instruments

Equity Instruments

Implied Return and Growth

Implied Return for Fixed-Income Instruments

Equity Instruments, Implied Return, and Implied Growth

Cash Flow Additivity

Implied Forward Rates Using Cash Flow Additivity

Forward Exchange Rates Using No Arbitrage

Option Pricing Using Cash Flow Additivity

Giá trị thời gian của tiền (TVM): Một đồng hôm nay giá trị hơn một đồng tương lai. Chúng ta sẽ áp dụng TVM cơ bản và liên tục để định giá công cụ nợ (Fixed-Income) và công cụ vốn (Equity).

4.1 Time Value of Money in Fixed Income and Equity

a | calculate and interpret the present value of fixed-income and equity instruments based on expected future cash flows

The relationship between a current or present value, PV_0, and future value, FV_t, of a cash flow—where r is the stated discount rate per period and t is the number of compounding periods—is:

$$FV_t = PV_0(1+r)^t \tag{4.1}$$

If the compounding frequency approaches infinity (continuous compounding), we calculate the future value as follows:

$$FV_t = PV_0 e^{r\times t} \tag{4.2}$$

Conversely, we can solve for the present value of a future cash flow under simple and continuous compounding:

$$PV_0 = FV_t(1+r)^{-t} = \frac{FV_t}{(1+r)^t} \tag{4.3}$$

$$PV_0 = FV_t e^{-r\times t} \tag{4.4}$$

Case Study: Annualizing Returns across Different Horizons

Annualized returns provide a standardized framework for comparing investments with different holding periods:

$$r_{\text{annualized}} = (1 + r_{\text{periodic}})^{\frac{365}{\text{Days}}} - 1 \tag{4.5}$$

For three securities in a portfolio held for different durations:

- **Security A:** Earned 6.2% over 100 days. Annualized return:

$$r_{\text{annual}} = (1 + 0.062)^{365/100} - 1 \approx 24.55\%$$

- **Security B:** Earned 2.0% over four weeks (28 days). Annualized return:

$$r_{\text{annual}} = (1 + 0.02)^{52/4} - 1 \approx 29.36\%$$

- **Security C:** Earned 5.0% over three months (90 days). Annualized return:

$$r_{\text{annual}} = (1 + 0.05)^{365/90} - 1 \approx 21.55\%$$

Quy đổi lợi nhuận về cùng năm (annualization) giúp chúng ta so sánh các kỳ hạn khách quan. Ở đây, Security B tuy suất sinh lời thấp nhất (2.0%) nhưng hiệu quả nhất năm (29.36%) nhờ nắm giữ ngắn (4 tuần).

4.1.1 Fixed-Income Instruments and the Time Value of Money

Cash flows associated with fixed-income instruments (debt) usually follow one of three general patterns:

1. **Discount (Zero-Coupon) Bonds:** Single payment representing both interest and principal repayment at maturity.
2. **Coupon Bonds:** Regular periodic interest coupon payments with the principal repaid at maturity.
3. **Annuities / Fully Amortizing Loans:** Equal periodic payments containing both interest and principal over the life of the instrument.

Discount Instruments

The cash flow pattern of a discount bond is illustrated in Exhibit 4.1.

The present value calculation for a discount bond with principal FV_t repaid fully after t periods with a periodic market discount rate of r per period is:

$$PV_{\text{Discount}} = \frac{FV_t}{(1 + r)^t} \tag{4.6}$$

Case Study: Discount Bonds under Positive and Negative Interest Rates

1. **Positive Rates (India STRIPS):** A 20-year stripped government bond with $FV = \text{INR } 100$ and required return $r = 6.70\%$:

$$PV_0 = \frac{\text{INR } 100}{(1 + 0.067)^{20}} \approx \text{INR } 27.33$$

If rates remain unchanged, after three years ($t = 17$ remaining):

$$PV_3 = \frac{\text{INR } 100}{(1 + 0.067)^{17}} \approx \text{INR } 33.21$$

If rates immediately rise to 7.70% at $t = 0$:

$$PV_0 = \frac{\text{INR } 100}{(1 + 0.077)^{20}} \approx \text{INR } 22.68 \quad \text{(reflects the inverse price-yield relationship)}$$

2. **Negative Rates (German Bund):** A 10-year sovereign bond with $FV = \text{EUR } 100$ issued at a negative yield $r = -0.05\%$:

$$PV_0 = \frac{\text{EUR } 100}{(1 - 0.0005)^{10}} \approx \text{EUR } 100.50$$

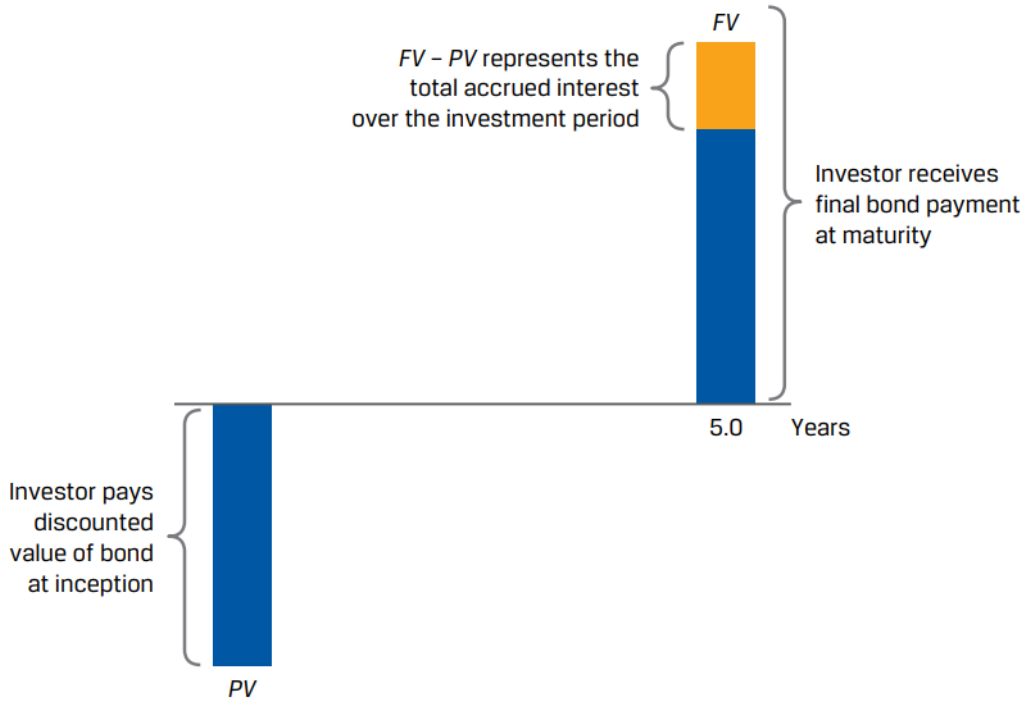


Figure 4.1: Discount Bond Cash Flows

Khi lãi suất dương, giá trái phiếu chiết khấu tăng dần (accretes) về mệnh giá. Khi lãi suất âm, nhà đầu tư trả phí giữ vốn an toàn, khiến giá trái phiếu giảm dần (amortizes) về mệnh giá tại ngày đáo hạn.

Coupon Instruments

The cash flow pattern of a coupon bond is illustrated in Exhibit 4.2.

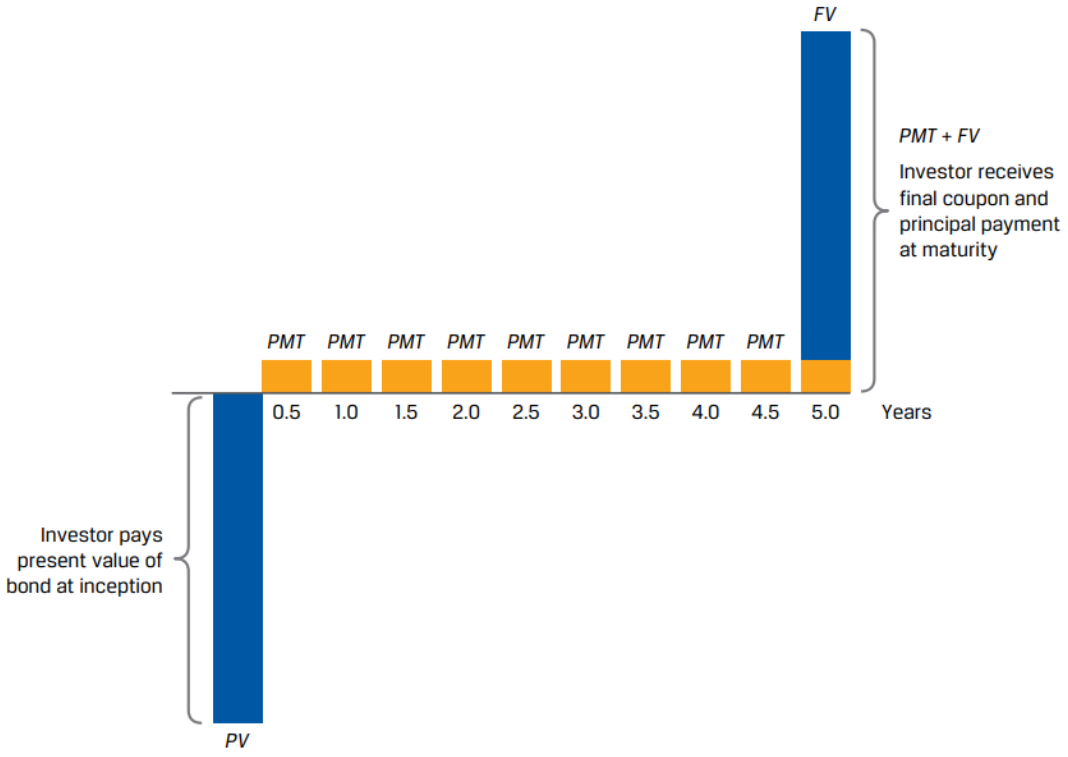


Figure 4.2: Coupon Bond Cash Flows

The price of a coupon bond is the sum of the present values of all periodic coupon payments and the present

value of the final principal repayment:

$$PV_{\text{Coupon}} = \sum_{t=1}^{T} \frac{PMT}{(1+r)^t} + \frac{FV}{(1+r)^T} \tag{4.7}$$

Using the sum of a geometric series, this simplifies to:

$$PV_{\text{Coupon}} = \frac{PMT}{r}\left[1 - \frac{1}{(1+r)^T}\right] + \frac{FV}{(1+r)^T} \tag{4.8}$$

Case Study: Coupon Pricing and Compounding Frequencies

1. **Annual Coupons (Greece):** A 7-year bond with a 2% annual coupon (FV = EUR 100) and $r = 2.00\%$ yield:

$$PV = \frac{\text{EUR } 2}{0.02}\left[1 - \frac{1}{(1.02)^7}\right] + \frac{\text{EUR } 100}{(1.02)^7} = \text{EUR } 100 \quad \text{(Par Bond)}$$

   - Coupon Rate = Yield $\implies$ $PV = FV$ (Par Bond)
   - Coupon Rate $>$ Yield $\implies$ $PV > FV$ (Premium Bond)
   - Coupon Rate $<$ Yield $\implies$ $PV < FV$ (Discount Bond)

2. **Semiannual Coupons (India):** A 20-year bond with a 6.70% coupon (FV = INR 100) paid semiannually. Periodic coupon PMT = INR 3.35, periodic discount rate $r = 3.35\%$, total periods $T = 40$:

$$PV = \frac{3.35}{0.0335}\left[1 - \frac{1}{1.0335^{40}}\right] + \frac{100}{1.0335^{40}} = \text{INR } 100$$

   If the yield-to-maturity immediately rises to 7.70% (periodic rate $r = 3.85\%$):

$$PV = \frac{3.35}{0.0385}\left[1 - \frac{1}{1.0385^{40}}\right] + \frac{100}{1.0385^{40}} \approx \text{INR } 89.88$$

Khi định giá trái phiếu, tần suất ghép lãi rất quan trọng. Khoản INR 100 ở năm 20: ghép năm có PV = INR 27.33, ghép nửa năm (40 kỳ) có PV = INR 26.77. Ghép lãi thường xuyên hơn làm giảm PV.

Floating-Rate and Perpetual Bonds

- **Floating-Rate Bonds:** Adjust their coupon payments periodically based on a reference rate (e.g., NIBOR, SOFR) plus a fixed credit margin.
- **Perpetual Bonds:** Have no stated maturity date ($T \to \infty$), paying coupons indefinitely. The valuation simplifies to:

$$PV_{\text{Perpetual}} = \frac{PMT}{r} \tag{4.9}$$

Examples: Floating-Rate and Perpetual Valuations

- **Floating-Rate (Sogn og Fjordane Energi A/S):** Norwegian hydropower green bond with quarterly coupon. Coupon Rate = 3-month NIBOR + 73 bps margin. If NIBOR is 4.52%, the quarterly periodic rate is:

$$r_{\text{periodic}} = \frac{4.52\% + 0.73\%}{4} = 1.3125\%$$

  The periodic coupon payment for a NOK 1 million face value is:

$$PMT = \text{NOK } 1,000,000 \times 1.3125\% = \text{NOK } 13,125$$

- **Perpetual Bond (KB Financial):** Perpetual debt with a 3.30% coupon paid quarterly (0.825% periodic coupon). If the required return is 3.40% quarterly compounded (periodic rate 0.85%):

$$PV = \frac{\text{KRW } 8.25}{0.0085} \approx \text{KRW } 970.59 \quad \text{(per KRW 1,000 face value)}$$

Annuities and Fully Amortizing Loans

The cash flow pattern for fully amortizing loans (mortgages) is illustrated in Exhibit 4.3.

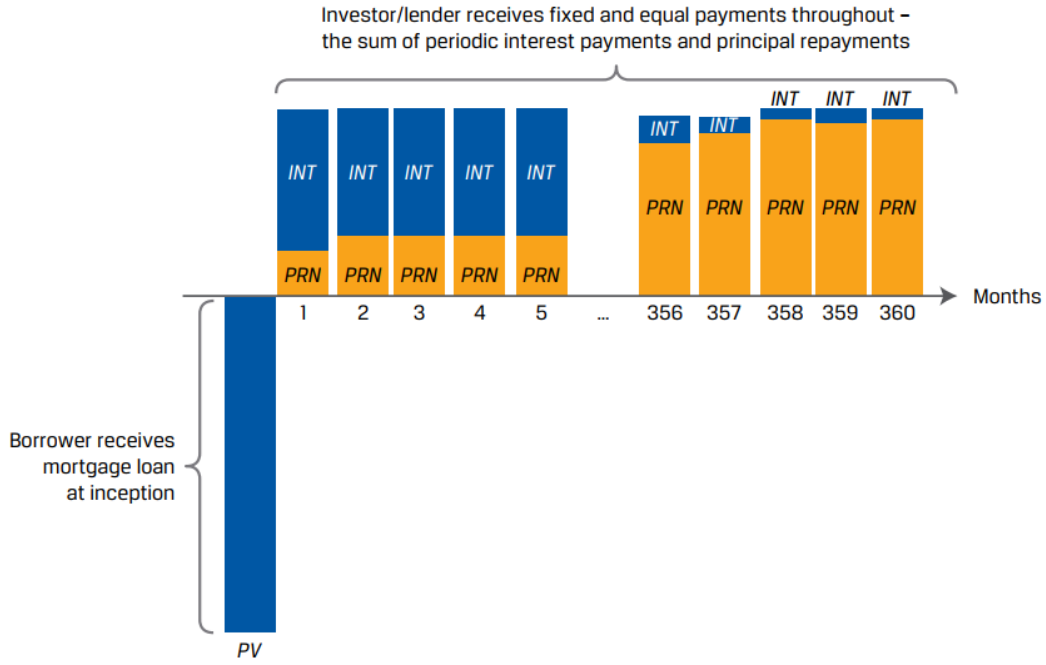


Figure 4.3: Mortgage Cash Flows

For level-payment instruments (like mortgages), the periodic level cash flow, PMT, containing both interest and principal repayment is:

$$PMT = \frac{r \times PV_0}{1 - (1+r)^{-t}} \tag{4.10}$$

Case Study: Mortgage Cash Flow Amortization

A 30-year residential mortgage is used to finance 80% of a USD $1,000,000$ building. Loan amount $PV_0 =$ USD $800,000$. Mortgage rate is 5.25% annual monthly compounded ($r = 0.4375\%$ per month, $t = 360$):

$$PMT = \frac{0.004375 \times \text{USD } 800,000}{1 - (1.004375)^{-360}} \approx \text{USD } 4,417.63 \text{ per month}$$

The amortization breakdown for the first two months is:

- **Month 1:**
  - Interest payment: USD $800,000 \times 0.4375\% =$ USD $3,500.00$
  - Principal repayment: USD $4,417.63 -$ USD $3,500.00 =$ USD 917.63
  - Remaining balance: USD $800,000 -$ USD $917.63 =$ USD $799,082.37$
- **Month 2:**
  - Interest payment: USD $799,082.37 \times 0.4375\% =$ USD $3,495.99$
  - Principal repayment: USD $4,417.63 -$ USD $3,495.99 =$ USD 921.64
  - Remaining balance: USD $799,082.37 -$ USD $921.64 =$ USD $798,160.73$

Dư nợ giảm dần: Dù tổng tiền trả hàng tháng cố định ở USD 4,417.63, cấu phần lãi giảm dần do dư nợ gốc giảm, và cấu phần gốc tăng dần.

4.2 Equity Instruments and the Time Value of Money

Equity shares have no maturity date and are assumed to remain outstanding indefinitely. Share valuation is based on discounting expected discretionary cash dividends.

4.2.1 Constant Dividends

The cash flow pattern for equity paying constant dividends is illustrated in Exhibit 4.4.

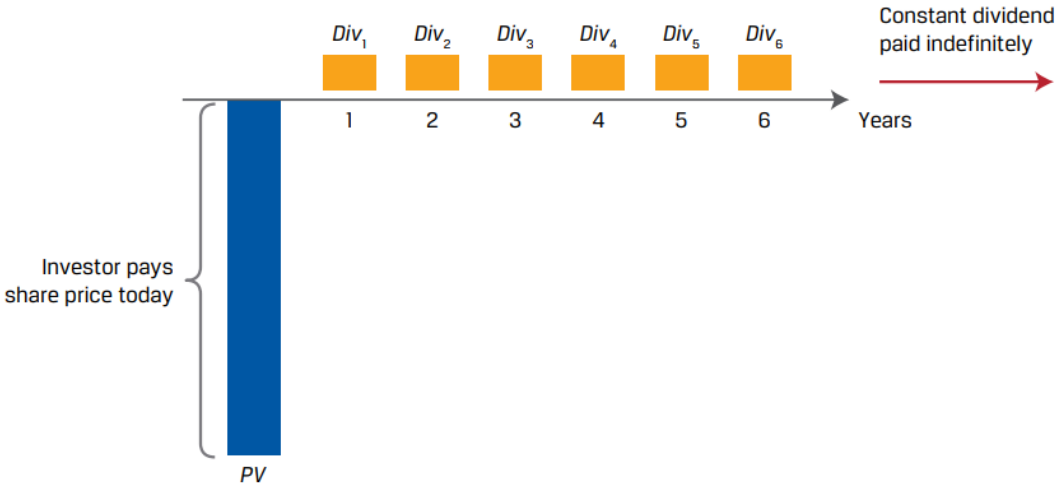


Figure 4.4: Equity Cash Flows with Constant Dividends

The valuation of equity paying constant dividends in perpetuity simplifies to:

$$PV_t = \sum_{i=1}^{\infty} \frac{Div_{t+1}}{(1+r)^i} = \frac{Div_{t+1}}{r} \tag{4.11}$$

Example: Shipline PLC Constant Dividend Valuation

Shipline PLC pays a constant annual dividend of GBP 1.50 indefinitely. With a required return $r = 15\%$:

$$PV_0 = \frac{\text{GBP } 1.50}{0.15} = \text{GBP } 10.00$$

4.2.2 Constant Dividend Growth (Gordon Growth Model)

The cash flow pattern for equity paying constant growth dividends is illustrated in Exhibit 4.5.

If dividends grow at a constant rate g in perpetuity:

$$PV_t = \frac{Div_t(1+g)}{r-g} = \frac{Div_{t+1}}{r-g} \tag{4.12}$$

where $r - g > 0$.

Example: Shipline PLC Constant Growth Valuation

If Shipline PLC's dividend grows at a constant $g = 6\%$ indefinitely, next year's dividend is:

$$Div_1 = \text{GBP } 1.50 \times (1 + 0.06) = \text{GBP } 1.59$$

The expected stock price is:

$$PV_0 = \frac{\text{GBP } 1.59}{0.15 - 0.06} \approx \text{GBP } 17.67$$

Chênh lệch $r - g$ ở mẫu số cực kỳ quan trọng. Khi chênh lệch này nhỏ (tốc độ tăng trưởng sát tỷ suất sinh lời yêu cầu), giá cổ phiếu cực kỳ nhạy cảm với bất kỳ thay đổi nhỏ nào của r hoặc g.

4.2.3 Two-Stage Dividend Growth Model

The cash flow pattern of a two-stage dividend growth model is illustrated in Exhibit 4.6.

For companies experiencing a rapid initial growth phase (g_s) for n periods, followed by a stable long-term growth rate (g_l):

$$PV_0 = \sum_{i=1}^{n} \frac{Div_0(1+g_s)^i}{(1+r)^i} + \frac{E[S_n]}{(1+r)^n} \tag{4.13}$$

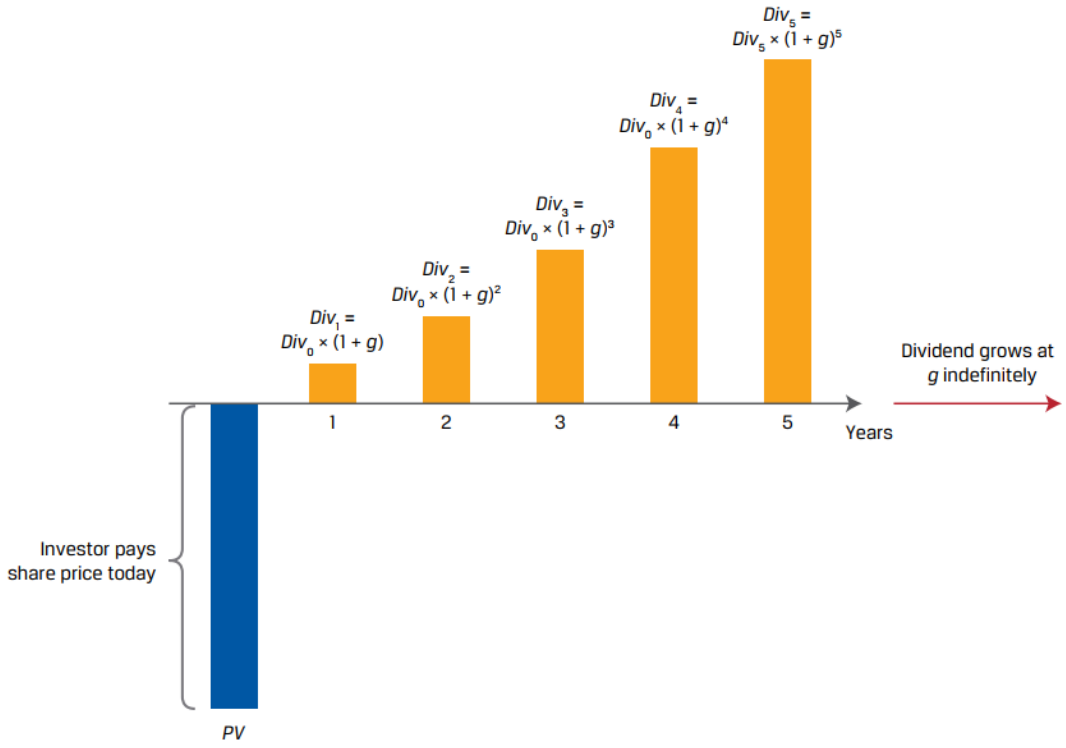


Figure 4.5: Equity Cash Flows with Constant Dividend Growth

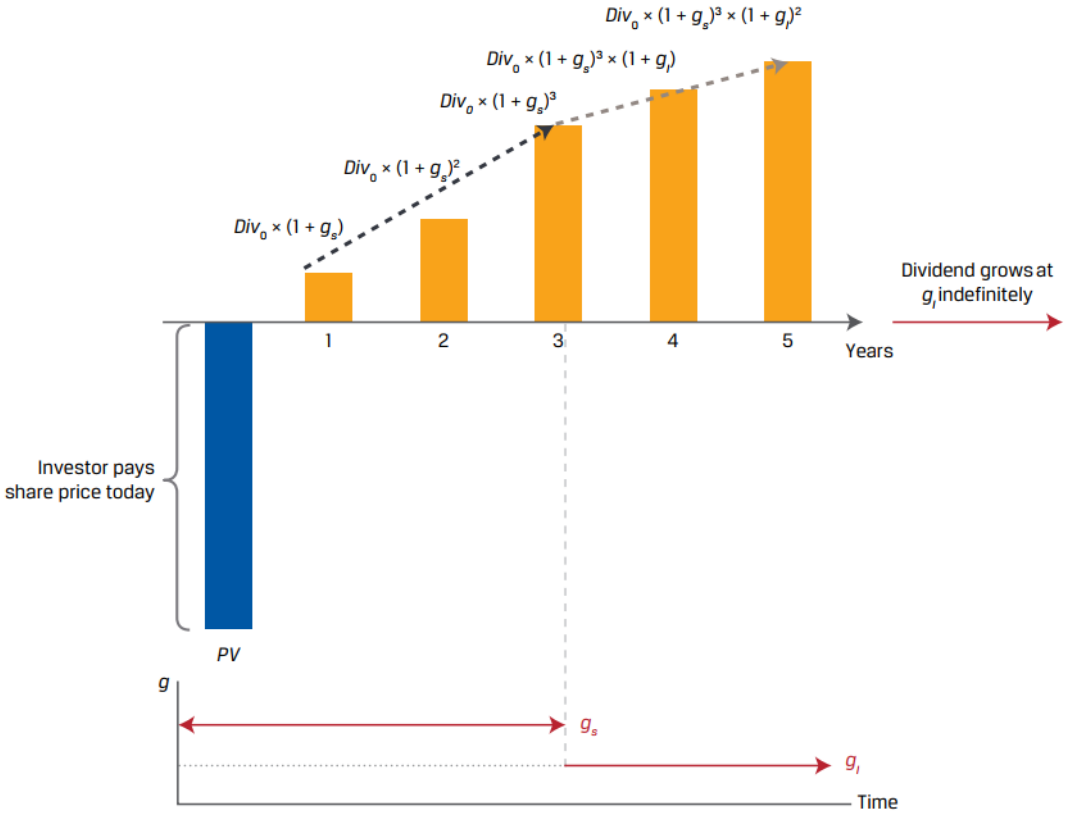


Figure 4.6: Equity Cash Flows with Two-Stage Dividend Growth

Where the expected terminal value at the end of the high-growth phase $E[S_n]$ is:

$$E[S_n] = \frac{Div_{n+1}}{r - g_l} \tag{4.14}$$

Example: Shipline PLC Two-Stage Dividend Growth Valuation

Assume Shipline grows its dividend by $g_s = 6\%$ for three years, followed by $g_l = 2\%$ constant growth. $r = 15\%$, $Div_0 = \text{GBP } 1.50$:

- **Step 1: PV of high-growth dividends (first 3 years):**

$$Div_1 = 1.5900 \implies PV(Div_1) = \frac{1.5900}{1.15} \approx 1.3826$$

$$Div_2 = 1.6854 \implies PV(Div_2) = \frac{1.6854}{1.15^2} \approx 1.2744$$

$$Div_3 = 1.7865 \implies PV(Div_3) = \frac{1.7865}{1.15^3} \approx 1.1747$$

$$\text{Sum of PV of high-growth phase} \approx 3.8317$$

- **Step 2: PV of terminal value at Year 3:**

$$Div_4 = Div_3 \times (1 + g_l) = 1.7865 \times 1.02 \approx \text{GBP } 1.8222$$

$$Terminal\ Value\ E[S_3] = \frac{1.8222}{0.15 - 0.02} \approx \text{GBP } 14.0169$$

$$PV\ of\ E[S_3] = \frac{14.0169}{1.15^3} \approx \text{GBP } 9.2163$$

- **Step 3: Total price:**

$$PV_0 = 3.8317 + 9.2163 = \text{GBP } 13.0480 \approx \text{GBP } 13.05$$

4.3 Implied Return and Growth

b	calculate and interpret the implied return of fixed-income instruments and required return and implied growth of equity instruments given their present value and cash flows

When market prices and expected cash flows are observed, analysts can calculate the implied rate of return and the implied growth rate to assess valuation.

4.3.1 Implied Return for Fixed-Income Instruments

The price-yield relationship of a discount bond and its implied return are illustrated in Exhibit 4.7.

For discount bonds, the implied return is:

$$r = \left(\frac{FV_t}{PV_0}\right)^{\frac{1}{t}} - 1 \tag{4.15}$$

Case Studies: Implied returns for zero-coupon and coupon bonds

1. **Zero-Coupon (German sovereign bond):** 10-year discount bond issued at EUR 100.50 per EUR 100 principal ($r = -0.05\%$). Six years later, it trades at EUR 95.72.
   - Annualized holding return for initial buyer (6 years):

$$r_{\text{hold}} = \left(\frac{95.72}{100.50}\right)^{1/6} - 1 \approx -0.81\%$$

   - Expected annualized return for buyer at Year 6 (4 years remaining):

$$r_{\text{buy}} = \left(\frac{100}{95.72}\right)^{1/4} - 1 \approx 1.10\%$$

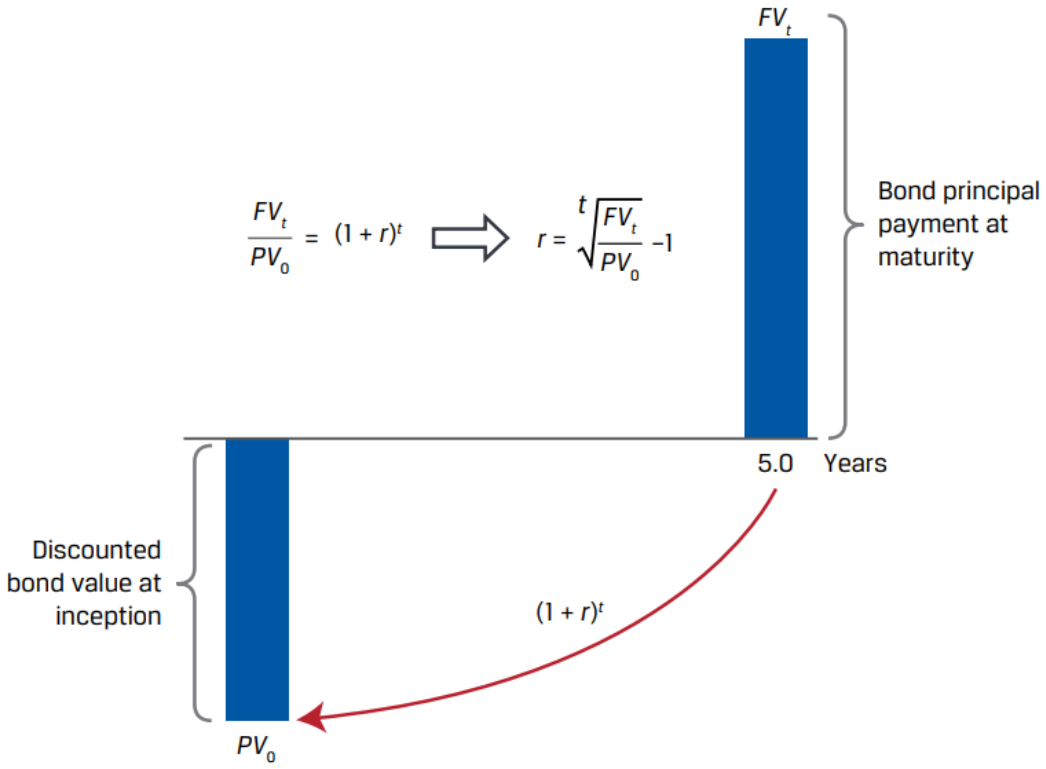


Figure 4.7: Discount Bond Implied Return

2. **Coupon-Paying (Greece 2% annual bond):** Issued at par (EUR 100, YTM 2%). Two years later, price drops to EUR 93.091.
   - Implied YTM for remaining 5 years: Solved iteratively using RATE function:

$$\text{Rate}(5, 2, -93.091, 100) \approx 3.5320\%$$

   - Two-year holding return if coupons are reinvested at 2.00%:

$$FV_2 = 2(1.02) + 2 + 93.091 = \text{EUR } 97.13 \implies r_{\text{hold}} = \left(\frac{97.13}{100}\right)^{1/2} - 1 \approx -1.445\%$$

Nhận xét: Ở ví dụ trái phiếu Hy Lạp, khoản lỗ vốn (từ EUR 100 xuống EUR 93.091) lớn hơn coupon tích lũy (EUR 4.04), khiến suất sinh lời thực tế bị âm trong 2 năm đầu dù coupon dương.

The relationship between a coupon bond's price and its implied return (YTM) is illustrated in Exhibit 4.8.

4.3.2 Equity Implied Return, Implied Growth, and Forward P/E Ratios

Rearranging the Gordon growth model, we can solve for implied return (r) and implied growth rate (g):

$$r = \frac{Div_{t+1}}{PV_0} + g \tag{4.16}$$

$$g = r - \frac{Div_{t+1}}{PV_0} \tag{4.17}$$

By dividing the valuation formula by expected earnings per share (E_{t+1}), we obtain the forward Price-to-Earnings (P/E) ratio:

$$\frac{PV_t}{E_{t+1}} = \frac{Div_{t+1}/E_{t+1}}{r - g} \tag{4.18}$$

where Div_{t+1}/E_{t+1} represents the expected dividend payout ratio.

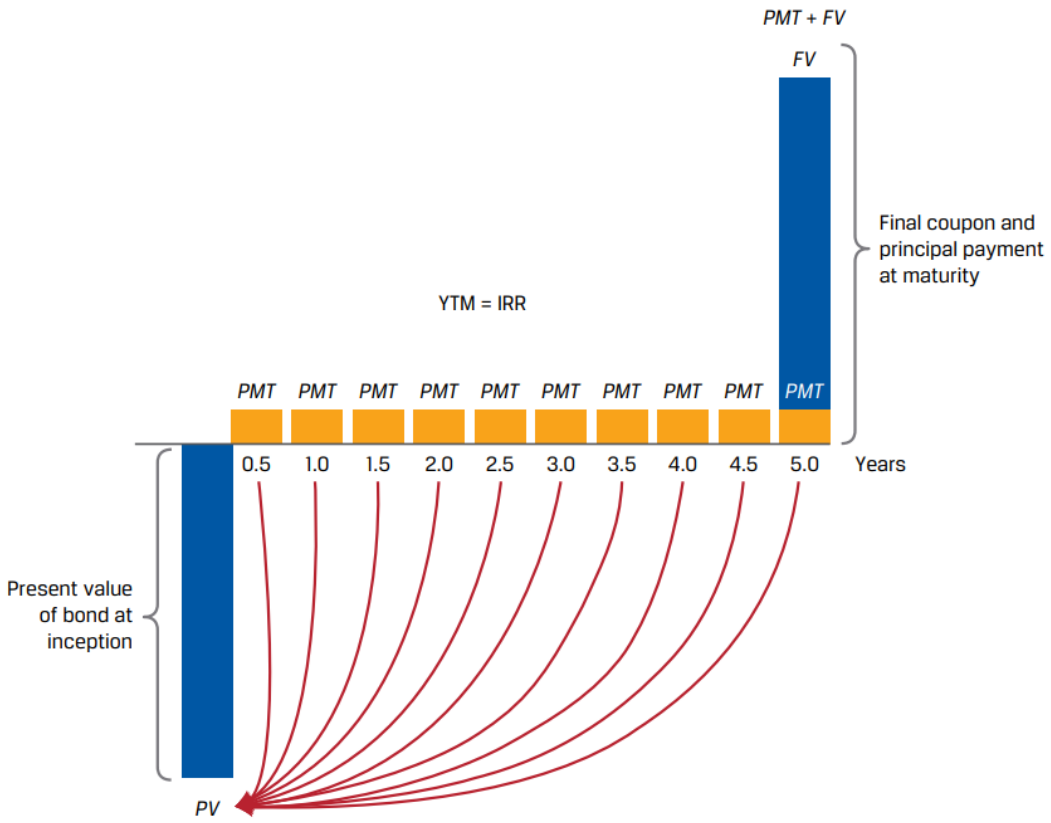


Figure 4.8: Coupon Bond Implied Return

Case Studies: Stock and index implied return and growth calculations

1. **Coca-Cola Stock:** Trades at USD 63.00, next year expected dividend is USD 1.76.
   - If $g = 4.00\%$ constant growth:
   $$r_{\text{implied}} = \frac{1.76}{63.00} + 0.04 \approx 6.79\%$$
   - If required return $r = 7.00\%$:
   $$g_{\text{implied}} = 0.07 - \frac{1.76}{63.00} \approx 4.21\%$$
2. **Coca-Cola Forward P/E:** Forward P/E is 28, payout ratio is 70%.
   - If required return $r = 9.00\%$ and growth $g = 4.50\%$:
   $$\text{Valuation P/E} = \frac{0.70}{0.09 - 0.045} \approx 15.56$$
   Since the market P/E (28) is much higher than the fundamental valuation (15.56), the stock is **overvalued** (recommend short position).
3. **Indian NIFTY 50 Index:** Forward P/E is 15, expected payout ratio is 50%, required return $r = 7.50\%$, expected growth $g = 4.50\%$:
   $$\text{Valuation P/E} = \frac{0.50}{0.075 - 0.045} \approx 16.67$$
   Since the market P/E (15) is lower than the valuation (16.67), the index is **undervalued** (recommend long position).

4.4 Cash Flow Additivity

c | explain the cash flow additivity principle and its importance for the condition of no arbitrage, and explain its use in calculating implied forward interest rates, forward exchange rates, and option values

The cash flow additivity principle states that the present value of any sum of future cash flows equals the sum of their individual present values. This principle is vital in maintaining the no-arbitrage condition (meaning it is impossible to make a riskless profit).

The comparison between two investment strategies under this principle is illustrated in Exhibit 4.9.

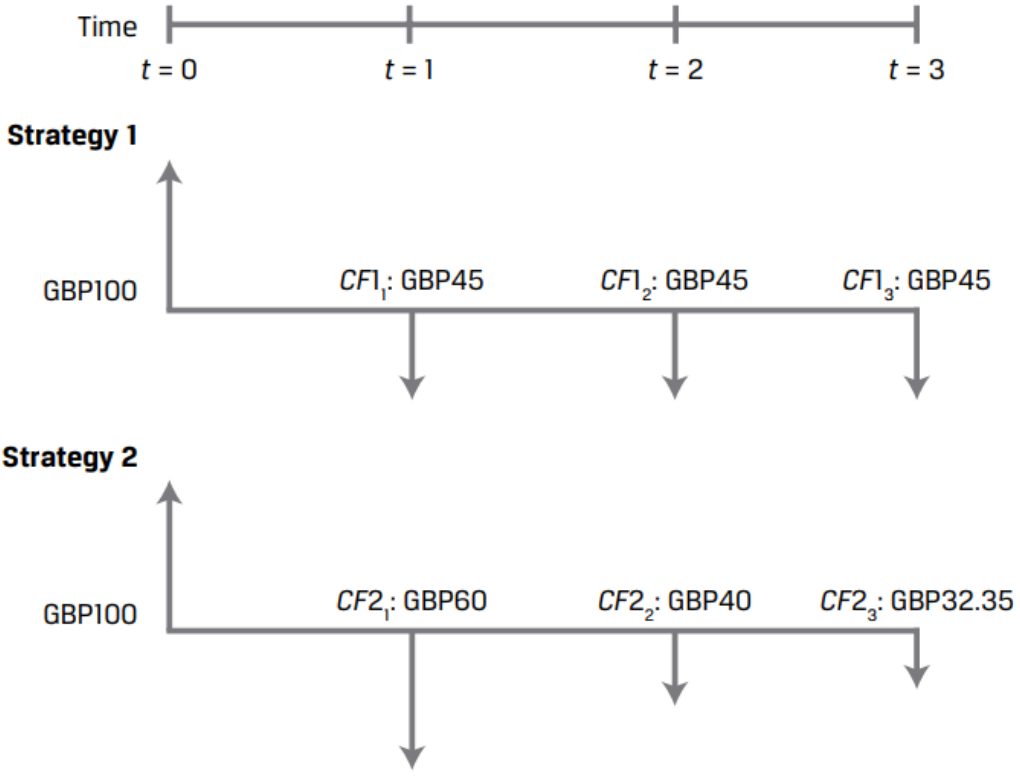


Figure 4.9: Two Investment Strategies

Case Study: Basic Cash Flow Additivity

Consider two strategies with an initial investment of GBP 100 and $r = 10\%$:

- **Strategy 1:** Cash flows are $-100, +45, +45, +45$.

$$PV_1 = -100 + \frac{45}{1.1} + \frac{45}{1.1^2} + \frac{45}{1.1^3} \approx \text{GBP } 11.91$$

- **Strategy 2:** Cash flows are $-100, +60, +40, +32.35$.

$$PV_2 = -100 + \frac{60}{1.1} + \frac{40}{1.1^2} + \frac{32.35}{1.1^3} \approx \text{GBP } 11.91$$

- **Incremental Cash Flow Diff:** Strategy 1 – Strategy 2 cash flows: $0, -15, +5, +12.65$.

$$PV_{\text{Diff}} = -\frac{15}{1.1} + \frac{5}{1.1^2} + \frac{12.65}{1.1^3} \approx 0$$

Tính cộng dồn dòng tiền (additivity): Hai danh mục tạo ra dòng tiền có PV tương đương phải giao dịch cùng giá để loại bỏ cơ hội arbitrage.

4.4.1 Implied Forward Interest Rates Using Cash Flow Additivity

The relationship between spot rates and implied forward rates is illustrated in Exhibit 4.10.

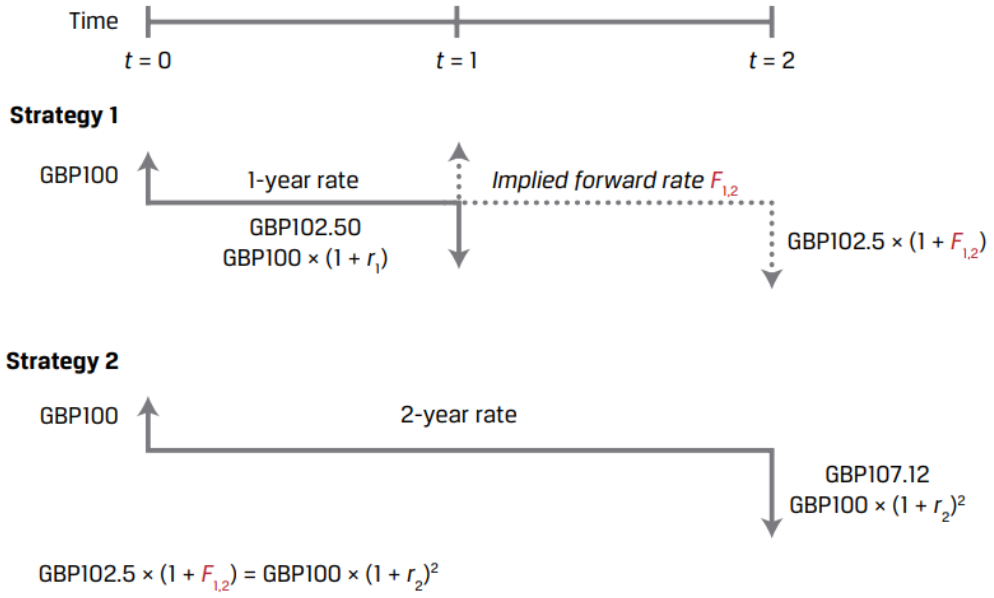


Figure 4.10: Implied Forward Rates

Under the no-arbitrage condition, investing for k periods must yield the same return as investing for j periods ($j < k$) and then reinvesting for the remaining $k - j$ periods at the forward rate:

$$(1 + S_k)^k = (1 + S_j)^j (1 + F_{j,k-j})^{k-j} \tag{4.19}$$

Case Study: Implied US Treasury forward rate calculations (STRIPS)

US Treasury STRIPS prices per USD 100 and implied spot rates:

Date	PV 1y	PV 2y	Spot S_1	Spot S_2
31 May 2022	98.028	95.109	2.012%	2.539%
15 Jun 2022	97.402	93.937	2.667%	3.177%

Using $(1 + S_2)^2 = (1 + S_1)(1 + F_{1,1})$:

- **May 31 Forward Rate:**
$$F_{1,1} = \frac{(1 + 0.02539)^2}{1 + 0.02012} - 1 \approx 3.069\%$$
- **June 15 Forward Rate:**
$$F_{1,1} = \frac{(1 + 0.03177)^2}{1 + 0.02667} - 1 \approx 3.689\%$$
- **Forward Rate Change:** $3.689\% - 3.069\% = 62$ bps.

The differences between spot and forward curves under rising rates are illustrated in Exhibit 4.11.

4.4.2 Forward Exchange Rates Using No Arbitrage

The no-arbitrage condition and the replication paths in the FX market are illustrated in Exhibit 4.12.

The forward exchange rate F (expressed as price/base) under continuous compounding interest rate parity is:

$$F = S \times e^{(r_{\text{price}} - r_{\text{base}})T} \tag{4.20}$$

Case Studies: Foreign Exchange forward rate calculations

1. **US/Japan (USD to JPY, 6 months):** Spot JPY/USD $= 134.40$. $r_{\text{JPY}} = 0.05\%$, $r_{\text{USD}} = 2.00\%$ (continuously compounded):
   - Strategy 1 (Domestic USD): USD $1,000 \times e^{0.02 \times 0.5} =$ USD $1,010.05$
   - Strategy 2 (Convert JPY): USD $1,000 \to$ JPY $134,400$. Grows to JPY $134,400 \times e^{0.0005 \times 0.5} \approx$ JPY $134,433.60$.

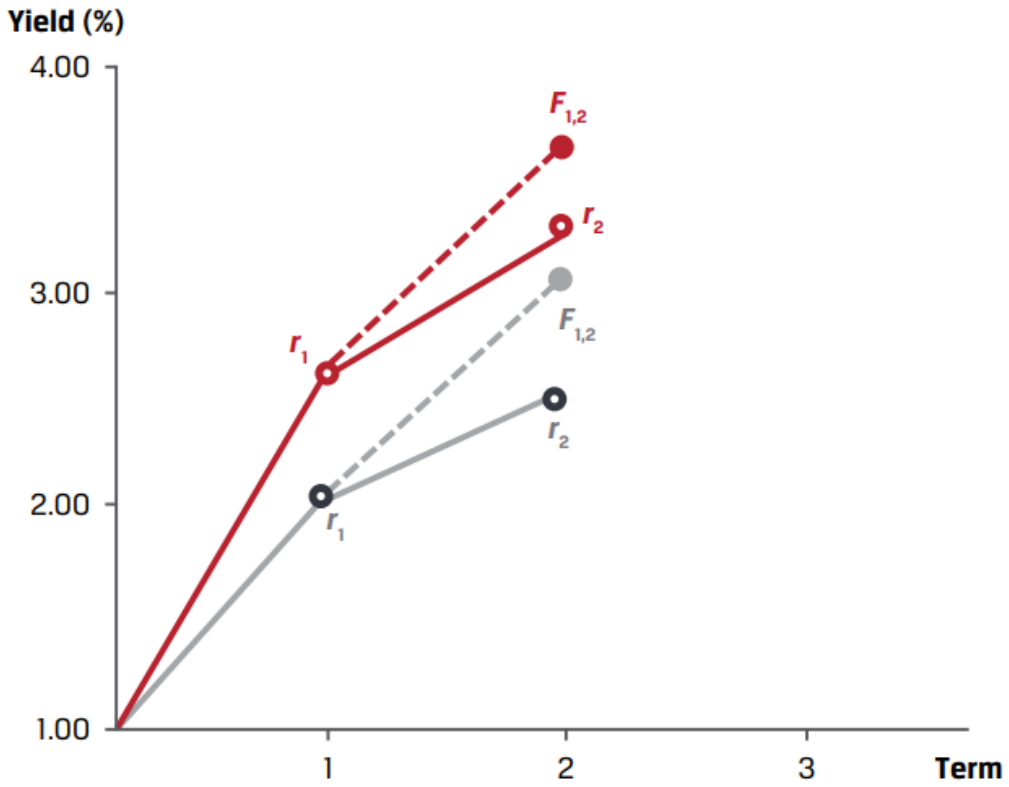


Figure 4.11: Forward Rate Differences

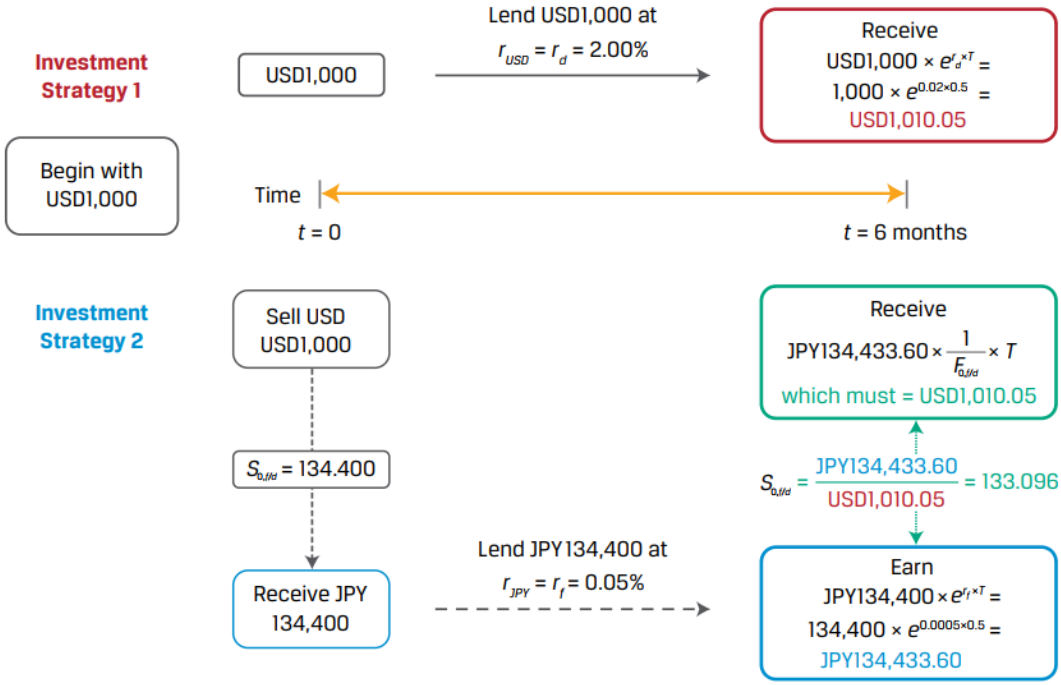


Figure 4.12: No-Arbitrage Condition in the Foreign Exchange Market

- Forward rate required:

$$F = \frac{\text{JPY } 134,433.60}{\text{USD } 1,010.05} \approx 133.096 \text{ JPY/USD}$$

2. **GBP/USD (Interest rate differential shift):** Spot GBP/USD = 1.2602 (USD per GBP):
   - **May 31:** $r_{\text{USD}} = 2.012\%$, $r_{\text{GBP}} = 1.291\% \implies F = 1.2602 \times e^{0.02012-0.01291} \approx 1.2693$
   - **June 15:** $r_{\text{USD}} = 2.667\%$, $r_{\text{GBP}} = 1.562\% \implies F = 1.2602 \times e^{0.02667-0.01562} \approx 1.2742$

Khi lãi suất Mỹ (r_{USD}) tăng mạnh hơn Anh (r_{GBP}), chênh lệch nới rộng khiến USD mất giá kỳ hạn (forward depreciation) so với GBP, tỷ giá kỳ hạn tăng từ 1.2693 lên 1.2742.

4.4.3 Option Pricing Using Cash Flow Additivity

The binomial tree model for the underlying asset's price is illustrated in Exhibit 4.13, and the contract value under the two scenarios is shown in Exhibit 4.14.

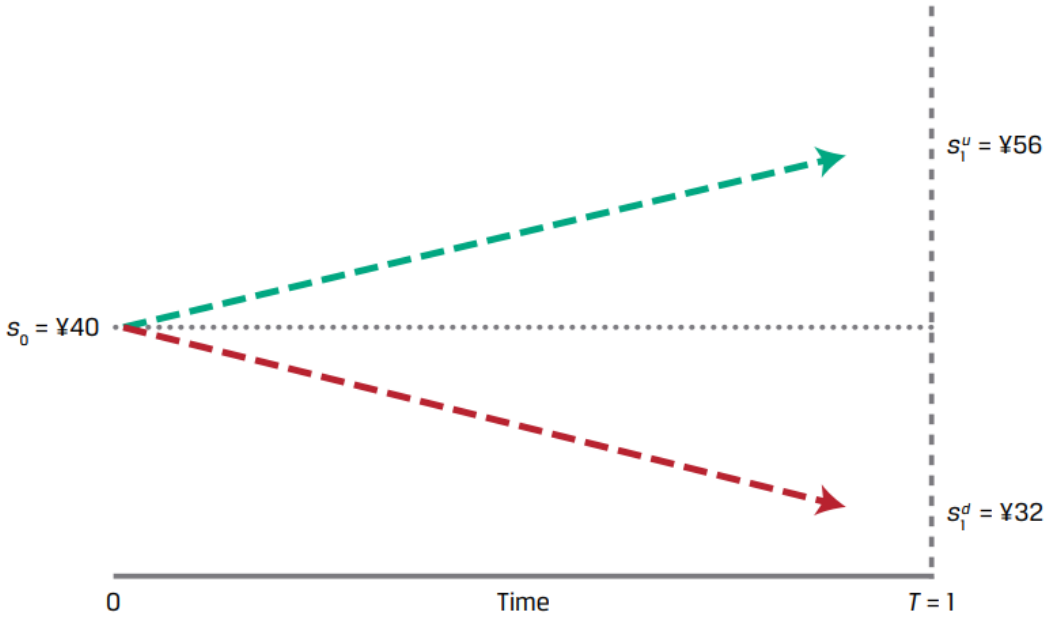


Figure 4.13: One-Period Binomial Tree for the Asset's Price

We can calculate the no-arbitrage price of derivative options using a replicating portfolio consisting of a position in the underlying asset (Δ shares) and risk-free lending/borrowing.

Case Study: Call and Put Option replication pricing

Stock price $S_0 = 40$. In one year, it can go up to $S_1^u = 56$ or down to $S_1^d = 32$. Risk-free rate is 5%. Option strike price $X = 50$.

1. **Call Option ($c_1^u = 6$, $c_1^d = 0$):** Replicate with Δ stock and short 1 call option.

$$\text{Hedge Ratio } \Delta = \frac{c_1^u - c_1^d}{S_1^u - S_1^d} = \frac{6-0}{56-32} = 0.25$$

   - Portfolio Value in up state: $0.25 \times 56 - 6 = 8$
   - Portfolio Value in down state: $0.25 \times 32 - 0 = 8$

   Since the portfolio is risk-free, its initial value must equal the discounted payoff of 8:

$$0.25(40) - c_0 = \frac{8}{1.05} \approx 7.62 \implies c_0 = 10 - 7.62 = \text{CNY } 2.38$$

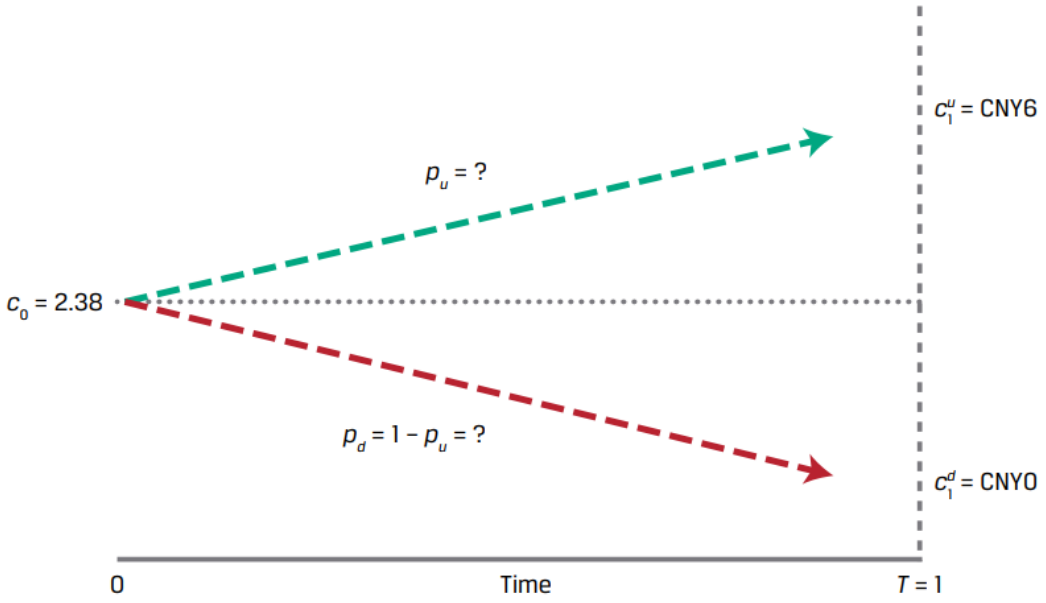


Figure 4.14: One-Period Binomial Tree for the Contract's Price

2. **Put Option ($p_1^u = 0$, $p_1^d = 18$):** Replicate with Δ stock and long 1 put option.

$$\text{Hedge Ratio } \Delta = \frac{p_1^d - p_1^u}{S_1^u - S_1^d} = \frac{18 - 0}{56 - 32} = 0.75$$

- Portfolio Value in up state: $0.75 \times 56 + 0 = 42$
- Portfolio Value in down state: $0.75 \times 32 + 18 = 42$

Setting initial value equal to discounted payoff of 42:

$$0.75(40) + p_0 = \frac{42}{1.05} = 40 \implies p_0 = 40 - 30 = \text{CNY } 10$$

The replication payoffs for the call option under the up and down scenarios are shown in Exhibit 4.15.

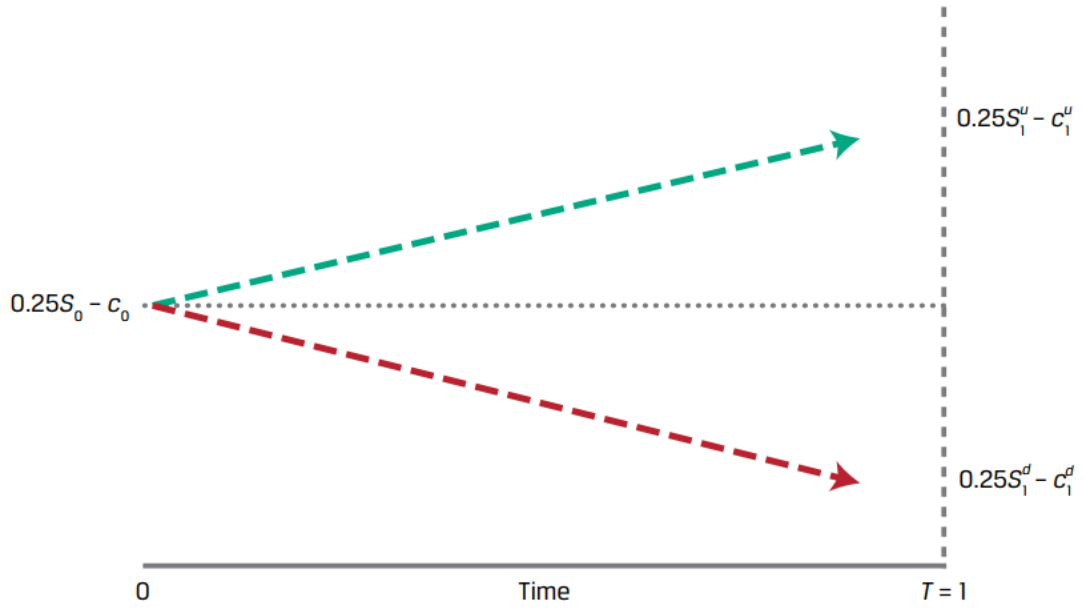


Figure 4.15: Call Option Replication

Case Study: Real Option Pricing in Project Valuation

A desalination project in Malaysia (MYR 10.5 million initial investment) has a lifetime of 20 years. In Year 10, the manager has the option to invest an additional MYR 0.5 million to lock in costs.

- **Project value with option exercised:** MYR 1.03 million (Case 1)
- **Project value without option exercised:** −MYR 2.12 million (Case 2)
- **Gross value of the option (at Year 10):**

$$\text{MYR } 1.03 \text{ million} - (-\text{MYR } 2.12 \text{ million}) = \text{MYR } 3.15 \text{ million}$$

- **PV of option cost (discounted 10 years at 6%):**

$$PV(\text{Cost}) = \frac{\text{MYR } 0.5 \text{ million}}{1.06^{10}} \approx \text{MYR } 0.28 \text{ million}$$

- **Net Value of Real Option:**

$$\text{MYR } 3.15 \text{ million} - \text{MYR } 0.28 \text{ million} = \text{MYR } 2.87 \text{ million}$$

Quyền chọn thực tế (Real Option) tạo sự linh hoạt trong đầu tư. Thực hiện quyền chọn ở năm 10 tăng giá trị ròng dự án thêm MYR 2.87 triệu so với không có quyền chọn.

The outcomes and decision tree of the desalination project with and without the real option are illustrated in Exhibit 4.16.

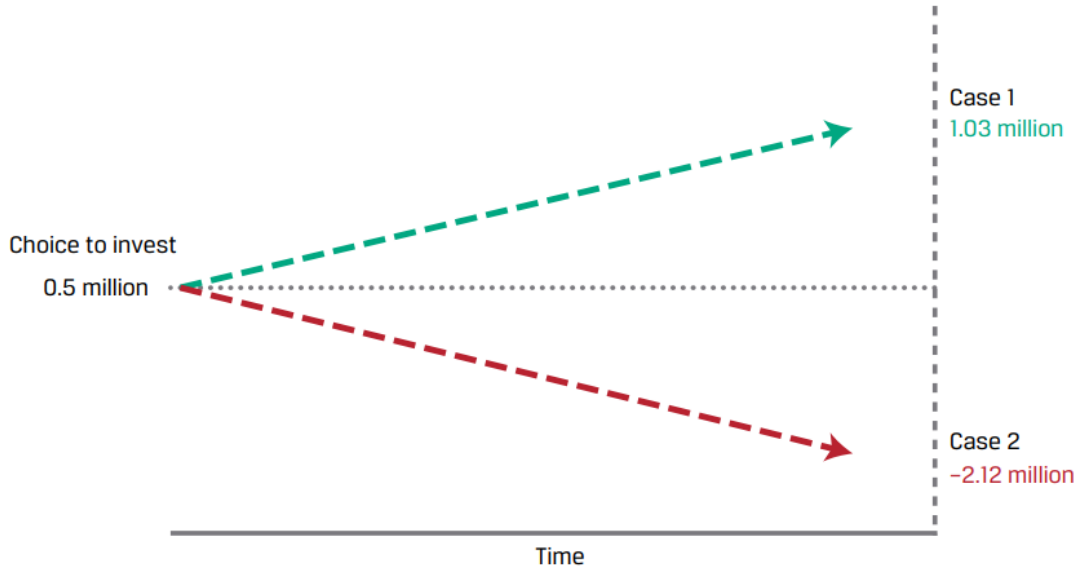


Figure 4.16: Project Outcomes

Module 5

Statistical Characteristics of Asset Returns

LEARNING OUTCOMES	
a	calculate, interpret, and evaluate various measures of (1) central tendency and location and (2) dispersion
b	describe, interpret, and evaluate measures of skewness and kurtosis
c	calculate, interpret, and evaluate covariance and correlation
d	calculate, interpret, and evaluate semi-deviation and coefficient of variation

Measures of Central Tendency and Dispersion

Measures of Central Tendency

Measures of Dispersion

Measures of Shape of a Distribution

Skewness

Kurtosis

Covariance and Correlation between Variables

Covariance

Correlation

Other Dispersion Measures

Semi-Deviation

Coefficient of Variation

Phân tích đặc trưng thống kê là bước đầu định lượng rủi ro và xây dựng danh mục. Chúng ta quan tâm toàn bộ hình dáng phân phối: độ lệch (skewness), đuôi dày (kurtosis) và sự đồng biến động giữa các tài sản.

5.1 Measures of Central Tendency and Dispersion

a	calculate, interpret, and evaluate various measures of (1) central tendency and location and (2) dispersion

To describe a dataset systematically, statisticians rely on two concepts: the **population** (the entire set of observations, such as the entire exchange rate history) and the **sample** (a subset of that population, such as daily exchange rates during a specific period).

5.1.1 Measures of Central Tendency

A measure of central tendency is a single numerical value that describes the center or typical position of a dataset.

1. **Arithmetic Mean:** The sum of all values divided by the number of observations.

$$\text{Population Mean} = \mu = \frac{\sum_{i=1}^{N} X_i}{N} \tag{5.1}$$

$$\text{Sample Mean} = \bar{X} = \frac{\sum_{i=1}^{n} X_i}{n} \tag{5.2}$$

   The arithmetic mean is measured in the same units as the underlying data. It is highly sensitive to outliers.

2. **Median:** The middle value in a dataset when the values are arranged in ascending or descending order.
   - If the number of observations n is odd, the median is the value at position $\frac{n+1}{2}$:

$$\text{Med} = X_{\left(\frac{n+1}{2}\right)} \tag{5.3}$$

   - If n is even, the median is the average of the two middle observations:

$$\text{Med} = \frac{X_{\left(\frac{n}{2}\right)} + X_{\left(\frac{n}{2}+1\right)}}{2} \tag{5.4}$$

   The median is robust to outliers, but it does not use all the information in the dataset.

3. **Mode:** The value that appears most frequently in a dataset. A dataset may be unimodal (one mode), multimodal (multiple modes), or have no mode at all.

Dealing with Outliers

Outliers can occur due to measurement errors, data entry issues, or natural variability. When outliers are valid, analysts use three primary strategies to manage them:

- **Leave the outliers unchanged:** Best if it is important to reflect the entire actual distribution.
- **Trimmed Mean:** Excludes a stated percentage of the lowest and highest values before computing the mean. For example, a 1% trimmed mean discards the bottom 0.5% and top 0.5% of values.
- **Winsorized Mean:** Substitutes extreme values with specified percentiles (e.g., setting the lowest 2.5% to the 2.5th percentile and the highest 2.5% to the 97.5th percentile value).

Table 5.1: Comparison of Mean, Trimmed Mean, and Winsorized Mean for Brent Crude Returns

Measure	**Original Data**	**Trimmed Data**		**Winsorized Data**	
		1%	**10%**	**1%**	**10%**
Mean	0.35%	0.83%	0.99%	0.80%	0.80%

5.1.2 Quantiles

Quantiles divide a sorted dataset into equal parts. Common quantile measures include quartiles (four parts), quintiles (five parts), deciles (ten parts), and percentiles (one hundred parts). The position L_y of the y-th percentile in a sorted dataset of size n is:

$$L_y = (n+1)\frac{y}{100} \tag{5.5}$$

If L_y is not an integer, we linearly interpolate between the two closest values. The **Box and Whisker Plot** visually summarizes these quartiles: the box spans the interquartile range ($\text{IQR} = Q_3 - Q_1$), a middle line marks the median, and the whiskers extend to the limits of normal data points. Outliers are typically identified as points beyond $1.5 \times \text{IQR}$ from the quartiles.

Table 5.2: Excel and Google Sheets Functions for Measures of Central Tendency

Calculation Type	Excel Formula	Google Sheets Formula
Mean	`=AVERAGE(A1:A5)`	`=AVERAGE(A1:A5)`
Median	`=MEDIAN(A1:A5)`	`=MEDIAN(A1:A5)`
Mode	`=MODE.SNGL(A1:A5)`	`=MODE(A1:A5)`
Trimmed Mean	`=TRIMMEAN(A1:A5, percent)`	`=TRIMMEAN(A1:A5, percent)`
1st Quartile	`=QUARTILE.INC(A1:A5, 1)`	`=QUARTILE(A1:A5, 1)`
3rd Quartile	`=QUARTILE.INC(A1:A5, 3)`	`=QUARTILE(A1:A5, 3)`
Percentile	`=PERCENTILE.INC(A1:A5, K)`	`=PERCENTILE(A1:A5, K)`

5.1.3 Measures of Dispersion

Dispersion quantifies the variability of data around its central tendency, serving as a measure of investment risk.

- **Range:** The difference between the highest and lowest values:

$$\text{Range} = \text{Maximum value} - \text{Minimum value} \tag{5.6}$$

- **Mean Absolute Deviation (MAD):** The arithmetic average of absolute deviations from the mean. It is less sensitive to outliers than standard deviation:

$$\text{MAD} = \frac{\sum_{i=1}^{n} |X_i - \bar{X}|}{n} \tag{5.7}$$

- **Variance (σ^2 or s^2):** The average of the squared deviations from the mean. Sample variance uses $n-1$ (Bessel's correction) to prevent underestimating the population variance:

$$\text{Population Variance} = \sigma^2 = \frac{\sum_{i=1}^{N} (X_i - \mu)^2}{N} \tag{5.8}$$

$$\text{Sample Variance} = s^2 = \frac{\sum_{i=1}^{n} (X_i - \bar{X})^2}{n-1} \tag{5.9}$$

- **Standard Deviation (σ or s):** The square root of variance ($s = \sqrt{s^2}$). It is measured in the same units as the underlying data.

Example: Calculating Sample Standard Deviation

Consider a sample of 10 observations: $-3, -11, 3, -18, 18, 20, -6, 9, 2, -16$.

1. **Calculate the sample mean $\bar{X}$:**

$$\bar{X} = \frac{-3 - 11 + 3 - 18 + 18 + 20 - 6 + 9 + 2 - 16}{10} = \frac{-2.00}{10} = -0.20$$

2. **Calculate the sum of squared differences $\sum (X_i - \bar{X})^2$:**

$$(-3 - (-0.2))^2 + (-11 - (-0.2))^2 + \cdots = 1563.60$$

3. **Divide by $n - 1 = 9$ and take the square root:**

$$s = \sqrt{\frac{1563.60}{9}} = 13.18$$

5.2 Measures of Shape of a Distribution

b | describe, interpret, and evaluate measures of skewness and kurtosis

To describe a distribution's shape completely, analysts evaluate its higher-order moments, particularly its asymmetry (**skewness**) and tail heaviness (**kurtosis**).

5.2.1 Skewness

Skewness is a measure of the asymmetry of a distribution.

- **Symmetric Distribution:** Skewness is zero. Mean, Median, and Mode are equal.
- **Positively Skewed (Right-skewed):** Has a long tail on the right side. Extreme positive values pull the mean upward, making Mean > Median > Mode.
- **Negatively Skewed (Left-skewed):** Has a long tail on the left side. Extreme negative values pull the mean downward, making Mean < Median < Mode.

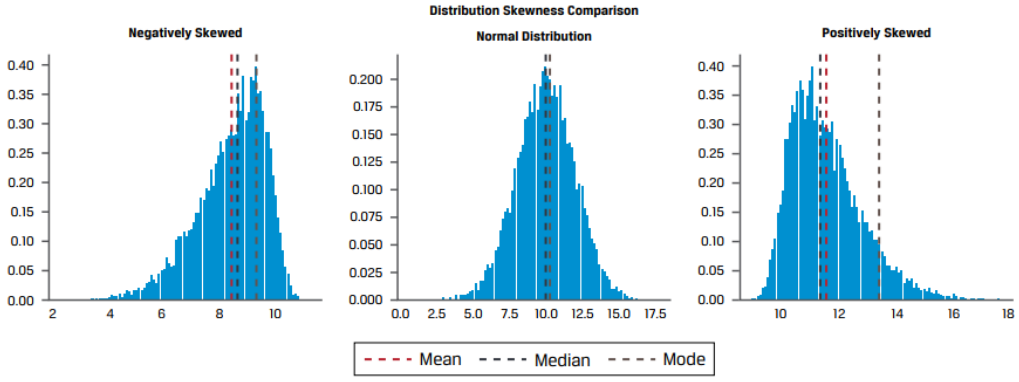


Figure 5.1: Skewness in Distributions

5.2.2 Kurtosis

Kurtosis measures the thickness or heaviness of the tails of a distribution relative to a normal distribution.

- **Leptokurtic (Fat-tailed):** Kurtosis > 3, Excess Kurtosis > 0. Taller peaks and fatter tails, representing higher probability of extreme outcomes.
- **Platykurtic (Thin-tailed):** Kurtosis < 3, Excess Kurtosis < 0. Flatter peaks and thinner tails.
- **Mesokurtic (Normal-like):** Kurtosis = 3, Excess Kurtosis = 0.

Excess Kurtosis is defined as:

$$\text{Excess Kurtosis} = \text{Kurtosis} - 3 \tag{5.10}$$

Skewness and Kurtosis Mathematical Approximations

For a sample of n observations, sample mean $\bar{X}$, and sample standard deviation s, the approximations for sample skewness and kurtosis are:

$$\text{Sample Skewness} \approx \frac{1}{n}\sum_{i=1}^{n}\left(\frac{X_i - \bar{X}}{s}\right)^3 \tag{5.11}$$

$$\text{Sample Kurtosis} \approx \frac{1}{n}\sum_{i=1}^{n}\left(\frac{X_i - \bar{X}}{s}\right)^4 \tag{5.12}$$

Chúng ta không cần ghi nhớ công thức chi tiết của Skewness và Kurtosis để thi. Đề thi CFA chỉ tập trung đọc hiểu ý nghĩa thực tế và so sánh chúng. Hãy tập trung vào bản chất thay vì công thức.

5.3 Covariance and Correlation

c | calculate, interpret, and evaluate covariance and correlation

To quantify the co-movements and linear relationships between two variables, analysts use covariance and correlation.

Hãy cẩn thận! Hệ số tương quan bằng 0 chỉ nghĩa là không có mối quan hệ TUYẾN TÍNH. Chúng ta đã nghe về Anscombe's Quartet chưa? Do đó, trực quan hóa dữ liệu bằng đồ thị (Scatter Plot) luôn là bước bắt buộc.

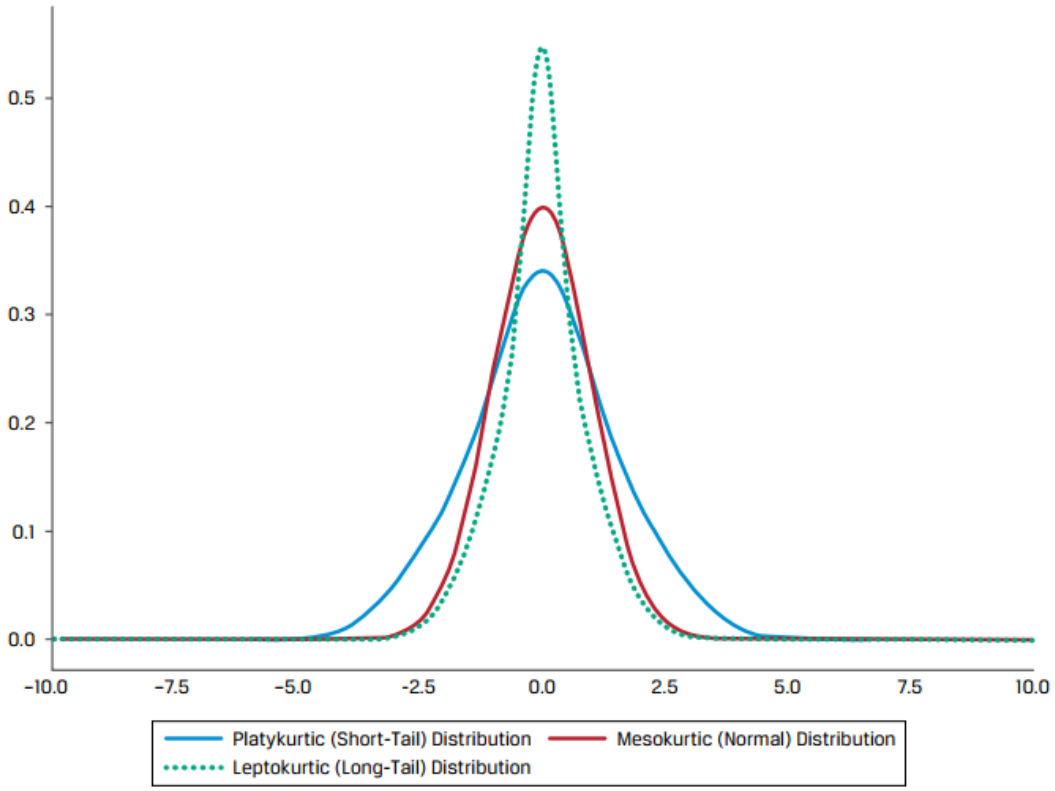


Figure 5.2: Types of Kurtosis

Table 5.3: The Four Moments of Statistical Distributions

Moment	**Moment Name / Description**	**Population Formula**	**Sample Approximation**
First	Mean (Central Tendency)	$\mu = \frac{\sum x_i}{N}$	$\bar{X} = \frac{\sum x_i}{n}$
Second	Variance (Dispersion)	$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$	$s^2 = \frac{\sum (x_i - \bar{X})^2}{n-1}$
Third	Skewness (Asymmetry)	$\text{Skew} = \frac{1}{N} \sum \left(\frac{x_i - \mu}{\sigma} \right)^3$	$\text{Skew} = \frac{1}{n} \sum \left(\frac{x_i - \bar{X}}{s} \right)^3$
Fourth	Kurtosis (Tail Heaviness)	$\text{Kurt} = \frac{1}{N} \sum \left(\frac{x_i - \mu}{\sigma} \right)^4$	$\text{Kurt} = \frac{1}{n} \sum \left(\frac{x_i - \bar{X}}{s} \right)^4$

Table 5.4: Excel and Google Sheets Functions for Skewness and Kurtosis

Calculation	**Excel Formula**	**Google Sheets Formula**
Population Skewness	`=SKEW.P(A1:A5)`	N/A
Sample Skewness	`=SKEW(A1:A5)`	`=SKEW(A1:A5)`
Excess Kurtosis	`=KURT(A1:A5)`	`=KURT(A1:A5)`

5.3.1 Covariance

Covariance measures the joint variability of two random variables, indicating how they move in relation to each other.

$$\sigma_{X,Y} = \frac{\sum_{i=1}^{N}(X_i - \mu_X)(Y_i - \mu_Y)}{N} \tag{5.13}$$

$$s_{X,Y} = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{n-1} \tag{5.14}$$

If covariance is positive, the variables tend to move in the same direction relative to their means. If negative, they tend to move in opposite directions.

5.3.2 Correlation

Correlation standardizes covariance into a dimensionless measure that ranges between -1 and $+1$:

$$\rho_{X,Y} = \frac{\sigma_{X,Y}}{\sigma_X \sigma_Y} \quad \text{or} \quad r_{X,Y} = \frac{s_{X,Y}}{s_X s_Y} \tag{5.15}$$

Table 5.5: Correlation Levels and Interpretation

Correlation Value	Correlation Type	Description
$\rho_{X,Y} = 0$	No correlation	Absence of any linear relationship.
$\rho_{X,Y} > 0$	Positive correlation	Moves in the same direction; $+1$ represents a perfect linear relationship.
$\rho_{X,Y} < 0$	Negative correlation	Moves in opposite directions; -1 represents a perfect inverse linear relationship.

5.3.3 Anscombe's Quartet

Anscombe's Quartet consists of four datasets that have nearly identical simple statistical properties (mean, variance, correlation), yet have completely different distributions and look completely different when graphed. This underscores the vital importance of visualizing data prior to performing correlation analysis.

5.4 Other Dispersion Measures

d | calculate, interpret, and evaluate semi-deviation and coefficient of variation

5.4.1 Semi-Deviation and Target Downside Deviation

While variance and standard deviation capture both upside and downside volatility, investors are primarily concerned with downside risk.

- **Semi-deviation (Semi-variance):** Measures variability for observations below the mean.
- **Target Downside Deviation:** Measures dispersion only for observations below a specific target or threshold B:

$$s_{\text{Target}} = \sqrt{\frac{\sum_{X_i \le B}(X_i - B)^2}{n-1}} \tag{5.16}$$

Example: Calculating Target Downside Deviation

Let's compute the sample target downside deviation for a portfolio with 12 monthly returns when the target return is $B = 3\%$:

$$5\%, 3\%, -1\%, -4\%, 4\%, 2\%, 0\%, 4\%, 3\%, 0\%, 6\%, 5\%$$

1. **Identify values below or equal to the target $B = 3\%$:**

$$3\%\ (0),\ -1\%\ (-4),\ -4\%\ (-7),\ 2\%\ (-1),\ 0\%\ (-3),\ 3\%\ (0),\ 0\%\ (-3)$$

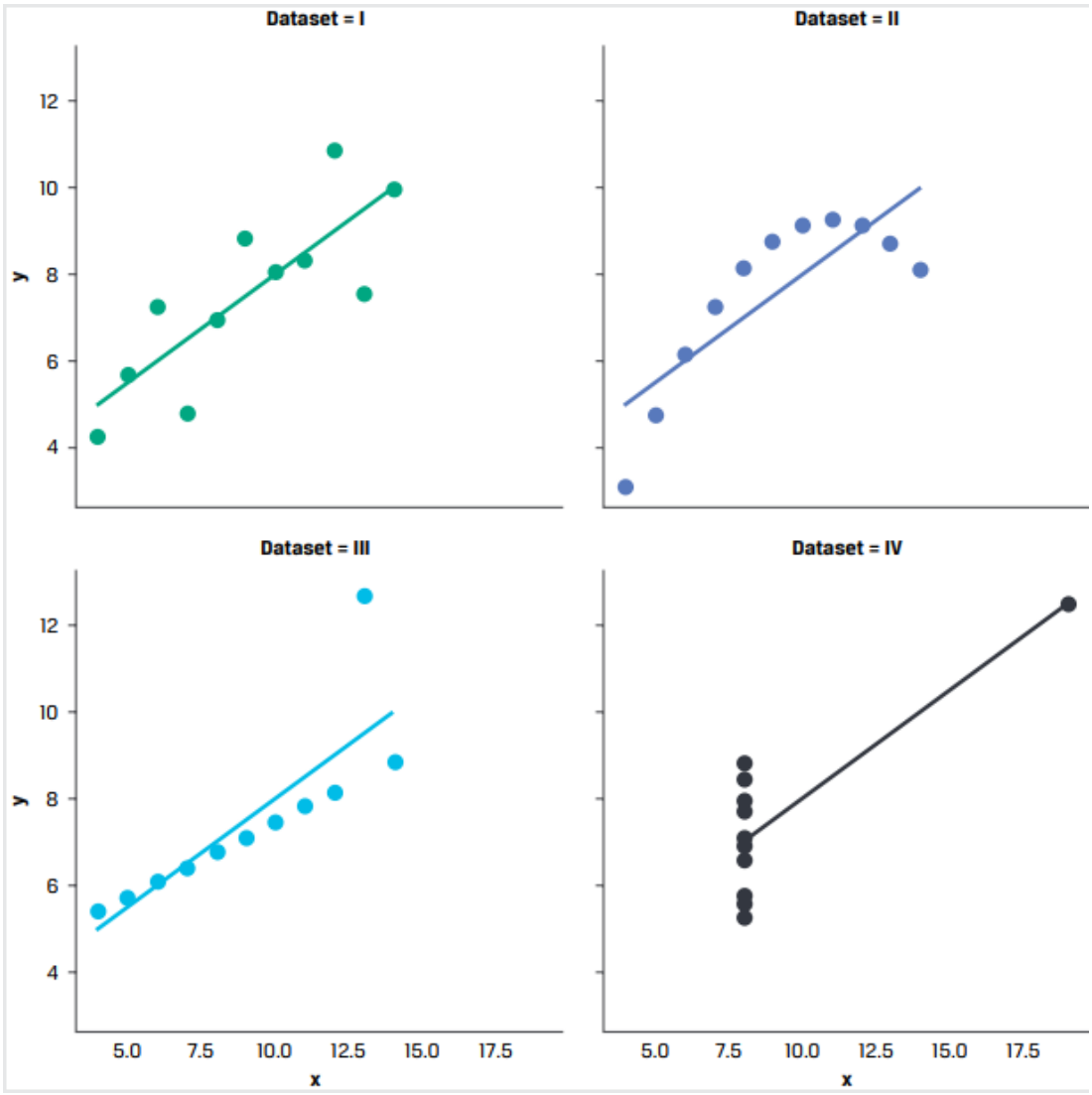


Figure 5.3: The Four Cases for Anscombe's Quartet Scatter Plots

2. **Calculate the squared deviations for these values:**

$$0^2 + (-4)^2 + (-7)^2 + (-1)^2 + (-3)^2 + 0^2 + (-3)^2 = 0 + 16 + 49 + 1 + 9 + 0 + 9 = 84$$

3. **Divide by total sample size $n - 1 = 11$ and take the square root:**

$$s_{\text{Target}} = \sqrt{\frac{84}{11}} \approx 2.76\%$$

5.4.2 Coefficient of Variation (CV)

The Coefficient of Variation measures the amount of risk per unit of return. It standardizes standard deviation, making it universally comparable:

$$\text{CV} = \frac{s}{\bar{X}} \tag{5.17}$$

A lower CV represents a more favorable risk-to-return ratio.

Module 6

Statistical Distributions for Financial Asset Prices and Returns

LEARNING OUTCOMES	
a	calculate, interpret, and evaluate unconditional expected values for mean, variance, and covariance
b	calculate, interpret, and evaluate the principal moments of key statistical distributions used in finance
c	calculate, interpret, and evaluate conditional expectations, variances, and covariances
d	formulate investment problems through Bayesian updating

Random Variables and Unconditional Expectations

Random Variables and Their Properties

Unconditional Expectations, Variance, and Covariance

Key Statistical Distributions in Finance

Discrete Statistical Distributions

Continuous Statistical Distributions

Practical Implementation

Conditional Expectations, Variances, and Covariances

Joint Distributions of Random Variables

Conditional Expectations

Conditional Means, Variances, and Covariances

Bayesian Updating

Mục tiêu ở Module 6 là giúp chúng ta hiểu cách dùng các phân phối (Nhị thức, Poisson, Chuẩn, Log-normal) để mô hình hóa giá tài sản, sinh lời, rủi ro. Đặc biệt lưu ý định lý Bayes vì tính thực tế và tần suất xuất hiện cao.

Lộ trình học đơn giản hóa: 1. Chúng ta chỉ cần hiểu xác suất biến liên tục là diện tích dưới đường cong PDF, không cần tính tích phân. 2. Tập trung vào biến rời rạc (kỳ vọng, phương sai qua xác suất) và định lý Bayes. 3. Ghi nhớ đặc trưng 4 moments các phân phối và lý do chọn (ví dụ: Log-normal cho giá ≥ 0, Poisson cho vỡ nợ).

6.1 Random Variables and Unconditional Expectations

a | calculate, interpret, and evaluate unconditional expected values for mean, variance, and covariance

This section outlines the characteristics of some of the most widely used statistical distributions to model prices and returns in finance, investments, and portfolio management.

6.1.1 Random Variables and Their Properties

A **random variable** is a function that assigns a number to each outcome of an experiment. In finance, random variables are often called **stochastic variables**, reflecting their inherent unpredictable nature. Random variables are categorized into two types:

- A **discrete random variable** has a countable number of distinct outcomes. For example, a fair coin toss results in heads or tails, each with a probability of $p = 0.50$. Similarly, rolling a fair six-sided die yields outcomes of $1, 2, 3, 4, 5$, or 6, each with a probability of $p = \frac{1}{6}$.
- A **continuous random variable** can take on an infinite number of possible values within a given range, such as picking a random number between 0 and 1 on the number line. When the number of observations is large, empirical distributions often converge to continuous distributions.

Biến rời rạc (Discrete) mô tả các sự kiện đếm được (vỡ nợ trái phiếu, số giao dịch). Ngược lại, tỷ suất sinh lời cổ phiếu dùng biến liên tục (Continuous) vì nhận bất kỳ giá trị thực nào.

A random variable, X, is fully described by its distribution functions:

- **Cumulative Distribution Function (CDF):** Gives the probability that the random variable X takes on a value less than or equal to x:

$$F(x) = P(X \leq x) \tag{6.1}$$

  The CDF is non-decreasing with $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to \infty} F(x) = 1$. The CDF applies to both discrete and continuous random variables.

- **Probability Density Function (PDF):** For continuous random variables, the PDF is the derivative of the CDF, representing the instantaneous rate of change of the CDF:

$$f_X(x) = \frac{dF_X(x)}{dx} \tag{6.2}$$

  The probability that X falls within an interval $[a, b]$ is the area under the PDF curve between a and b:

$$P(a \leq X \leq b) = \int_a^b f_X(x)dx = F(b) - F(a) \tag{6.3}$$

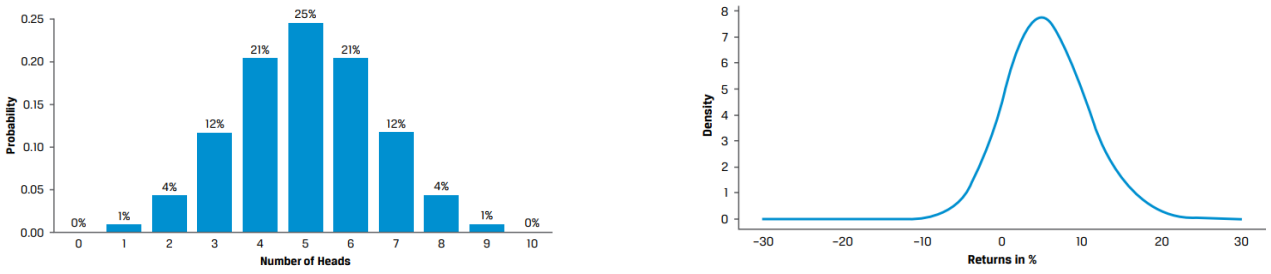
  The total area under the PDF curve equals 1:

$$\int_{-\infty}^{\infty} f_X(x)dx = 1 \tag{6.4}$$

- **Probability Mass Function (PMF):** For discrete random variables, the PMF gives the probability that X is exactly equal to a specific value x, $P(X = x)$. The probability that X lies within $[a, b]$ is:

$$P(a \leq X \leq b) = \sum_{x=a}^{b} P(X = x) \tag{6.5}$$

  The sum of probabilities for all possible outcomes equals 1: $\sum_x P(X = x) = 1$.



(a) Discrete PMF (Coin Tosses)

(b) Continuous PDF (Stock Returns)

Figure 6.1: Discrete vs. Continuous Probability Distributions

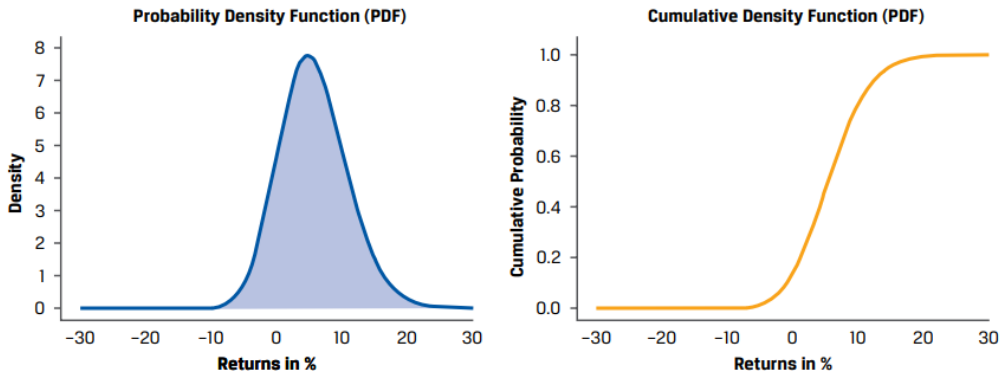


Figure 6.2: Theoretical PDF and CDF of Daily Stock Returns

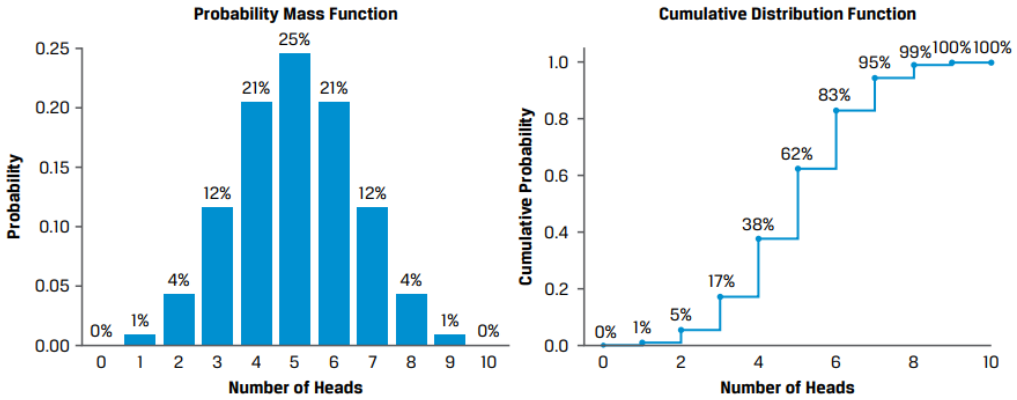


Figure 6.3: The PMF and CDF of the Coin Toss Experiment

Case Study: Roll of two fair dice

An economist analyzes the PMF and CDF of the sum of outcomes from rolling two fair six-sided dice. There are $6 \times 6 = 36$ possible combinations. The frequency and probability distributions are calculated in Table 6.1.

Table 6.1: Distribution of Outcomes for Rolling Two Six-Sided Dice

Outcome x	**Frequency** $n(x)$	**PMF** $P(X = x)$	**CDF** $P(X \leq x)$
2	1	0.0278	0.0278
3	2	0.0556	0.0834
4	3	0.0833	0.1667
5	4	0.1111	0.2778
6	5	0.1389	0.4167
7	6	0.1667	0.5834
8	5	0.1389	0.7223
9	4	0.1111	0.8334
10	3	0.0833	0.9167
11	2	0.0556	0.9723
12	1	0.0278	1.0000
Sum	**36**	**1.0000**	—

6.1.2 Unconditional Expectations, Variance, and Covariance

Unconditional expectations, variance, and covariance provide foundational tools for assessing the behavior of financial assets without requiring prior knowledge of specific market conditions.

- **Unconditional Mean (Expected Value):** The probability-weighted average of the outcomes of a ran-

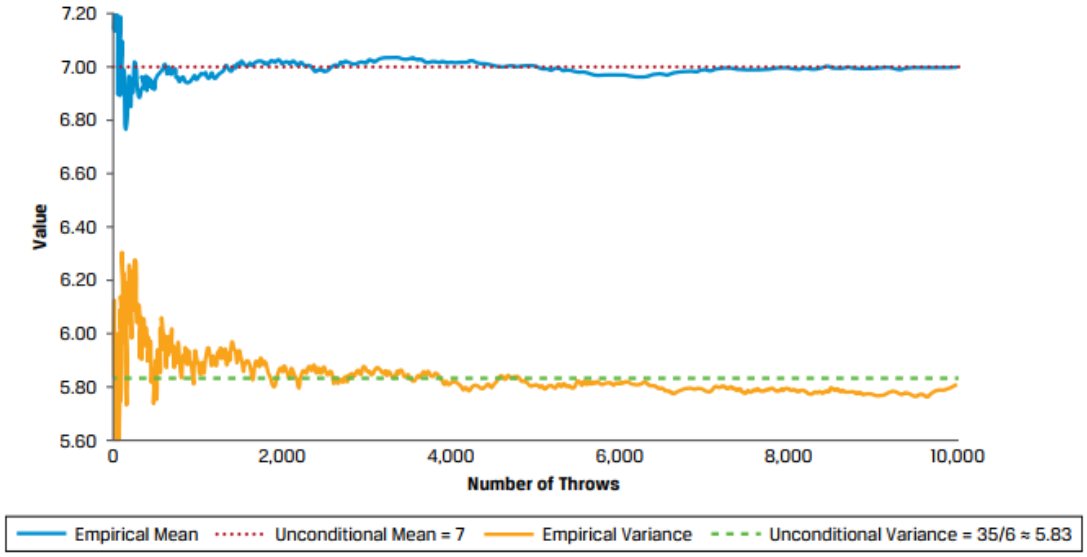


Figure 6.4: The PMF and CDF of Rolling Two Six-Sided Dice

dom variable over an infinite number of trials:

$$\text{Discrete: } \mu_X = E[X] = \sum_i x_i P(x_i) \tag{6.6}$$

$$\text{Continuous: } \mu_X = E[X] = \int_{-\infty}^{\infty} x f_X(x) dx \tag{6.7}$$

- **Unconditional Variance:** Measures the squared dispersion of a random variable around its mean:

$$\sigma_X^2 = \text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2 \tag{6.8}$$

- **Unconditional Standard Deviation:** $\sigma_X = \sqrt{\text{Var}(X)}$.
- **Unconditional Covariance:** Quantifies the linear relationship between two random variables X and Y:

$$\sigma_{XY} = \text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y] \tag{6.9}$$

Trong phòng thi, chúng ta rất dễ gặp tính kỳ vọng và phương sai biến rời rạc. Công thức: nhân từng kết quả với xác suất rồi cộng lại (Mean); lấy bình phương khoảng cách tới trung bình nhân xác suất (Variance).

Case Study: Expected Mean, Variance, and Covariances of Two Dice

For two fair dice, the expected sum is:

$$E[X] = 2\left(\frac{1}{36}\right) + 3\left(\frac{2}{36}\right) + \cdots + 12\left(\frac{1}{36}\right) = 7$$

To calculate variance, we find the expected value of X^2:

$$E[X^2] = 2^2\left(\frac{1}{36}\right) + 3^2\left(\frac{2}{36}\right) + \cdots + 12^2\left(\frac{1}{36}\right) = 54.84$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 54.84 - 7^2 = 5.84$$

Since rolls are independent, the covariance between the two individual dice is:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 12.25 - (3.5 \times 3.5) = 0$$

Case Study: Unconditional Mean, Variance, and Covariances of Two Assets

An analyst forecasts the performance of a stock index (X) and a bond index (Y) under different conditions:

Probability P	Stock Return X (%)	Bond Return Y (%)
0.2	-2.00	0.75
0.3	-1.00	0.50
0.3	0.00	0.25
0.2	1.00	0.00

Calculating expectations:

$$E[X] = (-2.00 \times 0.2) + (-1.00 \times 0.3) + (0 \times 0.3) + (1.00 \times 0.2) = -0.50\%$$
$$E[Y] = (0.75 \times 0.2) + (0.50 \times 0.3) + (0.25 \times 0.3) + (0 \times 0.2) = 0.375\%$$

Calculating variance:

$$E[X^2] = (0.2 \times (-2)^2) + (0.3 \times (-1)^2) + (0.3 \times 0^2) + (0.2 \times 1^2) = 1.30$$
$$\text{Var}(X) = 1.30 - (-0.50)^2 = 1.05 \implies \sigma_X = \sqrt{1.05} = 1.025\%$$
$$E[Y^2] = (0.2 \times 0.75^2) + (0.3 \times 0.50^2) + (0.3 \times 0.25^2) + 0 = 0.2063$$
$$\text{Var}(Y) = 0.2063 - 0.375^2 = 0.0656 \implies \sigma_Y = \sqrt{0.0656} = 0.2562\%$$

Calculating covariance:

$$E[XY] = (0.2 \times (-2 \times 0.75)) + (0.3 \times (-1 \times 0.50)) + 0 + 0 = -0.45$$
$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = -0.45 - (-0.50 \times 0.375) = -0.2625$$

The negative covariance of -0.2625 indicates that stocks and bonds tend to move in opposite directions, providing significant diversification benefits for a portfolio.

6.1.3 The Law of Large Numbers (LLN)

The **Law of Large Numbers** states that as the number of trials (sample size) increases, the empirical sample mean converges to the true population parameter (unconditional mean), assuming the sample is randomly drawn and representative of the population.

Case Study: The Law of Large Numbers and Dice Throws

An economist simulates 10,000 throws of a pair of fair dice. Table 6.2 summarizes the convergence of the sample statistics to the true population parameters ($\mu = 7.00$, $\sigma^2 = 5.83$).

Table 6.2: Convergence of Sample Mean and Variance to Unconditional Values

Throws of Dice	Sample Mean $\bar{X}$	Sample Variance s^2
1,000	6.96	5.89
2,000	7.01	5.86
5,000	7.00	5.80
10,000	7.00	5.81
Unconditional Expected	**7.00**	**5.83**

6.2 Key Statistical Distributions in Finance

b | calculate, interpret, and evaluate the principal moments of key statistical distributions used in finance

Probability distributions are required to model the random behavior of financial variables. This section reviews discrete and continuous distributions widely used in finance.

Để giúp học viên dễ dàng hình dung và phân loại, các phân phối xác suất chính trong chương trình CFA được hệ thống hóa qua Ma trận phân loại hai chiều dưới đây (phân biệt giữa biến ngẫu nhiên Rời rạc vs. Liên tục, và phân phối Đều vs. Không đều):

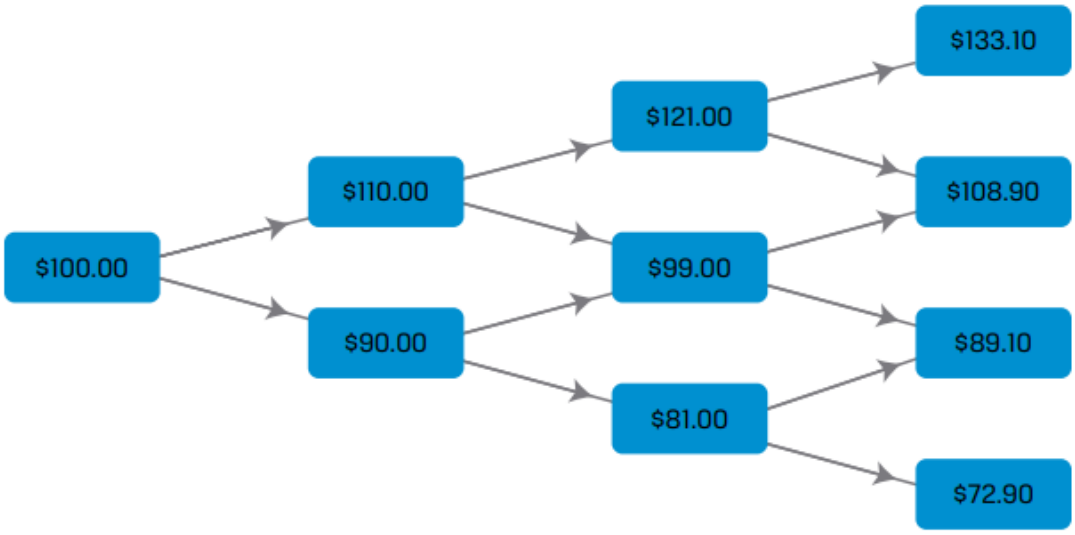


Figure 6.5: Convergence of Sample Mean and Variance to Unconditional Values in Dice Throws

Table 6.3: Classification Matrix of Statistical Distributions

	Uniform (Phân phối đều)	**Non-Uniform (Phân phối không đều)**
Discrete **(Biến rời rạc)**	**Discrete Uniform Distribution** - Mỗi kết quả có xác suất bằng nhau $(1/n)$. - Ví dụ: Gieo xúc xắc, tung đồng xu.	**Binomial Distribution** - Số lần thành công trong n phép thử độc lập. **Poisson Distribution** - Số sự kiện xảy ra trong một khoảng xác định.
Continuous **(Biến liên tục)**	**Continuous Uniform Distribution** - Hàm mật độ không đổi $f(x) = \frac{1}{b-a}$. - Ví dụ: Chọn ngẫu nhiên số trong $[0, 1]$.	**Normal Distribution** (Đối xứng, hình chuông) **Log-normal Distribution** (Chặn dưới bởi 0, lệch phải) **Logistic Distribution** (Đối xứng, đuôi dày hơn Normal)

6.2.1 Discrete Statistical Distributions

Discrete Uniform Distribution

A discrete uniform distribution consists of a finite number of equally spaced outcomes, each with equal probability of occurrence. For a variable X taking n outcomes in the range $[a, b]$, each outcome occurs with probability $\frac{1}{n}$.

$$\text{Mean: } \mu = \frac{a+b}{2} \tag{6.10}$$

$$\text{Variance: } \sigma^2 = \frac{(b-a+1)^2 - 1}{12} \tag{6.11}$$

Chứng minh phân phối đều rời rạc: - Giả định: $N = b - a + 1$ kết quả đều nhau trong $[a, b]$, xác suất bằng $\frac{1}{N}$. - Mean: Kỳ vọng là trung bình cộng cấp số cộng của a và b: $E[X] = \frac{a+b}{2}$. - Variance: Sử dụng biến chuẩn hóa $Y = X - a + 1$. Ta có $Var(X) = Var(Y) = E[Y^2] - (E[Y])^2$. Với các tổng chuẩn của dãy số, ta suy ra $Var(Y) = \frac{N^2-1}{12}$. Thay $N = b - a + 1$, ta được $\sigma^2 = \frac{(b-a+1)^2-1}{12}$.

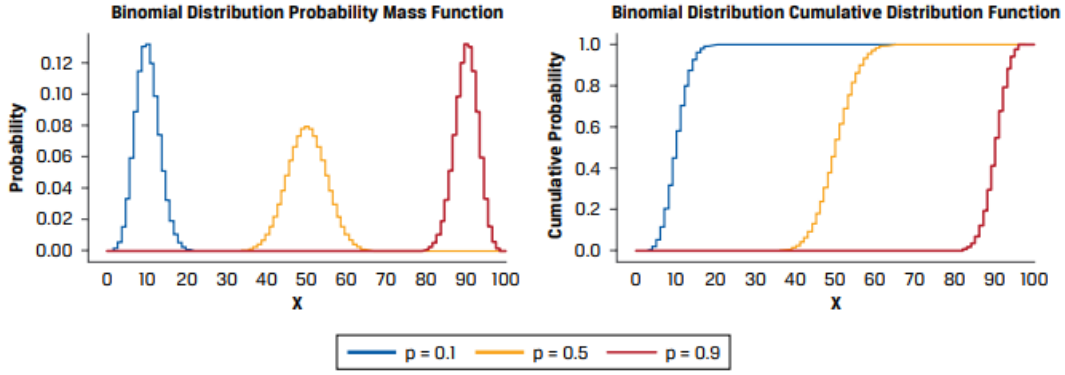


Figure 6.6: Discrete Uniform Distribution PMF and CDF

Binomial Distribution

The binomial distribution models the number of successes in a fixed number n of independent Bernoulli trials, where each trial has a success probability p and a failure probability $q = 1 - p$. The probability of observing exactly x successes is:

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x} \tag{6.12}$$

The mean and variance are:

$$\text{Mean: } \mu = n \times p \tag{6.13}$$

$$\text{Variance: } \sigma^2 = n \times p(1-p) \tag{6.14}$$

Chứng minh phân phối Nhị thức: - Giả định: n phép thử Bernoulli độc lập với xác suất thành công p. - Mean: Gọi Y_i là phép thử thứ i ($E[Y_i] = p$). Số lần thành công $X = \sum Y_i$. Theo tính tuyến tính: $E[X] = \sum E[Y_i] = n \times p$. - Variance: Do các phép thử độc lập, $Var(X) = \sum Var(Y_i)$. Mà $Var(Y_i) = E[Y_i^2] - E[Y_i]^2 = p - p^2 = p(1-p)$. Do đó $Var(X) = n \times p(1-p)$.

Thực tế: Phân phối nhị thức rất phổ biến, tiêu biểu là mô hình định giá quyền chọn nhị thức (BOPM) giả định giá đi lên/xuống cố định. Cây nhị thức giúp chúng ta định giá các hợp đồng tài chính phức tạp.

Case Study: Binomial Model for Stock Prices

Let S_0 be the initial price of a stock, u the up factor, and d the down factor. The stock price at node (i, j) after $n = i + j$ steps (where i is up movements and j is down movements) is:

$$S_{i,j} = S_0 \times u^i \times d^j$$

For $S_0 = 100$, $u = 1.1$, $d = 0.9$, and a path with $n = 3$ steps ($i = 2$ ups, $j = 1$ down), the price is:

$$S_{2,1} = 100 \times 1.1^2 \times 0.9^1 = 108.90$$

Exhibit 6.7 shows the tree of possible stock prices over three periods.

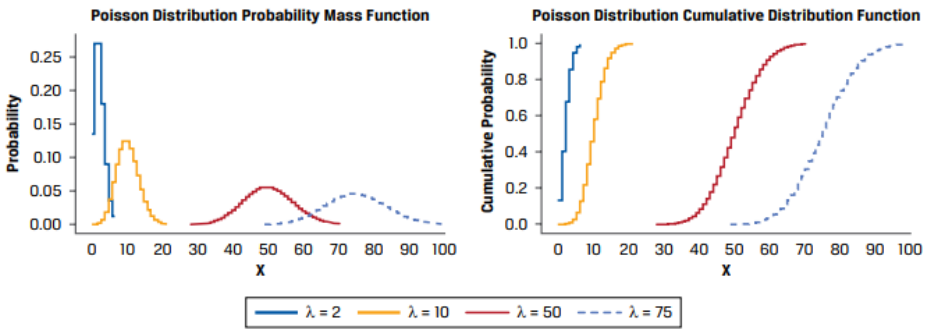


Figure 6.7: Binomial Stock Prices and Price Paths

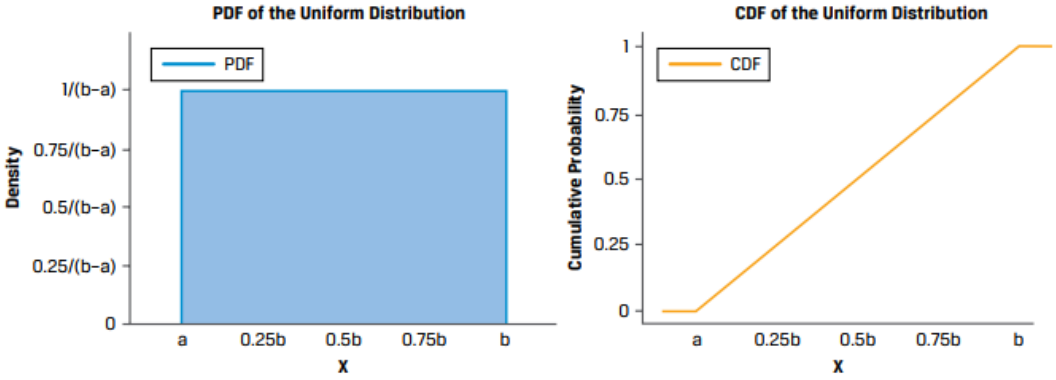


Figure 6.8: Binomial PMF and CDF Across Different Success Probabilities

If n is large, the binomial distribution can be approximated by a normal distribution:

$$X \sim N(np, np(1-p)) \tag{6.15}$$

This approximation is generally valid when both $np \geq 5$ and $n(1-p) \geq 5$.

Poisson Distribution

The Poisson distribution models the number of independent events occurring in a fixed interval of time or space at a constant average rate λ. The probability of observing k events is:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!} \tag{6.16}$$

The mean and variance of a Poisson distribution are identical:

$$\mu = \sigma^2 = \lambda \tag{6.17}$$

Chứng minh phân phối Poisson: - Giả định: Các sự kiện độc lập với tốc độ trung bình λ. Là giới hạn phân phối Nhị thức khi $n \to \infty, p \to 0, np = \lambda$. - Mean: Sử dụng khai triển Taylor cho e^λ: $E[X] = \sum k \frac{\lambda^k e^{-\lambda}}{k!} = \lambda e^{-\lambda} \sum \frac{\lambda^{k-1}}{(k-1)!} = \lambda$. - Variance: Tính $E[X(X-1)] = \sum k(k-1) \frac{\lambda^k e^{-\lambda}}{k!} = \lambda^2$. Suy ra $E[X^2] = \lambda^2 + \lambda$. Vậy $Var(X) = E[X^2] - E[X]^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$. Its skewness is $\frac{1}{\sqrt{\lambda}}$ and excess kurtosis is $\frac{1}{\lambda}$.

In credit risk modeling, the Poisson distribution is widely used in **reduced-form models** where default is modeled as a random event governed by a hazard rate (default intensity) λ.

- Probability of no default over time t: $P(\text{No default}) = e^{-\lambda t}$
- Probability of default over time t: $P(\text{Default}) = 1 - e^{-\lambda t}$
- Expected time to first default: $E[t_d] = \frac{1}{\lambda}$

Case Study: Predicting Corporate Default

Assume the hazard rate λ for a corporate bond is 0.02 per year.

- Probability of default within 1 year: $P(\text{Default}) = 1 - e^{-0.02\times1} \approx 1.98\%$
- Probability of no default within 1 year: $P(\text{No default}) = e^{-0.02\times1} \approx 98.02\%$
- Expected time until first default: $E[t_d] = \frac{1}{0.02} = 50$ years



Figure 6.9: Poisson Distributions for Different Values of Lambda

6.2.2 Continuous Statistical Distributions

Continuous Uniform Distribution

The continuous uniform distribution is defined over an interval $[a, b]$, where any value within this interval is equally likely to occur.

$$\text{Mean: } \mu = \frac{a+b}{2} \tag{6.18}$$

$$\text{Variance: } \sigma^2 = \frac{(b-a)^2}{12} \tag{6.19}$$

Chứng minh phân phối đều liên tục: - Giả định: X nhận giá trị trong đoạn $[a, b]$, hàm mật độ $f(x) = \frac{1}{b-a}$. - Mean: $E[X] = \int_a^b \frac{x}{b-a}dx = \frac{b^2-a^2}{2(b-a)} = \frac{a+b}{2}$. - Variance: Tính $E[X^2] = \int_a^b \frac{x^2}{b-a}dx = \frac{b^3-a^3}{3(b-a)} = \frac{a^2+ab+b^2}{3}$. Suy ra $Var(X) = E[X^2] - E[X]^2 = \frac{a^2+ab+b^2}{3} - \frac{a^2+2ab+b^2}{4} = \frac{(b-a)^2}{12}$. Its skewness is 0 and kurtosis is 1.8.

Continuous uniform distributions are key to generating Monte Carlo simulations. To simulate any continuous distribution, we first draw random numbers from a standard uniform distribution $U \sim [0, 1]$ and then map them to the target distribution using the inverse CDF method.

Case Study: Monte Carlo Simulation and Geometric Brownian Motion (GBM)

Stock price paths are often simulated using the GBM formula:

$$S_t = S_0 e^{\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t}$$

where μ is drift, σ is volatility, and W_t is Brownian motion (normally distributed noise). To simulate a single daily step ($t = 1/250$):

1. Draw a uniform random value from $[0, 1]$, say $U = 0.07$.
2. Convert it into a standard normal value: using the inverse standard normal CDF, $\Phi^{-1}(0.07) \approx -1.48$.
3. Substitute $W_t = -1.48$ into the formula to compute the next stock price S_1.

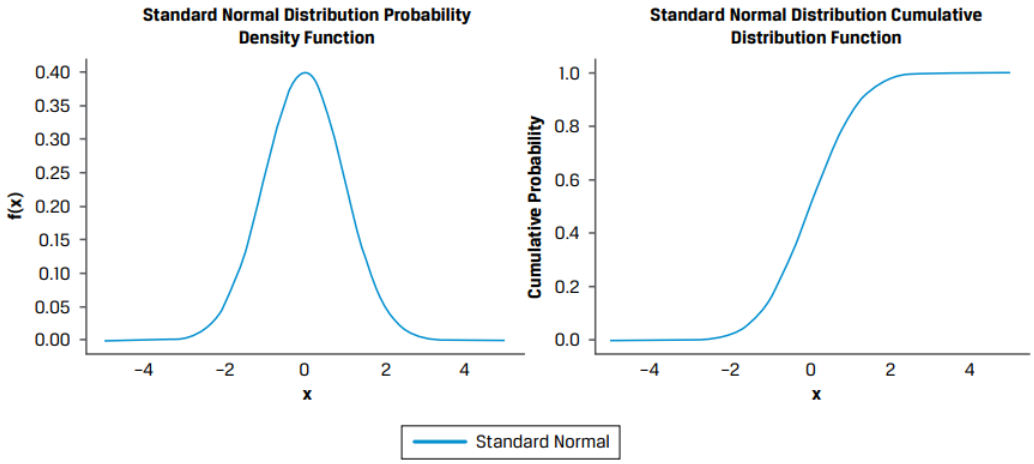


Figure 6.10: Continuous Uniform Probability Density between a and b

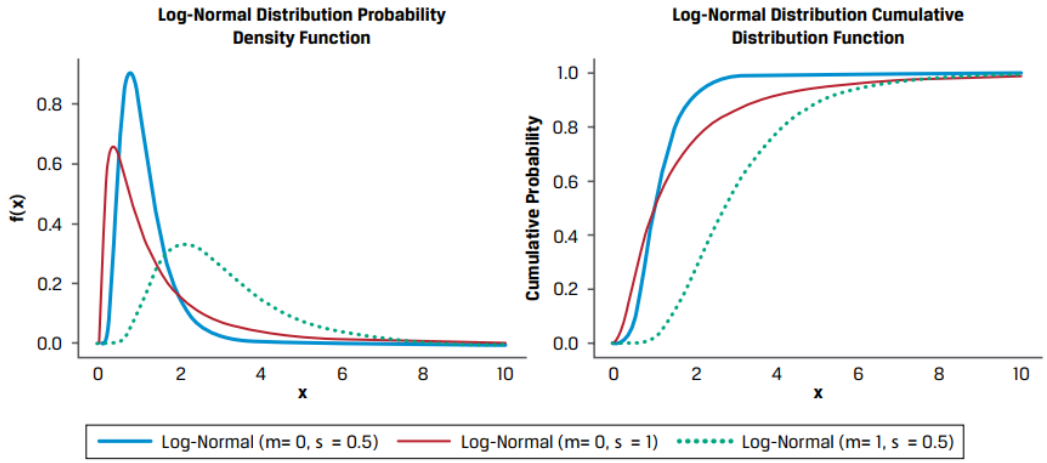


Figure 6.11: Monte Carlo Simulation Stock Price Paths

Normal Distribution

The normal distribution, or Gaussian distribution, is fully defined by its mean μ and variance σ^2, denoted as $N(\mu, \sigma^2)$. It has a symmetrical bell shape where Mean = Median = Mode. Its skewness is 0, kurtosis is 3, and excess kurtosis is 0.

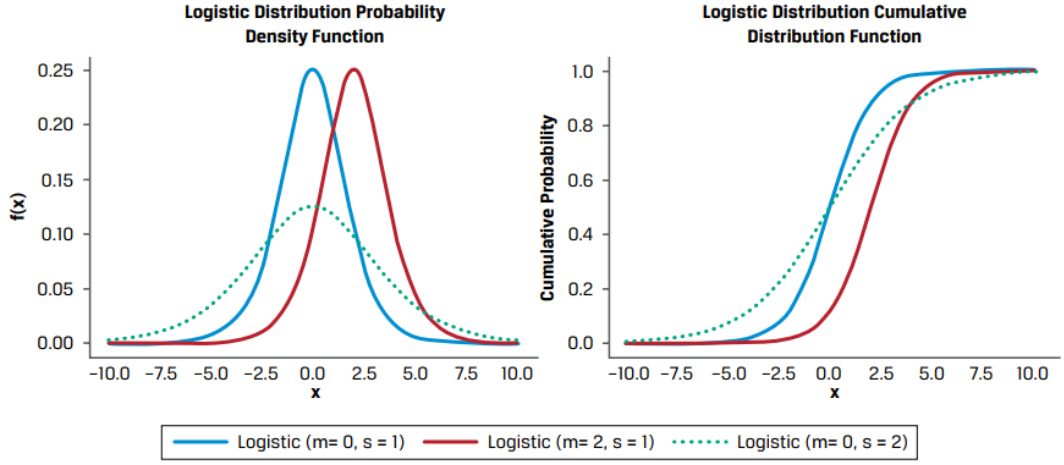


Figure 6.12: Standard Normal Distribution PDF and CDF

A standard normal distribution has $\mu = 0$ and $\sigma^2 = 1$. Z-scores standardize any normal variable: $Z = \frac{X-\mu}{\sigma}$. Key probability intervals (Empirical Rule):

- $\mu \pm 1\sigma$ contains approximately 68% of the observations.

- $\mu \pm 1.96\sigma$ contains approximately 95% of the observations (often rounded to $\mu \pm 2\sigma$).
- $\mu \pm 2.58\sigma$ contains approximately 99% of the observations.

Chứng minh phân phối Chuẩn: - Giả định: $Z \sim N(0,1)$ có hàm mật độ đối xứng $f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$. Biến tổng quát $X = \mu + \sigma Z$. - Mean: Do tích phân hàm lẻ đối xứng bằng 0, $E[Z] = 0$. Suy ra $E[X] = E[\mu + \sigma Z] = \mu$. - Variance: Tính $Var(Z) = E[Z^2]$ bằng tích phân từng phần (đặt $u = z, dv = ze^{-z^2/2}dz$), ta được $Var(Z) = 1$. Do đó $Var(X) = \sigma^2 \, Var(Z) = \sigma^2$.

Log-Normal Distribution

A random variable Y is log-normally distributed if $X = \ln(Y)$ is normally distributed. Because $Y = e^X$, log-normal variables can only take strictly positive values ($Y > 0$). The distribution is right-skewed. If $X \sim N(\mu_X, \sigma_X^2)$, the moments of Y are:

$$\text{Mean: } \mu_Y = e^{\mu_X + \frac{\sigma_X^2}{2}} \tag{6.20}$$

$$\text{Variance: } \sigma_Y^2 = (e^{\sigma_X^2} - 1)e^{2\mu_X + \sigma_X^2} \tag{6.21}$$

Chứng minh phân phối Log-normal: - Giả định: $Y = e^X$ với $X \sim N(\mu_X, \sigma_X^2)$. - Mean: Sử dụng hàm sinh mô-men $E[e^{tX}] = e^{\mu_X t + \sigma_X^2 t^2/2}$ với $t = 1$: $E[Y] = e^{\mu_X + \frac{\sigma_X^2}{2}}$. - Variance: Tính $E[Y^2] = E[e^{2X}]$ bằng cách đặt $t = 2$, ta có $E[Y^2] = e^{2\mu_X + 2\sigma_X^2}$. Suy ra $Var(Y) = E[Y^2] - E[Y]^2 = e^{2\mu_X + 2\sigma_X^2} - e^{2\mu_X + \sigma_X^2} = (e^{\sigma_X^2} - 1)e^{2\mu_X + \sigma_X^2}$.

Tại sao dùng Log-normal cho giá và Normal cho sinh lời? Vì giá cổ phiếu ≥ 0, phù hợp Log-normal. Tỷ suất sinh lời có thể âm, phù hợp phân phối Chuẩn. Ngoài ra, Log returns cộng dồn (additive) giúp đơn giản hóa tính toán.

Case Study: Modeling Asset Prices

Let $S_0 = 100$, logarithmic annual mean return $\mu = 8\%$, and volatility $\sigma = 20\%$. After 1 year, the logarithmic price is:

$$\ln(S_1) = \ln(S_0) + \left(\mu - \frac{\sigma^2}{2}\right) \times t = \ln(100) + \left(0.08 - \frac{0.04}{2}\right) \times 1 = 4.6652$$

The expected future price is $S_1 = e^{4.6652} = 106.19$.

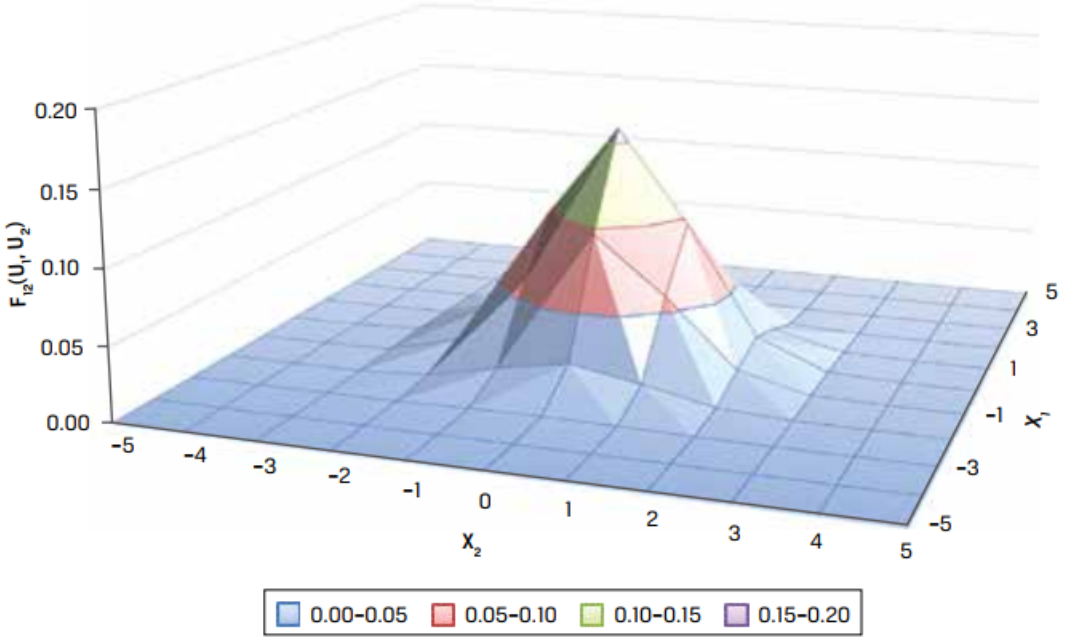


Figure 6.13: A Log-Normal Distribution Curve

Logistic Distribution

The logistic distribution is a continuous distribution defined by location μ and scale s. It resembles the normal distribution but has fatter tails.

$$\text{Mean: } \mu = \mu \tag{6.22}$$

$$\text{Variance: } \sigma^2 = \frac{s^2\pi^2}{3} \tag{6.23}$$

Chứng minh phân phối Logistic: - Giả định: Z chuẩn hóa có CDF $F(z) = \frac{1}{1+e^{-z}}$. Biến tổng quát $X = \mu + sZ$. - Mean: Do tính đối xứng hoàn hảo quanh μ, ta có $E[X] = \mu$. - Variance: Phương sai của Z được tính bằng tích phân Riemann-Zeta: $Var(Z) = \frac{\pi^2}{3}$. Do đó, $Var(X) = s^2\, Var(Z) = \frac{s^2\pi^2}{3}$. Its skewness is 0, kurtosis is 4.2, and excess kurtosis is 1.2. The logistic distribution is widely used in logistic regression to predict binary outcomes in finance (e.g., probability of default).

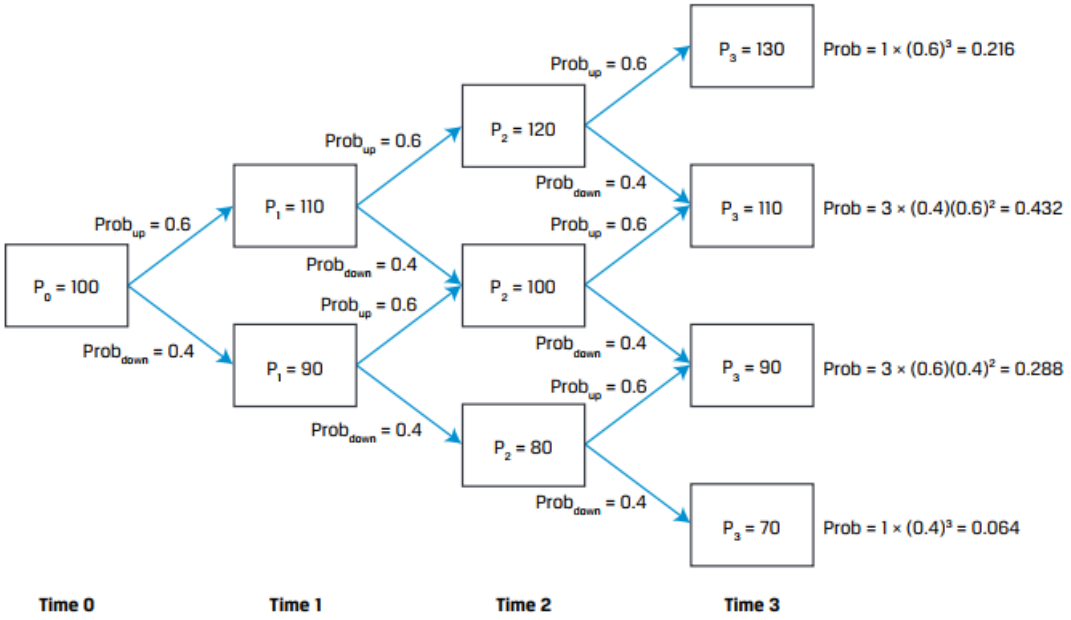


Figure 6.14: The PDF and CDF of Different Logistic Distributions

6.2.3 Practical Implementation

Table 6.4 summarizes the moments of all distributions covered in this module, and Table 6.5 shows their density/mass functions and cumulative distributions.

Table 6.4: The Four Statistical Moments of Distributions

Distribution	Mean	Variance	Skewness	Excess Kurtosis
Discrete Uniform	$\frac{a+b}{2}$	$\frac{(b-a+1)^2-1}{12}$	0	$-\frac{6(n^2+1)}{5(n^2-1)}$
Binomial	np	$np(1-p)$	$\frac{1-2p}{\sqrt{np(1-p)}}$	$\frac{1-6p(1-p)}{np(1-p)}$
Poisson	λ	λ	$\frac{1}{\sqrt{\lambda}}$	$\frac{1}{\lambda}$
Continuous Uniform	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	0	-1.2
Normal	μ	σ^2	0	0
Log-normal	$e^{\mu+\frac{\sigma^2}{2}}$	$(e^{\sigma^2}-1)e^{2\mu+\sigma^2}$	> 0	> 0
Logistic	μ	$\frac{s^2\pi^2}{3}$	0	1.2

Table 6.6 summarizes the Excel and Google Sheets formulas used to compute these distributions in practice.

6.3 Conditional Expectations, Variances, and Covariances

c	calculate, interpret, and evaluate conditional expectations, variances, and covariances

Table 6.5: CDFs, PMFs, and PDFs for Statistical Distributions

Distribution	PMF / PDF	CDF
Discrete Uniform	$\frac{1}{b-a+1}$	$\frac{x-a+1}{b-a+1}$
Binomial	$\binom{n}{x}p^x(1-p)^{n-x}$	$\sum_{i=0}^{x}\binom{n}{i}p^i(1-p)^{n-i}$
Poisson	$\frac{\lambda^x e^{-\lambda}}{x!}$	$e^{-\lambda}\sum_{i=0}^{x}\frac{\lambda^i}{i!}$
Continuous Uniform	$\frac{1}{b-a}$	$\frac{x-a}{b-a}$
Normal	$\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$	$\Phi\left(\frac{x-\mu}{\sigma}\right)$
Log-normal	$\frac{1}{x\sigma\sqrt{2\pi}}e^{-\frac{(\ln x-\mu)^2}{2\sigma^2}}$	$\Phi\left(\frac{\ln x-\mu}{\sigma}\right)$
Logistic	$\frac{e^{-(x-\mu)/s}}{s\left(e^{-(x-\mu)/s}+1\right)^2}$	$\frac{1}{e^{-(x-\mu)/s}+1}$

Table 6.6: Statistical Distribution Functions in Excel and Google Sheets

Distribution Function	Excel Formula	Google Sheets Formula
Uniform CDF	`BETA.DIST((x-a)/(b-a),1,1,TRUE)`	`BETA.DIST((x-a)/(b-a),1,1,TRUE)`
Uniform Random Draw	`=RAND()*(b-a)+a`	`=RAND()*(b-a)+a`
Binomial PMF	`=BINOM.DIST(x, n, p, FALSE)`	`=BINOM.DIST(x, n, p, FALSE)`
Binomial CDF	`=BINOM.DIST(x, n, p, TRUE)`	`=BINOM.DIST(x, n, p, TRUE)`
Poisson PMF	`=POISSON.DIST(x, lambda, FALSE)`	`=POISSON.DIST(x, lambda, FALSE)`
Normal PDF	`=NORM.DIST(x, mean, sd, FALSE)`	`=NORM.DIST(x, mean, sd, FALSE)`
Normal Random Draw	`=NORM.INV(RAND(), mean, sd)`	`=NORMINV(RAND(), mean, sd)`
Log-normal CDF	`=LOGNORM.DIST(x, mean, sd, TRUE)`	`=LOGNORM.DIST(x, mean, sd, TRUE)`

6.3.1 Joint Distributions of Random Variables

Joint distributions allow for examining the simultaneous distribution of two random variables. The bivariate cumulative distribution function (CDF) is defined as:

$$F_{12}(x_1, x_2) = P(X_1 \leq x_1, X_2 \leq x_2) \tag{6.24}$$

For continuous variables, the joint CDF is obtained by integrating the joint PDF $f_{12}(u_1, u_2)$:

$$F_{12}(x_1, x_2) = \int_{-\infty}^{x_1}\int_{-\infty}^{x_2} f_{12}(u_1, u_2)du_1du_2 \tag{6.25}$$

If the variables are independent:

$$f_{12}(u_1, u_2) = f_1(u_1) \times f_2(u_2) \tag{6.26}$$

The **marginal density** is the probability distribution of a single variable within a multivariate distribution. It is obtained by integrating out the other variables:

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} f_{X_1X_2}(x_1, u_2)du_2 \tag{6.27}$$

For discrete variables, joint and marginal distributions are summarized in contingency tables.

Case Study: Stock Market Returns and Technology Companies

An index consists of 500 firms, 100 of which are technology firms. Among the 100 technology firms, 60 had returns $> 10\%$ and 40 had returns $\leq 10\%$. Among the 400 non-tech firms, 100 had returns $> 10\%$ and 300 had returns $\leq 10\%$. Table 6.7 displays these frequencies.

6.3.2 Conditional Expectations

The conditional expectation $E[X|Y = y]$ is the expected value of X given that Y takes on the value y:

$$\text{Discrete: } E[X|Y = y] = \sum_x xP(X = x|Y = y) \tag{6.28}$$

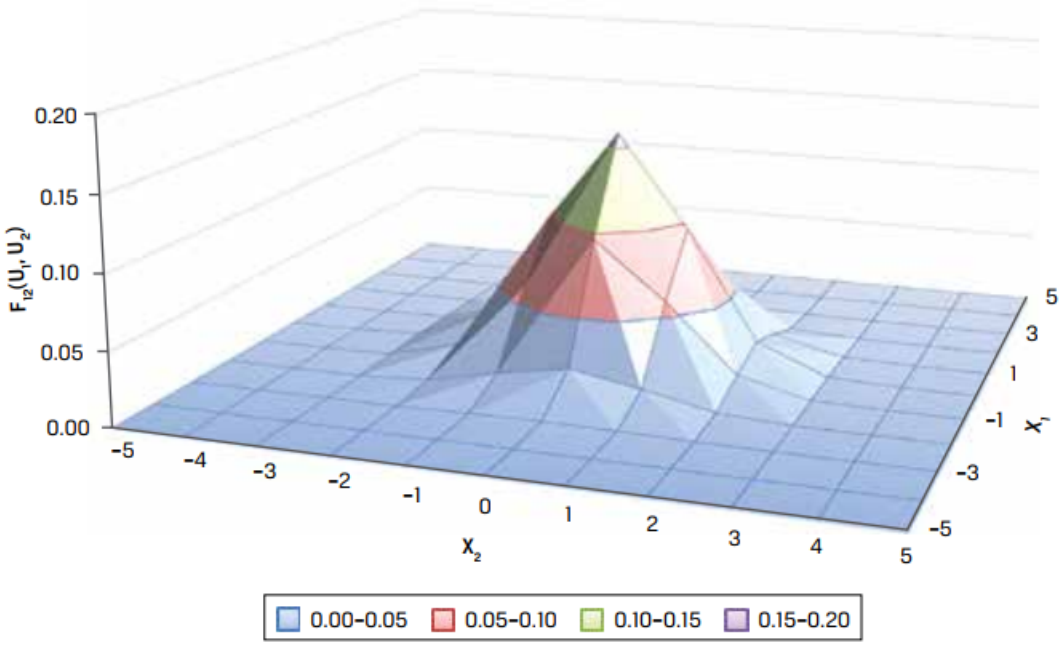


Figure 6.15: Joint Probability Density Surface Plot

Table 6.7: Returns for Technology and Non-Technology Firms

Rate of Return R	**Non-Technology**	**Technology**	**Total**
$R > 10\%$	100	60	160
$R \leq 10\%$	300	40	340
Total	**400**	**100**	**500**

$$\text{Continuous: } E[X|Y=y] = \int x f_{X|Y}(x|y)dx \tag{6.29}$$

The expected conditional variance is:

$$\text{Var}(X|Y=y) = E[X^2|Y=y] - (E[X|Y=y])^2 \tag{6.30}$$

The **Total Probability Rule for Expected Value** states:

$$E[X] = \sum_{i=1}^{n} E[X|S_i] \times P(S_i) \tag{6.31}$$

where $S_1, S_2, \ldots, S_n$ are mutually exclusive and exhaustive scenarios.

Case Study: The Effect of Tax Rebates on Spending

An economist analyzes value-added tax relief across two economic conditions:

- Strong economy ($Y = 1$, probability $P(Y = 1) = 0.30$)
- Weak economy ($Y = 0$, probability $P(Y = 0) = 0.70$)

Probabilities of scenarios are:

Scenario (Change in Spending)	**Strong Economy** $Y = 1$	**Weak Economy** $Y = 0$
Scenario 1: DKK 10,000	0.15	0.30
Scenario 2: DKK 1,000	0.80	0.40
Scenario 3: − DKK 2,000	0.05	0.30

The conditional expected changes in consumer spending are:

$$E[\text{Spending}|Y=1] = (0.15 \times 10,000) + (0.80 \times 1,000) + (0.05 \times -2,000) = \text{DKK } 2,200$$
$$E[\text{Spending}|Y=0] = (0.30 \times 10,000) + (0.40 \times 1,000) + (0.30 \times -2,000) = \text{DKK } 2,800$$

Case Study: Scenario Expected Returns and Variances

An analyst estimates portfolio return forecasts across three economic scenarios:

- **Depression** (prob 50%): −25% return (prob 50%), −10% return (prob 25%), −5% return (prob 15%), 0% return (prob 10%).
- **Recession** (prob 30%): −5% return (prob 30%), 0% return (prob 30%), 5% return (prob 30%), 15% return (prob 10%).
- **Expansion** (prob 40%? Wait, assuming scenarios are equally likely for simplicity, $P = 1/3$ each).

Conditional expected returns and variances are calculated as:

$$\begin{aligned}
E[R|\text{Depression}] &= 0.50(-0.25) + 0.25(-0.10) + 0.15(-0.05) + 0.10(0) = -15.75\% \\
\text{Var}[R|\text{Depression}] &= 0.50(-0.25 - (-0.1575))^2 + \cdots = 0.0093 \\
E[R|\text{Recession}] &= 0.30(-0.05) + 0.30(0) + 0.30(0.05) + 0.10(0.15) = 1.5\% \\
\text{Var}[R|\text{Recession}] &= 0.0035 \\
E[R|\text{Expansion}] &= 0.10(0.25) + 0.25(0.10) + 0.25(0.15) + 0.40(0.50) = 28.75\% \\
\text{Var}[R|\text{Expansion}] &= 0.0317
\end{aligned}$$

If scenarios are equally likely, the unconditional expected return is:

$$E[R] = \frac{-15.75\% + 1.50\% + 28.75\%}{3} = 4.83\%$$

The unconditional variance is:

$$\sigma^2 = \frac{1}{3}(-0.1575 - 0.0483)^2 + \frac{1}{3}(0.015 - 0.0483)^2 + \frac{1}{3}(0.2875 - 0.0483)^2 = 0.0234$$

6.3.3 Conditional Means, Variances, and Covariances

Incorporating conditioning information updates expected values:

- **Conditional Mean:** $E[X|S] = \sum_{i=1}^{n} x_i P(X_i|S)$
- **Conditional Variance:** $\text{Var}(X|S) = E[(X - E[X|S])^2|S]$
- **Conditional Covariance:**

$$\text{Cov}(X, Y|S) = E[(X - E[X|S])(Y - E[Y|S])|S] \tag{6.32}$$

Case Study: Three-Period Asset Price Evolution

Let $P_0 = 100$. Each period, the price rises by 10 with probability 0.60 or falls by 10 with probability 0.40. At Time 3, possible outcomes follow a binomial tree (Exhibit 6.16):

- $P_3 = 130$ (prob $0.6^3 = 0.216$)
- $P_3 = 110$ (prob $3 \times 0.6^2 \times 0.4 = 0.432$)
- $P_3 = 90$ (prob $3 \times 0.6 \times 0.4^2 = 0.288$)
- $P_3 = 70$ (prob $0.4^3 = 0.064$)

The unconditional price expectation at Time 3 is:

$$E[P_3] = 130(0.216) + 110(0.432) + 90(0.288) + 70(0.064) = 106$$

The unconditional variance is:

$$\text{Var}(P_3) = E[P_3^2] - (E[P_3])^2 = 11,524 - 106^2 = 288$$

Suppose at Time 1, the price is observed to be 110 ($P_1 = 110$, Exhibit 6.17). The conditional probabilities for Time 3 update to:

- $P_3 = 130$ (prob $0.6^2 = 0.36$)
- $P_3 = 110$ (prob $2 \times 0.6 \times 0.4 = 0.48$)
- $P_3 = 90$ (prob $0.4^2 = 0.16$)

The conditional expectation is:

$$E[P_3|P_1 = 110] = 130(0.36) + 110(0.48) + 90(0.16) = 114$$

The conditional variance is:

$$\text{Var}(P_3|P_1 = 110) = E[P_3^2|P_1 = 110] - E[P_3|P_1 = 110]^2 = 15,568 - 114^2 = 192$$

Suppose at Time 2, the price is 100 ($P_2 = 100$, Exhibit 6.18). The conditional statistics update to:

$$E[P_3|P_2 = 100] = 110(0.60) + 90(0.40) = 102$$

$$\text{Var}(P_3|P_2 = 100) = (110^2 \times 0.60 + 90^2 \times 0.40) - 102^2 = 96$$

Table 6.8 summarizes the updates:

Table 6.8: Expectation and Variance Under Different Conditions

Conditionality	**Expected Price**	**Variance**
Unconditional	106	288
Conditional on $P_1 = 110$	114	192
Conditional on $P_2 = 100$	102	96

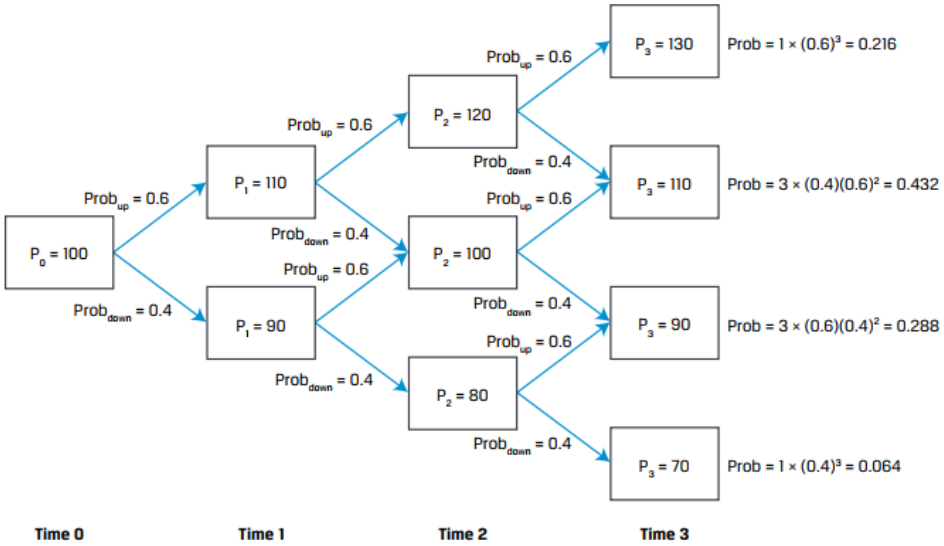


Figure 6.16: Potential Asset Price Evolution over Three Periods

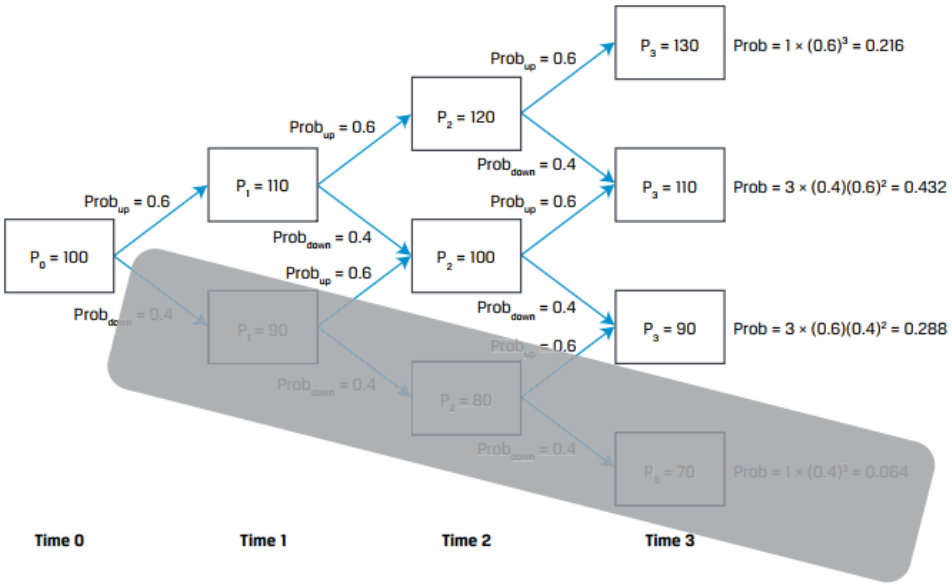


Figure 6.17: Price Evolution Given Time 1 Outcome $P_1 = 110$

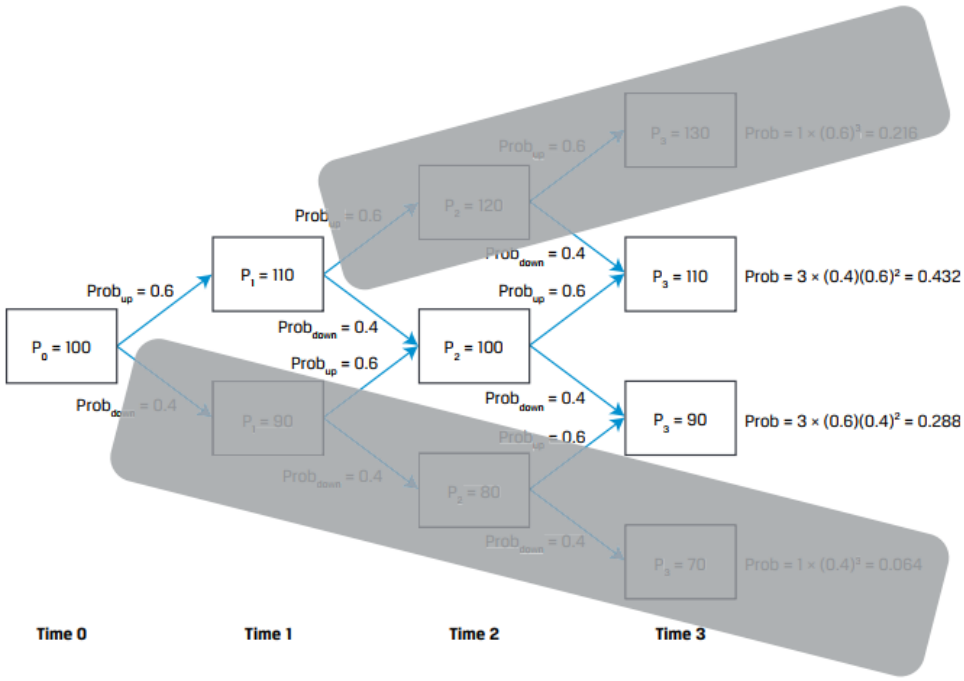


Figure 6.18: Price Evolution Given Time 2 Outcome $P_2 = 100$

6.4 Bayesian Updating

d | formulate investment problems through Bayesian updating

Bayes' theorem provides a formal framework for updating the probability of a hypothesis (prior probability) in light of additional data or evidence to form an updated probability (posterior probability).

- **A Priori Probability (Prior):** The probability of an event formulated before new data is considered.
- **A Posteriori Probability (Posterior):** The updated probability of the event after incorporating new observations.

Bayes' formula is stated as:

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)} \tag{6.33}$$

where:

- $P(A|B)$ is the posterior probability of A occurring given B is true.
- $P(B|A)$ is the likelihood of observing B given A is true.
- $P(A)$ is the prior probability of A.
- $P(B)$ is the marginal probability of observing B, computed using the total probability rule:

$$P(B) = \sum_i P(B|A_i) \times P(A_i) \tag{6.34}$$

*Thực tế: Định lý Bayes giúp chúng ta cập nhật nhận định khi có tin mới (ví dụ xác suất vỡ nợ, tăng trưởng cổ phiếu). Công thức: Posterior = (Conditional Likelihood / Total Probability) * Prior.*

Case Study: Technology Companies and Stock Returns

Using the contingency table for technology firms in Table 6.7, we calculate:

- $P(\text{Tech}) = 0.20$ and $P(\text{Non-tech}) = 0.80$
- $P(R > 10\%|\text{Tech}) = 0.60$ and $P(R > 10\%|\text{Non-tech}) = 0.25$

The overall probability of a firm having a return $R > 10\%$ is:

$$P(R > 10\%) = (0.60 \times 0.20) + (0.25 \times 0.80) = 0.32$$

Using Bayes' formula, the probability that a firm is a technology company given its return exceeds 10% is:

$$P(\text{Tech}|R > 10\%) = \frac{P(R > 10\%|\text{Tech}) \times P(\text{Tech})}{P(R > 10\%)} = \frac{0.60 \times 0.20}{0.32} = 37.5\%$$

This represents an increase from the unconditional prior probability of 20%.

Case Study: Sustainability Investments and Earnings Expectations

EcoMed A/S will release its annual earnings next week. Prior probabilities for consensus EPS outcomes are:

- $P(\text{Exceeded Consensus}) = 0.45$
- $P(\text{Met Consensus}) = 0.30$
- $P(\text{Fell Short}) = 0.25$

Before the release, EcoMed announces expanding sustainable practices. The likelihood of this announcement under each earnings scenario is estimated as:

- $P(\text{Sustainability Announcement}|\text{Exceeded}) = 0.75$
- $P(\text{Sustainability Announcement}|\text{Met}) = 0.20$
- $P(\text{Sustainability Announcement}|\text{Fell Short}) = 0.05$

First, we find the unconditional probability of the sustainability announcement:

$$P(\text{Sustainability}) = (0.75 \times 0.45) + (0.20 \times 0.30) + (0.05 \times 0.25) = 0.41$$

We update the probabilities using Bayes' formula:

$$P(\text{Exceeded}|\text{Sustainability}) = \frac{0.75 \times 0.45}{0.41} = 82.32\%$$
$$P(\text{Met}|\text{Sustainability}) = \frac{0.20 \times 0.30}{0.41} = 14.63\%$$
$$P(\text{Fell Short}|\text{Sustainability}) = \frac{0.05 \times 0.25}{0.41} = 3.05\%$$

Note that the updated posterior probabilities sum to exactly 1.0000:

$$0.8232 + 0.1463 + 0.0305 = 1.0000$$

Module 7

Estimation and Hypothesis Testing

LEARNING OUTCOMES	
a	explain the central limit theorem and the application of confidence intervals and sampling methodologies
b	explain hypothesis testing and its components, including statistical significance, Type I and Type II errors, and the power of a test; construct appropriate hypothesis tests; and interpret the results
c	compare and contrast parametric and non-parametric tests, describe situations in which each is the more appropriate type of test, construct appropriate hypothesis tests, and interpret the results

7.1 The Central Limit Theorem, Confidence Intervals, and Sampling

a	explain the central limit theorem and the application of confidence intervals and sampling methodologies

Chúng ta lưu ý: (1) CLT giải thích tại sao trung bình mẫu xấp xỉ phân phối chuẩn khi $n \geq 30$ với mọi biến gốc; (2) Khoảng tin cậy (Confidence Interval) lượng hóa độ chính xác ước lượng; (3) Sampling là bước nền tảng tránh thiên lệch dữ liệu.

7.1.1 The Central Limit Theorem (CLT)

The law of large numbers provides context for using the sample mean, $\bar{X}$, to estimate the population mean, μ. The central limit theorem (CLT) is a fundamental concept in statistics. It states that if random samples are taken from any distribution with a finite mean μ and finite variance σ^2, then the sample means, $\bar{X}$, will converge to a normal distribution as the sample size n increases.

Specifically:

- The expected value of the sample mean is equal to the population mean:

$$E[\bar{X}] = \mu \tag{7.1}$$

- The variance of the sampling distribution is:

$$\sigma^2_{\bar{X}} = \frac{\sigma^2}{n} \tag{7.2}$$

- The standard error of the sample mean (the standard deviation of the sample mean) is:

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} \quad \text{or} \quad s_{\bar{X}} = \frac{s}{\sqrt{n}} \tag{7.3}$$

The standardized format of the sample mean converges in distribution to the standard normal distribution $N(0,1)$ as sample size $n \to \infty$:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1) \quad \text{as } n \to \infty \tag{7.4}$$

For symmetric and unimodal distributions, a sample size as small as 4 or 5 might provide an adequate normal approximation. For skewed distributions, a sample size of at least 25–30 is typically necessary, whereas extremely skewed distributions may require even larger sample sizes.

Case Study: Die Rolling and the Central Limit Theorem

Assume a six-sided unfair die, biased to return a 6 more often than a fair die, is rolled 10,000 times. We create different samples of these rolls and calculate their means. As shown in Exhibit 7.1, as the sample size n increases from 1 to 50:

- The distribution of the sample mean converges to a normal distribution.
- The distribution centers more tightly around the true population mean (marked by the vertical line).
- The variability (standard deviation) of the sample means decreases.

An estimator $\hat{\theta}$ is:

- **Unbiased** if $E[\hat{\theta}] = \theta$. The sample mean is an unbiased estimator of the population mean ($E[\bar{X}] = \mu$).
- **Efficient** if it has the smallest possible standard error among all unbiased estimators. A minimum-variance unbiased estimator (MVUE) is both unbiased and efficient. The best linear unbiased estimator (BLUE) is both unbiased and maximally efficient among all linear estimators.
- **Consistent** if it converges to the true parameter value as the sample size increases. The sample mean $\bar{X}$ is a consistent estimator.
- **Robust** if it provides accurate parameter estimates even when the data include outliers or violate distribution assumptions (e.g., using trimmed means).

7.1.2 Confidence Intervals

Confidence intervals are probabilistic ranges that are likely to contain the true parameter value with a specified level of confidence $(1 - \alpha)$. For large sample sizes, the confidence interval for the population mean μ is:

$$\bar{X} \pm Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} \tag{7.5}$$

Where $Z_{\alpha/2}$ is the critical Z-score from the normal distribution, and $\sigma/\sqrt{n}$ is the standard error. Exhibit 7.2 illustrates the normal distribution curve and confidence intervals.

The margin of error, $Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$, is the half-width of the confidence interval. The two most frequently used confidence intervals are:

- **95% Confidence Interval:** $Z_{0.025} = 1.9600$, leaving 2.5% in each tail.
- **99% Confidence Interval:** $Z_{0.005} = 2.5760$, leaving 0.5% in each tail.

Case Study: Estimating the Monthly Return on Gold

Based on 34 years of monthly returns on gold ($n = 408$), the average monthly return is 0.49% and the sample standard deviation is 4.38%.

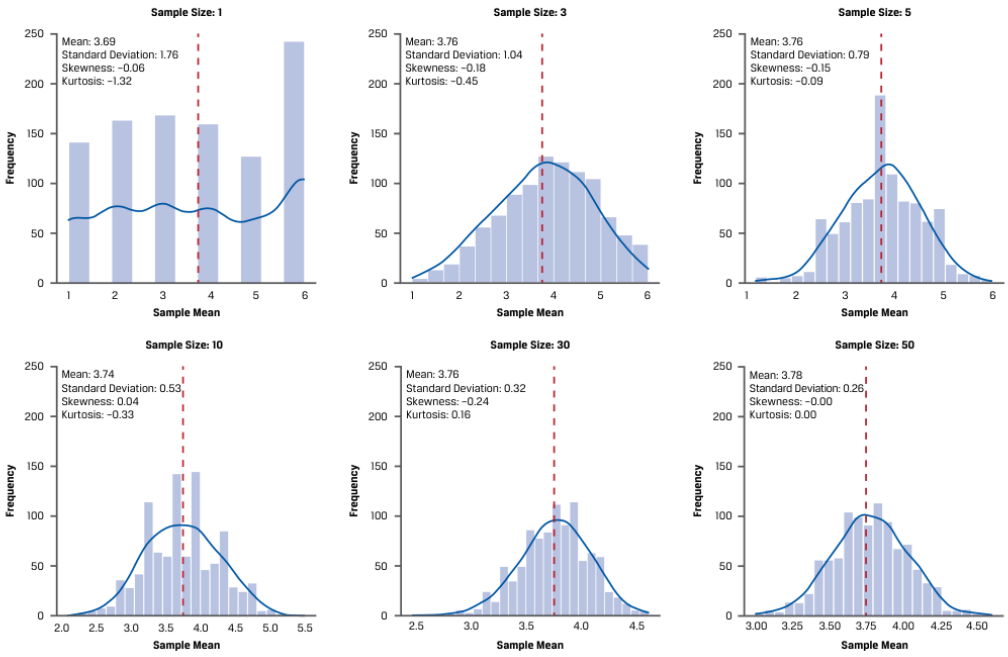
1. **Standard Error:** $SE = \frac{s}{\sqrt{n}} = \frac{0.0438}{\sqrt{408}} \approx 0.0022$ (or 0.22%).
2. **95% Confidence Interval:** $0.49\% \pm 1.9600 \times 0.22\% = 0.49\% \pm 0.43\% \implies [0.06\%, 0.92\%]$.
3. **99% Confidence Interval:** $0.49\% \pm 2.5760 \times 0.22\% = 0.49\% \pm 0.57\% \implies [-0.08\%, 1.06\%]$.

These intervals are shown in Exhibit 7.3. Higher confidence levels result in wider confidence intervals.

Case Study: Estimating the Average Price of Gold

Based on 8,820 weekly gold price data points, the average is GBP 867.83 and standard deviation is GBP 566.60 (Exhibit 7.4). The distribution of prices is bimodal and highly skewed. This violates the normal distribution assumption of the CLT, making absolute price confidence intervals unreliable. Thus,

Exhibit 1: The Central Limit Theorem



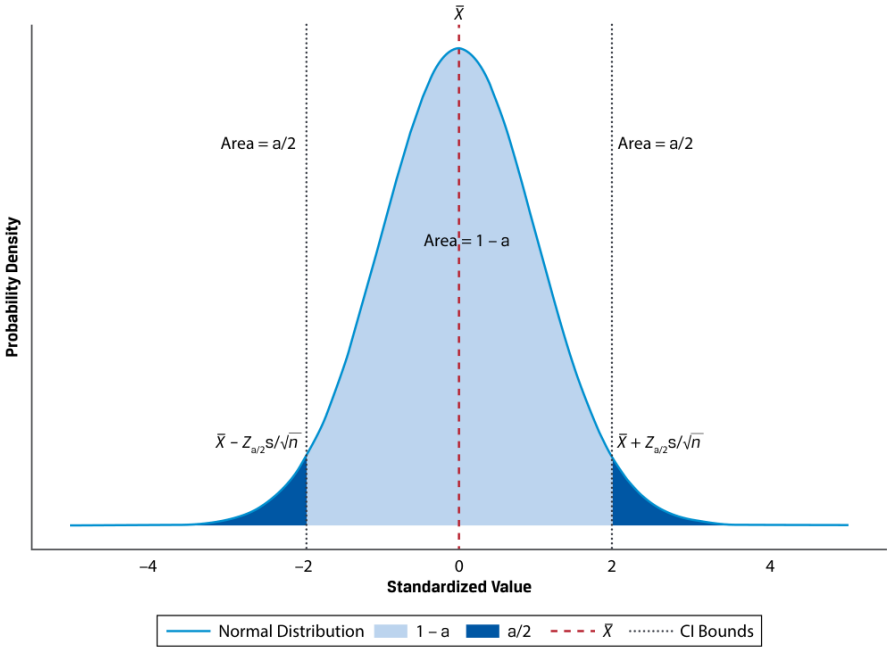
These graphs demonstrate the convergence of the sample mean to the true population mean with the decreasing variability of the mean when increasing the number of trials, illustrating the central limit theorem:

- As the sample size, n, increases, the distribution of sample means becomes more normally distributed and converges to the normal distribution.
- The distribution centers more tightly around the true population mean, indicated by the vertical line, with larger sample sizes.
- The variability of the sample means decreases as the sample size grows.

The practical implications are as follows: For symmetric and unimodal distributions, a sample size as small as 4 or 5 might provide an adequate approximation of the mean. For skewed distributions, a sample size of at least 25–30 is typically necessary for a reasonable approximation, but extremely skewed distributions may require even larger sample sizes for accurate estimation.

Figure 7.1: Histograms Showing the Convergence of the Sample Mean under CLT

Exhibit 2: The Normal Distribution Curve and Confidence Intervals



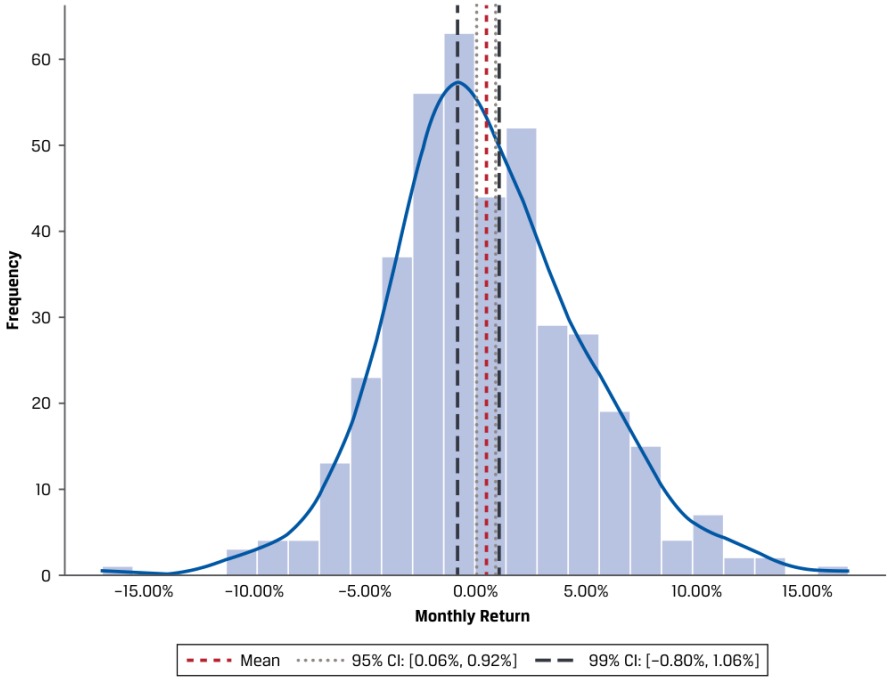
The **margin of error**, $z_{\frac{\alpha}{2}} \times \frac{\sigma}{\sqrt{n}}$, defines the range within which the true population parameter is expected, with a certain level of confidence, to fall on each side of the sample mean. It represents the sampling error and is the half-width of the confidence interval.

A particular observation's critical Z-score represents the number of standard deviations the observation is from the mean in a standard normal distribution. The two most frequently used confidence intervals are 95% and 99%:

- For a 95% confidence interval, a Z-score of 1.9600 indicates that 95% of the data sample observations fall within 1.9600 standard deviations of the mean, leaving 2.5% in each tail of the distribution.
- For a 99% confidence interval, a Z-score of 2.5760 indicates that 99% of the sample observations falls within 2.5760 standard deviations of the mean, leaving 0.5% in each tail of the distribution.

Figure 7.2: The Normal Distribution Curve and Confidence Intervals

Exhibit 3: Distribution of the Sample Mean with 95% and 99% Confidence Intervals



Based on historical data, we can state with 95% confidence that the average one-month return for gold in British pounds falls between 0.06% and 0.92%. With 99% confidence, this range widens to between −0.80% and 1.06%. It is important to note that higher confidence levels result in wider confidence intervals.

In theory, these estimates and intervals enable reliable statistical inference about the true mean, μ, based on the sample data and sample mean, $\bar{x}$, due to the normal approximation provided by the CLT. However, this approach assumes a finite variance and limited dispersion around the mean.

In financial data, a large dispersion of values or returns can result from either outliers or trends. This is typical for asset prices, which often show changing average values and variances, complicating forecasts using absolute prices. To address this, we often transform price data or build models that account for trends and changing variances. Calculating percentage changes can mitigate some of these issues by normalizing data and providing a more stable basis for forecasting and statistical inference.

CASE STUDY



Estimate the Average Price of Gold

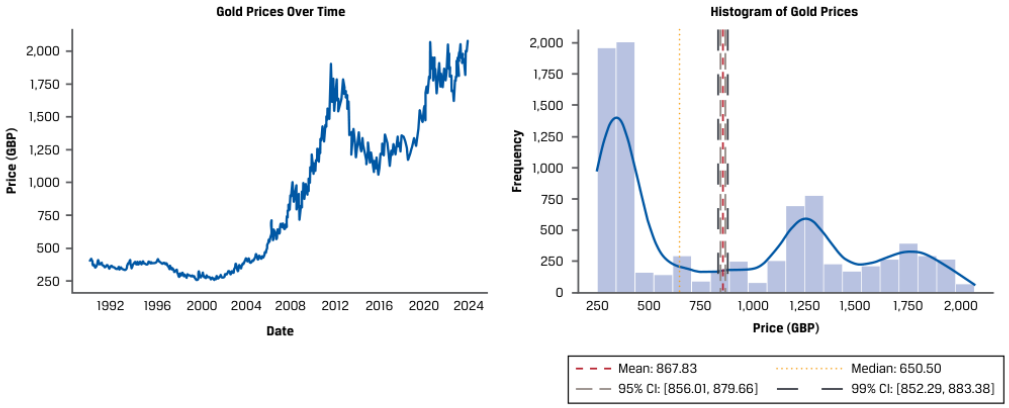
The analyst is now focusing on the weekly gold price. Based on 8,820 data points of the weekly gold prices in British pounds per troy ounce (approximately 31.1035 grams), the following descriptive statistics have been calculated:

- Average: GBP867.83
- Standard deviation: GBP566.60
- Minimum: GBP252.55
- Maximum: GBP2,077.49

Figure 7.3: Distribution of the Sample Mean with 95% and 99% Confidence Intervals

transforming data into returns (percentage changes) is necessary to stabilize the variance.

Exhibit 4: Gold Prices and Their Distribution over Time



Gold prices can be highly variable and exhibit trends, making their utility in forecasting limited. Confidence intervals also become less meaningful in this context. For example, when gold prices exceed GBP2,000, the historical mean offers little predictive power, even after accounting for confidence intervals.

The CLT assumes the population has a finite variance and is not highly skewed. Significant skewness or extreme values, such as those seen with gold prices, violate these assumptions, making the resulting confidence intervals less reliable. Such non-stationarity causes both the historical mean and confidence interval to lose their reliability because they fail to capture long-term trends and high variance.

An important example of using confidence intervals in finance is to quantify an investment's **value at risk (VaR)**. VaR provides a statistical measure of potential losses in a portfolio over a specific time horizon at a given confidence level. VaR represents the lower bound in a one-sided confidence interval around the expected value. VaR considers the investment's expected value, its variance, the entire return distribution, and the chosen confidence level.

VaR can be expressed in two primary ways:

- *Percentage of returns*: VaR can measure the percentage loss in returns that a portfolio might experience. A one-day VaR of 5% at a 95% confidence level means there is a 95% chance that the portfolio will not lose more than 5% of its value in one day.
- *Dollars or similar monetary amount*: VaR can express the actual amount that a portfolio might lose. A one-day VaR of GBP1 million at a 95% confidence level means there is a 95% chance that the portfolio will not lose more than GBP1 million in one day.

To illustrate, consider the VaR of holding gold. If the one-month VaR at the 99% one-sided confidence interval is computed as 1 million, it implies the following:

- $P(\text{Loss} > \text{GBP1 million}) = 1\%$.
- $P(\text{Loss} \leq \text{GBP1 million}) = 99\%$.

Figure 7.4: Gold Prices and Their Bimodal Distribution over Time

Exam Tip

Value at Risk (VaR) as a One-Sided Confidence Interval
VaR provides a statistical measure of potential losses in a portfolio over a specific time horizon at a given confidence level. It represents the lower bound of a one-sided confidence interval.

- $P(\text{Loss} > \text{VaR}) = \alpha$
- $P(\text{Loss} \leq \text{VaR}) = 1 - \alpha$

Assuming normally distributed asset returns and a mean return of zero, the parametric VaR is:

$$\text{VaR} = Z_\alpha \times \sigma \times P \tag{7.6}$$

Where Z_α corresponds to the one-sided percentile (e.g., 1.65 for 95%, 2.33 for 99%), σ is standard deviation, and P is the current position value (Exhibit 7.5).

Case Study: Estimating the Weekly VaR of Gold

An analyst forecasts potential loss of a spot gold position worth GBP 10 million. Monthly standard deviation is 5.02%. Assuming a mean return of 0% and normal distribution, the 99% one-month VaR is:

$$\text{VaR} = 2.33 \times 5.02\% \times \text{GBP } 10,000,000 = \text{GBP } 1,169,660$$

There is a 1% probability that the position will lose more than GBP 1,169,660 over the next month.

7.1.3 Sampling Methodologies

To extract information about a population, two sampling approaches can be used:

1. **Probability Sampling:** Every member of the population has a known, non-zero chance of selection.
   - **Simple Random Sampling:** Each element has an equal selection chance.
   - **Systematic Sampling:** Selects every k-th member of a population.
   - **Stratified Random Sampling:** Divides the population into strata based on criteria, then draws random samples proportionally from each stratum. This increases precision.
   - **Cluster Sampling:** Divides the population into small clusters, randomly selects clusters, and samples all members in selected clusters.
2. **Non-Probability Sampling:** Relies on judgment or convenience; does not guarantee representativeness, so the CLT does not reliably apply.
   - **Convenience Sampling:** Selecting accessible elements.
   - **Judgmental Sampling:** Selecting elements based on expertise.

Case Study: Sampling of Daily Returns

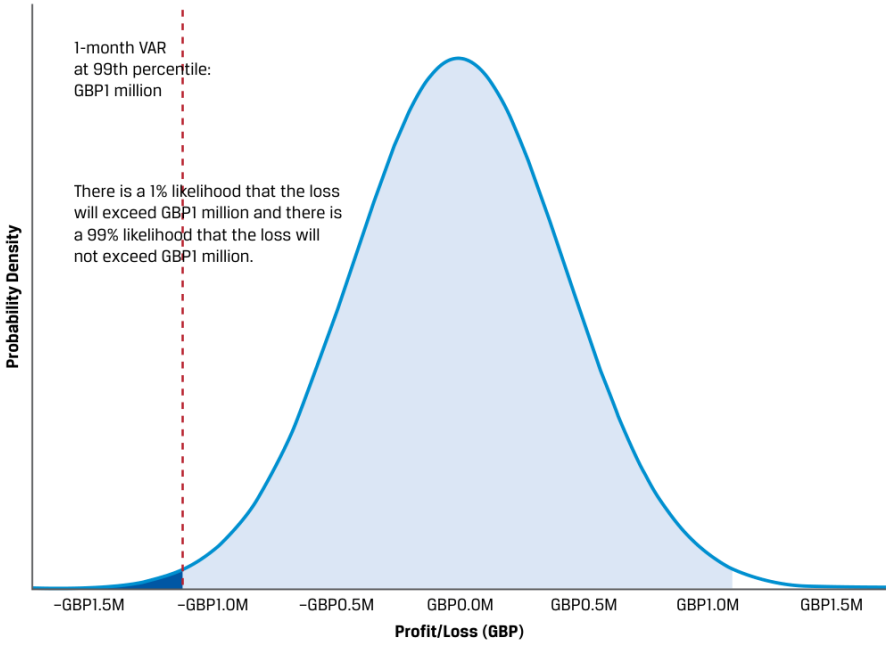
Based on 1,258 daily return observations of a global small-cap stock index, the true population mean is 0.035%. As shown in Exhibit 7.6, mean absolute errors decrease rapidly as sample size n increases. Stratified sampling (stratified by year) generally yields smaller errors than simple random sampling, although the advantage decreases as sample size grows.

7.2 Hypothesis Testing and Parametric Testing

b	explain hypothesis testing and its components, including statistical significance, Type I and Type II errors, and the power of a test; construct appropriate hypothesis tests; and interpret the results

Quy trình kiểm định giả thuyết gồm 4 bước (Exhibit 7.7). Chúng ta đặc biệt lưu ý lỗi Loại I (α - bác bỏ H_0 đúng), lỗi Loại II (β - không bác bỏ H_0 sai) và lực lượng kiểm định ($1 - \beta$). Nhớ phân phối tương ứng: kiểm định trung bình (t), một phương sai (χ^2), hai phương sai (F).

Exhibit 5: Monthly Gold Return Distribution and 99% VaR



Value at risk can be calculated using the parametric variance–covariance method, historical simulation, or the Monte Carlo simulation method. Our discussion will focus on the parametric variance–covariance method, which assumes normally distributed asset returns. The VaR, assuming a mean return of zero, is

$$\text{VaR} = Z \times \sigma \times P, \tag{5}$$

where

Z = the Z-score corresponding to the desired confidence level

σ = the standard deviation of the asset's or portfolio's returns

P = the current value of the asset or portfolio

Using a portfolio's current market value, the VaR estimate yields the potential value loss of the portfolio, providing a practical measure of potential value loss for financial professionals.

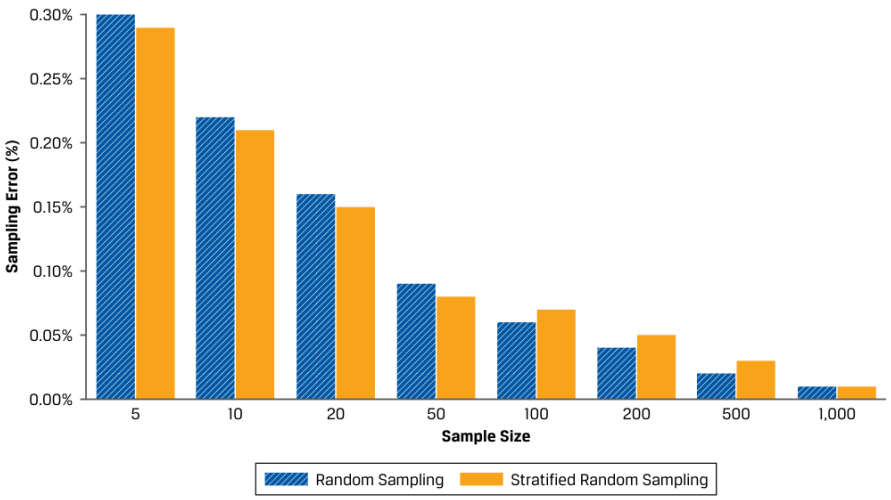
CASE STUDY

Estimate the Weekly VaR of Holding Gold

An analyst is forecasting the potential loss of a physical or spot gold position worth GBP10 million. The analysis is based on 44 years of monthly returns

Figure 7.5: Monthly Gold Return Distribution and 99% VaR

Exhibit 6: Mean Absolute Error of Two Sampling Approaches



Stratified sampling generally produces smaller mean absolute errors because it more accurately reflects population characteristics. This advantage decreases as the sample size increases. A minimum sample size is needed to limit sample error and achieve accuracy. Beyond around 200 observations, the incremental benefit from adding more observations is typically limited.

QUESTION SET 1

1. The central limit theorem states that the:

   A. population distribution mean will always be normally distributed regardless of the sample size.

   B. sample mean will always be an unbiased estimator of the population mean, regardless of the sample size.

   C. sampling distribution of the sample mean converges to a normal distribution as the sample size increases.

   Solution:

   The correct response is C. The CLT ensures that the distribution of the sample mean, $\bar{x}$, converges to a normal distribution as the sample size increases, regardless of the population's underlying actual distribution, provided the actual distribution meets a few conditions, such as fixed mean and finite variance.

   A is incorrect: The CLT says the distribution of the sample mean, $\bar{x}$ (not the the population mean, μ), converges to a normal distribution as the sample size increases.

Figure 7.6: Mean Absolute Error of Random vs Stratified Sampling

7.2.1 The Hypothesis Testing Process

Hypothesis testing involves four steps, as outlined in Exhibit 7.7:

1. State the null hypothesis (H_0) and alternative hypothesis (H_A).
2. Identify the test statistic and its probability distribution.
3. Specify the level of significance (α).
4. Establish the decision rule, calculate the statistic, and make the decision.

Hypothesis tests can be formulated in three ways (Exhibit 7.8 & Exhibit 7.1):

- **Right-tail test:** Tests whether a parameter θ is significantly greater than θ_0 ($H_0 : \theta \leq \theta_0$ vs $H_A : \theta > \theta_0$).
- **Left-tail test:** Tests whether a parameter θ is significantly less than θ_0 ($H_0 : \theta \geq \theta_0$ vs $H_A : \theta < \theta_0$).
- **Two-tailed test:** Tests whether a parameter θ is significantly different from θ_0 ($H_0 : \theta = \theta_0$ vs $H_A : \theta \neq \theta_0$).

Table 7.1: General Formulation of Hypotheses

Test Type	Null Hypothesis (H_0)	Alternative (H_A)	Rejection Region
Right-tail test	$\theta \leq \theta_0$	$\theta > \theta_0$	Right tail
Left-tail test	$\theta \geq \theta_0$	$\theta < \theta_0$	Left tail
Two-tailed test	$\theta = \theta_0$	$\theta \neq \theta_0$	Both tails

Exhibit 7.9 depicts the rejection areas for the three types of hypothesis tests.

7.2.2 Type I and Type II Errors

The decision-making process in hypothesis testing involves two types of errors:

- **Type I Error (False Positive):** Rejecting a true null hypothesis. The probability of a Type I error is the level of significance, α.
- **Type II Error (False Negative):** Failing to reject a false null hypothesis. The probability of a Type II error is denoted by β.

Table 7.2: Decisions in Hypothesis Testing

Decision	Null Hypothesis (H_0) is True	Null Hypothesis (H_0) is False
Fail to reject H_0	Correct Decision ($1 - \alpha$)	Type II Error (β, False Negative)
Reject H_0	Type I Error (α, False Positive)	Correct Decision ($1 - \beta$, Power of the Test)

The relation between α and β is shown in Exhibit 7.10. Reducing α increases β for a fixed sample size. To decrease both errors simultaneously, we must increase the sample size n.

7.2.3 Parametric Tests for a Single Variable

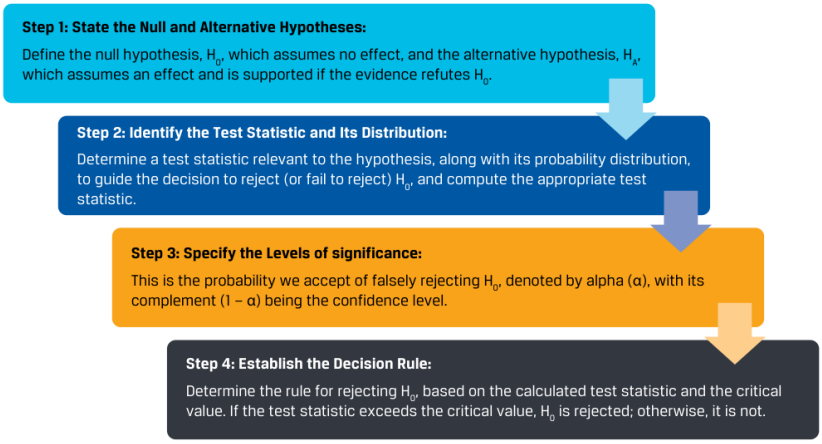
We utilize annual return data from 1979 to 2023 ($n = 44$) for the MSCI World Value and Growth Indexes to illustrate parametric tests (Exhibit 7.11 & Table 7.3).

Test of a Single Mean

When the population standard deviation is unknown, the t-test statistic with $n - 1$ degrees of freedom is:

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \tag{7.7}$$

Exhibit 7: Steps of the Hypothesis Testing Process



State the Null and Alternative Hypotheses

The first step in hypothesis testing is to state the two mutually exclusive and collectively exhaustive complementary hypotheses:

- The **null hypothesis**, H_0, is a statement about a population parameter, assumed to be true or equal to a specific parameter, unless refuted by sufficient evidence from the sample.
- The **alternative hypothesis**, H_A, is supported if sufficient evidence from the sample indicates the rejection of the null hypothesis.

Typically, the null hypothesis represents the status quo, suggesting no difference or no effect. It is formulated to state that a parameter is equal to, less than or equal to, or greater than or equal to a specified value. Conversely, the alternative hypothesis is the claim to be proven, indicating the presence of an effect, a difference, or a relationship:

- Null hypothesis, H_0: Parameter θ is equal to, less than or equal to, or greater than or equal to a specified value $\hat{\theta}$.
- Alternative hypothesis, H_A: Parameter θ differs from, is greater than, or is less than the specified value $\hat{\theta}$.

Consider a scenario where we want to test whether a new investment strategy leads to higher returns compared to the current strategy. In this case, the null and alternative hypotheses would be formed as follows:

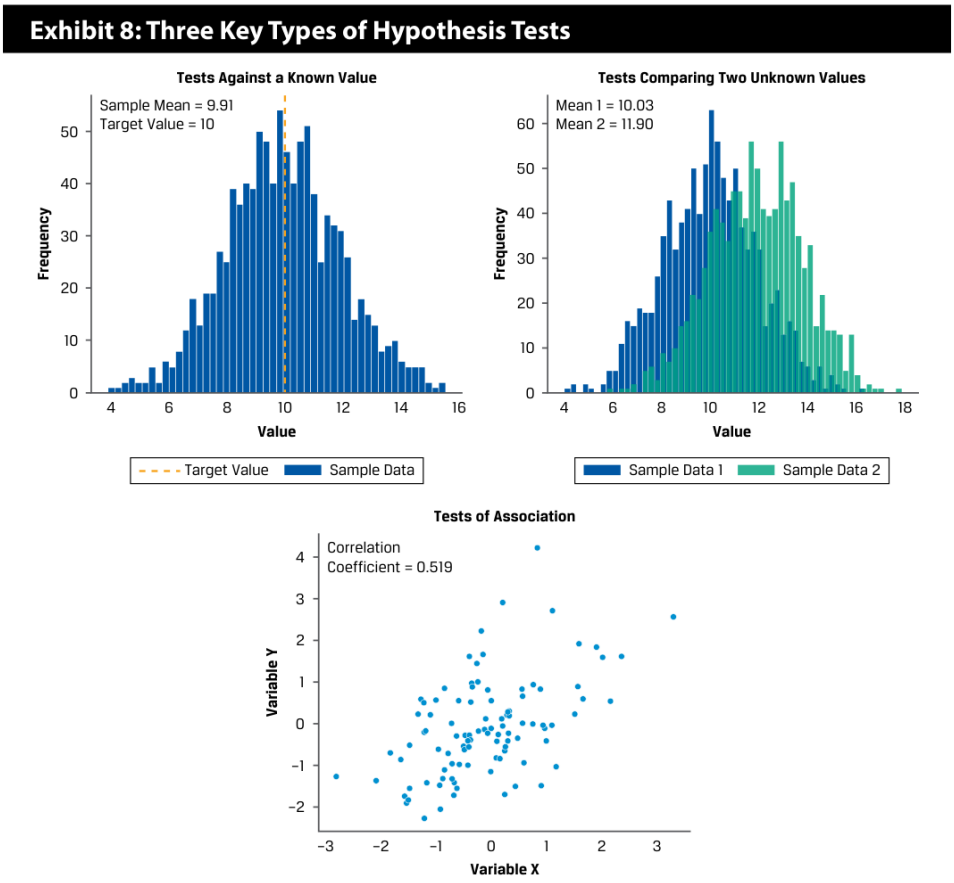
- Null hypothesis, H_0: The new investment strategy does not lead to higher returns; that is, the new strategy's returns are less than or equal to the current strategy's returns:

$$\mu_{new} \leq \mu_{current}.$$

- Alternative hypothesis, H_A: The new investment strategy leads to higher returns compared to the current strategy:

$$\mu_{new} > \mu_{current}.$$

Figure 7.7: Steps of the Hypothesis Testing Process



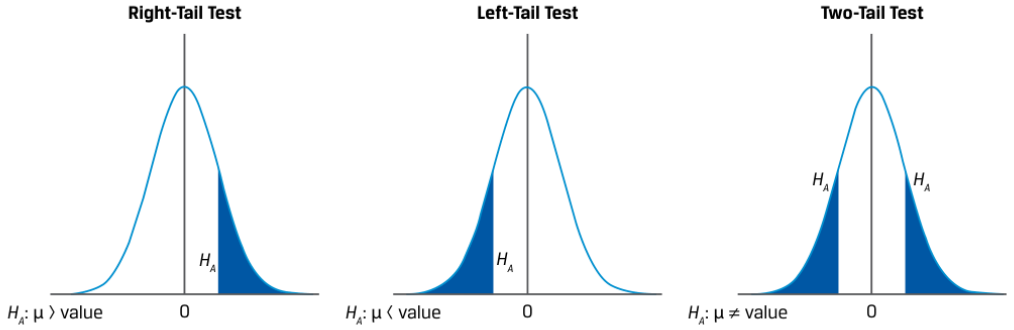
- *Tests against a known value*: These tests compare a sample's mean or variance with respect to a known value, such as a theoretical or empirical statistical benchmark. For instance, a fund manager might test whether the average portfolio return meets or exceeds a 10% target return. The null hypothesis, H_0, states that the portfolio return is equal to or lower than the target return, while the alternative hypothesis, H_A, states that the return is higher than the target.
- *Tests comparing two unknown values*: These tests compare the means or variances of two different financial datasets. For instance, comparing the average returns of two different investment portfolios or the volatility of two investments helps investors evaluate options. The null hypothesis, H_0, states that there is no significant difference between the returns or variances, while the alternative hypothesis, H_A, states that there is a difference.

Figure 7.8: Three Key Types of Hypothesis Tests

Table 7.3: Descriptive Statistics for MSCI World, Value, and Growth Indexes

Metric	MSCI World	Growth Index	Value Index
Average annual return ($\bar{X}$)	7.91%	8.81%	7.17%
Sample variance (s^2)	0.0381	0.0456	0.0362
Sample standard deviation (s)	0.1952	0.2134	0.1902
Number of observations (n)	44	44	44

Exhibit 10: The Rejection Areas of Three Types of Hypothesis Test



Identify the Test Statistic and Its Distribution

The second step in hypothesis testing is to identify the appropriate test statistic. This value is calculated from a sample and, along with a decision rule, is used to decide whether to reject the null hypothesis. The choice of test statistic depends on what is being tested, as summarized in Exhibit 11.

The first four tests are order-independent: The order of the data does not matter. For example, rearranging the dates in a return time series will not change the test statistics. Testing a single mean or variance involves comparing the parameter to a specific number. Testing the difference in means or variances involves comparing two parameters.

The last three tests are order-dependent: These tests depend on the order of the data because they capture relationships, such as paired differences, correlations, and other relationships.

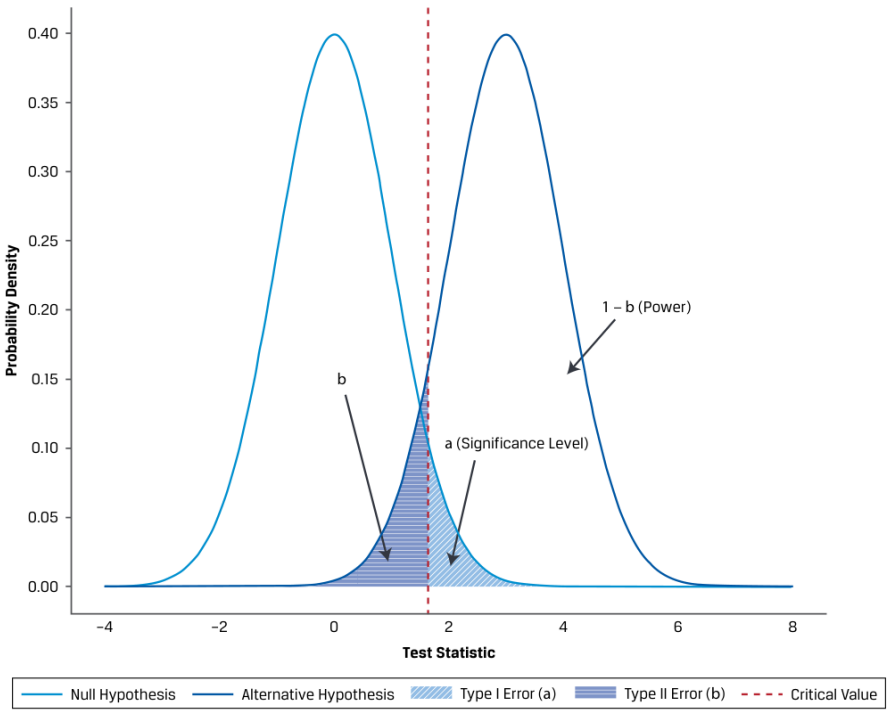
The sampling distribution of the mean return on an asset is *t*-distributed when the population standard deviation is unknown and normally distributed. It is *Z*-distributed when the population standard deviation is known. Since the population standard deviation is unknown in almost all cases, our focus is on using the *t*-distributed test statistic. However, when n is large, the *t*-distribution converges to the normal distribution.

Specify the Levels of Significance

The third step in hypothesis testing is to specify the **levels of significance**, or α, which reflects the strength of evidence needed to reject the null hypothesis. This choice depends on the nature of the hypothesis and the severity of potential errors. The levels of significance determine the critical value against which the test statistic

Figure 7.9: Rejection Regions for One-Tailed and Two-Tailed Tests

Exhibit 13: Type I and Type II Errors



Establish the Decision Rule

The fourth and final step in hypothesis testing is to establish the decision rule that compares the calculated test statistics to the critical values. The choice of distribution for this comparison depends on the specific parameters being tested and the characteristics of the sample:

- The *normal Z-distribution* is used when the sample size is large ($n \geq 30$), the population standard deviation is known, or the population is normally distributed.
- The *t-distribution* is used when the sample size is small ($n < 30$) or the population standard deviation is unknown.

Note that the sample size standards mentioned for mean testing do not apply to chi-square distributions and *F*-distributions.

After selecting the appropriate distribution, we determine the critical values. These can be derived from statistical tables but are more commonly computed using software such as Excel, R, Python, MATLAB, or Mathematica. We will later provide a list of Excel and Google Sheets built-in functions for hypothesis testing.

In hypothesis testing, we often use the principle "big is bad for the null." This means that a large test statistic value is disadvantageous to the null hypothesis: It suggests that the observed data are inconsistent with the null hypothesis and support the alternative hypothesis, which is the hypothesis of change or effect. The decision rule can be stated as follows:

- If the absolute value of the test statistic exceeds the critical value, we reject the null hypothesis in favor of the alternative hypothesis, considering the

Figure 7.10: The Trade-Off Between Type I and Type II Errors

Exhibit 14: Comparative Performance of MSCI World, Growth, and Value Indexes, 1979–2023

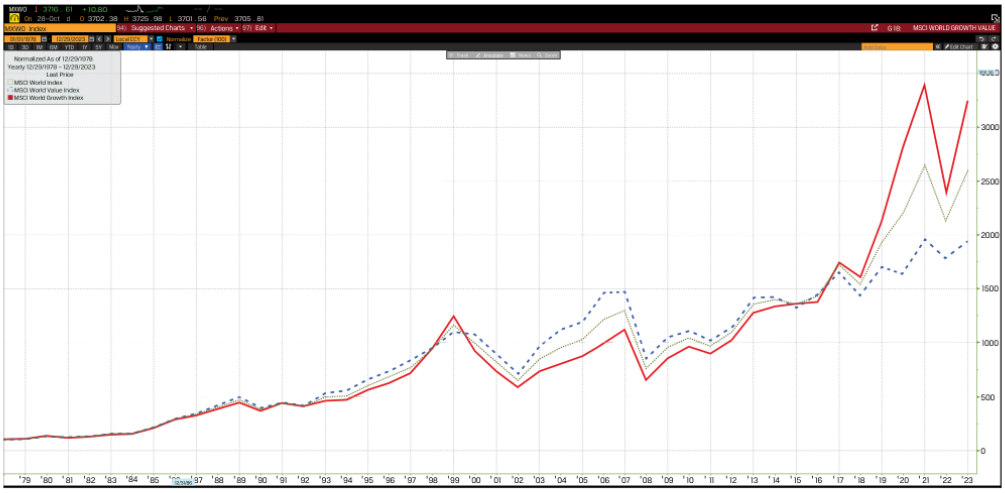


Exhibit 15 shows descriptive statistics for the annual returns for the performance of these indexes. The information in this exhibit will be used throughout the learning module as we test for the difference in performance between the value and growth

Figure 7.11: Performance of MSCI World, Growth, and Value Indexes, 1979–2023

Case Study: Testing MSCI World Value Index Mean Return

We test whether the average return of global value stocks is greater than zero ($H_0 : \mu \leq 0$ vs $H_A : \mu > 0$).

1. **Calculate Test Statistic:** $t = \frac{0.0717-0}{0.1902/\sqrt{44}} \approx 2.5001$.
2. **Decision Rule ($df = 43$):** One-tailed critical values are $t_{0.05} = 1.6811$ and $t_{0.01} = 2.4163$ (Exhibit 7.12).
3. **Conclusion:** Since $2.5001 > 2.4163$, we reject H_0 at both the 5% and 1% levels. The average return is significantly greater than zero.

Case Study: Testing MSCI World Growth Index Mean Return

We test whether the average return of global growth stocks is different from zero ($H_0 : \mu = 0$ vs $H_A : \mu \neq 0$).

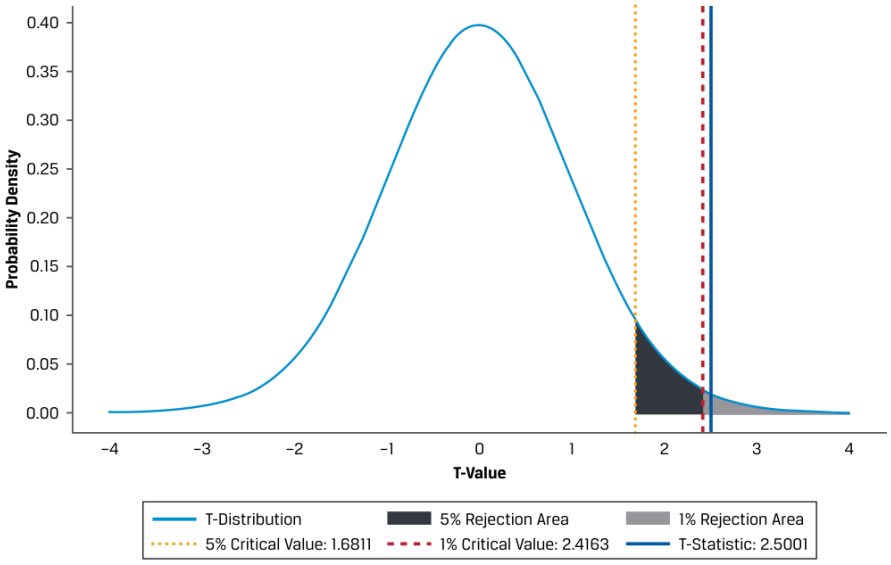
1. **Calculate Test Statistic:** $t = \frac{0.0881-0}{0.2134/\sqrt{44}} \approx 2.7373$.
2. **Decision Rule ($df = 43$):** Two-tailed critical values are ± 2.0167 (at 5%) and ± 2.6951 (at 1%) (Exhibit 7.13).
3. **Conclusion:** Since $2.7373 > 2.6951$, we reject H_0 at both levels. Growth stock returns are significantly different from zero.

Test of a Single Variance

To test the variance of a single normally distributed population, we use the Chi-square (χ^2) test statistic with $n-1$ degrees of freedom (Exhibit 7.14):

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} \tag{7.8}$$

Exhibit 16: Critical Values of a One-Sided Hypothesis Test for Means



2. Based on the annual returns for the MSCI World Growth Index in Exhibit 15, develop a set of hypotheses and test whether the average annual return from holding an index of global growth stocks is greater than zero at the 95% and 99% confidence levels.

   Solution:

   Step 1: State the hypotheses.

   In the hypothesis testing framework, we can directly define the following:

   - *Null hypothesis*, H_0: The annual return of the MSCI World Growth Index is not greater than zero ($\mu \leq 0$).
   - *Alternative hypothesis*, H_A: The annual return of the MSCI World Growth Index is greater than zero ($\mu > 0$).

   Step 2: Identify the test statistic and distribution.

   Given the data in Exhibit 15:

   - The average annual return is $\bar{x} = 8.81\% = 0.0881$.
   - The sample standard deviation is $s = 21.34\% = 0.2134$.
   - The sample size is $n = 44$.
   - The test statistic used is the t-statistic, which follows a t-distribution with $n - 1 = 43$ degrees of freedom, and it is calculated as

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{0.0881 - 0}{0.2134/\sqrt{44}} \approx 2.7373.$$

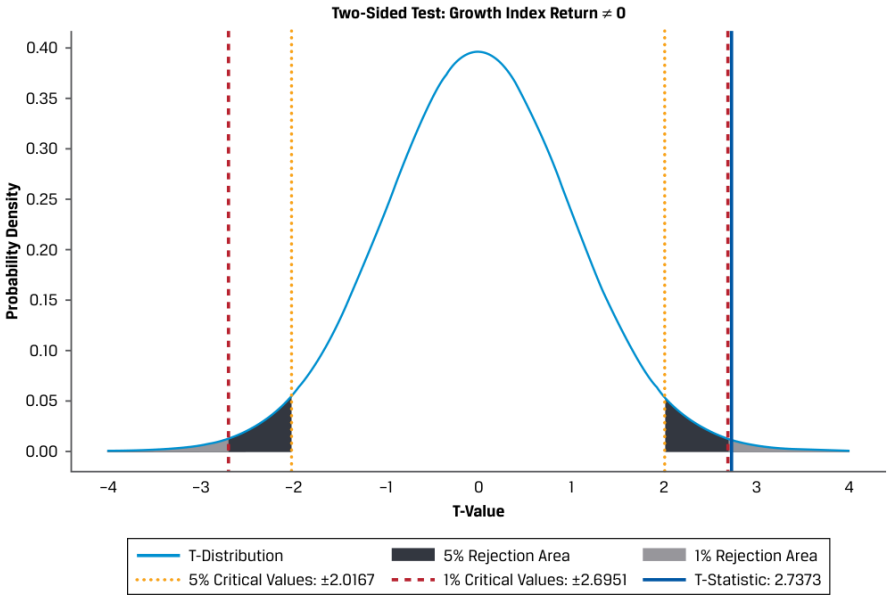
   Step 3: Specify the levels of significance.

   At the 95% and 99% confidence levels, the levels of significances, α, are 5% and 1%, respectively. The levels of significance, α, determine the probability threshold for rejecting H_0. Common values for α include 0.05 (5%) and 0.01 (1%), corresponding to confidence levels of 95% and 99%, respectively.

   Step 4: Establish the decision rule.

Figure 7.12: One-Sided Mean Test Critical Values and Rejection Regions

Exhibit 17: Critical Values of a Two-Sided Hypothesis Test for Means



Thus, based on our analysis, we have sufficient evidence to conclude that the average annual return of the MSCI World Growth Index is significantly different from zero at both the 1% and 5% significance levels.

Based on the results of the hypotheses tests in the previous examples, we can conclude that there is sufficient evidence at both the 1% and 5% significance levels that the average annual return from holding an index of

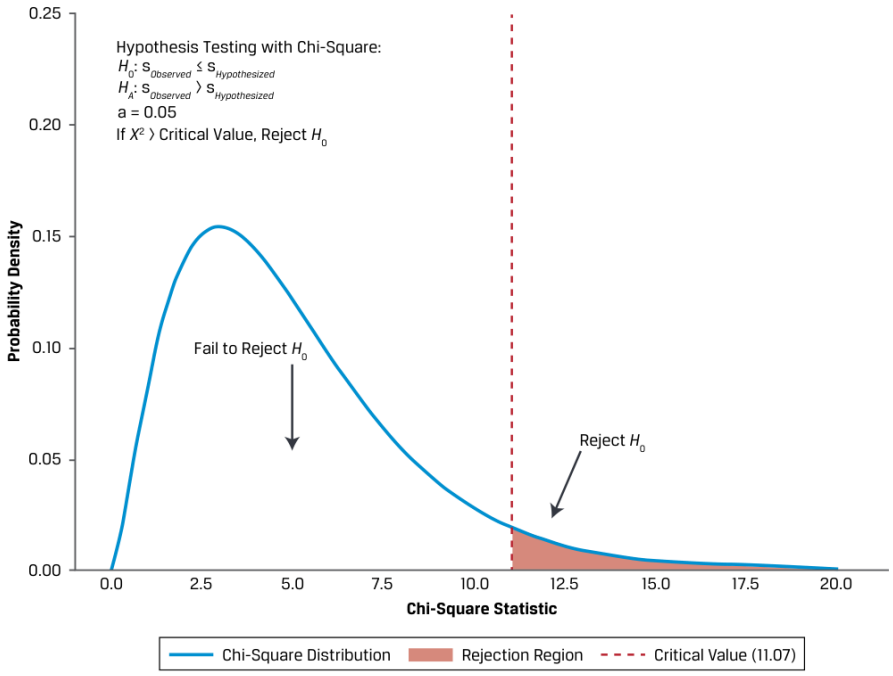
- global value stocks is greater than zero,
- global growth stocks is greater than zero, and
- global growth stocks is different from zero.

Note that the second conclusion implies the third conclusion, but we conducted the test for completeness and comprehensiveness.

These results suggest that both value and growth stocks have historically provided positive returns over the long term, which can be valuable information for investors considering investment choices and portfolio allocation strategies. The statistical significance of these findings at both the 1% and 5% levels provides strong support for the reliability of these conclusions. Note, however, that past performance does not guarantee future results, and investors should consider multiple factors when

Figure 7.13: Two-Sided Mean Test Critical Values and Rejection Regions

Exhibit 18: Hypothesis Testing with a Chi-Square Distribution



CASE STUDY

Variance of Long-Run Returns

1. Based on the variance of the annual returns for the MSCI Value Index in Exhibit 15, develop a set of hypotheses and test whether the observed variance is greater than a hypothesized variance of 0.0381.

   Solution:

   The steps of the hypothesis testing process are as follows:

   Step 1: State the hypotheses.

   We establish two competing hypotheses:

   - *Null hypothesis*, H_0: The sample variance of the returns from the MSCI World Value Index returns is less than or equal to the hypothesized variance, 0.0381: $\sigma^2_{Value} \leq \sigma^2_0$.
   - *Alternative hypothesis*, H_A: The sample variance of the returns from the MSCI World Value Index is greater than the hypothesized variance, 0.0381: $\sigma^2_{Value} > \sigma^2_0$.

   Step 2: Identify the test statistic and distribution.

   Given the data:

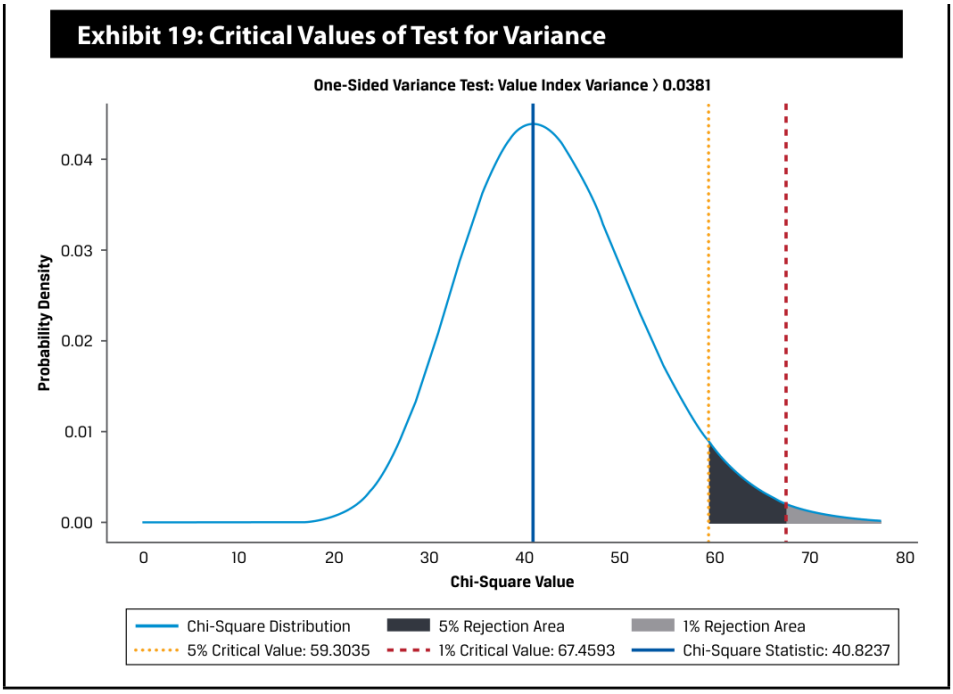
   - The variance for the MSCI Value Index returns is 0.0362.
   - The hypothesized variance is 0.0381.
   - The sample size is $n = 44$.

Figure 7.14: Hypothesis Testing with a Chi-Square Distribution ($df = 5$)

Case Study: Testing MSCI Value Index Variance

We test whether the variance of the Value Index is greater than 0.0381 ($H_0 : \sigma^2 \leq 0.0381$ vs $H_A : \sigma^2 > 0.0381$).

1. **Calculate Test Statistic ($df = 43$):** $\chi^2 = \frac{43 \times 0.0362}{0.0381} \approx 40.8237$.
2. **Decision Rule:** Right-tailed critical values are $\chi^2_{0.05} = 59.3035$ and $\chi^2_{0.01} = 67.4593$ (Exhibit 7.15).
3. **Conclusion:** Since $40.8237 < 59.3035$, we fail to reject H_0. The variance is not significantly greater than 0.0381.



Parametric Tests for Multiple Variables

We often need to evaluate two sets of data to determine whether there is a statistically significant relationship between them. In finance, we frequently seek to compare returns and variances from different investments, portfolios, or trading strategies. There are

Figure 7.15: One-Sided Variance Test Critical Values and Rejection Regions

7.2.4 Parametric Tests for Multiple Variables

We compare the means and variances of two populations.

Test of Equality of Two Unknown Means (Independent Samples)

Assuming equal population variances, the t-test statistic with $n_1 + n_2 - 2$ degrees of freedom is:

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - \Delta_0}{\sqrt{\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2}}} \quad \text{where} \quad s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} \tag{7.9}$$

Case Study: Comparing Growth vs Value Average Returns

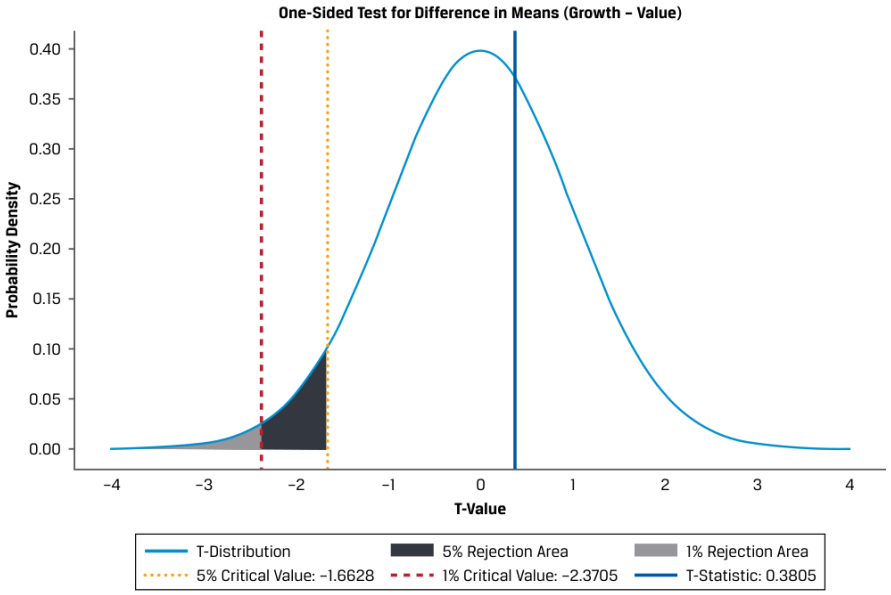
We test whether Growth average returns exceed Value returns ($H_0 : \mu_G - \mu_V \leq 0$ vs $H_A : \mu_G - \mu_V > 0$).

1. **Calculate Test Statistic ($df = 86$):**

$$t = \frac{0.0881 - 0.0717 - 0}{\sqrt{\frac{0.0456}{44} + \frac{0.0362}{44}}} = \frac{0.0164}{0.0431} \approx 0.3805$$

2. **Decision Rule:** One-tailed critical values are $t_{0.05} = 1.6628$ and $t_{0.01} = 2.3705$ (Exhibit 7.16).
3. **Conclusion:** Since $0.3805 < 1.6628$, we fail to reject H_0. The difference in average returns is not statistically significant.

Exhibit 20: Critical Values for Testing the Equality of Two Unknown Means



The evidence is insufficient to conclude that the mean difference in annual returns for the MSCI World Growth Index exceeds that of the MSCI World Value Index at both the 5% and the 1% significance levels.

Tests of the Differences in Variances

Performing a test of the differences between two population variances requires specifying the null hypothesis that the variances are equal. This is the baseline for comparison.

We can formulate hypotheses to test whether the variance of one population is equal to the variance of another population or whether they differ, as shown in the following example:

- *Null hypothesis*, H_0: The variance of the first populations is less than or equal to the variance of the second population ($\sigma_1^2 \leq \sigma_2^2$).
- *Alternative hypothesis*, H_A: The variance of the first populations is greater than that of the second population ($\sigma_1^2 > \sigma_2^2$).

The *F*-test statistic is the appropriate choice to compare variances of two normally distributed populations. Typically, the population with the larger sample variance is designated as σ_1^2 unless there is a strong theoretical reason to assign labels differently.

The formula for the *F*-test statistic is

$$F = \frac{s_1^2}{s_2^2}, \tag{9}$$

where

s_1^2 = the sample variance of the first population

s_2^2 = the sample variance of the second population

Figure 7.16: Independent Means Test Critical Values and Rejection Regions

Test of Difference in Two Variances (F-test)

To compare variances of two independent, normally distributed populations, the F-test statistic is:

$$F = \frac{s_1^2}{s_2^2} \quad (\text{where } s_1^2 \geq s_2^2) \tag{7.10}$$

With $df_1 = n_1 - 1$ and $df_2 = n_2 - 1$ (Exhibit 7.17).

Case Study: Comparing Growth vs Value Variances

We test whether Growth variance is greater than Value variance ($H_0 : \sigma_G^2 \leq \sigma_V^2$ vs $H_A : \sigma_G^2 > \sigma_V^2$).

1. **Calculate Test Statistic ($df_1 = 43, df_2 = 43$):** $F = \frac{0.0456}{0.0362} \approx 1.2595$.
2. **Decision Rule:** Critical values are $F_{0.05} = 1.6607$ and $F_{0.01} = 2.0569$ (Exhibit 7.18).
3. **Conclusion:** Since $1.2595 < 1.6607$, we fail to reject H_0. Growth variance is not significantly larger.

Test of Paired Differences (Dependent Samples)

For dependent observations, the t-test statistic with $n - 1$ degrees of freedom is:

$$t = \frac{\bar{d} - \mu_{d0}}{s_d/\sqrt{n}} \tag{7.11}$$

Where $\bar{d}$ is the sample mean difference, and s_d is the sample standard deviation of differences.

Case Study: Paired Difference of Growth and Value Returns

We test whether the mean difference in paired returns is different from zero ($H_0 : \mu_d = 0$ vs $H_A : \mu_d \neq 0$). We calculate $\bar{d} = 1.64\%$ and $s_d = 10.62\%$.

1. **Calculate Test Statistic ($df = 43$):** $t = \frac{0.0164 - 0}{0.1062/\sqrt{44}} \approx 1.0243$.
2. **Decision Rule:** Two-tailed critical values are ± 2.0167 (at 5% significance, Exhibit 7.19).
3. **Conclusion:** Since $1.0243 < 2.0167$, we fail to reject H_0. The paired differences are not statistically significant.

Parametric Test of Correlations

To test whether the Pearson correlation coefficient between X and Y is zero ($H_0 : \rho = 0$ vs $H_A : \rho \neq 0$), we use the t-test statistic with $n - 2$ degrees of freedom:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \tag{7.12}$$

As sample size n increases, the standard error decreases, causing the power of the test to increase. Thus, in very large datasets, even minor correlations are rejected as statistically different from zero (Exhibit 7.20).

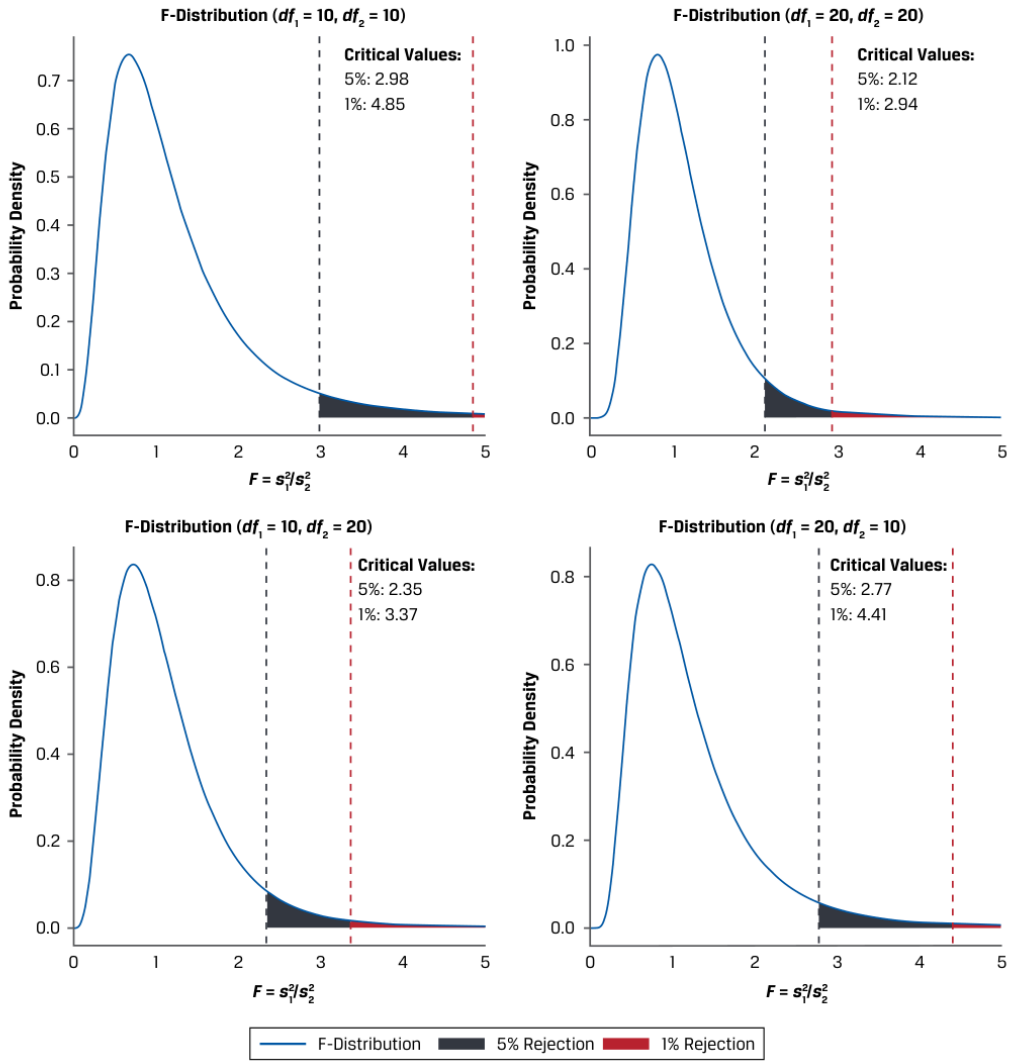
Case Study: Fund Correlations and STOXX50 Index

Table 7.4 presents correlations of four funds with the STOXX50 index ($n = 36$, $df = 34$). The critical t-value at the 5% significance level is ± 2.032. For Fund 3 and Fund 4:

$$t = \frac{0.3102\sqrt{34}}{\sqrt{1 - 0.3102^2}} \approx 1.903$$

Since $1.903 < 2.032$, the correlation between Fund 3 and Fund 4 is not statistically significant. However, all other correlations (e.g., Fund 2 and STOXX50, $r = 0.8223 \implies t = 8.426$) are statistically significant, suggesting limited diversification benefits except between Fund 3 and 4.

Exhibit 21: *F*-Distributions with Different df_1 and df_2 Combinations



CASE STUDY

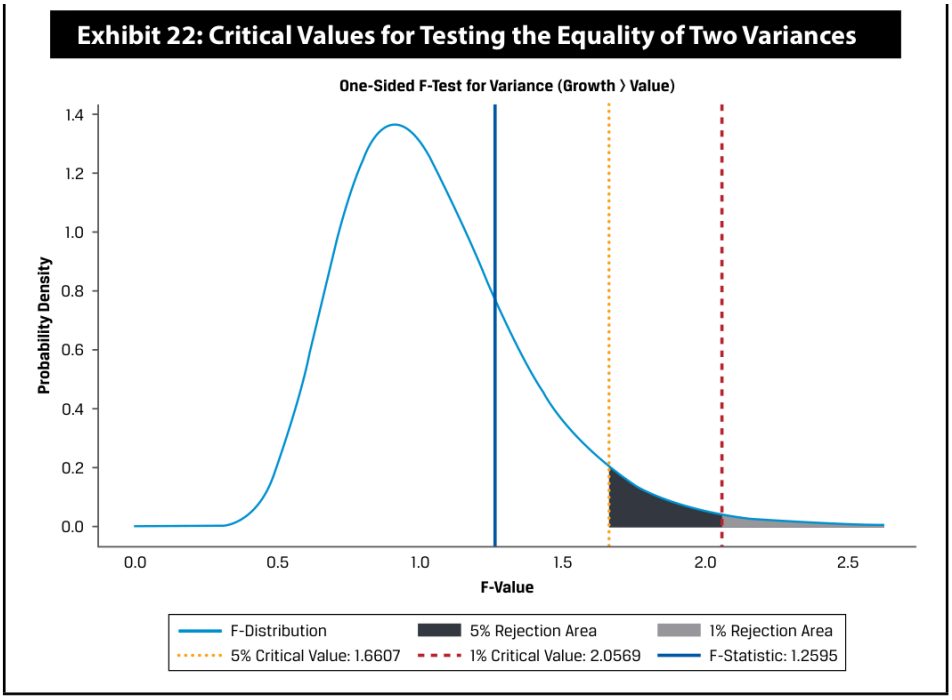
Difference in Variances

1. Based on the variances of the annual returns for the MSCI World Growth Index and the Value Index in Exhibit 15, develop a set of hypotheses and test whether the variance of MSCI World Growth Index is greater than that of the MSCI World Value Index at the 95% and 99% confidence levels.

   Solution:

   The steps of the hypothesis testing process are as follows:

Figure 7.17: *F*-Distributions with Different Degrees of Freedom



Tests of Paired Difference

The test of paired differences is appropriate for dependent observations, such as before-and-after measurements. This test examines the difference between paired observations to determine whether there is a significant change. When performing a paired t-test, we specify the hypothesized mean difference, μ_0, which serves as the baseline for comparison. Since we analyze a series of paired differences, the order of the observations matters.

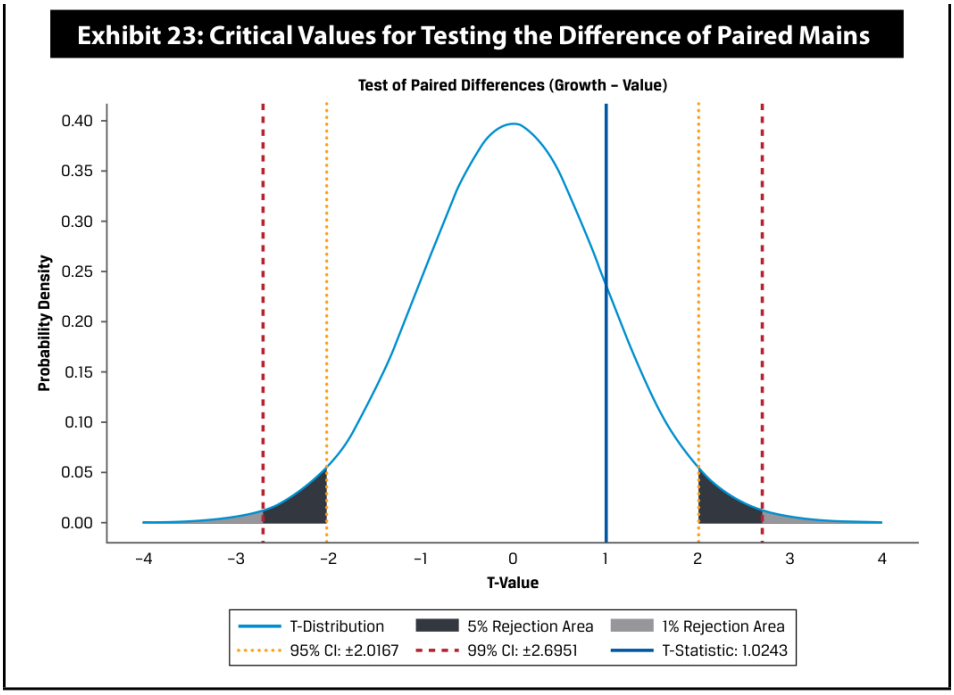
The hypotheses for paired observations are as follows:

- *Null hypothesis*, H_0: The mean difference is equal to the hypothesized mean difference, $\mu_D = \mu_0$.
- *Alternative hypothesis*, H_A: The mean difference is not equal to the hypothesized mean difference, $\mu_D \neq \mu_0$.

The paired t-test compares the mean of the differences to the hypothesized mean difference, assuming the differences are approximately normally distributed. It is different from the test of differences in means, which we covered earlier. These two distinct statistical tests are used in different scenarios.

- *Test of differences in means*, H_0: $\mu_1 = \mu_2$, where μ_1 and μ_2 are hypothesized means: This test is used for independent samples and tests whether the means of two independent samples are equal. The order of observations does not matter, because each sample is drawn independently—for example, comparing monthly returns of two different assets or the same asset before and after a significant event, such as an earnings announcement. Here, the test of differences in means compares two separate groups that are not inherently related. This test is appropriate when the samples are drawn independently from their respective populations and there is no natural pairing between observations in the two groups.
- *Test of paired differences*, H_0: $\mu_d = 0$, where μ_d is the mean difference: This test is used for dependent observations and tests the mean of the differences

Figure 7.18: Equality of Variances F-Test Critical Values



Parametric Test of Correlations

The parametric pairwise correlation coefficient, often called the Pearson correlation or just correlation, measures the relationship between two variables, X and Y. For example, we might be interested in testing whether the true population correlation between two variables is different from zero, resulting in the following hypothesis set:

- *Null hypothesis*, H_0 : The true population correlation, $\rho_{X,Y}$, is equal to zero ($\rho_{X,Y} = 0$).
- *Alternative hypothesis*, H_A: The true population correlation, $\rho_{X,Y}$, is not equal to zero ($\rho_{X,Y} \neq 0$).

The test statistic for the Pearson correlation coefficient is calculated as

Figure 7.19: Paired Means Test Critical Values and Rejection Regions

Table 7.4: Fund Correlations Matrix ($n = 36$)

Asset	Fund 1	Fund 2	Fund 3	Fund 4	STOXX50
Fund 1	1.0000	—	—	—	—
Fund 2	0.9231	1.0000	—	—	—
Fund 3	0.4771	0.4156	1.0000	—	—
Fund 4	0.7111	0.7238	0.3102	1.0000	—
STOXX50	0.8277	0.8223	0.5791	0.7515	1.0000

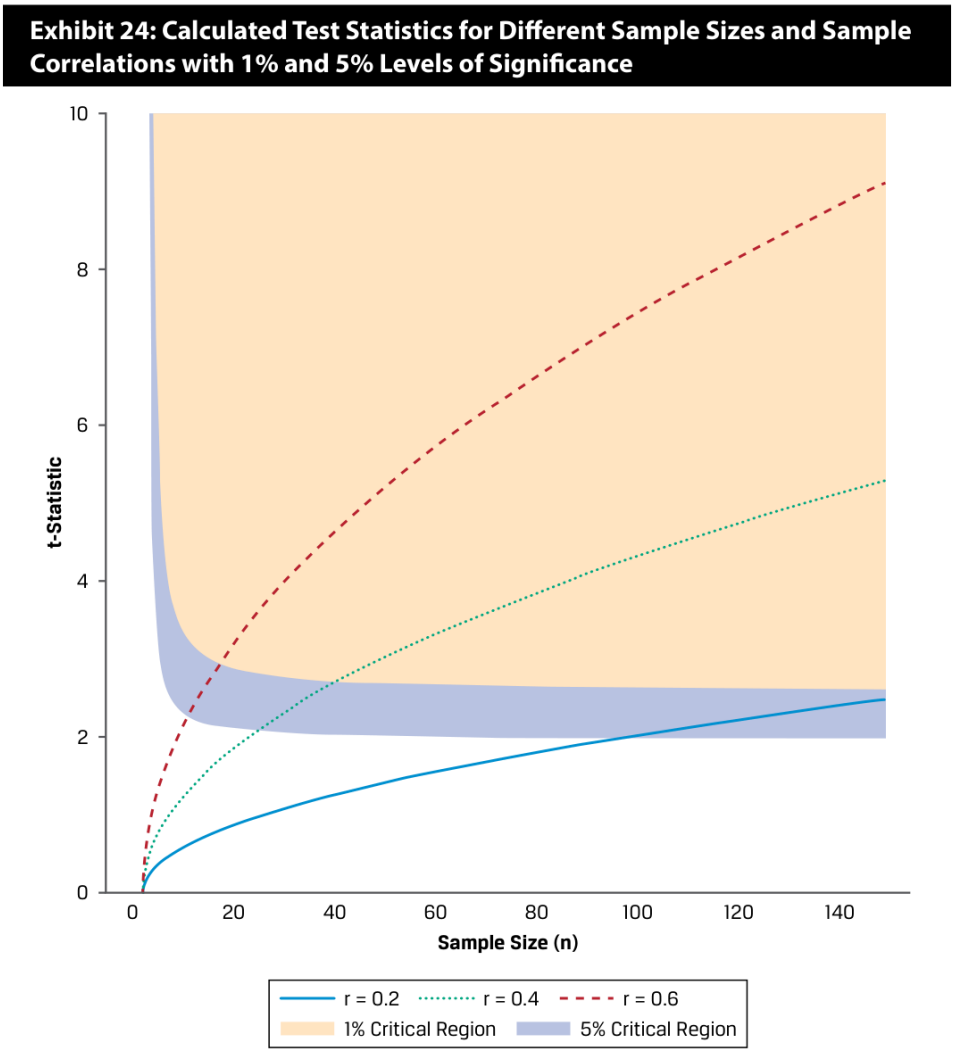


Figure 7.20: t-Statistic Values for Various Sample Sizes and Correlation Coefficients

7.2.5 P-Value and P-Hacking

The p-value is the smallest level of significance at which we reject the null hypothesis:

- **Decision Rule:** Reject H_0 if p-value $< \alpha$; otherwise, fail to reject.

For the MSCI Value Index mean test ($t = 2.5001$, $df = 43$), the one-tailed p-value is 0.0081, which is less than both 5% and 1%, leading to rejection of H_0.

Exam Tip

Beware of P-Hacking
p-hacking refers to the practice of manipulating data or models (such as trimming time horizons, selectively removing outliers, or testing dozens of models) until a statistically significant result ($p < 0.05$) is obtained. This inflates the Type I error rate. To prevent p-hacking, analysts must use out-of-sample validation and strict methodologies.

7.2.6 Practical Implementation (Excel and Google Sheets)

Table 7.5 lists functions to compute critical values for hypothesis testing.

Table 7.5: Built-In Excel and Google Sheet Functions for Hypothesis Testing

Function	Excel Formula	Google Sheets Formula
Inverse Student's t	`=T.INV(prob, df)`	`=T.INV(prob, df)`
Two-tailed t Inverse	`=T.INV.2T(prob, df)`	`=T.INV.2T(prob, df)`
Right-tailed t Dist	`=T.DIST.RT(x, df)`	`=T.DIST.RT(x, df)`
Inverse Standard Normal	`=NORM.S.INV(prob)`	`=NORM.S.INV(prob)`
Inverse Normal Cumulative	`=NORM.INV(prob, mean, sd)`	`=NORM.INV(prob, mean, sd)`
Inverse Chi-Square	`=CHISQ.INV.RT(prob, df)`	`=CHISQ.INV.RT(prob, df)`
Inverse Left-tail F	`=F.INV(prob, df1, df2)`	`=F.INV(prob, df1, df2)`
Inverse Right-tail F	`=F.INV.RT(prob, df1, df2)`	`=F.INV.RT(prob, df1, df2)`

7.3 Non-Parametric Tests

c | compare and contrast parametric and non-parametric tests, describe situations in which each is the more appropriate type of test, construct appropriate hypothesis tests, and interpret the results

Thực tế: Khi dữ liệu không chuẩn (đuôi béo, lệch) hoặc ở dạng định danh/thứ bậc, kiểm định phi tham số (Non-parametric) như Spearman hoặc Chi-square test of independence là bắt buộc để tránh kết luận sai lệch.

7.3.1 Comparison of Parametric and Non-Parametric Tests

- **Parametric Tests:** Perform testing using parameters (mean, variance) and require specific distribution assumptions (e.g., normality). They are more powerful if assumptions are met.
- **Non-Parametric Tests:** Do not make distribution assumptions. They are suitable for ordinal/categorical data, small sample sizes, or when outliers skew parametric estimations. However, they are less powerful when normality assumptions hold.

7.3.2 Non-Parametric Test for Correlation (Spearman Rank Correlation)

The Spearman rank correlation coefficient (r_s) evaluates monotonic (not necessarily linear) relationships by analyzing the ranks of variables:

$$r_s = 1 - \frac{6 \sum_{i=1}^{n} d_i^2}{n(n^2 - 1)} \tag{7.13}$$

Where d_i is the difference in ranks for each observation pair. For $n > 30$, the test statistic follows a t-distribution with $n - 2$ degrees of freedom:

$$t = \frac{r_s \sqrt{n-2}}{\sqrt{1 - r_s^2}} \tag{7.14}$$

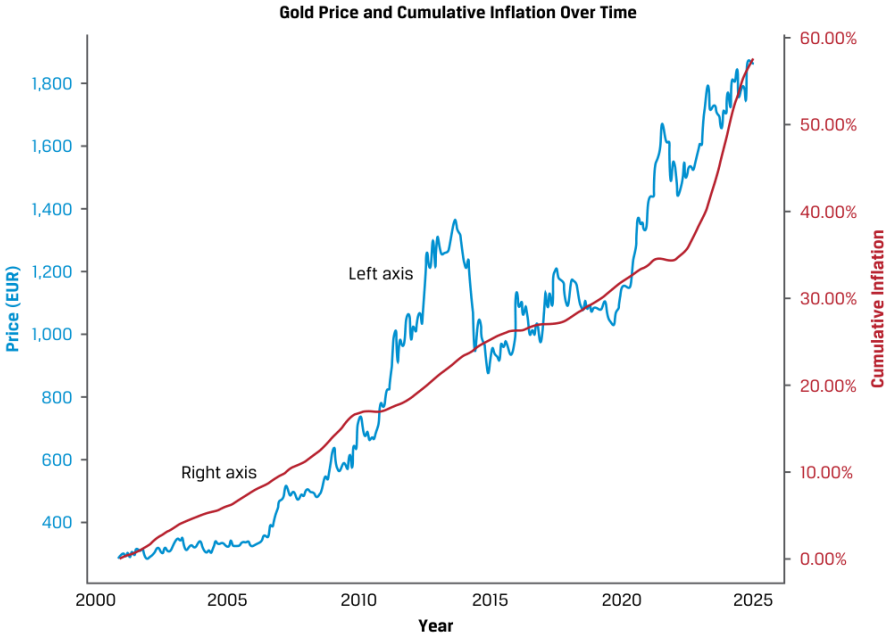
Case Study: Is Gold an Inflation Hedge?

Using 24 years of monthly observations ($n = 288$), we analyze German inflation and gold price returns in EUR (Exhibit 7.21 & Exhibit 7.22). Inflation is highly skewed (2.2507) and leptokurtic (5.8089), violating normality.

- **Pearson Correlation:** $r = -0.0086 \implies t = -0.1454$. Critical value at 5% is ± 1.96. Not statistically significant.
- **Spearman Rank Correlation:** $r_s = 0.0480 \implies t = \frac{0.0480\sqrt{286}}{\sqrt{1-0.0480^2}} \approx 0.8127$.

Since $0.8127 < 1.96$, we fail to reject the null hypothesis of no monotonic correlation. There is insufficient evidence that gold is a good inflation hedge in Germany.

Exhibit 26: Cumulative Monthly German Consumer Inflation and Physical Gold Prices in Euros, 1 January 2000–31 December 2023



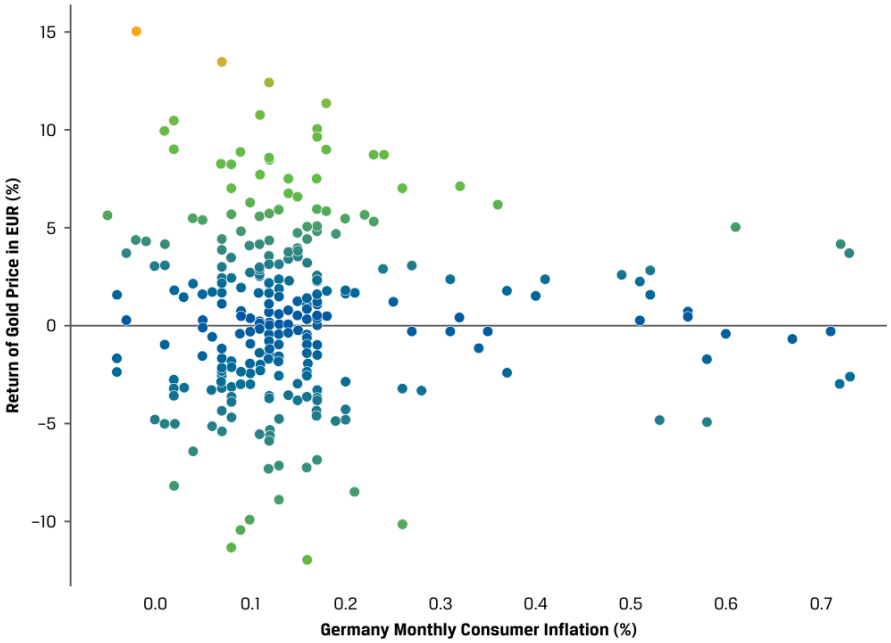
The monthly data over 24 years with 288 monthly observations are summarized in Exhibit 27.

Exhibit 27: German Monthly Consumer Inflation Rate and Return on Physical Gold Prices in Euros

	German Monthly Consumer Inflation Rate	Return on Physical Gold Price in Euros
Average	0.1600%	0.6500%
Standard Deviation	0.1400%	4.3400%
Skewness	2.2507	0.1917
Kurtosis	5.8089	0.6206
Pearson correlation, ρ		–0.0086
Spearman rank correlation, r_S		0.0480

Figure 7.21: German Consumer Inflation and Physical Gold Prices (EUR)

Exhibit 28: Scatterplot of German Monthly Consumer Inflation Rate and the Monthly Return on Physical Gold Price in Euros



The −0.0086 Pearson correlation coefficient, ρ, suggests almost no linear relationship or correlation. To formally test whether it is indeed statistically insignificantly different from zero, we formulate the following hypotheses:

- The null hypothesis, H_0:ρ = 0.
- The alternative hypothesis, H_A : ρ ≠ 0.

Using the sample correlation r with $n = 288$ observations, the t-statistic using the formula is calculated as

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{-0.0086\sqrt{288-2}}{\sqrt{1-0.0086^2}} \approx -0.1454.$$

Given that the degrees of freedom ($df = 286$), are large, the critical value for a two-tailed test at the 0.05 significance level is approximately ±1.96. Since the computed t-statistic of −0.1454 is well within the range of −1.96 to 1.96, the result is not significant. Thus, we fail to reject the null hypothesis, suggesting that there is no significant linear relationship between the monthly euro-denominated price appreciation in gold and the monthly German consumer inflation rate.

Figure 7.22: Scatterplot of German Inflation and Gold returns

7.3.3 Chi-Square Test of Independence

For categorical data, we test whether classifications are independent using a Chi-square test statistic with $df = (r-1)(c-1)$:

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{(O_{ij} - E_{ij})^2}{E_{ij}} \tag{7.15}$$

Where O_{ij} is the observed frequency and E_{ij} is the expected frequency assuming independence:

$$E_{ij} = \frac{(\text{Total Row } i) \times (\text{Total Column } j)}{\text{Overall Total } N} \tag{7.16}$$

Standardized residuals help identify which cells deviate significantly (values exceeding ± 1.96 are significant):

$$\text{Standardized Residual} = \frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}}} \tag{7.17}$$

Case Study: Comparing Environmental and Governance Ratings

We examine the independence of ESG ratings of 500 firms (Table 7.6). Expected frequency E_{22} (Average/Average) is $\frac{260\times230}{500} = 119.6$. Summing scaled squared deviations (Exhibit 7.23), we obtain $\chi^2 \approx 35.75$. With $df = (3-1)(3-1) = 4$, the critical value at 5% is $\chi^2_{0.05} = 9.49$. Since $35.75 > 9.49$, we reject the null hypothesis of independence. Environmental and governance ratings are related.

Table 7.6: ESG Ratings Contingency Table ($N = 500$)

Environmental Rating	**Progressive (GR1)**	**Average (GR2)**	**Poor (GR3)**	**Total**
Progressive (ER1)	35	40	5	80
Average (ER2)	80	130	50	260
Poor (ER3)	40	60	60	160
Total	155	230	115	500

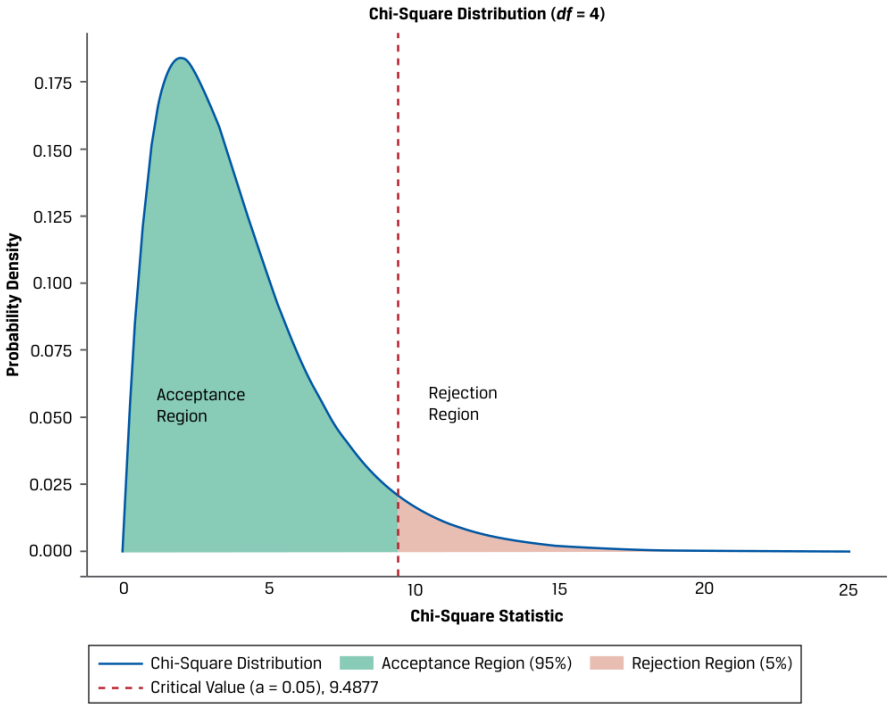
Case Study: Fund Size and Investment Style of 1,594 ETFs

We test whether ETF size and investment style are independent (Table 7.7). Calculated expected frequencies and scaled squared deviations yield $\chi^2 = 32.07$ (Exhibit 7.24). With $df = 4$, the critical value at 1% is 13.28. Since $32.07 > 13.28$, we reject the null hypothesis of independence. Analyzing standardized residuals (Exhibit 7.25), the cell for Medium-Growth has a residual of $+3.69$ (significantly higher than expected), and Large-Growth has a residual of -2.72 (significantly lower than expected).

Table 7.7: ETFs Classifications Contingency Table ($N = 1,594$)

Investment Type	**Small**	**Medium**	**Large**	**Total**
Value	50	110	343	503
Growth	42	122	202	366
Blend	56	149	520	725
Total	148	381	1,065	1,594

Exhibit 31: Chi-Square Distribution Curve for 4 Degrees of Freedom

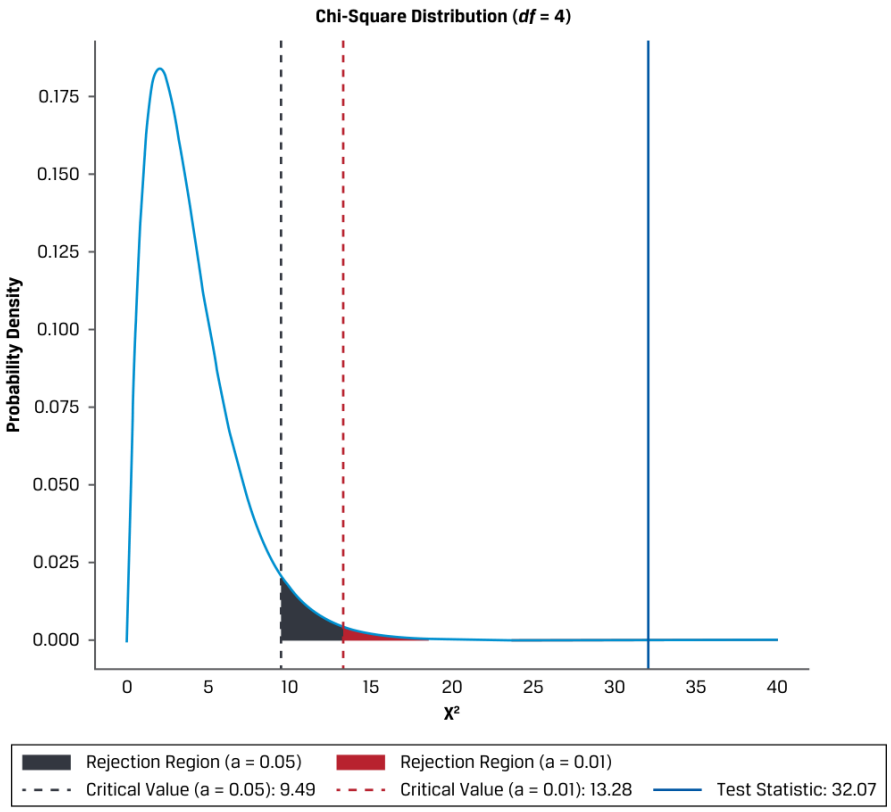


Then, the steps of the hypothesis testing process are as follows:

Step 1: State the hypotheses.

Figure 7.23: Chi-Square Distribution Curve with 4 Degrees of Freedom

Exhibit 35: Chi-Square Distribution Curve for 4 Degrees of Freedom



The standardized residuals for each cell in the contingency table calculated in Exhibit 36 help identify where the observed frequencies deviate significantly from the expected frequencies. Large positive or negative residuals indicate cells where the observed frequency is significantly higher or lower than expected under the null hypothesis of independence.

Figure 7.24: Chi-Square Rejection Regions for ETF Independence Test ($df = 4$)

Exhibit 36: Standardized Residuals

Standardized Residuals, $\frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}}}$

	Size Based on Market Capitalization		
Investment Type	**Small**	**Medium**	**Large**
Value	0.48	–0.93	0.38
Growth	1.38	3.69	–2.72
Blend	–1.38	–1.85	1.62

A standardized residual of 3.69 indicates there are significantly more medium-size growth ETFs than expected if size and investment type were independent. Similarly, a standardized residual of –2.72 suggests there are significantly fewer large growth ETFs than expected. Both these deviations are considered statistically significant because, at a 95% confidence level, residuals beyond ±1.96 indicate a significant departure from expected values. Since the residuals of 3.69 and –2.72 exceed this threshold, they highlight meaningful deviations from the assumption of independence.

QUESTION SET 3



1. Non-parametric tests are:

   A. more robust with ordinal data or when the data distribution is unknown.

   B. always more accurate than parametric tests regardless of data distribution.

   C. more suitable for data that conform to a specific distribution, such as a normal distribution.

   Solution:

   The correct response is A. Non-parametric tests do not require the data to conform to any particular distribution, and as such, these tests are more flexible and are suitable for ordinal data or when the measurement scales do not meet the stringent requirements of parametric testing. Moreover, they are more robust and effective even when the data do not meet normality assumptions.

   For ordinal data, non-parametric tests are suitable and are more robust.

   B is incorrect: While non-parametric tests are more robust for non-normal data, they may be less effective at detecting true effects compared to parametric tests if the latter's assumptions are met. Comparing the accuracy of the two tests is difficult because they are used for populations with different distribution characteristics. When the distribution is known, parametric tests are more powerful than non-parametric tests.

   C is incorrect: Non-parametric tests are used when data do not meet the stringent requirements of parametric tests. When the data conform to the normal distribution, parametric tests are used.

Figure 7.25: Standardized Residuals plot for Size vs Style of ETFs

Module 8

The Return and Risk of a Financial Portfolio

LEARNING OUTCOMES	
a	calculate, interpret, and evaluate the expected return, variance, standard deviation, covariance, and correlation of portfolio returns
b	describe, calculate, and interpret the minimum-variance portfolio and portfolios that lie on the efficient frontier
c	explain the selection of an optimal portfolio, given an investor's risk aversion and the capital allocation line, and how this extends to the market portfolio and the capital market line

8.1 Calculating Portfolio Statistics

a	calculate, interpret, and evaluate the expected return, variance, standard deviation, covariance, and correlation of portfolio returns

Portfolio statistics, such as expected returns, standard deviations, and correlations, are fundamental for effective portfolio management and asset allocation. These statistics can be historical (factoring in past returns) or forward-looking (using model-based forecasts).

8.1.1 Returns of a Portfolio

The return of a portfolio is the weighted average of the returns of its individual assets. Historical portfolio returns are calculated by summing the products of the beginning-of-period asset weights and their corresponding realized returns:

$$R_{P,t} = \sum_{i=1}^{N} w_{i,t-1} R_{i,t} \tag{8.1}$$

where N is the number of assets, $w_{i,t-1}$ is the weight of asset i at the end of period $t-1$ (beginning of period t), and $R_{i,t}$ is the realized return of asset i in period t, defined as:

$$R_{i,t} = \frac{P_{i,t} - P_{i,t-1} + \text{Div}_{i,t}}{P_{i,t-1}} \tag{8.2}$$

To evaluate returns over multiple periods from $t = 1$ to T, if the weights are held constant at their initial values $w_{i,0}$ (which implies continuous rebalancing), the aggregate portfolio return is:

$$R_{P,T} = \sum_{i=1}^{N} w_{i,0} R_{i,T} \tag{8.3}$$

where $R_{i,T}$ is the cumulative return of asset i over the entire period T. The arithmetic average return of the portfolio across these periods is:

$$\bar{R}_P = \frac{1}{T} \sum_{t=1}^{T} R_{P,t} \tag{8.4}$$

When compounding returns over multiple periods without rebalancing, the asset weights will naturally shift as a result of varying asset performance. This phenomenon is known as **portfolio drift**. The cumulative return over T periods with drifting weights is:

$$R_{P,T} = \prod_{t=1}^{T} \left(1 + \sum_{i=1}^{N} w_{i,t-1} R_{i,t}\right) - 1 \tag{8.5}$$

The geometric average return, or geometric return, is defined as:

$$\bar{R}_{g,P} = \left[\prod_{t=1}^{T}(1 + R_{P,t})\right]^{\frac{1}{T}} - 1 = (1 + R_{P,T})^{\frac{1}{T}} - 1 \tag{8.6}$$

Portfolio drift làm lệch tỷ trọng mục tiêu (style drift) do tài sản tăng trưởng không đều. Chúng ta cần đặt câu hỏi: Nếu không tái cân bằng (rebalancing) định kỳ, rủi ro danh mục sẽ thay đổi thế nào? Có tiệm cận rủi ro của tài sản sinh lời cao nhất?

Example 1: Calculating Portfolio Return

Consider a portfolio with two assets, A and B, each with a 50% initial weight ($w_A = w_B = 0.5$). The weights are held constant through rebalancing.

- Year 1: Asset A returns +10%, Asset B returns −5%.
- Year 2: Asset A returns −10%, Asset B returns +20%.

Calculate the arithmetic average return and geometric average return of the portfolio.
Solution:

1. **Year 1 Portfolio Return (Arithmetic Return):** $R_{P,1} = 0.5(10\%) + 0.5(-5\%) = 2.5\%$.
2. **Year 2 Portfolio Return (Arithmetic Return):** $R_{P,2} = 0.5(-10\%) + 0.5(20\%) = 5.0\%$.
3. **Arithmetic Average Portfolio Return:** $\bar{R}_P = \frac{2.5\%+5\%}{2} = 3.75\%$.
4. **Geometric Average Portfolio Return:** $\bar{R}_{g,P} = \sqrt{(1+0.025)(1+0.05)} - 1 = 3.74\%$.

Exam Tip

BA II Plus Keystrokes for Geometric Average Return:

1. Press **1.025** → [*] → **1.05** → [=] (screen displays **1.07625**).
2. Press [**2nd**] → [**Vx**] to take the square root (screen displays **1.03742**).
3. Press [**-**] → **1** → [=] (screen displays **0.03742** or **3.74%**).

Example 2: Portfolio Drift over Three Periods

An investor holds a portfolio of two assets, A and B, starting with EUR 1 invested in each ($w_{A,0} = w_{B,0} = 0.5$, total portfolio value = EUR 2). The portfolio is left untouched (no rebalancing) over three periods. The returns are:

Period	Asset A Return ($R_{A,t}$)	Asset B Return ($R_{B,t}$)
1	+20%	+5%
2	−10%	+15%
3	+25%	−5%

Calculate the portfolio returns and weights at each step, and determine the cumulative arithmetic and geometric returns.
Solution:

- **Period 1:**
  - Portfolio return: $R_{P,1} = 0.5(20\%) + 0.5(5\%) = 12.5\%$.

- Ending value: Asset A = EUR 1.20; Asset B = EUR 1.05. Total value = EUR 2.25.
- Ending weights for Period 2: $w_{A,1} = \frac{1.20}{2.25} \approx 53.33\%$; $w_{B,1} = \frac{1.05}{2.25} \approx 46.67\%$.

- **Period 2:**
  - Portfolio return: $R_{P,2} = 0.5333(-10\%) + 0.4667(15\%) = 1.67\%$.
  - Ending value: Asset A = EUR 1.08; Asset B = EUR 1.2075. Total value = EUR 2.2875.
  - Ending weights for Period 3: $w_{A,2} = \frac{1.08}{2.2875} \approx 47.21\%$; $w_{B,2} = \frac{1.2075}{2.2875} \approx 52.79\%$.
- **Period 3:**
  - Portfolio return: $R_{P,3} = 0.4721(25\%) + 0.5279(-5\%) = 9.16\%$.
  - Ending value: Asset A = EUR 1.35; Asset B = EUR 1.1471. Total value = EUR 2.4971.
  - Ending weights: $w_{A,3} = \frac{1.35}{2.4971} \approx 54.06\%$; $w_{B,3} = \frac{1.1471}{2.4971} \approx 45.94\%$.
- **Cumulative Returns:**
  - Cumulative arithmetic return: $R_{P,T} = R_{P,1} + R_{P,2} + R_{P,3} = 12.5\% + 1.67\% + 9.16\% = 23.33\%$.
  - Arithmetic average return: $\bar{R}_P = \frac{23.33\%}{3} \approx 7.78\%$.
  - Cumulative geometric return: $R_{P,T} = (1.125 \times 1.0167 \times 1.0916) - 1 \approx 24.86\%$.
  - Geometric average return: $\bar{R}_{g,P} = (1.125 \times 1.0167 \times 1.0916)^{\frac{1}{3}} - 1 \approx 7.68\%$.

Exam Tip

BA II Plus Keystrokes for Cumulative Geometric Return:

1. Calculate cumulative factor: Press **1.125** → [*] → **1.0167** → [*] → **1.0916** → [=] (screen displays **1.2486**).
2. Calculate cube root: Press [yx] → [(] → **1** → [/] → **3** → [)] → [=] (screen displays **1.0768**).
3. Subtract 1: Press **-** → **1** → [=] (screen displays **0.0768** or **7.68%**).

Table 8.1: Portfolio Drift Characteristics by Weighting Scheme

Weighting Method	Impact of Drift	Resulting Portfolio Behavior
Equal-Weighted	Weights drift as top performers grow larger	Potential bias toward high growth and higher-volatility assets
Non-Equally Weighted	Large-allocation assets drift and dominate portfolio performance	Momentum-driven returns with increased concentration risk

8.1.2 Variance and Standard Deviation of Portfolio Returns

The risk of a portfolio is measured by its variance and standard deviation. The variance of portfolio returns (σ_P^2) incorporates both individual asset variances and the covariances between all asset pairs:

$$\sigma_P^2 = \sum_{i=1}^{N} \sum_{j=1}^{N} w_i w_j \sigma_{i,j} \tag{8.7}$$

where $\sigma_{i,j}$ represents the covariance between asset i and asset j. Note that $\sigma_{i,i}$ is the variance of asset i (σ_i^2). Standard deviation (σ_P) is the square root of variance:

$$\sigma_P = \sqrt{\sum_{i=1}^{N} \sum_{j=1}^{N} w_i w_j \sigma_{i,j}} \tag{8.8}$$

If covariances are modeled as random variables, the expected variance of portfolio returns is:

$$E[\sigma_P^2] = \sum_{i=1}^{N} \sum_{j=1}^{N} w_i w_j E[\sigma_{i,j}] \tag{8.9}$$

Covariance is linked to the correlation coefficient ($\rho_{i,j}$) by:

$$\sigma_{i,j} = \rho_{i,j}\sigma_i\sigma_j \tag{8.10}$$

Due to the non-linear interaction of volatility and correlation, the expected covariance $E[\sigma_{i,j}] \neq E[\rho_{i,j}] \times E[\sigma_i] \times E[\sigma_j]$ in the presence of joint dependencies.

For a two-asset portfolio, the portfolio variance formula is:

$$\sigma_P^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\sigma_{1,2} = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{1,2}\sigma_1\sigma_2 \tag{8.11}$$

Example 3: Portfolio Allocation (Two-Asset)

A portfolio allocates 80% to the S&P 500 Index ($w_A = 0.8$) and 20% to the MSCI Emerging Markets Index ($w_B = 0.2$). Expected returns are 9.93% and 18.20%, and standard deviations are 16.21% and 33.11%, respectively. The covariance between the two is 0.0050. Calculate the expected portfolio return, variance, and standard deviation.

Solution:

1. **Expected Portfolio Return:** $E[R_P] = 0.8(9.93\%) + 0.2(18.20\%) = 11.58\%$.
2. **Portfolio Variance:**

$$E[\sigma_P^2] = (0.8^2 \times 0.1621^2) + (0.2^2 \times 0.3311^2) + (2 \times 0.8 \times 0.2 \times 0.0050)$$

$$E[\sigma_P^2] = 0.01684 + 0.00439 + 0.00160 \approx 0.02283$$

3. **Portfolio Standard Deviation:** $\sigma_P = \sqrt{0.02283} \approx 15.11\%$.

Exam Tip

BA II Plus Keystrokes for Portfolio Standard Deviation:

1. Calculate Asset A variance and store in memory: Press **0.8** → [x^2] → [*] → **0.1621** → [x^2] → [=] → [**STO**] → **1** (stores ≈ 0.016839).
2. Calculate Asset B variance and add to memory: Press **0.2** → [x^2] → [*] → **0.3311** → [x^2] → [=] → [+] → [**RCL**] → **1** → [=] → [**STO**] → **1** (stores sum ≈ 0.021224).
3. Calculate covariance term and add to memory: Press **2** → [*] → **0.8** → [*] → **0.2** → [*] **0.005** → [=] → [+] → [**RCL**] → **1** → [=] (variance displays ≈ 0.022824).
4. Take square root to get standard deviation: Press [**2nd**] → [**Vx**] (screen displays ≈ 0.151076 or **15.11%**).

Table 8.2: Portfolio Return and Risk under Different Weights (S&P 500 vs. MSCI EM)

Portfolio	S&P 500 Weight (%)	MSCI EM Weight (%)	Expected Return (%)	Risk / SD (%)
1	0	100	18.20	33.11
2	10	90	17.37	29.85
3	20	80	16.55	26.70
4	30	70	15.72	23.71
5	40	60	14.89	20.93
6	50	50	14.07	18.47
7	60	40	13.24	16.47
8	70	30	12.41	15.12
9	**80**	**20**	**11.58**	**14.59**
10	90	10	10.76	14.98
11	100	0	9.93	16.21

Tương quan S&P 500 và MSCI EM thấp tạo đồ thị rủi ro - sinh lời cong lõm (concave), giúp Portfolio 9 có độ lệch chuẩn (14.59%) thấp hơn cả S&P 500 (16.21%). Chúng ta hãy suy nghĩ: Nếu tương quan bằng +1.0, liệu có thể giảm rủi ro danh mục? Tại sao tương quan thấp lại là c̈hén thánh?

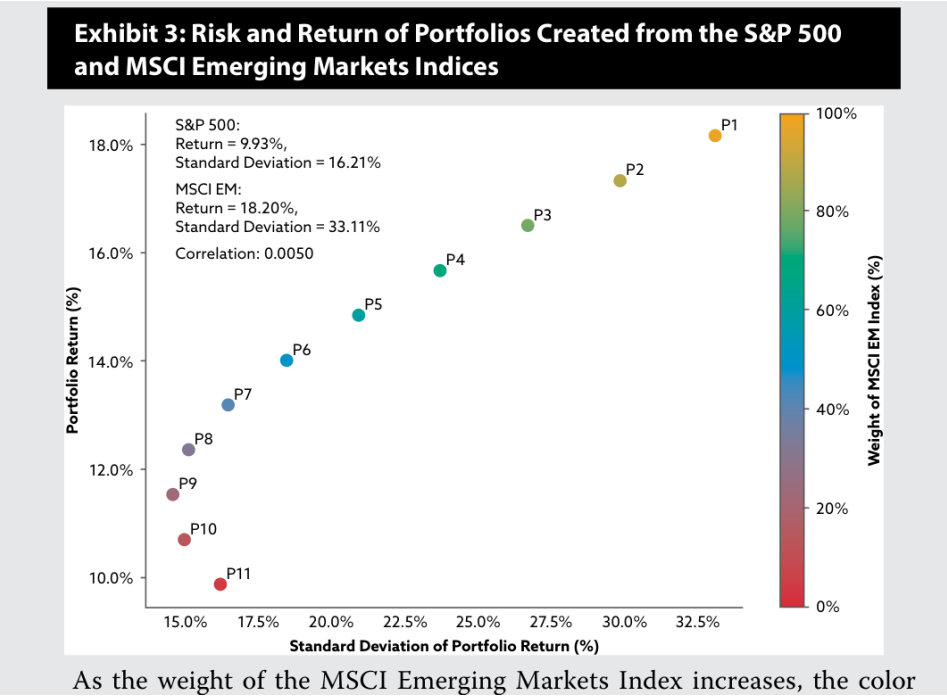


Figure 8.1: Feasible Set of Portfolios from S&P 500 and MSCI EM (Covariance = 0.0050)

Example 4: Varying Correlations

Asset 1 has expected return 7.0% and standard deviation 12.0%. Asset 2 has expected return 15.0% and standard deviation 25.0%. Compare the portfolio risk (σ_P) for different asset weights under correlations (ρ) of +1.0, +0.5, +0.2, and −1.0.

Solution: The calculated portfolio standard deviation (σ_P) for each scenario is summarized in Table 8.3.

Table 8.3: Portfolio Return and Risk Under Different Correlations

Weight of Asset 1 (%)	**Portfolio Return (%)**	**Portfolio Standard Deviation (%) with ρ of:**			
		+1.0	+0.5	+0.2	−1.0
0	15.0	25.0	25.0	25.0	25.0
20	13.4	22.4	21.3	20.6	17.6
40	11.8	19.8	17.9	16.6	10.2
50	11.0	18.5	16.3	14.9	6.5
60	10.2	17.2	15.0	13.4	2.8
80	8.6	14.6	12.9	11.7	4.6
100	7.0	12.0	12.0	12.0	12.0

As correlation declines, the frontier curves further to the left, which indicates a greater reduction in portfolio risk for a given expected return. When $\rho = -1$, the portfolio risk can be reduced to zero.

Matrix Notation of Portfolio Risk

For a portfolio with N assets, the number of unique covariances that must be computed is:

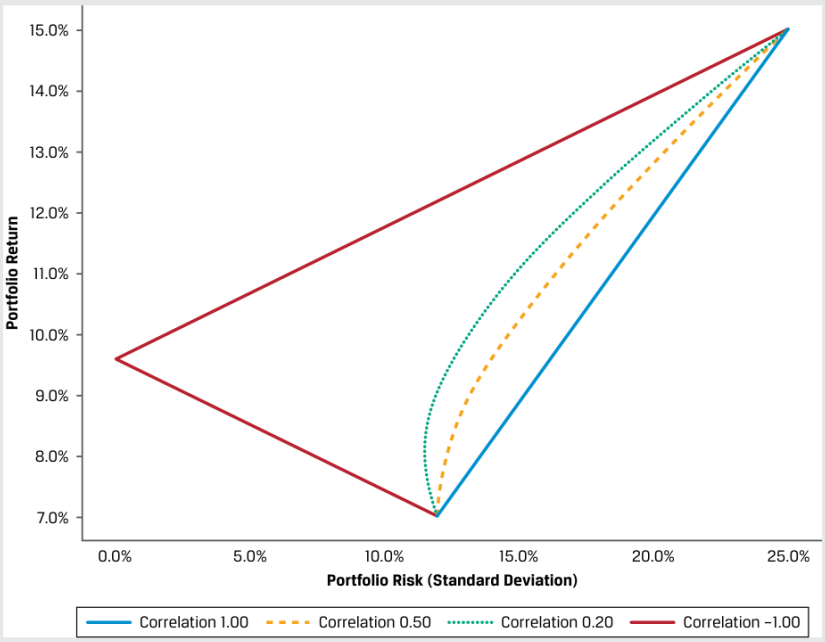
$$\text{Unique Covariances} = \frac{N(N-1)}{2} \tag{8.12}$$

Along with the N individual variances, the total parameters needed for an N-asset portfolio is $\frac{N(N+1)}{2}$. For example, a 100-security portfolio requires 4,950 unique covariances.

This structure is organized into the symmetric **variance-covariance matrix V**:

$$\mathbf{V} = \begin{bmatrix} \sigma_1^2 & \sigma_{1,2} & \sigma_{1,3} & \cdots & \sigma_{1,N} \\ \sigma_{2,1} & \sigma_2^2 & \sigma_{2,3} & \cdots & \sigma_{2,N} \\ \sigma_{3,1} & \sigma_{3,2} & \sigma_3^2 & \cdots & \sigma_{3,N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sigma_{N,1} & \sigma_{N,2} & \sigma_{N,3} & \cdots & \sigma_N^2 \end{bmatrix} \tag{8.13}$$

Exhibit 5: Relationship Between Portfolio Risk and Return with Varying Correlation Coefficients



Portfolio standard deviation decreases with each successive decrease in the correlation coefficient and is minimized when $\rho = -1$. Exhibit 5 shows that the risk–return relationship is a straight line when $\rho = 1$, and it consists of two straight lines when $\rho = -1$. As the correlation declines, the trade-off between lower risk and lower return improves, so that any given decline in risk comes at the expense of less of a decline in the portfolio expected return. Differently put, the frontier becomes more convex as the correlation between the assets falls. The curvilinear nature of a portfolio of assets is recognizable in all investment opportunity sets.

Figure 8.2: Portfolio Opportunity Frontier for Different Correlation Coefficients

The matching normalized matrix is the **correlation matrix $\boldsymbol{\rho}$**:

$$\boldsymbol{\rho} = \begin{bmatrix} 1 & \rho_{1,2} & \rho_{1,3} & \cdots & \rho_{1,N} \\ \rho_{2,1} & 1 & \rho_{2,3} & \cdots & \rho_{2,N} \\ \rho_{3,1} & \rho_{3,2} & 1 & \cdots & \rho_{3,N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho_{N,1} & \rho_{N,2} & \rho_{N,3} & \cdots & 1 \end{bmatrix} \tag{8.14}$$

Thực tế: Bloomberg thường nhân covariance matrix với 10^4 để hiển thị dưới dạng $\%^2$. Chúng ta hãy lưu ý: Khi đổi hệ số (310.07) về số thập phân để tính, phải chia cho bao nhiêu? (Đáp án: 10^4). Nhầm lẫn này gây kết quả lệch hàng ngàn lần!

Example 5: A Portfolio with Three Assets (Covariance Matrix)

A portfolio contains three assets. The expected returns, standard deviations, and correlation matrix are given below:

- Expected Returns: $E[R_1] = 10\%$, $E[R_2] = 12\%$, $E[R_3] = 8\%$
- Volatilities: $\sigma_1 = 15\%$, $\sigma_2 = 20\%$, $\sigma_3 = 10\%$
- Correlation Matrix $\boldsymbol{\rho}$:

$$\boldsymbol{\rho} = \begin{bmatrix} 1 & 0.2 & -0.3 \\ 0.2 & 1 & 0.4 \\ -0.3 & 0.4 & 1 \end{bmatrix}$$

Construct the variance-covariance matrix $\mathbf{V}$.
Solution: Using the relationship between covariance and correlation, $\sigma_{i,j} = \sigma_i \sigma_j \rho_{i,j}$:

1. **Asset Variances (main diagonal of the covariance matrix):**
   - $\sigma_1^2 = 0.15^2 = 0.0225$
   - $\sigma_2^2 = 0.20^2 = 0.0400$
   - $\sigma_3^2 = 0.10^2 = 0.0100$
2. **Asset Covariances (off-diagonal elements):**
   - $\sigma_{1,2} = \sigma_{2,1} = 0.15 \times 0.20 \times 0.2 = 0.0060$
   - $\sigma_{1,3} = \sigma_{3,1} = 0.15 \times 0.10 \times (-0.3) = -0.0045$
   - $\sigma_{2,3} = \sigma_{3,2} = 0.20 \times 0.10 \times 0.4 = 0.0080$

Therefore, the variance-covariance matrix $\mathbf{V}$ is:

$$\mathbf{V} = \begin{bmatrix} 0.0225 & 0.0060 & -0.0045 \\ 0.0060 & 0.0400 & 0.0080 \\ -0.0045 & 0.0080 & 0.0100 \end{bmatrix}$$

Example 6: Return and Risk of a Three-Asset Portfolio

Using the data from Example 5, assume the portfolio weights are $w_1 = 0.4$, $w_2 = 0.3$, $w_3 = 0.3$. Calculate the expected portfolio return and standard deviation.
Solution:

1. **Expected Portfolio Return:**

$$E[R_P] = (0.4 \times 10\%) + (0.3 \times 12\%) + (0.3 \times 8\%) = 10\%$$

2. **Expected Portfolio Variance:**

$$\sigma_P^2 = \sum_{i=1}^{3} \sum_{j=1}^{3} w_i w_j \sigma_{i,j}$$

$$\sigma_P^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + w_3^2\sigma_3^2 + 2w_1w_2\sigma_{1,2} + 2w_1w_3\sigma_{1,3} + 2w_2w_3\sigma_{2,3}$$

$$\sigma_P^2 = (0.4^2 \times 0.0225) + (0.3^2 \times 0.04) + (0.3^2 \times 0.01) + 2(0.4)(0.3)(0.006) + 2(0.4)(0.3)(-0.0045) + 2(0.3)(0.3)(0.008)$$

$$\sigma_P^2 = 0.0036 + 0.0036 + 0.0009 + 0.00144 - 0.00108 + 0.00144 = 0.0099$$

3. **Portfolio Standard Deviation:** $\sigma_P = \sqrt{0.0099} \approx 9.95\%$.

8.1.3 Expected Return and Variance under Economic Scenarios

Expected returns and covariances can be derived from joint probabilities over M economic scenarios. Under scenario M occurring with probability p_M, the portfolio return is:

$$R_{P,M} = \sum_{i=1}^{N} w_i R_{i,M} \tag{8.15}$$

The expected portfolio return over all scenarios is:

$$E[R_P] = \sum_{M=1}^{M} p_M R_{P,M} = \sum_{M=1}^{M} p_M \left(\sum_{i=1}^{N} w_i R_{i,M} \right) \tag{8.16}$$

The portfolio variance is the weighted sum of squared deviations from the expected return:

$$\sigma_P^2 = \sum_{M=1}^{M} p_M \left(R_{P,M} - E[R_P] \right)^2 \tag{8.17}$$

Example 7: Expected Portfolio Returns and Scenarios

Consider three assets, A, B, and C, with weights $w_A = 0.4$, $w_B = 0.3$, $w_C = 0.3$. Their scenario returns are:

Scenario	Probability	Asset A (%)	Asset B (%)	Asset C (%)
1 (Expansion)	30%	15	10	5
2 (Steady)	50%	10	12	8
3 (Recession)	20%	-5	20	12

Calculate the expected portfolio return and standard deviation.
Solution:

1. **Scenario Returns:**
   - Scenario 1 (Expansion): $R_{P,1} = 0.4(15) + 0.3(10) + 0.3(5) = 10.5\%$.
   - Scenario 2 (Steady): $R_{P,2} = 0.4(10) + 0.3(12) + 0.3(8) = 10.0\%$.
   - Scenario 3 (Recession): $R_{P,3} = 0.4(-5) + 0.3(20) + 0.3(12) = 7.6\%$.
2. **Expected Portfolio Return:**
$$E[R_P] = 0.3(10.5\%) + 0.5(10.0\%) + 0.2(7.6\%) = 9.67\%$$
3. **Expected Portfolio Variance:**
   - Scenario 1 contribution: $0.3 \times (10.5\% - 9.67\%)^2 = 0.3 \times (0.83\%)^2 \approx 0.0000207$
   - Scenario 2 contribution: $0.5 \times (10.0\% - 9.67\%)^2 = 0.5 \times (0.33\%)^2 \approx 0.0000054$
   - Scenario 3 contribution: $0.2 \times (7.6\% - 9.67\%)^2 = 0.2 \times (-2.07\%)^2 \approx 0.0000857$
$$\sigma_P^2 = 0.0000207 + 0.0000054 + 0.0000857 = 0.0001118$$
4. **Portfolio Standard Deviation:** $\sigma_P = \sqrt{0.0001118} \approx 1.06\%$.

Table 8.4: Comparison of Summation and Matrix Notation

Concept	Summation Notation	Matrix Notation
Asset Weights Vector	w_i	$\mathbf{w} = [w_1, w_2, \ldots, w_N]^T$
Expected Returns Vector	$E[R_i]$	$\boldsymbol{\mu} = [E[R_1], E[R_2], \ldots, E[R_N]]^T$
Portfolio Return	$E[R_P] = \sum_{i=1}^{N} w_i E[R_i]$	$E[R_P] = \mathbf{w}^T \boldsymbol{\mu}$
Covariance Matrix	$\sigma_{i,j} = \rho_{i,j}\sigma_i\sigma_j$	$\mathbf{V}$ (symmetric $N \times N$ matrix)
Portfolio Variance	$\sigma_P^2 = \sum_{i=1}^{N}\sum_{j=1}^{N} w_i w_j \sigma_{i,j}$	$\sigma_P^2 = \mathbf{w}^T\mathbf{V}\mathbf{w}$

Summation vs. Matrix Notation Comparison

For portfolios with a large number of assets, matrix notation provides a compact, computationally efficient mathematical representation.

Example 8: The Three-Asset Portfolio Revisited (Matrix Form)

Using the data from Example 5, confirm the return and variance using matrix multiplication.

$$\mathbf{w} = \begin{bmatrix} 0.4 \\ 0.3 \\ 0.3 \end{bmatrix}, \quad \boldsymbol{\mu} = \begin{bmatrix} 0.10 \\ 0.12 \\ 0.08 \end{bmatrix}, \quad \mathbf{V} = \begin{bmatrix} 0.0225 & 0.0060 & -0.0045 \\ 0.0060 & 0.0400 & 0.0080 \\ -0.0045 & 0.0080 & 0.0100 \end{bmatrix}$$

Solution:

1. **Expected Return:**

$$E[R_P] = \mathbf{w}^T\boldsymbol{\mu} = \begin{bmatrix} 0.4 & 0.3 & 0.3 \end{bmatrix} \begin{bmatrix} 0.10 \\ 0.12 \\ 0.08 \end{bmatrix} = 0.4(0.10) + 0.3(0.12) + 0.3(0.08) = 10\%$$

2. **Portfolio Variance ($\mathbf{w}^T\mathbf{V}\mathbf{w}$):** First, compute the matrix-vector product $\mathbf{V}\mathbf{w}$:

$$\mathbf{V}\mathbf{w} = \begin{bmatrix} 0.0225(0.4) + 0.006(0.3) - 0.0045(0.3) \\ 0.006(0.4) + 0.040(0.3) + 0.008(0.3) \\ -0.0045(0.4) + 0.008(0.3) + 0.010(0.3) \end{bmatrix} = \begin{bmatrix} 0.00945 \\ 0.01680 \\ 0.00360 \end{bmatrix}$$

Next, multiply the row vector $\mathbf{w}^T$ by the column vector ($\mathbf{V}\mathbf{w}$):

$$\sigma_P^2 = 0.4(0.00945) + 0.3(0.0168) + 0.3(0.0036) = 0.00378 + 0.00504 + 0.00108 = 0.0099$$

This yields $\sigma_P^2 = 0.0099$, which perfectly matches the result from the summation formula in Example 6.

8.1.4 The Benefits of Diversification

The power of diversification lies in combining assets that are not perfectly positively correlated to reduce overall portfolio risk. Consider an equally weighted portfolio ($w_i = 1/N$) where all assets have the same variance σ^2 and the average correlation coefficient is $\bar{\rho}$. The portfolio variance is:

$$\sigma_P^2 = \frac{\sigma^2}{N} + \left(1 - \frac{1}{N}\right)\bar{\rho}\sigma^2 \tag{8.18}$$

As the number of assets $N \to \infty$:

- The contribution of the individual asset variance, $\frac{\sigma^2}{N}$, converges to 0.
- The multiplier $\left(1 - \frac{1}{N}\right)$ approaches 1.

This leaves only the average covariance term to determine the portfolio's risk:

$$\lim_{N\to\infty} \sigma_P^2 = \bar{\rho}\sigma^2 = \text{Average Covariance} \tag{8.19}$$

Rủi ro danh mục đa dạng hóa cao chỉ phụ thuộc covariance trung bình giữa các tài sản (rủi ro hệ thống) chứ không phụ thuộc rủi ro riêng lẻ (phi hệ thống). Chúng ta hãy tự hỏi: Tại sao thị trường chỉ bù đắp lợi nhuận cho rủi ro hệ thống mà không bù đắp rủi ro phi hệ thống?

Example 9: Approximating Standard Deviation of a Portfolio with Many Assets

An equally weighted portfolio has 50 assets ($N = 50$). The average asset variance is $\sigma^2 = 0.0400$ and the average correlation coefficient is $\bar{\rho} = 0.3$. Calculate the exact portfolio standard deviation and compare it to the large-N approximation.
Solution:

1. **Exact Portfolio Variance using Equation 8.18:**

$$\sigma_P^2 = \frac{0.04}{50} + \left(1 - \frac{1}{50}\right)(0.3)(0.04) = 0.0008 + 0.98(0.012) = 0.01256$$

   Exact standard deviation: $\sigma_P = \sqrt{0.01256} \approx 11.21\%$.

2. **Approximated Variance using Equation 8.19:**

$$\sigma_P^2 \approx \bar{\rho}\sigma^2 = 0.3(0.04) = 0.0120$$

   Approximated standard deviation: $\sigma_P \approx \sqrt{0.0120} \approx 10.95\%$.

The approximation is very close to the exact value, demonstrating that for a sufficiently large portfolio, the portfolio variance is dominated by the average covariance between the assets.

EXAMPLE 9

Approximating Standard Deviation of a Portfolio with Many Assets

Calculate the portfolio standard deviation for a portfolio with the following characteristics:

- Average variance of individual assets $\sigma^2 = 0.04$
- Average correlation between the assets $\bar{\rho} = 0.3$
- Number of assets in the portfolio $N = 50$

Figure 8.3: Portfolio Risk Reduction as the Number of Assets (N) Increases

Example 10: Portfolio Size and Portfolio Variance (GBM Simulation)

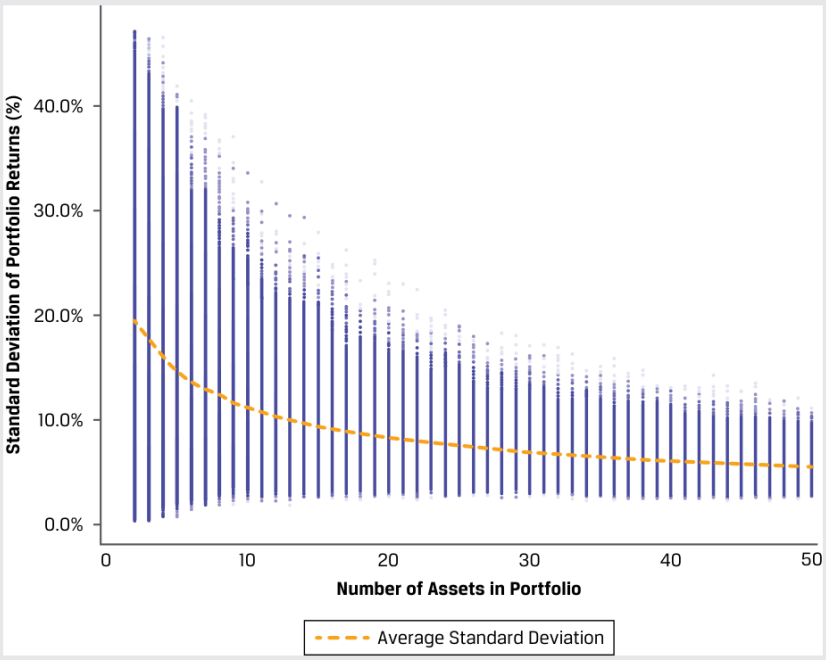
A simulation generates 250 daily returns for 100 randomly selected assets using geometric Brownian motion (GBM). Portfolios ranging from $N = 2$ to 50 assets are constructed. As shown in Exhibit 8.4, the standard deviation of portfolio returns decreases sharply as N increases from 2 to 20, but the marginal benefit of adding more assets diminishes beyond $N = 30$.
Key Takeaway: A portfolio of 20 to 30 assets captures most of the benefits of diversification, provided the assets have low correlations. Adding assets beyond this threshold yields negligible reduction in portfolio risk, as the remaining risk is systematic risk, which cannot be diversified away.

8.2 The Minimum-Variance Portfolio and the Efficient Frontier

b | describe, calculate, and interpret the minimum-variance portfolio and portfolios that lie on the efficient frontier

Portfolio optimization involves finding the optimal weights of assets to minimize risk for a given target return, or maximize expected return for a given risk level.

Exhibit 12: The Relationship Between the Dispersion of Portfolio Standard Deviation and the Number of Assets



From a portfolio-management perspective, this simulation indicates that well-diversified portfolios should contain multiple assets. Typically, the bulk of the available diversification benefits are achieved with a relatively small number of assets. A common rule of thumb is that a portfolio of 20 to 30 assets can capture most of the benefits of diversification, specifically when the assets have a low correlation. A diversified portfolio reduces the risk specific to individual assets, the unsystematic risk, but the overall risk inherent in the market, the systematic risk, remains and cannot be diversified away. We will discuss extensively the relationship between unsystematic risk, systematic risk, and portfolio risk in later learning modules.

Figure 8.4: GBM Simulation of Portfolio Standard Deviation vs. Number of Assets

8.2.1 The Feasible Set and Efficient Frontier

The **feasible set** (or investment opportunity set) represents all possible portfolios that can be constructed from a given group of assets. When there are multiple risky assets, the feasible set forms a cloud of portfolios in risk-return space.

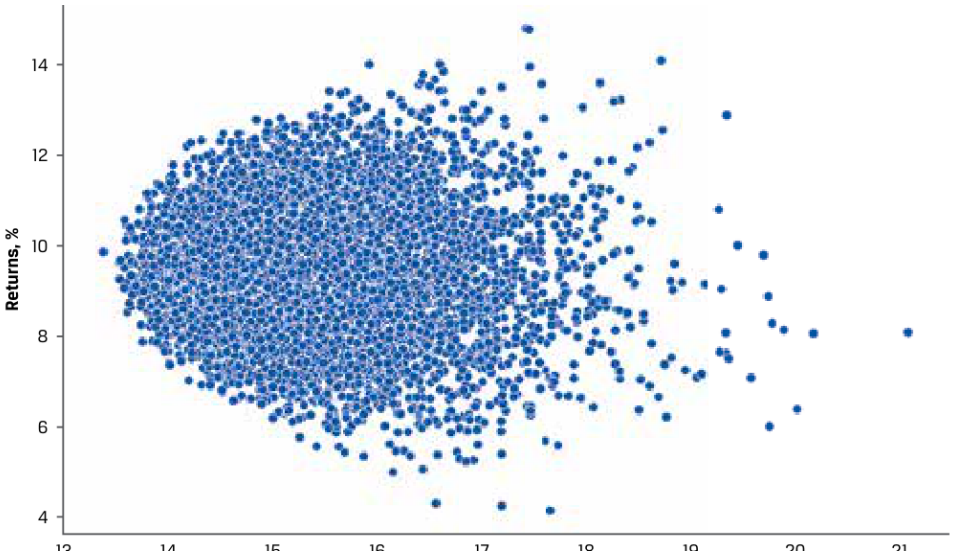


Figure 8.5: The Cloud of Feasible Portfolios in Risk-Return Space

The boundary of this cloud on the upper-left represents the optimal trade-off between risk and return:

- The **Minimum-Variance Frontier** is the boundary that minimizes variance for each level of expected return.
- The **Global Minimum-Variance Portfolio** is the single portfolio on the frontier that has the lowest possible risk.
- The **Markowitz Efficient Frontier** is the portion of the minimum-variance frontier that lies above the Global Minimum-Variance portfolio. Risk-averse, utility-maximizing investors will only choose portfolios on this segment.

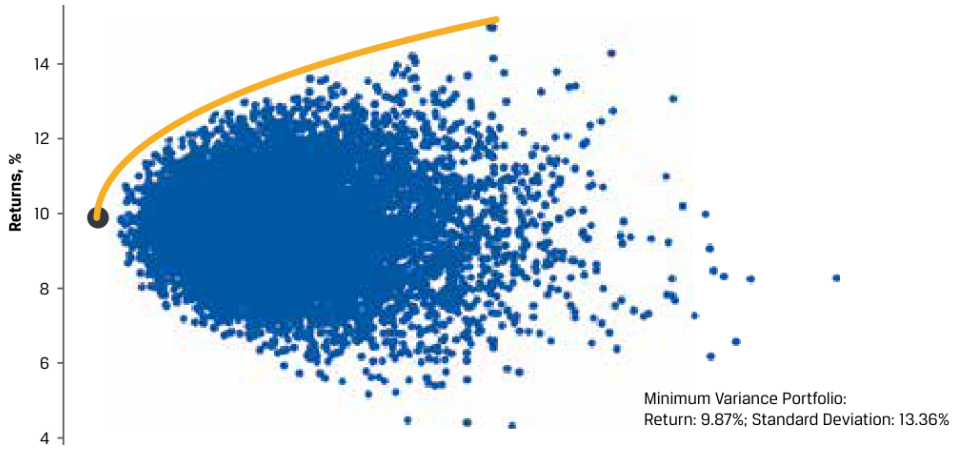


Figure 8.6: The Minimum-Variance Frontier and the Efficient Frontier

Mathematically, the optimization problem is formulated as:

$$\text{Minimize } \sigma_P^2 = \sum_{i=1}^{N}\sum_{j=1}^{N} w_i w_j \sigma_{i,j} \quad \text{subject to } \sum_{i=1}^{N} w_i = 1, \quad w_i \geq 0, \quad E[R_P] = k \tag{8.20}$$

where k is the target expected return. Under matrix form:

$$\text{Minimize } \sigma_P^2 = \mathbf{w}^T \mathbf{V} \mathbf{w} \quad \text{subject to } \mathbf{w}^T \mathbf{1} = 1, \quad \mathbf{w}^T \boldsymbol{\mu} = k, \quad \mathbf{w} \geq \mathbf{0} \tag{8.21}$$

The constraint $w_i \geq 0$ represents the **no-short-selling** constraint.

8.2.2 Derivation of the Minimum-Variance Portfolio Weights

For a two-asset portfolio, we can find the weights that minimize variance without complex optimization. Let w_A be the weight of Asset A and $w_B = 1 - w_A$ be the weight of Asset B.

$$\sigma_P^2 = w_A^2\sigma_A^2 + (1 - w_A)^2\sigma_B^2 + 2w_A(1 - w_A)\sigma_{A,B}$$

Taking the first derivative of σ_P^2 with respect to w_A and setting it equal to zero:

$$\frac{d\sigma_P^2}{dw_A} = 2w_A\sigma_A^2 - 2(1 - w_A)\sigma_B^2 + 2(1 - 2w_A)\sigma_{A,B} = 0$$

$$w_A(\sigma_A^2 + \sigma_B^2 - 2\sigma_{A,B}) - \sigma_B^2 + \sigma_{A,B} = 0$$

Solving for the optimal weight w_A^* gives the global minimum-variance weight:

$$w_A^* = \frac{\sigma_B^2 - \sigma_{A,B}}{\sigma_A^2 + \sigma_B^2 - 2\sigma_{A,B}} \tag{8.22}$$

Correspondingly, the weight for Asset B is:

$$w_B^* = 1 - w_A^* = \frac{\sigma_A^2 - \sigma_{A,B}}{\sigma_A^2 + \sigma_B^2 - 2\sigma_{A,B}} \tag{8.23}$$

Example 11: Calculating Minimum-Variance Weights

We revisit Example 3 where Asset A (S&P 500) has standard deviation $\sigma_A = 16.21\%$ ($\sigma_A^2 = 0.0263$), Asset B (MSCI EM) has standard deviation $\sigma_B = 33.11\%$ ($\sigma_B^2 = 0.1096$), and the covariance is $\sigma_{A,B} = 0.0107$ (corresponding to a 20% correlation). Calculate the weights, return, and standard deviation of the global minimum-variance portfolio.

Solution:

1. **Optimal Weights using Equation 8.22:**

$$w_A^* = \frac{0.1096 - 0.0107}{0.0263 + 0.1096 - 2(0.0107)} = \frac{0.0989}{0.1145} \approx 86.4\%$$

   Weight of Asset B:

$$w_B^* = 1 - 0.864 = 13.6\%$$

2. **Expected Portfolio Return:** $E[R_P] = 0.864(9.93\%) + 0.136(18.20\%) = 11.05\%$.

3. **Portfolio Standard Deviation (Risk):**

$$\sigma_P = \sqrt{(0.864^2 \times 0.0263) + (0.136^2 \times 0.1096) + 2(0.864)(0.136)(0.0107)}$$

$$\sigma_P = \sqrt{0.01963 + 0.00203 + 0.00251} = \sqrt{0.02417} \approx 15.55\%$$

Example 12: Selected Frontier Portfolios

We examine selected portfolios on the efficient frontier established between the S&P 500 and MSCI EM under the covariance assumption of $\sigma_{A,B} = 0.0050$ (as shown in Exhibit 8.8).

- Portfolio P6 (Global MVP): 71.8% S&P 500, 28.2% MSCI EM. Expected return: 12.26%, Risk: 14.96%.
- Portfolio P4: 68.5% S&P 500, 31.5% MSCI EM. Expected return: 12.53%, Risk: 15.27%.
- Portfolio P5: 49.8% S&P 500, 50.2% MSCI EM. Expected return: 14.08%, Risk: 18.52%.
- Portfolio P1: 16.3% S&P 500, 83.7% MSCI EM. Expected return: 16.85%, Risk: 27.85%.

Portfolios lying below P6 (offering lower expected returns for the same level of risk) are inefficient portfolios.

MVP phân chia đường biên tối thiểu làm hai phần. Chúng ta hãy suy nghĩ: Tại sao nhà đầu tư duy lý không chọn danh mục ở nửa dưới (dưới MVP)? Có phải vì cùng mức rủi ro, họ luôn tìm được danh mục ở nửa trên có tỷ suất sinh lời cao hơn?

Exhibit 16: The Minimum-Variance Portfolio and the Minimum-Variance Frontier

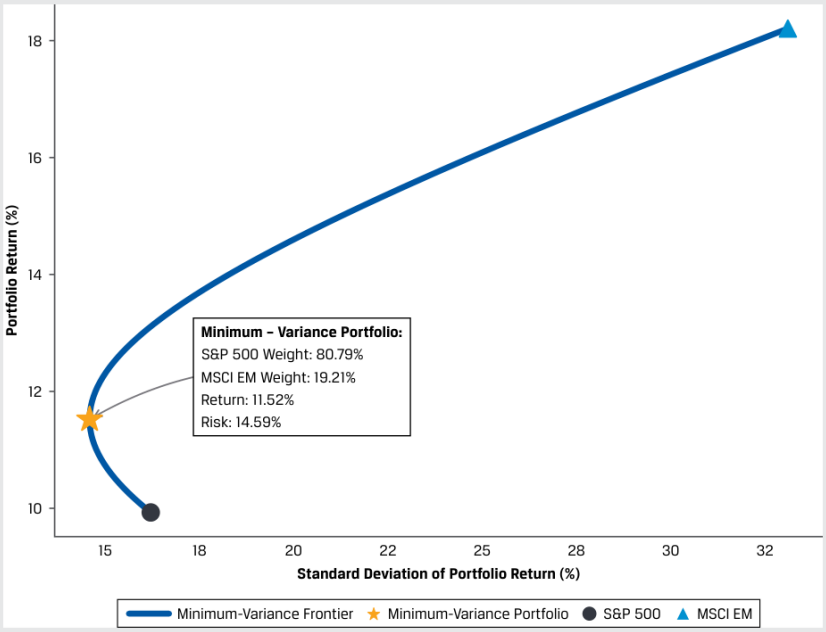


Exhibit 16 illustrates various points. An investor who seeks high returns and is willing to take on more risk will opt for a portfolio further to the upper right of the efficient frontier. Likewise, an investor who prefers lower risk will choose a portfolio that lies closer to the lower left of the efficient frontier. And an investor who wants a balance of risk and return will choose a portfolio around the center of the efficient frontier. The segment of the curved line that lies below the minimum-variance portfolio is not used, as there are portfolios on the efficient frontier that offer a higher expected return for the same level of risk. Finally, Exhibit 16 shows the benefit of diversification: Combining different assets can reduce portfolio risk at any given level of expected return, particularly if the assets are negatively or weakly correlated. In other words a combination of two risky assets can be less risky than either of them.

Figure 8.7: The Minimum-Variance Frontier and the S&P 500 / MSCI EM Indices

EXAMPLE 13

Combining the Optimal Risky Portfolio with a Risk-Free Asset

A portfolio manager creates an optimal risky portfolio by combining the risky portfolio, proxied by the MSCI World Index, and a risk-free asset to achieve a target return of 9%:

- Expected return of MSCI World Index, $r_{\text{MSCI}} = 10.4\%$
- Volatility of MSCI World Index, $\sigma_{\text{MSCI}} = 14.8\%$
- Risk-free rate, $r_f = 1\%$
- Risk-free volatility, $\sigma_f = 0\%$

What are the optimal weights and standard deviation for the portfolio?

Step 1: Calculating Portfolio Weights

To achieve the target return, r_p, we use the expected return constraint:

$$w_{\text{MSCI}} \times r_{\text{MSCI}} + w_f \times r_f = r_p$$

along with the constraint that the weights must sum to 1:

$$w_f = 1 - w_{\text{MSCI}}.$$

Substituting for w_f into the expected return constraint, we solve for w_{MSCI}:

$$w_{\text{MSCI}} = \frac{r_p - r_f}{r_{\text{MSCI}} - r_f}.$$

Figure 8.8: The Efficient Frontier and Selected Opportunity Portfolios

Example 13: Capital Allocation weights

An analyst creates a target portfolio return of 9.0% by combining the MSCI World Index (expected return 10.4%, standard deviation 14.80%) and a risk-free asset returning 1.0% ($\sigma_f = 0.0\%$). Calculate the weights and portfolio standard deviation.
Solution:

1. **Weights:** Let w_M be the weight allocated to the MSCI World Index:

$$R_P = w_M R_M + (1 - w_M) R_f \implies 9\% = 10.4\% w_M + 1\%(1 - w_M)$$

$$8\% = 9.4\% w_M \implies w_M = \frac{8\%}{9.4\%} \approx 85.1\%$$

   Weight allocated to the risk-free asset: $w_f = 1 - 0.851 = 14.9\%$.

2. **Portfolio Standard Deviation:** Since the risk-free asset has zero variance and covariance:

$$\sigma_P = w_M \sigma_M = 0.851 \times 14.80\% \approx 12.59\%$$

Example 14: Portfolio Optimization with Several Assets

An analyst aims to optimize a traditional 60/40 stock/bond portfolio ($R_{60/40} = 10\%$, $\sigma_{60/40} = 13\%$) using five assets: Global Developed Equities (12%, 20%), Emerging Market Equities (16%, 33%), Long-Term Bonds (5%, 8%), International Bonds (7%, 15%), and Short-Term Bills (3%, 1%).
Solution:

1. **Return-Constrained Optimization:** Minimize portfolio risk for a target expected return of 11%.
2. **Risk-Constrained Optimization:** Maximize expected return for a target portfolio risk of 12%.

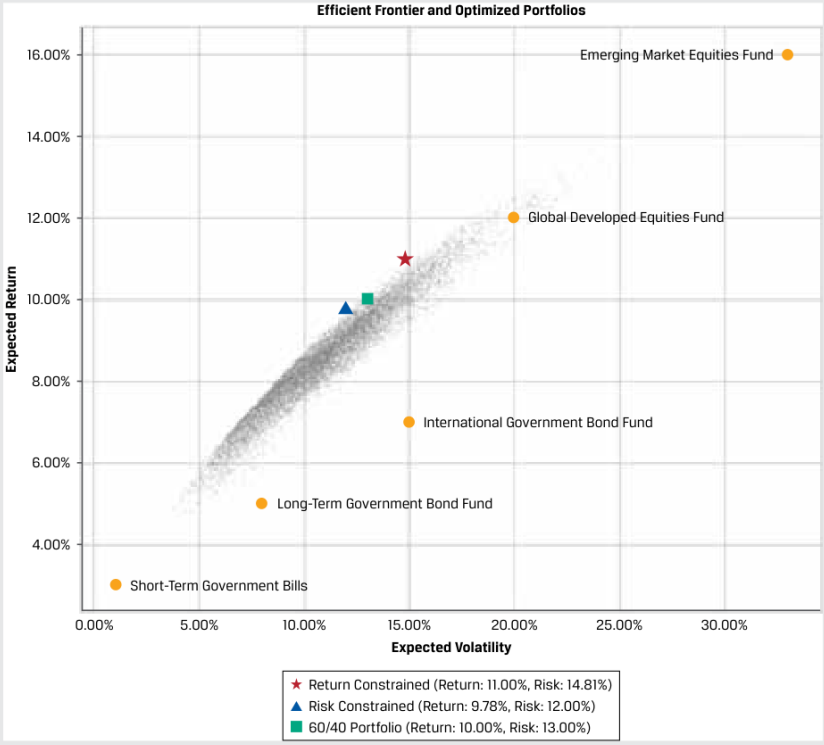
The resulting optimal asset weights are detailed in Table 8.5.

Table 8.5: Optimal Portfolio Weights vs. Traditional 60/40 Portfolio

Asset / Metric	Return-Constrained	Risk-Constrained
Global Developed Equities	53.45%	48.64%
Emerging Market Equities	14.75%	5.33%
Long-Term Government Bonds	0.00%	6.61%
International Government Bonds	31.80%	39.42%
Short-Term Government Bills	0.00%	0.00%
Expected Portfolio Return	**11.00%**	**9.78%**
Expected Portfolio Risk	**14.81%**	**12.00%**

Both portfolios lie on the multi-asset efficient frontier (Exhibit 8.9), outperforming the traditional 60/40 allocation.

Exhibit 21: Efficient Frontier with Assets and Constrained Portfolios



Asset allocation can enhance portfolio performance beyond the 60/40 equity–bond strategy. The return-constrained portfolio achieved an 11% return with a 14.8% standard deviation. The risk-constrained strategy achieved a 9.8% return with a 12% risk. Note that the risk-constrained portfolio achieves close to the same return, 9.8% vs. 10%, as the original 60/40 portfolio split.

Figure 8.9: Efficient Frontier and Optimized Constrained Portfolios

8.3 Optimal Portfolio Selection

c | explain the selection of an optimal portfolio, given an investor's risk aversion and the capital allocation line, and how this extends to the market portfolio and the capital market line

Optimal portfolio selection balances expected returns with risk, matching the opportunity set against an investor's personal preferences.

8.3.1 Investor Utility and Indifference Curves

The utility function represents an investor's satisfaction or ranking of portfolios based on risk and expected return. For a risk-averse investor, utility is formulated as:

$$U = E[R] - \frac{1}{2}A\sigma^2 \tag{8.24}$$

where $E[R]$ is expected return, σ^2 is variance of returns, and A is the **coefficient of risk aversion**.

- $A > 0$: Risk-averse investor. Utility decreases with higher risk.
- $A = 0$: Risk-neutral investor. Utility depends solely on expected return.
- $A < 0$: Risk-seeking investor. Utility increases with risk.

An **indifference curve** maps the combinations of risk and expected return that yield the same level of utility.

Table 8.6: Risk Preferences and Indifference Curves

Preference	Aversion (A)	Indifference Slope	Utility Curve Shape
Risk-seeking	$A < 0$	Downward/Concave	Convex utility in wealth
Risk-neutral	$A = 0$	Flat/Horizontal	Linear utility in expected return
Risk-averse	$A > 0$	Upward/Convex	Concave utility in wealth

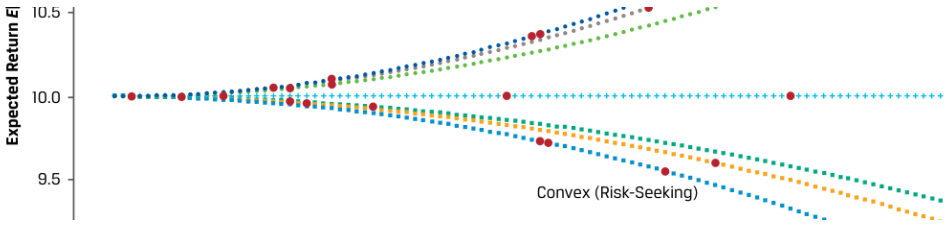


Figure 8.10: Indifference Curves for Risk-Averse, Risk-Neutral, and Risk-Seeking Investors

Steeper indifference curves represent higher risk aversion (A), meaning the investor requires a much higher expected return to accept a small increase in risk.

Knowledge Check: Utility calculations

An investor evaluates three investments:

Investment	Expected Return $E[R]$ (%)	Standard Deviation σ (%)
A	12	30
B	15	35
C	21	40

Determine the optimal choice for:

1. An investor with $A = 4$.
2. An investor with $A = 2$.
3. A risk-seeking investor with $A = -4$.

Solution: Using $U = E[R] - \frac{1}{2}A\sigma^2$:

1. **Case 1 ($A = 4$):**
   - $U_A = 0.12 - 0.5(4)(0.09) = -0.060$
   - $U_B = 0.15 - 0.5(4)(0.1225) = -0.095$
   - $U_C = 0.21 - 0.5(4)(0.16) = -0.110$

The investor chooses **Investment A** (highest utility / lowest negative utility).

2. **Case 2** ($A = 2$)**:**
   - $U_A = 0.12 - 0.5(2)(0.09) = 0.0300$
   - $U_B = 0.15 - 0.5(2)(0.1225) = 0.0275$
   - $U_C = 0.21 - 0.5(2)(0.16) = 0.0500$

   The investor chooses **Investment C**. Despite higher risk, the expected return compensation is sufficient.

3. **Case 3** ($A = -4$)**:**
   - $U_A = 0.12 - 0.5(-4)(0.09) = 0.300$
   - $U_B = 0.15 - 0.5(-4)(0.1225) = 0.395$
   - $U_C = 0.21 - 0.5(-4)(0.16) = 0.530$

   The investor chooses **Investment C**. Utility increases with both return and risk.

8.3.2 The Capital Allocation Line (CAL) and Sharpe Ratio

When a risk-free asset (with return R_f and standard deviation $\sigma_f = 0$) is introduced, investors can combine it with a risky portfolio P. The expected return of the combined portfolio is:

$$E[R_c] = w_f R_f + (1 - w_f) E[R_P] \tag{8.25}$$

The standard deviation is:

$$\sigma_c = (1 - w_f)\sigma_P \implies (1 - w_f) = \frac{\sigma_c}{\sigma_P} \tag{8.26}$$

Substituting $(1 - w_f)$ into the expected return equation yields the **Capital Allocation Line (CAL)**:

$$E[R_c] = R_f + \left(\frac{E[R_P] - R_f}{\sigma_P} \right) \sigma_c \tag{8.27}$$

The slope of the CAL is the **Sharpe ratio (SR)** of the risky portfolio:

$$\text{Sharpe Ratio} = \frac{E[R_P] - R_f}{\sigma_P} \tag{8.28}$$

The optimal risky portfolio (tangency portfolio) is the point on the efficient frontier that maximizes the Sharpe ratio, creating the steepest possible CAL.

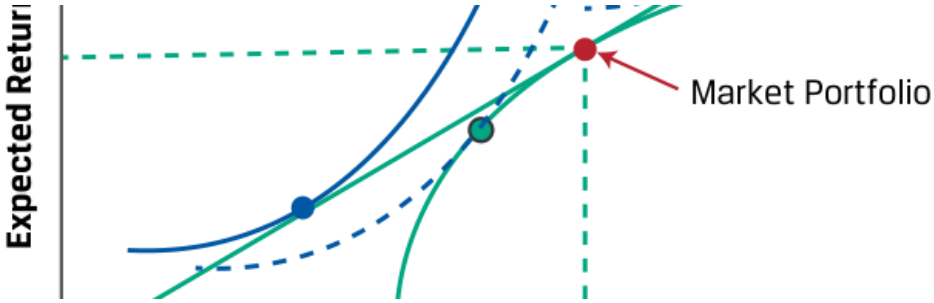


Figure 8.11: Optimal Portfolio Selection via Tangency between Indifference Curves and CAL

The optimal portfolio for a specific investor lies at the point of tangency between their highest attainable indifference curve and the CAL:

- More risk-averse investors select portfolios closer to the risk-free asset (left on the CAL).
- Less risk-averse investors select portfolios closer to or beyond the tangency portfolio P, utilizing leverage (borrowing at the risk-free rate) to buy more of the risky portfolio.

Example 15: Calculating the Sharpe Ratio

Assuming a risk-free rate of 1.0%, compare:

- Portfolio A: expected return 9.0%, weight in MSCI World 85.1%, standard deviation 12.59%.
- Portfolio B: expected return 12.0%, weight in MSCI World 117.1%, standard deviation 17.32%.

Solution:

- $\text{SR}_A = \frac{9\%-1\%}{12.59\%} \approx 0.635$
- $\text{SR}_B = \frac{12\%-1\%}{17.32\%} \approx 0.635$
- Both portfolios lie on the same Capital Allocation Line (CAL) and have the same Sharpe ratio. Portfolio B is a leveraged portfolio, borrowing 17.1% at the risk-free rate to increase investment in the risky asset portfolio.

8.3.3 The Capital Market Line (CML)

If all investors share identical assumptions and hold mean-variance efficient portfolios, the optimal risky portfolio is the **Market Portfolio** (M). The Market Portfolio theoretically contains all risky assets in the universe, weighted by their market values. The Capital Allocation Line using the Market Portfolio as the risky portfolio is called the **Capital Market Line (CML)**:

$$E[R_P] = R_f + \left(\frac{E[R_M] - R_f}{\sigma_M}\right)\sigma_P \tag{8.29}$$

where $E[R_M]$ and σ_M are the expected return and standard deviation of the market portfolio.

According to the **mutual fund theorem**, all rational investors can achieve optimal asset allocation by holding only two assets: the risk-free asset and the market portfolio.

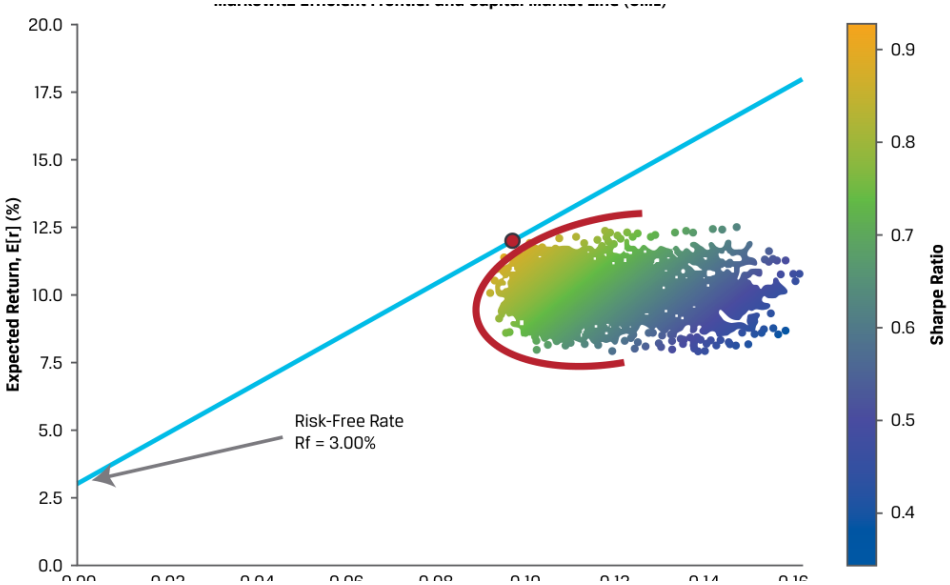


Figure 8.12: The Capital Market Line (CML) and the Markowitz Efficient Frontier

8.3.4 The Capital Asset Pricing Model (CAPM) and Beta

Total risk consists of two parts:

- **Unsystematic Risk** (diversifiable / idiosyncratic risk): Unique to a specific company or sector. Can be eliminated in a well-diversified portfolio.
- **Systematic Risk** (undiversifiable / market risk): Driven by macroeconomic factors that affect all assets. Cannot be diversified away.

Since investors can eliminate unsystematic risk at no cost through diversification, they are not compensated for bearing it. The expected return of an asset depends solely on its systematic risk.

Systematic risk is measured by **Beta** (β), which quantifies the sensitivity of an asset's returns to movements in the market portfolio:

$$\beta_i = \frac{\sigma_{i,M}}{\sigma_M^2} = \frac{\rho_{i,M}\sigma_i\sigma_M}{\sigma_M^2} = \rho_{i,M}\frac{\sigma_i}{\sigma_M} \tag{8.30}$$

where $\sigma_{i,M}$ is the covariance between asset i and the market, and σ_M^2 is the variance of the market portfolio.

- $\beta = 1.0$: Asset has the same systematic risk as the market.
- $\beta > 1.0$: Asset is more volatile than the market (aggressive asset).
- $\beta < 1.0$: Asset is less volatile than the market (defensive asset).

The **Capital Asset Pricing Model (CAPM)** states that the expected return of asset i is:

$$E[R_i] = R_f + \beta_i \left(E[R_M] - R_f\right) \tag{8.31}$$

where $(E[R_M] - R_f)$ is the **market risk premium (MRP)**.

The graphical representation of CAPM is the **Security Market Line (SML)**, which plots expected return against Beta. In contrast to the CML (which uses total risk / standard deviation and applies only to efficient portfolios), the SML uses systematic risk / Beta and applies to **all** assets, whether efficient or individual.

$$E[R_i] = R_f + \left(\frac{E[R_M] - R_f}{\sigma_M^2}\right)\sigma_{i,M} \tag{8.32}$$

CML dùng tổng rủi ro (σ) cho danh mục hiệu quả. SML (CAPM) dùng rủi ro hệ thống (β) cho mọi tài sản. Chúng ta hãy thảo luận: Điểm khác biệt cốt lõi về rủi ro giữa CML và SML là gì? Tại sao cổ phiếu đơn lẻ có độ lệch chuẩn lớn không nằm trên CML nhưng vẫn nằm trên SML?

Example 16: CAPM Return Calculation

The risk-free rate is 3.0% and the expected market return is 8.0%. Calculate the expected return of an asset with a Beta of 1.2.

Solution: Using the Capital Asset Pricing Model (CAPM) in Equation 8.31:

$$E[R_i] = 3.0\% + 1.2(8.0\% - 3.0\%) = 3.0\% + 1.2(5.0\%) = 3.0\% + 6.0\% = 9.0\%$$

Knowledge Check: The Capital Asset Pricing Model

Consider: $R_f = 2.0\%$, $E[R_M] = 8.0\%$, $\sigma_M = 15.0\%$, and an investor's optimal portfolio has standard deviation $\sigma_P = 10.0\%$.

1. Calculate the expected return of the portfolio using CML.
2. Calculate the Beta of the portfolio.
3. Calculate the expected return using CAPM and confirm it matches the CML.
4. Calculate the Sharpe ratio of the market portfolio.

Solution:

1. **CML expected return:**

$$E[R_P] = R_f + \left(\frac{E[R_M] - R_f}{\sigma_M}\right)\sigma_P = 2\% + \left(\frac{8\% - 2\%}{15\%}\right)(10\%) = 2\% + 0.4(10\%) = 6\%$$

2. **Beta:** For an efficient portfolio on the CML, the correlation with the market is $\rho = 1.0$:

$$\beta_P = \frac{\sigma_P}{\sigma_M} = \frac{10\%}{15\%} \approx 0.6667$$

3. **CAPM return:**

$$E[R_P] = R_f + \beta_P(E[R_M] - R_f) = 2\% + 0.6667(8\% - 2\%) = 2\% + 4\% = 6\%$$

   The results match exactly, confirming the consistency between CML and CAPM for efficient portfolios.

4. **Sharpe ratio:** $\text{SR}_M = \frac{8\%-2\%}{15\%} = 0.40$.

Estimating Beta using Linear Regression

In practice, Beta is estimated using ordinary least squares (OLS) linear regression of the asset's excess returns against the market's excess returns. This is known as the **Market Model**:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i \left(R_{M,t} - R_{f,t}\right) + \epsilon_{i,t} \tag{8.33}$$

where α_i is the intercept, β_i is the slope, and $\epsilon_{i,t}$ is the error term. Under CAPM, the expected alpha (α_i) should be exactly 0 in equilibrium.

Example 17: Estimating LVMH's Beta via Bloomberg

An analyst regresses monthly returns of LVMH Moët Hennessy Louis Vuitton against the CAC 40 index over 3 years (36 data points). The Bloomberg regression output shows:

- Regression line: $y = 0.939x + 1.318$ ($\beta = 0.939$, $\alpha = 1.318\%$)
- Correlation coefficient $R = 0.850$ ($R^2 = 0.722$)
- Standard error of Beta: 0.101

The expected market return is 9.577% and the risk-free rate is 2.560%. Calculate LVMH's expected return.

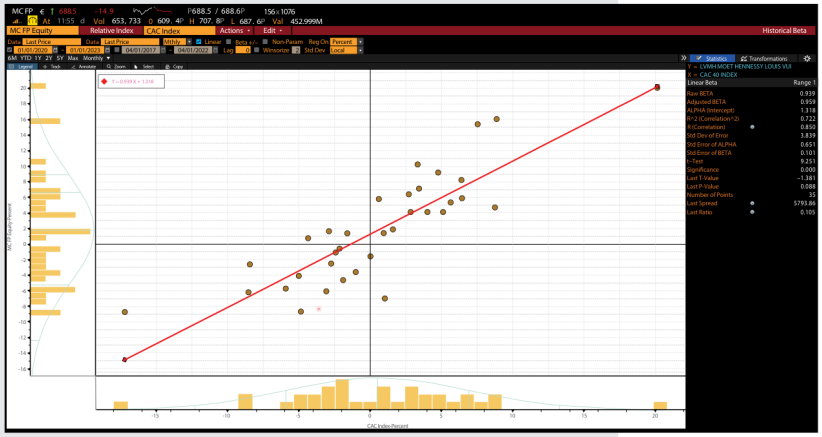
Solution:

1. **Beta Interpretation:** A beta of 0.939 indicates that LVMH has slightly lower systematic risk than the French equity market average (CAC 40 Index).
2. **Expected Return (CAPM):**

$$E[R_{\text{LVMH}}] = R_f + \beta(E[R_M] - R_f) = 2.56\% + 0.939(9.577\% - 2.56\%)$$

$$E[R_{\text{LVMH}}] = 2.56\% + 0.939(7.017\%) = 2.56\% + 6.589\% \approx 9.15\%$$

Exhibit 26: Bloomberg Screenshot of LVMH Stock Returns vs. Returns of the CAC 40 Index



Source: Bloomberg.

Exhibit 27 shows another Bloomberg screenshot, EQRP, or equity risk premium, for LVMH Moët Hennessy Louis Vuitton, where:

- the expected market return, $E[r_m]$, is 9.577%;
- the risk-free rate, RFR, r_f, is 2.560%; and
- the country/region risk premium, CRP, $E[r_m] - r_f$, is 7.017%.

As of 29 December 2023, LVMH Moët Hennessy Louis Vuitton has an estimated β of 0.939 and a risk premium of $\beta\left(E[r_m] - r_f\right) = 6.589\%$.

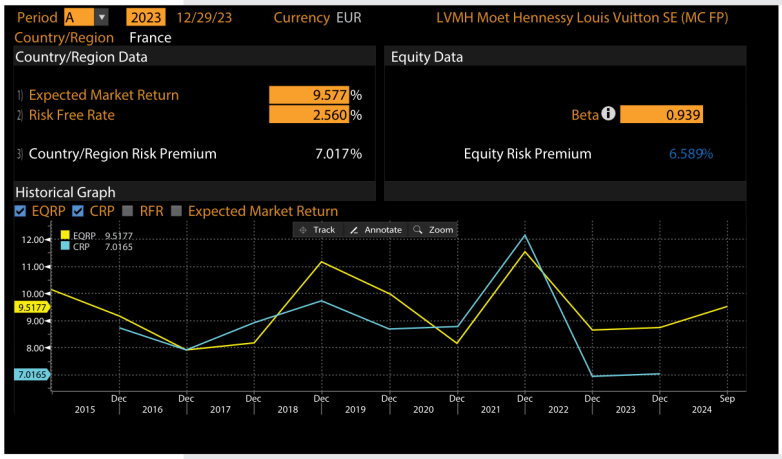
As Bloomberg's methodology is proprietary, we do not know exactly how it estimates the expected return of the market. Even so, it is useful to see how the expected return of an asset can be derived from that of the market.

Figure 8.13: Bloomberg Regression Output (BETA) for LVMH vs. CAC 40 Index

8.3.5 Limitations of Portfolio Theory and CAPM

Several assumptions underlying Markowitz optimization and CAPM do not hold in real-world markets:

Exhibit 27: LVMH Moët Hennessy Louis Vuitton Expected Return and Risk Premium



Source: Bloomberg.

Under the CAPM, the expected annual return on LVMH Moët Hennessy Louis Vuitton would be calculated as:

$$E[r_i] = r_f + \beta(E[R_m] - Rr_f)$$
$$= 2.560\% + 0.939 \times (9.577\% - 2.560\%)$$
$$= 2.560\% + 0.939 \times 7.017\%$$
$$= 2.560\% + 6.589\% = 9.149\%.$$

Figure 8.14: Bloomberg Risk Premium Screen (EQRP) for LVMH Moët Hennessy Louis Vuitton

- **Distributional Assumptions:** Financial returns are rarely normally distributed, exhibiting fat tails (high kurtosis) and skewness.
- **Time-Varying Volatility and Correlation:** Variances and correlations are not constant but fluctuate, especially rising during systemic market crises.
- **Market Inefficiencies:** Information asymmetries exist, leading to mispricings and non-zero alphas.
- **Estimation Errors:** Small errors in the expected return and covariance inputs can result in dramatically different and highly concentrated optimal weights (input sensitivity).
- **Historical limits:** Historical relationships may not persist in the future.
- **Transaction Costs and Taxes:** Real-world investors must pay brokerage fees, spreads, and taxes. Capital distributions (dividends) and capital gains face differing tax structures, which reduces the after-tax efficient frontier.

Module 9

Simulation of Financial Asset Prices and Returns

LEARNING OUTCOMES	
a	describe historical simulation and explain how it can be used in investment applications
b	describe bootstrap resampling, and explain how it can be used in investment applications
c	describe Monte Carlo simulation and explain how it can be used in investment applications

9.1 Historical Simulation

a | describe historical simulation and explain how it can be used in investment applications

Mô phỏng lịch sử (Historical Simulation) là phương pháp phi tham số đơn giản nhất: thay vì giả định phân phối xác suất, chúng ta "tái hiện" lại quá khứ bằng cách áp dụng trực tiếp các biến động lịch sử lên danh mục hiện tại. Ưu điểm lớn nhất là gì? Đó là bảo toàn nguyên vẹn các hiện tượng thực tế như đuôi béo (fat tails) hay tương quan tài sản mà không cần mô hình hóa phức tạp. Nhưng liệu quá khứ có luôn lặp lại đúng như vậy không?

9.1.1 Overview of Simulation Methods

Simulation predicts potential outcomes for a system based on theoretical scenarios by defining objectives, selecting variables, assigning values, evaluating scenarios, and interpreting results. Figure 9.1 outlines the key steps.

analyzing the resulting scenarios, and then compiling and interpreting the outcomes. The key steps of the simulation process are outlined in Exhibit 1.

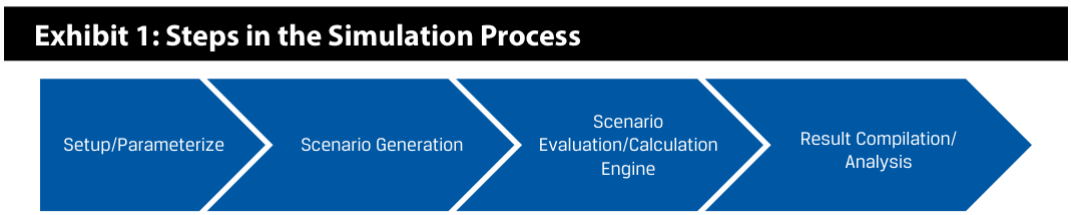


Figure 9.1: Steps in the Simulation Process

The simulation process comprises:

- **Step 1: Setup/parameterize.** Define the target output variable (e.g., risk-adjusted return) and identify the necessary inputs and mathematical relationships.
- **Step 2: Scenario generation.** Generate input values (scenarios) representing different market conditions.

- **Step 3: Scenario evaluation/calculation engine.** Process scenarios through a mathematical model to calculate outputs.
- **Step 4: Result compilation/analysis.** Aggregate outputs into a probability distribution to assess risks.

Three common simulation approaches differ primarily in how they generate scenarios (Step 2):

- **Historical simulation** applies actual past market movements directly to current portfolio factors.
- **Bootstrap simulation** resamples observed historical data with replacement to generate synthetic datasets.
- **Monte Carlo simulation** generates random scenarios based on assumed probability distributions.

Table 9.1 compares these approaches.

Table 9.1: Comparing the Steps for Different Simulation Approaches

Simulation Approach	Step 1: Setup/ Parameterize	Step 2: Scenario Generation	Step 3: Scenario Evaluation	Step 4: Result Analysis
Historical Simulation	Use historical data to initialize model parameters.	Incorporate actual past observations for scenario inputs.	Run the model using each historical scenario.	Examine performance under real-world conditions encountered in the past.
Bootstrap Simulation	Adopt historical data to define parameters (with replacement).	Formulate synthetic scenarios by resampling the observed dataset.	Conduct calculations for each resampled scenario.	Assess variability while preserving the original data structure.
Monte Carlo Simulation	Specify probability distributions for model inputs.	Draw random samples from these distributions to shape scenario inputs.	Execute calculations using each random input vector.	Evaluate the resulting distribution of outputs to determine risks and probabilities.

The three simulation approaches differ primarily in Step 2 (scenario generation); the other steps remain functionally identical.

9.1.2 Historical Simulation Overview

Historical simulation uses historical prices and returns to model future outcomes under the assumption that past patterns approximate future risks. This non-parametric approach captures empirical features like fat tails, kurtosis, and volatility clustering without distributional assumptions. However, it relies on the assumption that past patterns represent the future, which fails during structural shifts.

9.1.3 Implementing Historical Simulation

In historical simulation, scenarios are assembled from observed historical changes in each variable, as shown in Figure 9.2.

To construct a robust simulation, key valuation and risk factors need to be identified, which serve as the fundamental inputs in the simulation process. For equities, the primary variables are usually continuously compounded (logarithmic) price returns:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) \tag{9.1}$$

For fixed-income instruments, the simulation models changes in the yield curve. The price of a bond is determined by discounting future cash flows using simulated yields:

$$PV_B = \sum_{t=1}^{T} \frac{CF_t}{(1+y_t)^t} \tag{9.2}$$

Historical simulation is commonly used to calculate portfolio Value at Risk (VaR). The case study below applies this method to a long-only portfolio.

historical market movements. For each input variable and each historical data point, we assign the corresponding historical change to that variable, as illustrated in Exhibit 3.

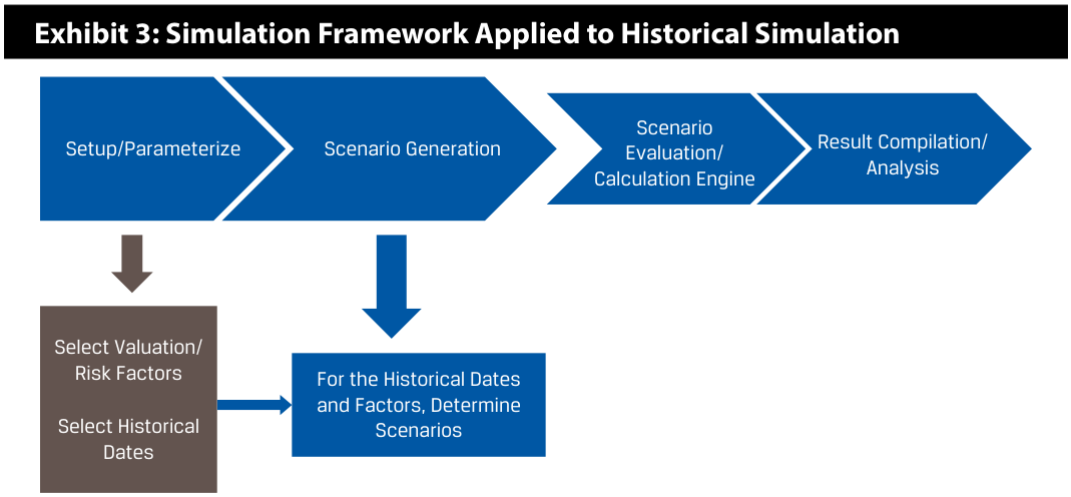


Figure 9.2: Simulation Framework Applied to Historical Simulation

Case Study: Historical Simulation of a Long-Only Portfolio

A portfolio manager wants to determine the daily Value at Risk (VaR) of a portfolio at the 95

- HSI (Hang Seng Index): HKD 55,000,000
- CSI 300 Index: HKD 40,000,000

The manager uses a one-year history of daily log returns (246 trading days).
Step 1: Calculate log returns.
Daily log returns for both indexes are calculated as: $r = \ln(P_1/P_0)$. For example, on Day 1:

$$r_{HS,1} = \ln\left(\frac{20,145.29}{19,781.41}\right) = 1.82\%$$

Returns are shown in Figure 9.3 and Table 9.2.

The calculated daily log returns for the 246 trading days are depicted in Exhibit 4 and compiled into a table in Exhibit 5.

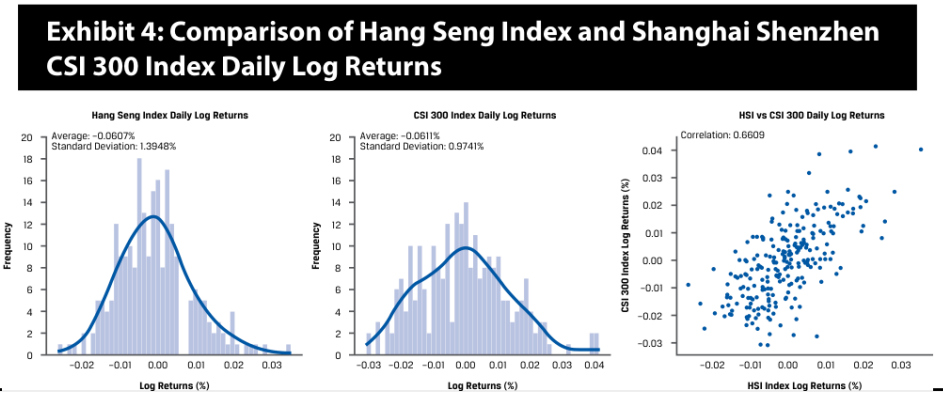


Figure 9.3: Comparison of Hang Seng Index and Shanghai Shenzhen CSI 300 Index Daily Log Returns

Table 9.2: Hang Seng Index and CSI 300 Daily Log Returns

Scenario No.	Hang Seng Index in HKD	CSI 300 in HKD	Hang Seng Index Log Return	CSI 300
0, $t = 0$	19,781.41	4,381.91	–	
1	20,145.29	4,391.77	1.82%	0.
2	20,793.11	4,416.16	3.17%	0.
3	21,052.17	4,504.74	1.24%	1.
4	20,991.64	4,543.39	-0.29%	0.
...	...	...	...	
244	16,624.84	3,648.65	1.73%	-0
245	17,043.53	3,752.58	2.49%	2.
246	17,047.39	3,772.51	0.02%	0.

Step 2: Calculate daily profit and loss (P/L).
For each day, the P/L is computed as: $P/L = \text{Position value} \times (e^r - 1)$. For Day 1:

$$\text{Total P/L} = 55\text{M} \times (e^{1.82\%} - 1) + 40\text{M} \times (e^{0.22\%} - 1) = \text{HKD } 1,098,262$$

The P/L across all scenarios are compiled in Table 9.3.

Table 9.3: Portfolio P/L for the Long-Only Portfolio

Scenario	HSI Log Return	CSI 300 Log Return	HSI P/L (HKD)	CSI 300 P/L (HKD)	Total Por
1	1.82%	0.22%	1,010,165	88,097	
2	3.17%	0.55%	1,771,429	220,606	
3	1.24%	1.99%	686,246	803,973	
4	-0.29%	0.85%	-159,269	341,449	
...	...	...	...	...	
244	1.73%	-0.14%	959,778	-55,961	
245	2.49%	2.81%	1,386,693	1,139,941	
246	0.02%	0.53%	11,001	212,563	

Step 3: Create the P/L distribution.
Daily P/L results generate the empirical distribution shown in Figure 9.4.

bution of potential daily portfolio total profit and loss outcomes. This distribution can be visualized using a histogram, a P/L curve, as depicted in Exhibit 7.

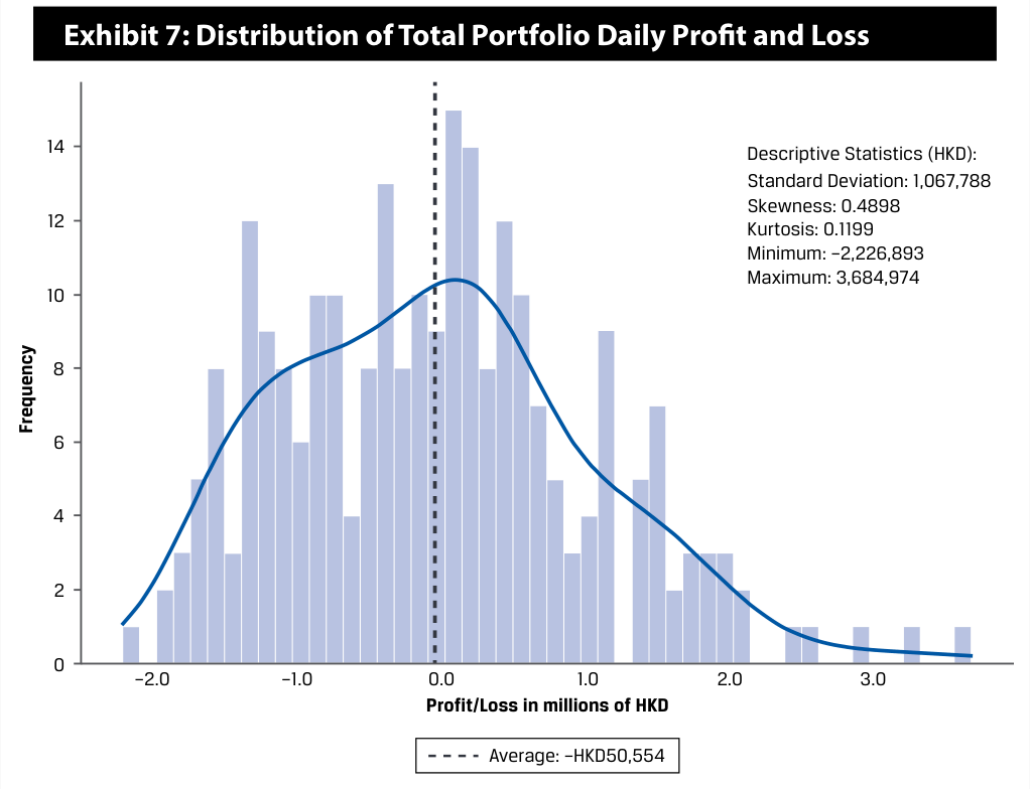


Figure 9.4: Distribution of Total Portfolio Daily Profit and Loss

Step 4: Determine the VaR.
The daily P/L estimates are ranked from lowest to highest (Table 9.4).

Table 9.4: Ranking of Total Portfolio Daily Profit and Loss

Rank	Day	Total P/L (HKD)
1	262	-2,226,893
2	157	-1,949,811
3	39	-1,888,929
...	...	...
11	49	-1,651,318
12	43	-1,624,801
13	25	-1,607,739
...	...	...
246	176	3,684,974

For 246 observations, the 5

$$\text{VaR}_{95\%} = 0.3 \times 1,607,739 + 0.7 \times 1,624,801 = \text{HKD } 1,619,682$$

For comparison, a parametric VaR is calculated assuming a joint normal distribution with parameters in Table 9.5.

Table 9.5: Descriptive Statistics

Statistic	Hang Seng Index	CSI 300 (in HKD)
Average Log Return, μ	-0.0607%	-0.0611%
Standard Deviation, σ	1.3948%	0.9741%
Correlation, ρ	0.6609	–

Assuming zero mean returns for daily calculations, the portfolio value variance (for $D_{HS} = 55\text{M}$ and $D_{SS} = 40\text{M}$) is:

$$\sigma_p^2 = D_{HS}^2\sigma_{HS}^2 + D_{SS}^2\sigma_{SS}^2 + 2D_{HS}D_{SS}\rho\sigma_{HS}\sigma_{SS} = 1.1354\,(\text{million HKD})^2 \tag{9.3}$$

The standard deviation is $\sigma_p = \text{HKD } 1.0656$ million. The 95

$$\text{VaR}_{95\%} = 1.96 \times 1.0656 = \text{HKD } 2.0885 \text{ million}$$

The difference arises because parametric VaR relies on the normality assumption, whereas historical VaR directly reflects empirical tail behavior and any inherent drift.

Chúng ta lưu ý lý do thực tế tại sao VaR tham số dùng $z = 1.96$ ở đây lớn hơn VaR lịch sử: giả định phân phối chuẩn phân bổ đều xác suất về hai phía đuôi, trong khi dữ liệu thực tế ít biến động cực đại hơn. Điều này phản ánh rõ: VaR tham số phụ thuộc vào mô hình toán, còn VaR lịch sử phụ thuộc vào mẫu dữ liệu cụ thể.

Case Study: Modeling Investment Scenarios for a Sustainable Energy Startup

A venture capital firm is evaluating an investment with a five- to seven-year horizon. It analyzes 50 past investments to estimate risks by reviewing deviations between projected and realized revenues.
Step 1: Determine the historical distribution of returns.
Figure 9.5 and Table 9.6 show the distribution of revenue forecast deviations.

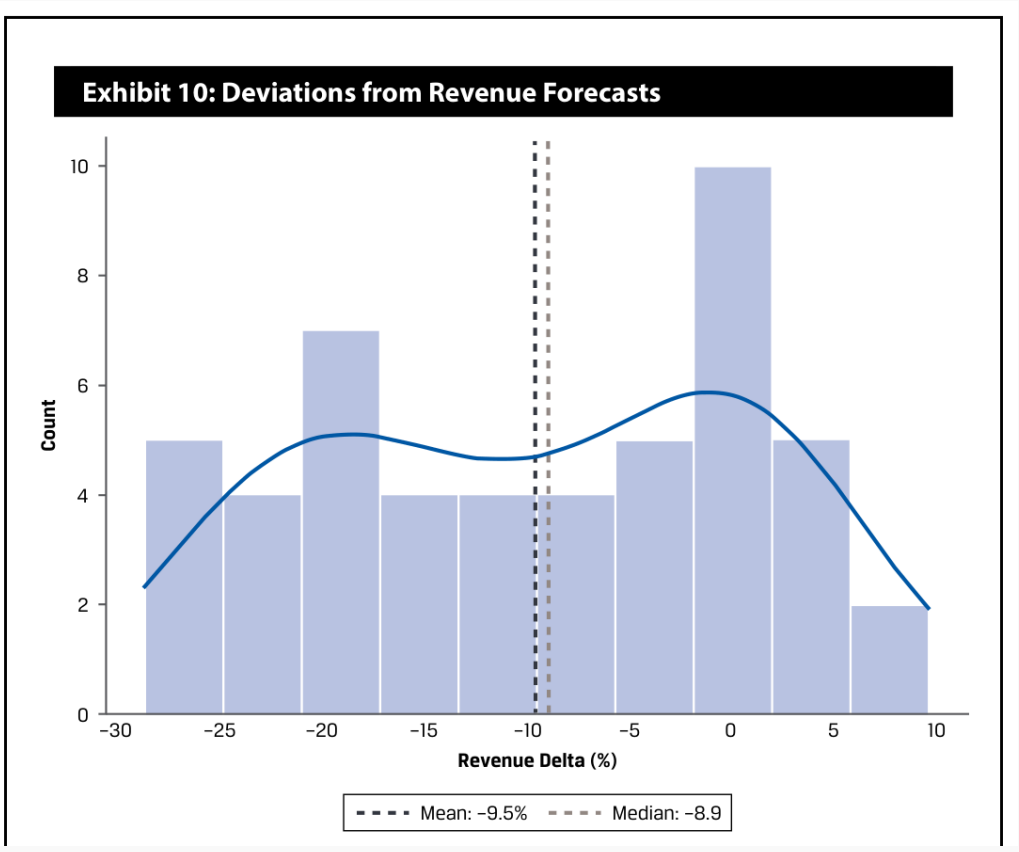


Figure 9.5: Deviations from Revenue Forecasts

Table 9.6: The Two Tail Ends of the Revenue Forecast Deviations

Investment	Return Deviation (%)
1	-28.56
2	-25.60
3	-25.56
...	...
48	5.49
49	7.30
50	9.69

The deviations range from -28.56% to 9.69%, suggesting optimistic forecasts and a heavily left-skewed, high-variance distribution.

Step 2: Establish the return deviation distribution.

To find the 2.5th and 97.5th percentiles of the return deviations, we interpolate linearly:

- **2.5th percentile** (between ranks 2 and 3):

$$x = -25.60 + \frac{2.5 - 2}{3 - 2} \times (-25.56 - (-25.60)) = -25.58\% \tag{9.4}$$

- **97.5th percentile** (between ranks 48 and 49):

$$x = 5.49 + \frac{97.5 - 96}{98 - 96} \times (7.30 - 5.49) = 6.89\%$$

Thus, the 2.5th percentile deviation is -25.58%, and the 97.5th percentile is 6.89%.

Step 3: Calculate the Net Present Value (NPV) range.

The baseline NPV is GBP 1,000,000. Every 1% deviation in revenue changes NPV by GBP 123,897.22. Extreme scenario NPVs are calculated as:

- **Downside (2.5th percentile):**

$$\text{NPV} = 1,000,000 + (-25.58 \times 123,897.22) \approx -\text{GBP } 2,170,000$$

- **Upside (97.5th percentile):**

$$\text{NPV} = 1,000,000 + (6.89 \times 123,897.22) \approx \text{GBP } 1,854,000$$

The potential NPV ranges from $-\text{GBP } 2,170,000$ to GBP $1,854,000$.

Using empirical data to define realistic outcomes allows the firm to adjust expectations and mitigate overconfidence in forecast models.

9.1.4 Features of Historical Simulation

Historical simulation is a valuable tool for financial analysis, but its implementation requires attention to data quality and availability.

Missing data in global investment portfolios may be caused by:

- **Asynchronous global markets:** Different trading holidays and time zones.
- **Non-existence of securities:** A security listing after the simulation window begins.
- **Thinly traded securities:** Sparse or stale pricing due to low volume.

To compensate for missing data, analysts often use a **proxy**—a substitute financial instrument or index representing the missing security's risk factor.

For fixed-income securities, applying historical price changes directly can be misleading. Instead, simulations model changes in benchmark yields (e.g., zero-coupon curves at specific maturities) rather than the bond prices directly.

Strengths of historical simulation:

- **Simplicity:** Straightforward implementation without complex statistical models.
- **No distribution assumptions:** Preserves empirical distribution features like fat tails and kurtosis.

- **Implicit correlation:** Preserves natural historical correlations between assets.
- **Low computational cost:** Scenarios are sourced directly from past observations.

Disadvantages of historical simulation:

- **Backward looking:** Relies on the assumption that past risk distributions represent the future.
- **Limited scenario variety:** Cannot simulate extreme events that have not occurred in the historical sample.
- **Data dependency:** Results depend entirely on the quality and completeness of historical datasets.

9.2 Bootstrapping

b | describe bootstrap resampling, and explain how it can be used in investment applications

Bootstrapping (tái lấy mẫu lặp) cho phép chọn mẫu có thay thế (with replacement) từ tập mẫu ban đầu để tạo ra vô số kịch bản tổng hợp. Điểm khác biệt mấu chốt là: bootstrapping coi mẫu hiện tại là đại diện cho toàn bộ quần thể, còn mô phỏng lịch sử coi dữ liệu quá khứ là cố định và duy nhất.

9.2.1 Bootstrapping Overview

Resampling repeatedly draws samples from observed data to make statistical inferences. It is particularly useful when population distributions are unknown or sample sizes are small.

Bootstrapping is a resampling method that generates a simulated distribution by resampling with replacement from the observed sample (treating the sample as a proxy for the population).

Resampling with replacement means each data point has a probability of $1/n$ (where n is sample size) of being selected in each draw, allowing the same value to be selected multiple times. For example, from sample $[2, 5, 8]$, bootstrap samples could be $[5, 5, 8]$ or $[2, 2, 5]$.

Bootstrapping differs from historical simulation:

- **Bootstrapping** resamples with replacement to create synthetic datasets, preserving the empirical distribution structure while generating new combinations.
- **Historical simulation** applies the actual chronological sequence of past market movements directly as fixed scenarios.

Table 9.7 compares these two non-parametric approaches.

Table 9.7: Comparison of Historical Simulation and Bootstrapping

Feature	Historical Simulation	Bootstrapping
Source of data	Historical dataset	Observed sample
Sampling method	Simple sampling of historical data	Resampling with replacement from the sample
Assumptions	Assumes past data reflects future	Assumes sample represents population
Primary use cases	Risk management, financial stress tests	Estimating confidence intervals, standard errors
Computational intensity	Low to moderate	Moderate

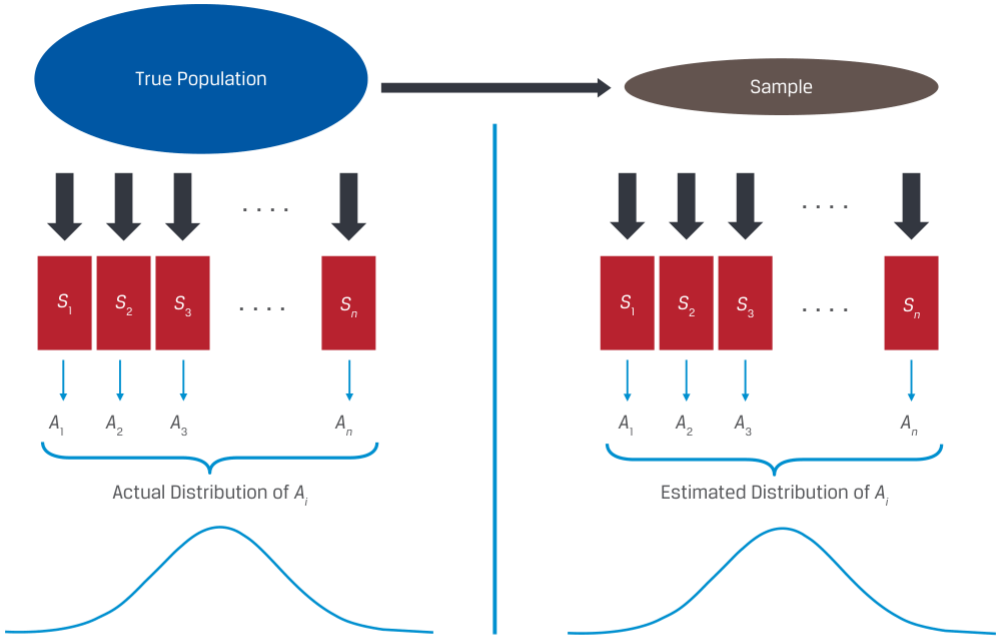
By using the observed sample as a proxy for the unknown population, bootstrapping eliminates the need for external data or theoretical models, making it extremely flexible and widely applicable in situations where we have little prior information about the population distribution.

9.2.2 Implementing Bootstrapping

Bootstrapping is non-parametric, flexible, and simple to implement. Its accuracy depends on how well the sample represents the true distribution; larger samples are always preferred.

The only functional change when using the bootstrap method is the source of scenarios (Step 2), as shown in Figure 9.6.

Exhibit 13: Bootstrap Resampling



The next case study presents the application of bootstrapping techniques to value a European call option. The example highlights the non-parametric nature of bootstrapping.

Figure 9.6: Bootstrap Resampling

Case Study: Bootstrapping Techniques to Value a European-Style Call Option on Brighton Rock plc Stock

We value a 10-week European call option on Brighton Rock plc stock using bootstrapping. The strike price is equal to the current stock price ($K = P_0 =$ GBP 141.25).

Step 1: Setup/parameterization

- Option Payoff: $V_{\text{Call option}} = \max[P_T - K, 0]$ where $K =$ GBP 141.25. - Steps: $n = 10$ weeks. - Interest rate: $r = 5.0\%$ annualized (continuously compounded). - Sample data: 25 weekly log returns ($t = 1$ to $t = 25$) calculated from 26 weeks of historical price data (Table 9.8).

Table 9.8: Weekly Closing Prices and Log Returns for Brighton Rock plc

Week	Price (GBP)	Log Return (r_i)	Week	Price (GBP)	Log Return (r_i)
0, $t = 0$	100.00	NA	13	114.07	-0.0056
1, $t = 1$	100.26	0.0026	14	115.83	0.0153
2	102.36	0.0207	15	123.01	0.0601
3	102.92	0.0055	16	128.71	0.0453
4	103.55	0.0061	17	131.07	0.0182
5	103.75	0.0019	18	131.88	0.0062
6	109.44	0.0534	19	132.15	0.0020
7	112.79	0.0302	20	129.54	-0.0199
8	109.26	-0.0318	21	133.28	0.0285
9	108.96	-0.0027	22	134.02	0.0055
10	110.69	0.0158	23	135.02	0.0074
11	111.57	0.0079	24	137.09	0.0152
12	114.71	0.0278	25, $t = 25$	141.25	0.0299
Average		**0.0138**	**Cumulative**		**0.3454**

We run 500 simulations to estimate the option value.

Step 2: Scenario generation.

We construct simulated price paths by drawing 10 random integers between 1 and 25 (with replacement) for each scenario and assigning the corresponding historical return. Table 9.9 lists Scenario 1's details.

Table 9.9: Random Numbers and Corresponding Log Returns for a Scenario

Pull (Random Number)	Corresponding Log Return (r_i)
21	0.0285
25	0.0299
4	0.0061
13	-0.0056
16	0.0453
8	-0.0318
7	0.0302
8	-0.0318
15	0.0601
10	0.0158
Total return	**0.1467**

Step 3: Scenario evaluation/calculation engine.

Log returns are additive, so for Scenario 1:

$$r_{\text{Scenario 1}} = \sum_{i=1}^{10} r_i = 0.1467$$

The terminal stock price is:

$$P_T = \text{GBP } 141.25 \times e^{0.1467} = \text{GBP } 163.57$$

The option payoff at maturity is:

$$\text{Payoff} = \max(163.57 - 141.25, 0) = \text{GBP } 22.32$$

Figure 9.7 shows the price paths of 500 simulations.

We run 500 simulations in Step 3 to get a distribution of potential outcomes. Exhibit 17 depicts selected share price paths for the 500 simulations and shows the first five simulated price paths, the mean price path, and confidence intervals.

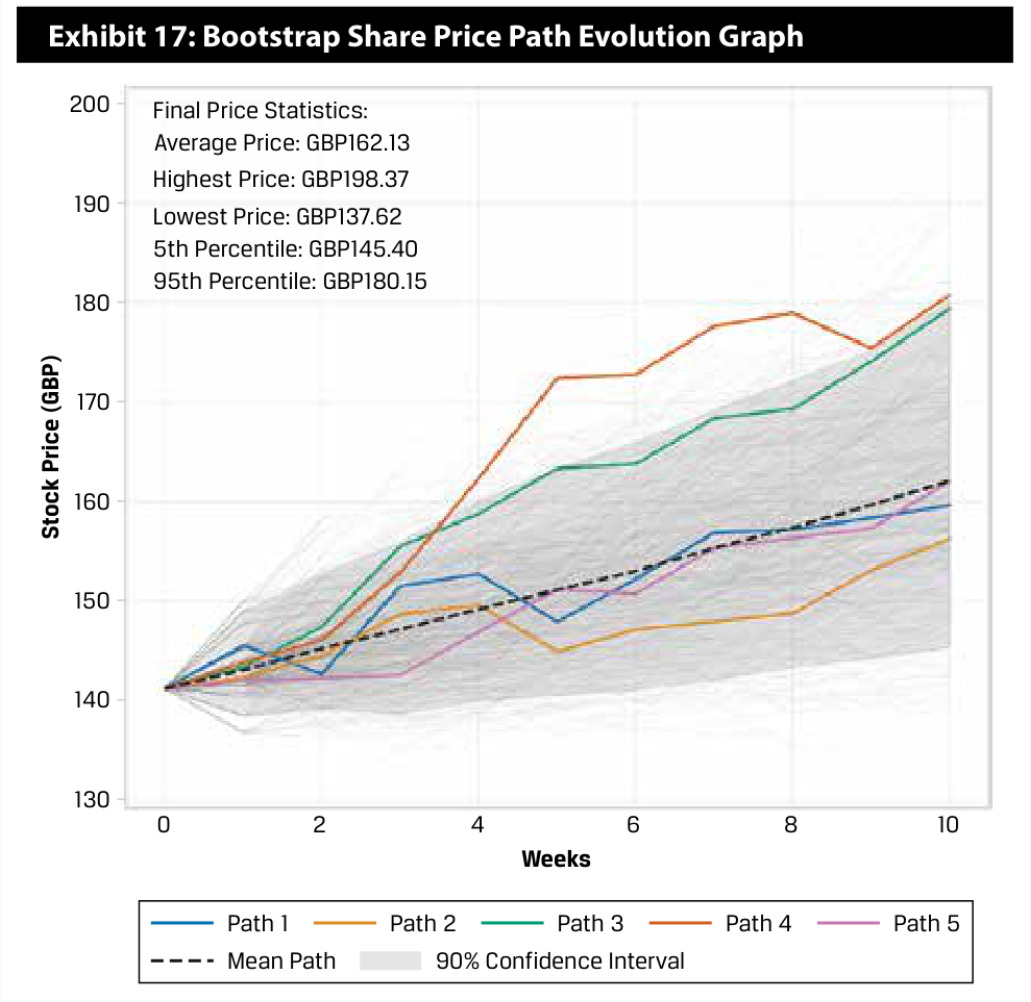


Figure 9.7: Bootstrap Share Price Path Evolution Graph

Step 4: Result compilation/analysis.
Figure 9.8 shows the resulting histograms for 500 simulations (mean option value is $\approx$ GBP 20.90).

After performing 500 simulations, we obtain the histograms for the Brighton Rock plc share price and the final option price in Exhibit 18.

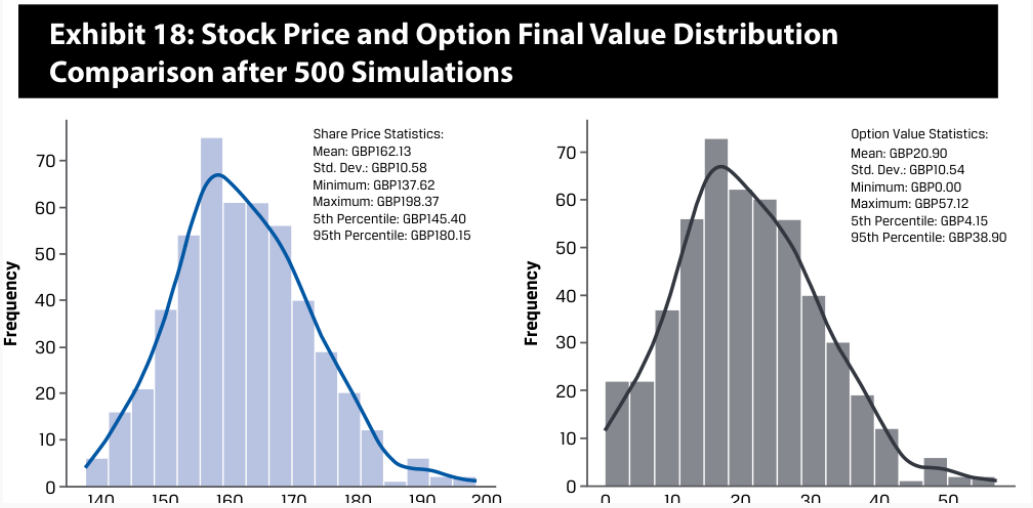
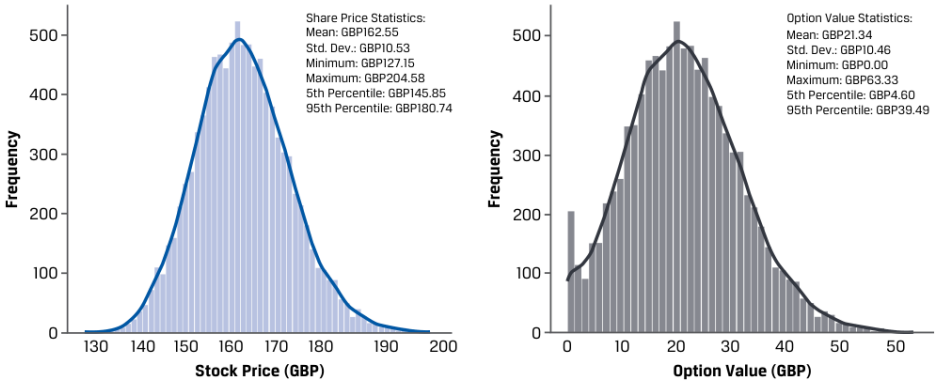


Figure 9.8: Stock Price and Option Final Value Distribution Comparison after 500 Simulations

Figure 9.9 shows the histograms for 10,000 simulations (mean option value is $\approx$ GBP 21.34), showing smoother tails.

For comparison, we also performed 10,000 simulations which yielded the histograms presented in Exhibit 19.

Exhibit 19: Stock Price and Option Value Distribution Comparison after 10,000 Simulations



Comparing the results of the 10,000-run simulations and the 500-run simulations, we observe smoother distribution curves in the 10,000-run graphs, as confirmed by the lower standard deviation. There is also a slight change in average ending values: The average price change increased slightly, from GBP162.13 to GBP162.55, and the average option value increased from GBP20.90 to GBP21.34. The minor changes in the 5th and 95th percentile estimates suggest more precise estimation of the distribution tails.

Note that the bootstrapped value should not be expected to exactly match the traded option price. Under the assumption of a 5% risk-free interest rate, the **closed-form solution** for this option is less than GBP5. The discrepancy between the two approaches lies in the implicitly large drift in the underlying price assumed in the bootstrapped approach.

Bootstrapping can also be used to set confidence intervals for specific parameters of interest, beyond prices or returns.

Figure 9.9: Stock Price and Option Value Distribution Comparison after 10,000 Simulations

Under the standard risk-free BSM model, this option is worth $<$ GBP 5. The bootstrapped price ($\approx$ GBP 21) is much higher because bootstrapping uses historical returns, incorporating the large positive drift (1.38% weekly) observed in the sample data, whereas BSM assumes a risk-neutral drift.

Chúng ta chú ý sự chênh lệch lớn này: định giá quyền chọn bằng bootstrapping ra khoảng GBP 21, khi công thức BSM chỉ ra kết quả dưới GBP 5! Lý do là vì bootstrapping sử dụng tăng trưởng lịch sử trung bình tuần 1.38% (hơn 70% một năm) làm drift, trong khi lý thuyết định giá quyền chọn chuẩn tắc bắt buộc sử dụng suất sinh lời phi rủi ro (5.0% một năm) làm drift. Đây là minh chứng rõ ràng: mô hình đúng số liệu nhưng sai nguyên lý tài chính sẽ dẫn đến kết quả sai lệch nghiêm trọng.

Case Study: Analyzing Portfolio Volatility of a Target Date Fund Using Bootstrapping

A company evaluates a target date retirement fund with only 24 months of available return data. To estimate its volatility over various horizons, bootstrapping is used to generate robust standard error estimates.

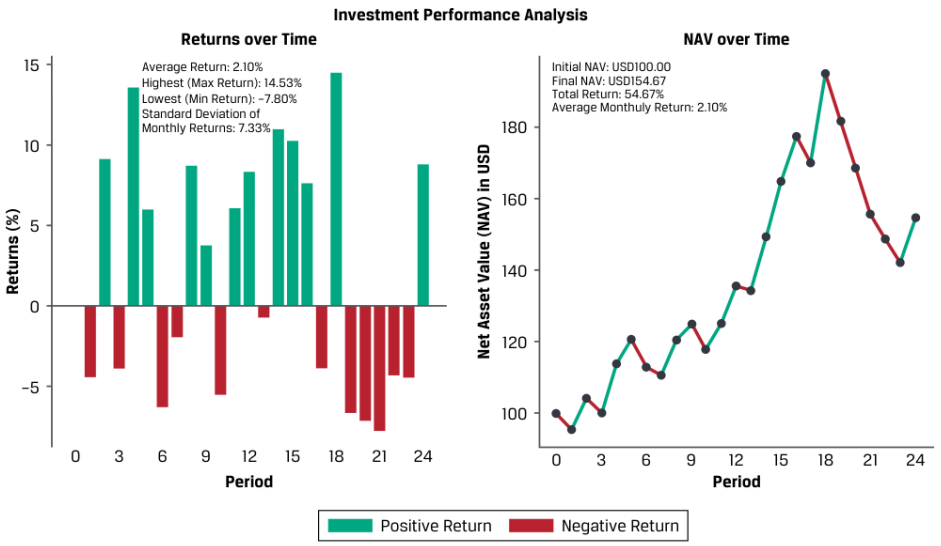
Step 1: Setup and parameterization.

Table 9.10 lists the observed monthly returns and corresponding NAVs starting from USD 100. Figure 9.10 illustrates their evolution.

Table 9.10: Monthly Returns and NAVs

Month	Monthly Return (%)	NAV (USD)	Month	Monthly Return (%)	NAV (USD)
0	0.00%	100.00	13	-0.77%	134.50
1	-4.46%	95.54	14	11.00%	149.30
2	9.16%	104.29	15	10.26%	164.62
3	-3.94%	100.18	16	7.64%	177.20
4	13.55%	113.75	17	-3.94%	170.22
5	6.02%	120.60	18	14.53%	194.95
6	-6.33%	112.97	19	-6.70%	181.89
7	-1.97%	110.74	20	-7.20%	168.79
8	8.71%	120.39	21	-7.80%	155.62
9	3.75%	124.90	22	-4.34%	148.87
10	-5.56%	117.96	23	-4.50%	142.17
11	6.09%	125.14	24	8.79%	154.67

Exhibit 21: Returns and NAV of the Fund



This initial setup creates a basis for simulating future return scenarios.

Step 2: Scenario generation with bootstrapping

To estimate volatility over longer periods, 1,000 simulated 50-month return scenarios are created using bootstrapping. Each scenario involves the following:

- *Random sampling:* Drawing 50 random numbers between 1 and 24 (inclusive) with replacement, where each number corresponds to one of the observed monthly returns
- *Assigning returns:* Replacing the random numbers with their corresponding monthly returns from the dataset
- *Repeating:* Repeating the process for all 1,000 bootstrap samples

An example scenario is developed:

- *Random sampling:* Generate random numbers: $[8, 7, 22, ..., 3, 10, 18]$.
- *Assigning returns:* Find and use corresponding monthly returns: [0.0871, −0.0197, −0.0434, ..., −0.0394, −0.0556, 0.1453], and compute the standard deviation for this scenario: 7.61%.

This random sampling ensures each scenario uses variations of the observed returns to simulate potential outcomes.

Step 3: Scenario evaluation and calculation

For each of the 1,000 simulated scenarios, the standard deviation of the 50 returns is calculated to measure volatility. These values are then ranked from lowest to highest volatility. Exhibit 22 shows the distribution of monthly return standard deviations.

Figure 9.10: Returns and NAV of the Fund

Step 2: Scenario generation with bootstrapping.
We construct 1,000 scenarios, each simulating a 50-month path by randomly drawing 50 returns from the 24-month sample (with replacement). For example: - Random draws: $[8, 7, 22, \ldots, 18]$. - Assigned returns: $[8.71\%, -1.97\%, -4.34\%, \ldots, 14.53\%]$. The sample standard deviation is 7.61%.
Step 3: Scenario evaluation and calculation.

For each scenario, the monthly returns' standard deviation is computed. The resulting frequency distribution is shown in Figure 9.11.

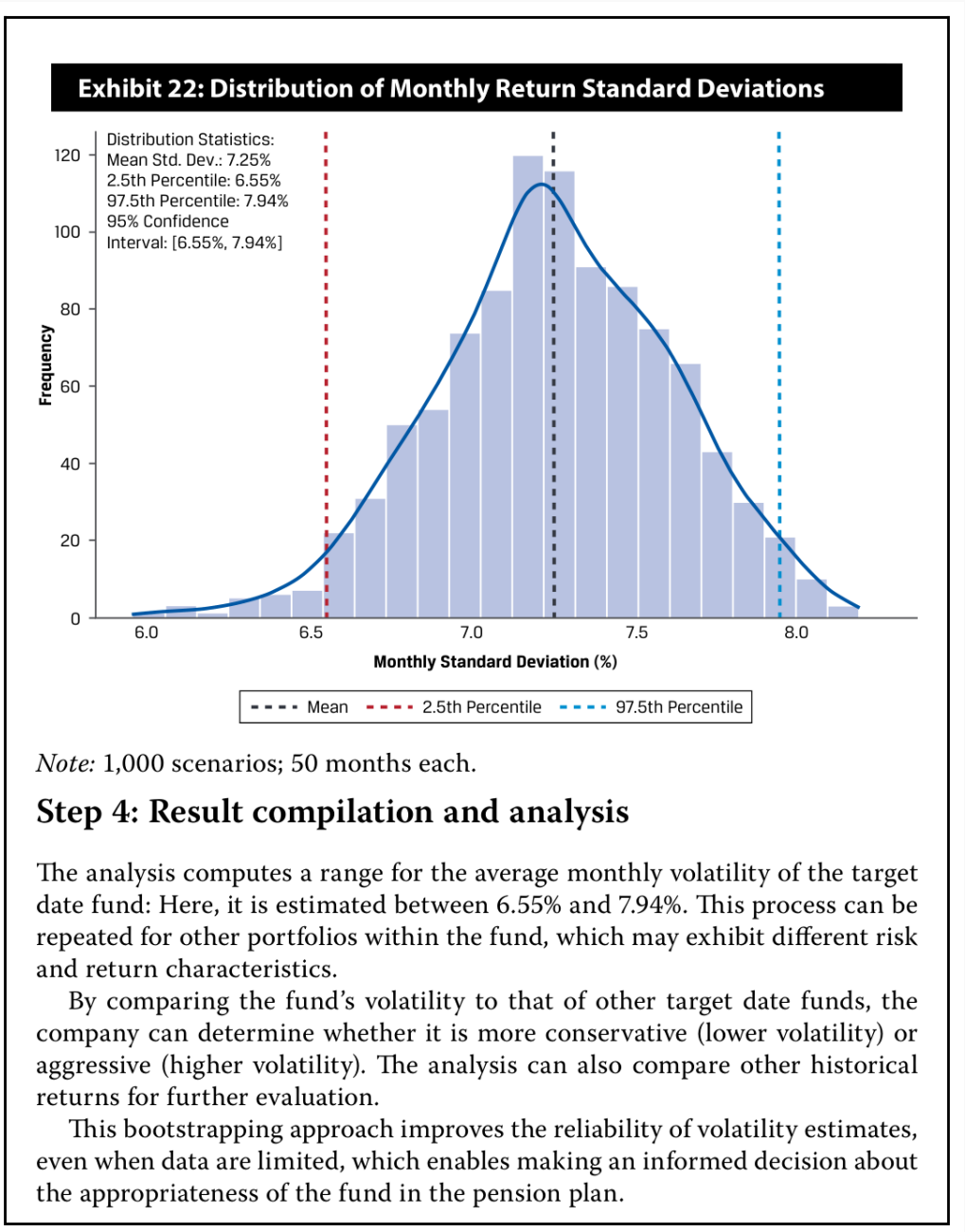


Figure 9.11: Distribution of Monthly Return Standard Deviations

Step 4: Result compilation and analysis.
The 95

9.2.3 Features of Bootstrapping Simulations

Bootstrapping avoids parametric distribution assumptions, allows constructing confidence intervals, and assesses estimator stability. Its effectiveness depends on sample representativeness. Larger sample sizes yield more accurate results.

Strengths of bootstrapping:

- **No distribution assumptions:** Works directly with the empirical sample distribution.
- **Small sample efficiency:** Provides robust estimates of standard errors and confidence intervals for limited datasets.

- **Versatility:** Applicable to a wide range of statistical estimators and valuation tasks.

Disadvantages of bootstrapping:

- **Sample dependency:** Highly sensitive to sample quality; cannot generate values outside the sample range.
- **Time-series limitations:** Simple bootstrapping breaks down autocorrelation in time-series data.
- **Computational cost:** Requires running many iterations to build the simulated distribution.

9.3 Monte Carlo Simulation

c | describe Monte Carlo simulation and explain how it can be used in investment applications

Mô phỏng Monte Carlo là phương pháp mô phỏng tham số: thay vì dùng dữ liệu quá khứ, chúng ta giả định phân phối toán học (như chuẩn, log-normal) và chạy hàng chục ngàn kịch bản ngẫu nhiên. Đây là công cụ hữu ích nhất để định giá tài sản phi tuyến tính hoặc phụ thuộc đường đi (path-dependent) như quyền chọn kiểu Á hay chứng khoán bảo đảm bằng thế chấp (MBS).

9.3.1 Monte Carlo Simulation Overview

A Monte Carlo simulation generates a large number of random samples from a specified probability distribution or distributions to produce a simulated frequency distribution of outcomes.

Unlike historical simulation and bootstrapping, Monte Carlo simulation is a parametric method that requires explicit assumptions about the input variables' probability distributions. This allows analysts to evaluate the sensitivity of output variables to changes in parameters like mean, variance, and correlation.

In finance, Monte Carlo simulation is widely used to value complex securities and derivatives (e.g., path-dependent contingent claims) that lack analytical closed-form pricing formulas.

Case Study: European Call Option Valuation Using Monte Carlo Simulation

We evaluate a European call option using Monte Carlo simulation and compare the results to the analytic BSM formula:

- Option Type: European Call, Strike (K): CAD 90
- Current Price (P_0): CAD 85, Time (T): 1 Year
- Volatility (σ): 25

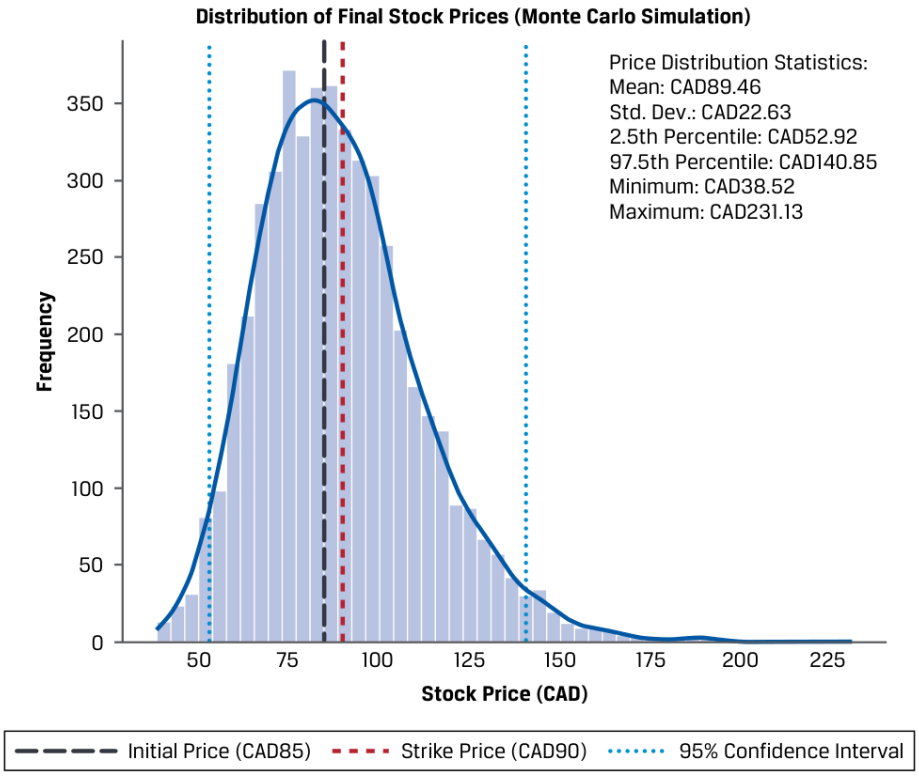
Step 1: Setup/parameterization
- Valuation: $V_{\text{Call option}} = \max[P_T - K, 0]$ where $K =$ CAD 90. - Under risk-neutral pricing, stock price paths are assumed to follow a log-normal distribution.
Step 2: Scenario generation
The log return return is simulated using Geometric Brownian Motion over one step ($s = 1.0$):

$$r_t = \left(r_f - \frac{\sigma^2}{2}\right) \times s + \sigma\sqrt{s}Z \tag{9.5}$$

where $Z \sim N(0,1)$ and the drift is: $0.05 - (0.25^2/2) = 0.01875$. The ending stock price is $P_T = P_0 \times e^{r_t}$.
Step 3: Scenario evaluation/calculation engine
We run 5,000 simulations to generate ending prices (Figure 9.12).

are only concerned with the final outcome, we can do a one-step simulation. Accordingly, we obtain the following distribution of potential prices in Exhibit 23.

Exhibit 23: Distribution of Stock Prices, Simulation Result



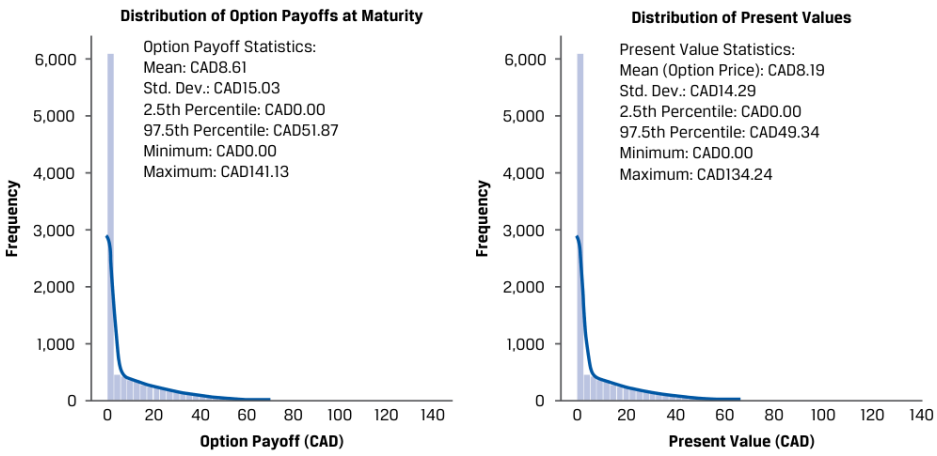
Step 4: Result compilation/analysis

We can now analyze the results. Exhibit 24 shows the distribution of option payoffs derived from the scenarios executed in Step 3.

Figure 9.12: Distribution of Stock Prices, Simulation Result

Step 4: Result compilation/analysis
The resulting option value distribution is shown in Figure 9.13.

Exhibit 24: Distribution of European Call Option Values for Simulation



The mean value of this distribution is CAD8.61, which discounted at the 5.0% risk-free rate is CAD8.19. This can be contrasted with the result obtained from the Black–Scholes–Merton equation of CAD8.18. As the number of simulations increases, there will be less variability in the result and we will have a convergence to CAD8.18.

We showed how Monte Carlo simulation replicates the Black–Scholes–Merton formula for a European call option, where the payoff depends only on the stock price at maturity. However, many options, such as **Asian-style options**, are path dependent, with payoffs influenced by the entire price trajectory (Asian-style options are based upon the average value over a certain period of time). Traditional models struggle with such complexity, making Monte Carlo simulation essential. The next case study explores this by valuing an Asian option, showcasing its use in handling path-dependent scenarios.

Figure 9.13: Distribution of European Call Option Values for Simulation

The mean option payoff is CAD 8.61. Discounted at the risk-free rate of 5.0

$$V_0 = 8.61 \times e^{-0.05 \times 1} = \text{CAD } 8.19$$

This closely matches the BSM formula value of CAD 8.18.

Chúng ta lưu ý: Mô phỏng Monte Carlo hội tụ về giá trị của công thức BSM khi tăng số lượng kịch bản. Cả hai cho kết quả giống nhau vì cùng giả định chuyển động Brown hình học (Geometric Brownian Motion): phân phối log-normal, biến động không đổi và drift bằng lãi suất phi rủi ro. Nhưng nếu tài sản phụ thuộc vào đường đi giá (như quyền chọn kiểu Á) thì sao?

Case Study: Asian Option Valuation Using Monte Carlo Simulation

An Asian call option's payoff depends on the average price of the underlying asset: $V_{\text{Payoff}} = \max[P_T - \text{Average price}, 0]$. We value a 21-day Asian option on a stock denominated in Swiss francs (CHF).

Step 1: Setup/Parameterization

- Objective: Estimate $V_0 = E[\max[P_{21} - \text{Average price (Days 1–21)}, 0]] \times e^{-rT}$. - Input parameters: $P_0 = \text{CHF } 25.00$, $T = 21$ days, $\sigma = 25\%$, $r = 5.0\%$.

Step 2: Scenario Generation

Daily stock returns follow a log-normal distribution:

$$\text{Log price change}_t = \left(0.05 - \frac{0.25^2}{2}\right) \times \left(\frac{1}{252}\right) + 0.25 \times \sqrt{\frac{1}{252}} \times Z \tag{9.6}$$

where $Z \sim N(0,1)$. Prices evolve as $P_t = P_{t-1} \times e^{r_t}$. Table 9.11 shows one simulated run.

Table 9.11: Asian Option Valuation: Daily Log Returns and Prices for One Simulated Run

Day	Return (%)	Price (CHF)	Day	Return (%)	Price (CHF)
1	-2.32%	24.43	12	0.22%	24.80
2	-0.70%	24.26	13	-1.43%	24.45
3	1.45%	24.61	14	4.87%	25.67
4	0.49%	24.73	15	-2.80%	24.96
5	-1.01%	24.48	16	-1.25%	24.65
6	1.15%	24.76	17	-1.54%	24.27
7	1.16%	25.05	18	0.24%	24.33
8	0.46%	25.16	19	-4.29%	23.31
9	-0.58%	25.01	20	0.41%	23.41
10	0.12%	25.04	21	-1.28%	23.11
11	-1.17%	24.75			
Average					**24.54**

For this run, the ending price is CHF 23.11, which is below the average price of CHF 24.54, yielding a payoff of zero. Figure 9.14 shows the price path of this run.

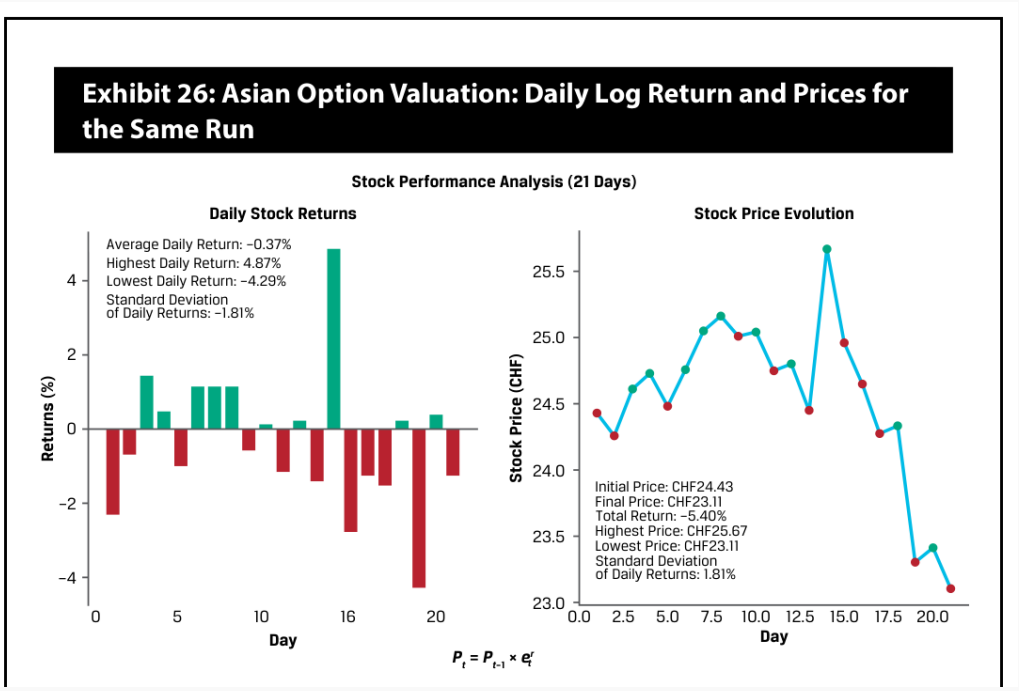


Figure 9.14: Asian Option Valuation: Daily Log Return and Prices for the Same Run

Step 3: Scenario evaluation/calculation engine.
We run 500 price path simulations (Figure 9.15).

levels, calculating the average price over 21 days, and then determining the payoff. Exhibit 27 shows the 500 simulated stock price paths.

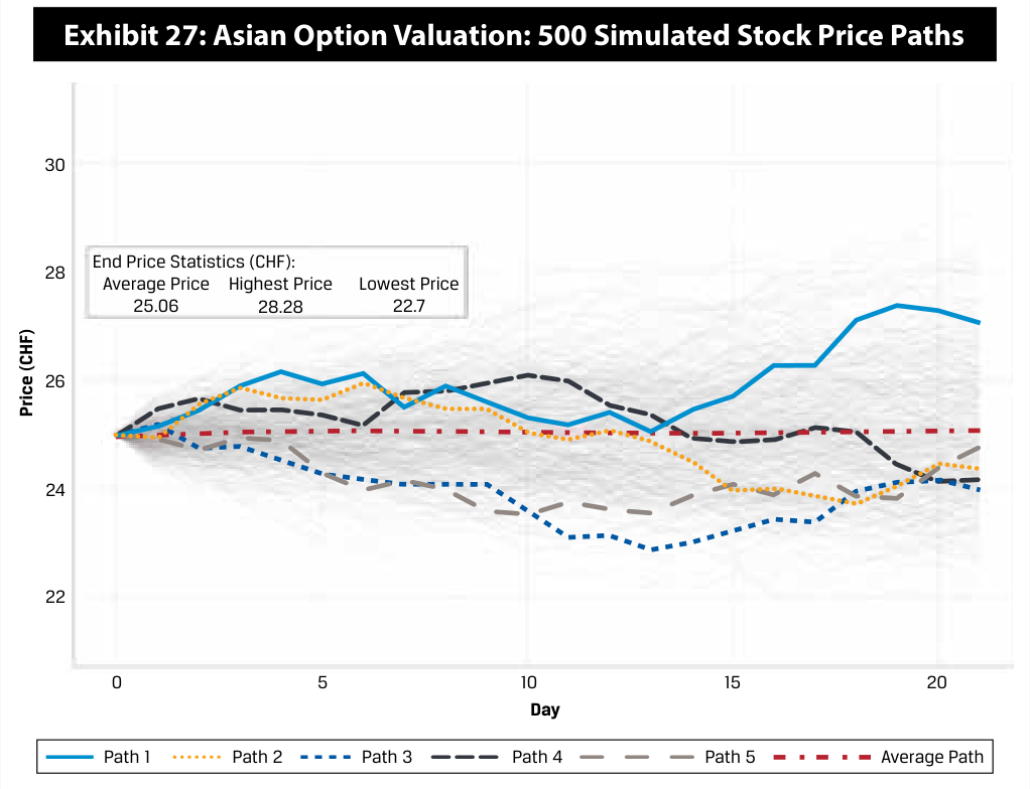


Figure 9.15: Asian Option Valuation: 500 Simulated Stock Price Paths

Step 4: Result compilation/analysis.
The distribution of payoffs is shown in Figure 9.16.

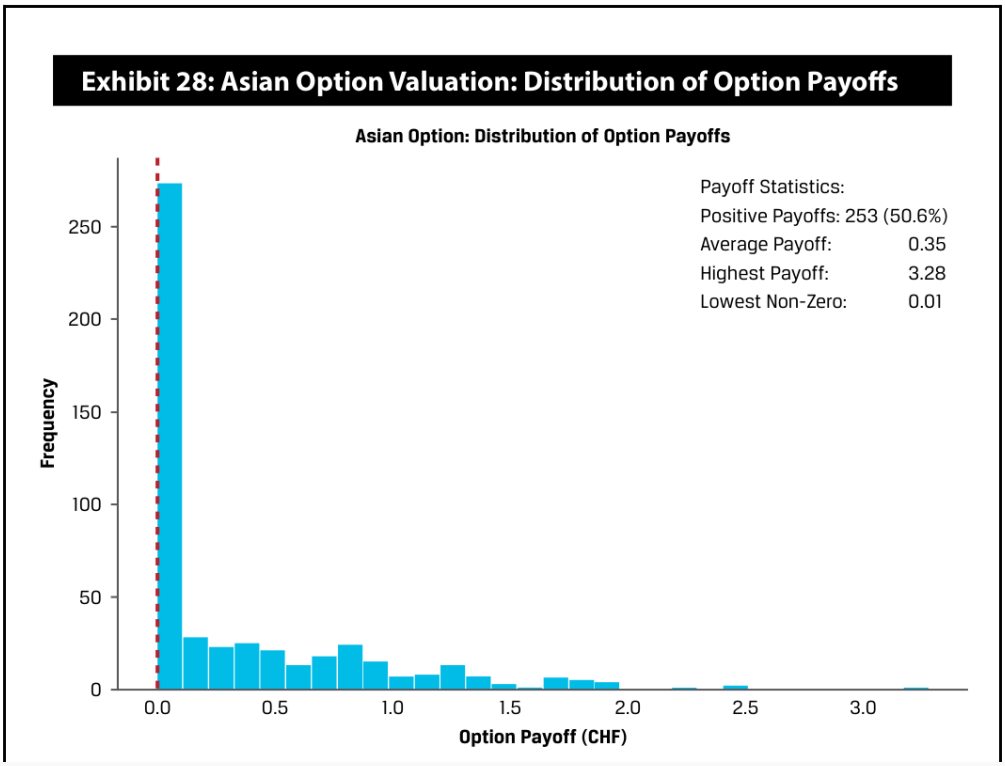


Figure 9.16: Asian Option Valuation: Distribution of Option Payoffs

The average payoff across all 500 simulations was CHF 0.35. Discounting using the risk-free rate yields:

$$V_0 = 0.35 \times e^{-0.05 \times \frac{21}{252}} = 0.35 \times 0.9958 = \text{CHF } 0.3485$$

9.3.2 Multivariate Monte Carlo Simulation

Simulating multiple variables simultaneously is necessary to capture complex correlations and dependencies in financial portfolios or multi-asset derivatives.

Correlated variables are generated by drawing independent standard normal variables and transforming them using the Cholesky decomposition—a matrix factorization technique that decomposes a covariance or correlation matrix. The framework is shown in Figure 9.17.

The process to simulate random variables X_1 and X_2 with a correlation of ρ is shown in Exhibit 29.

Exhibit 29: Simulation Framework Applied to Multivariate Monte Carlo Simulation (Two Variables)

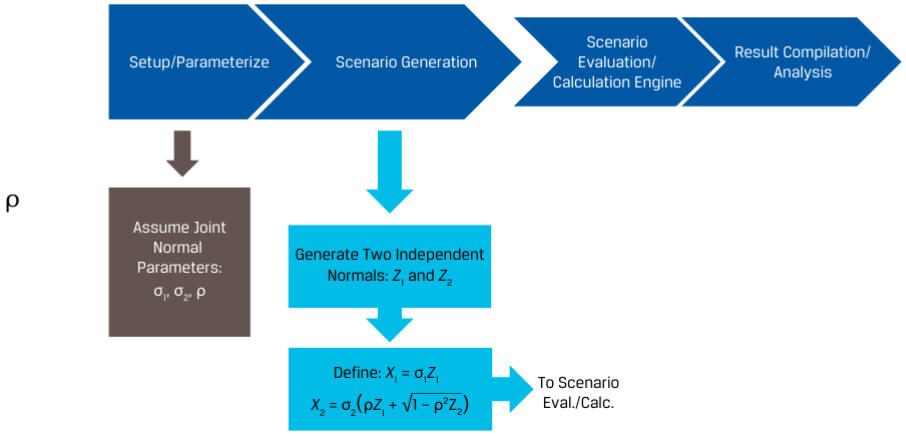


Figure 9.17: Simulation Framework Applied to Multivariate Monte Carlo

Step 1: Setup/parameterize

We assume a joint normal distribution for the two variables with standard deviations σ_1 and σ_2 and correlation ρ.

Step 2: Scenario generation

The Cholesky decomposition transforms independent standard normal variables Z_1 and Z_2 into correlated variables X_1 and X_2:

$$X_1 = \sigma_1 Z_1 \tag{9.7}$$

$$X_2 = \sigma_2 \left(\rho Z_1 + \sqrt{1-\rho^2} Z_2\right) \tag{9.8}$$

This approach ensures that the final variables, X_1 and X_2, adhere to our calibration:

$$E[X_1] = \sigma_1 E[Z_1] = 0 \tag{9.9}$$

$$E[X_2] = \sigma_2 E[\rho Z_1 + \sqrt{1-\rho^2} Z_2] = 0 \tag{9.10}$$

The variances are:

$$E[X_1^2] = \sigma_1^2; \quad E[X_2^2] = \sigma_2^2 \tag{9.11}$$

Since Z_1 and Z_2 are independent, the covariance is:

$$\sigma_{X_1,X_2} = E[X_1 X_2] = \rho \sigma_1 \sigma_2 \tag{9.12}$$

Knowledge Check

An analyst is reviewing intermediate outputs from a Monte Carlo simulation designed to model log price changes. The simulation involves two normally distributed variables, both of which have a mean of 0

and standard deviations of 1.5% and 2.0%, respectively. The analyst has specified a correlation of 30% between the two variables, and the simulations are conducted using the Cholesky decomposition method. In a specific scenario, the random draws from independent standard normal distributions are $Z_1 = 0.4$ and $Z_2 = -1.2$. The first variable, x_1, is calculated using:

$$x_1 = \sigma_1 \times Z_1 = 1.5\% \times 0.4 = 0.60\%$$

The value of the second variable, x_2, is closest to:

A. -2.40%.

B. -2.05%.

C. -1.80%.

Solution:
B là phương án đúng. Áp dụng công thức liên kết hai biến số tương quan bằng Cholesky decomposition:

$$x_2 = \sigma_2 \left(\rho Z_1 + \sqrt{1-\rho^2} Z_2\right)$$

$$x_2 = 2.0\% \times \left(0.3 \times 0.4 + \sqrt{1-0.3^2} \times (-1.2)\right)$$

$$x_2 = 2.0\% \times (0.12 + 0.9539 \times (-1.2)) = 2.0\% \times (0.12 - 1.1447) = 2.0\% \times (-1.0247) \approx -2.05\%$$

Phương án A tính sai khi bỏ qua yếu tố tương quan và nhân trực tiếp: $2.0\% \times Z_2 = -2.40\%$.
Phương án C sai vì dùng sai tỷ lệ của x_1.

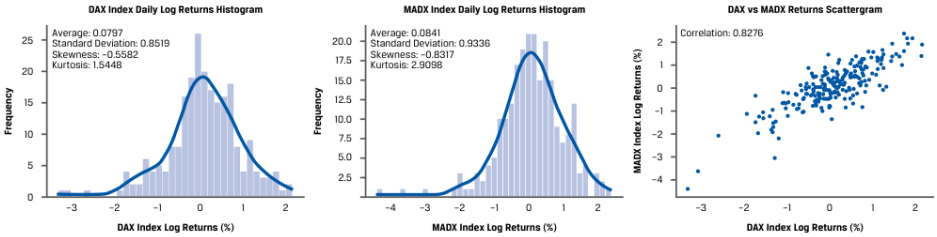
Chúng ta lưu ý: Biến đổi Cholesky duy trì cấu trúc hiệp phương sai trong mô hình Monte Carlo đa biến. Thiếu bước này, các biến ngẫu nhiên sẽ độc lập, làm mất đi tính tương quan thực tế giữa các tài sản và tác động đa dạng hóa (portfolio diversification).

Case Study: Analyzing the Potential Distribution of a Long–Short Strategy

A manager implements a daily long–short strategy, investing EUR 10 million (long) in the DAX (Germany) and shorting EUR 10 million in the MADX (Spain) (Figure 9.18). We simulate the portfolio's P/L distribution.

DAX versus the MADX to understand their correlation, which can be seen in Exhibit 30. Analyzing this relationship is essential for modeling their joint distribution in a Monte Carlo simulation. A strong positive correlation indicates that the indexes move in sync, while a weak or negative correlation suggests they move independently or inversely.

Exhibit 30: The Distribution of the Daily Log Returns for the DAX and MADX over One Year and the Correlation between These Two Index Returns



We make the following *general assumptions*:

- *Model.* We use a bivariate normal distribution applied to the log returns of the DAX and MADX to model their joint behavior.
- *Time horizon.* The simulation considers a one-day holding period.
- *Mean returns.* These are assumed to be zero, which simplifies the simulation and focuses on the variability and correlation between indexes. Additionally, over time, the average daily changes are typically close to zero.
- *Standard deviations.* The daily standard deviation of the DAX is 0.82%, and the daily standard deviation of the MADX is 0.86%.
- *Correlation coefficient.* The correlation between the DAX and MADX is 0.8276, indicating that the indexes tend to move in the same direction.

Figure 9.18: The Distribution of the Daily Log Returns for the DAX and MADX over One Year

We make the following parameters: - Model: Bivariate normal distribution on log returns. - Standard deviations: DAX $\sigma_1 = 0.82\%$, MADX $\sigma_2 = 0.86\%$. - Correlation coefficient: $\rho = 0.8276$. - Expected returns: Assumed to be zero.
We generate independent standard normals Z_1, Z_2 and apply Cholesky decomposition:

$$X_1 = \sigma_{\text{DAX}} \times Z_1 = 0.0082 Z_1$$

$$X_2 = 0.0086 \times \left(0.8276 Z_1 + \sqrt{1 - 0.8276^2} Z_2\right) = 0.0071 Z_1 + 0.0048 Z_2$$

The portfolio return is $R_P = X_1 - X_2$ (due to shorting MADX). Using 250 simulations, the resulting P/L distribution is shown in Figure 9.19.

The analyst conducts 250 simulations and obtains the distribution illustrated in Exhibit 31.

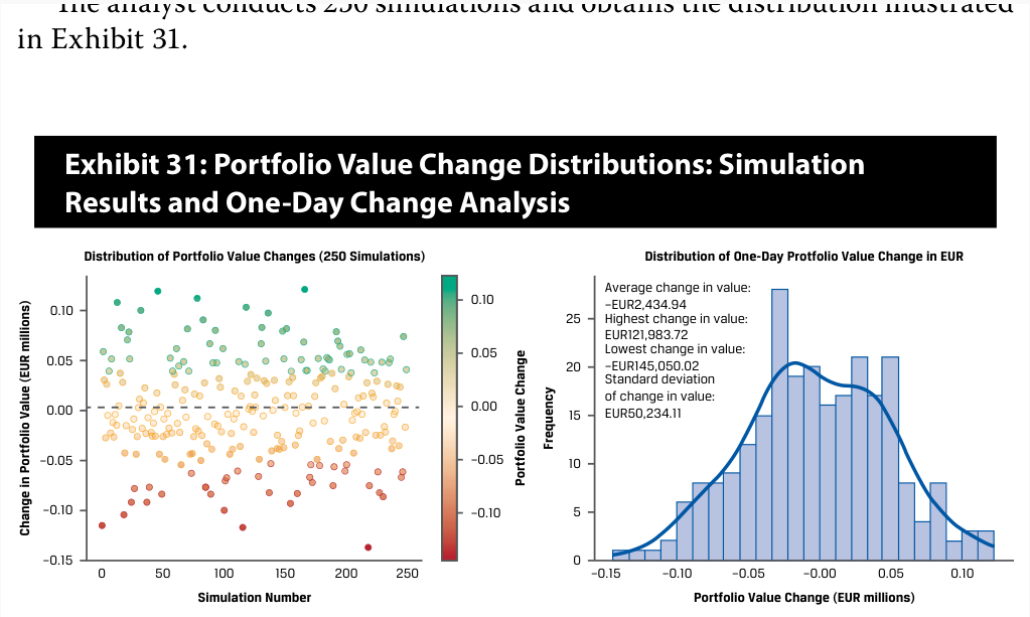


Figure 9.19: Portfolio Value Change Distributions: Simulation Results

The simulation results show an average value change of -EUR 2,434.94 (theoretically zero) and standard deviation of EUR 50,234.11. Increasing the number of simulations would reduce simulation noise and move the mean closer to the theoretical zero.

9.3.3 Features of Monte Carlo Simulation

Monte Carlo simulation is highly flexible because the modeler controls the probability distributions, parameters, and relationships of the inputs.

Strengths of Monte Carlo simulation:

- **Flexibility:** Can simulate complex, non-linear, and path-dependent cash flows.
- **Sensitivity analysis:** Allows testing portfolio behavior under different distributional parameters and stress scenarios.
- **No historical data constraint:** Can be applied to new instruments by defining pricing parameters.

Disadvantages of Monte Carlo simulation:

- **Model risk:** Results depend entirely on input distribution and correlation assumptions ("garbage in, garbage out").
- **High computational cost:** Requires simulating thousands of trials, which can be computationally intensive.
- **Complexity:** Harder to design, calibrate, and explain compared to historical simulation.

Table 9.12 compares the three simulation approaches.

9.3.4 Practical Implementation of Simulation Approaches Using Excel and Other Tools

Excel and Google Sheets provide functions to implement these techniques. Table 9.13 summarizes these functions.

Table 9.14 provides more detail on Excel functions used in simulations.

Implementing Historical Simulation in Excel:

- Input historical price or return data in Column A.
- Calculate returns in Column B.
- Sort returns in Column C: `=SORT(B:B, 1, TRUE)`.

Table 9.12: Comparison of Simulation Approaches

Feature	**Historical Simulation**	**Bootstrapping**	**Monte Carlo Simulation**
Source of data	Historical dataset	Observed sample	Assumed distribution
Sampling method	Sampling from historical data	Resampling with replacement	Random sampling from predefined distribution
Assumptions	Assumes past reflects future	Assumes sample represents population	Assumes generation process can be modeled statistically
Flexibility	Limited to historical patterns	Limited by sample size	Highly flexible
Computational intensity	Low to moderate	Moderate	High

Table 9.13: Functions Used for Simulation

Technique	**Excel Functions**	**Google Sheets**	**Description**
Monte Carlo	RAND(), NORM.INV(), LOGNORM.INV(), RANDBETWEEN(), IF(), SUM(), AVERAGE(), STDEV(), CONFIDENCE.NORM()	RAND(), NORMINV(), LOGNORMINV(), RANDBETWEEN(), IF(), SUM(), AVERAGE(), STDEV(), CONFIDENCE()	Generate random values from normal, lognormal, or uniform distributions.
Historical	INDEX(), SORT(), PERCENTILE.EXC(), PERCENTILE.INC(), LARGE(), SMALL(), AVERAGE(), STDEV(), IFERROR()	INDEX(), SORT(), PERCENTILE.EXC(), PERCENTILE.INC(), LARGE(), SMALL(), AVERAGE(), STDEV(), IFERROR()	Sort, rank, and calculate percentiles of historical return series.
Bootstrapping	RAND(), INDEX(), MATCH(), COUNTIF(), SUM(), AVERAGE(), STDEV(), LARGE(), SMALL(), CONFIDENCE.NORM()	RAND(), INDEX(), MATCH(), COUNTIF(), SUM(), AVERAGE(), STDEV(), LARGE(), SMALL(), CONFIDENCE()	Resample with replacement using RAND() and INDEX().

Table 9.14: Detail on Excel Functions in Simulations

Function	Use Case	Example
RAND()	Generates uniform random numbers between 0 and 1	=RAND()
NORM.INV()	Converts random numbers to normally distributed values	=NORM.INV(0.5,100,15)
LOGNORM.INV()	Converts random numbers to log-normally distributed values	=LOGNORM.INV(0.5,0,1)
RANDBETWEEN()	Generates random integers for discrete simulations	=RANDBETWEEN(1,100)
IF()	Performs conditional logic	=IF(A1>100,"Profit","Loss")
SUM()	Adds values; aggregates returns	=SUM(A1:A10)
AVERAGE()	Calculates the mean of a dataset	=AVERAGE(A1:A10)
STDEV()	Estimates the standard deviation (volatility)	=STDEV(A1:A10)
CONFIDENCE.NORM()	Calculates confidence intervals for normal distribution	=CONFIDENCE.NORM(...)
INDEX()	Retrieves a value from a specified row in a range	=INDEX(A1:A10,5)
SORT()	Sorts a range of values	=SORT(A1:A10,1,TRUE)
PERCENTILE.EXC()	Calculates percentiles, excluding extreme values	=PERCENTILE.EXC(A1:A10,0.05)
PERCENTILE.INC()	Calculates percentiles, inclusive of extreme values	=PERCENTILE.INC(A1:A10,0.05)
LARGE()	Returns the k-th largest value	=LARGE(A1:A10,2)
SMALL()	Returns the k-th smallest value	=SMALL(A1:A10,3)
IFERROR()	Returns an alternative value if an error occurs	=IFERROR(A1/B1,"Error")
MATCH()	Finds the position of a specified value in a range	=MATCH(100,A1:A10,0)
COUNTIF()	Counts cells that meet a condition	=COUNTIF(A1:A10,">100")

- Calculate VaR using percentile functions: `=PERCENTILE.EXC(B:B, 0.05)`.

Implementing Bootstrapping in Excel:

- Input historical data in Column A and calculate returns in Column B.
- Resample returns with replacement in Column C: `=INDEX(B:B, RANDBETWEEN(2, N))`, which randomly selects a row.
- Replicate the formula across multiple rows and columns to generate paths.

Implementing Monte Carlo in Excel:

- Define initial stock price, volatility, risk-free rate, and time steps.
- Generate standard normal variables: `=NORM.INV(RAND(), 0, 1)`.
- Simulate price paths using geometric Brownian motion:

$$P_{t+\Delta t} = P_t \times e^{(r-0.5\sigma^2)\Delta t + \sigma\sqrt{\Delta t}Z} \tag{9.13}$$

  In Excel, this is implemented as: `=P_t * EXP((r - 0.5 * o^2) * dt + o * SQRT(dt) * NORM.INV(RAND(), 0, 1))` (where `o` is volatility σ).

Module 10

Applications of Simple Linear Regression in Finance

LEARNING OUTCOMES	
a	describe, interpret, and explain simple linear regression, including coefficient estimation using the least squares criterion
b	describe and compare the assumptions of simple linear regression, identify violations through analyzing residuals, evaluate the estimated model's goodness-of-fit and regression coefficients, and results of ANOVA estimates
c	calculate and interpret predicted values, the standard error of the estimate, and prediction intervals for the dependent variable in a simple linear regression model and describe different functional forms
d	calculate and interpret the variable estimates of the capital asset pricing model (CAPM)

10.1 Linear Regression Estimation

a	describe, interpret, and explain simple linear regression, including coefficient estimation using the least squares criterion

Hồi quy tuyến tính (Linear Regression) là nền tảng của mọi mô hình định lượng trong tài chính. Bản chất của nó là gì? Đó là đi tìm một đường thẳng mô tả tốt nhất mối quan hệ nhân quả giữa hai biến số. Bằng cách sử dụng tiêu chuẩn OLS (Bình phương tối thiểu), chúng ta có thể tối thiểu hóa sai số dự báo. Nhưng hãy lưu ý: mối quan hệ hồi quy thu được có thực sự mang tính nhân quả, hay chỉ đơn thuần là một sự tương quan ngẫu nhiên trong quá khứ? Chúng ta cần rà soát kỹ lưỡng các giả định đằng sau mô hình này.

10.1.1 Linear Regression Analysis

Regression analysis estimates the relationship between a dependent variable (Y), whose variation we want to explain, and an independent variable (X), which is hypothesized to influence or explain changes in Y.

In simple linear regression, the population model is expressed as:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \quad i = 1, \ldots, n \tag{10.1}$$

where:

- Y_i is the dependent variable (or response variable) for observation i.
- X_i is the independent variable (or explanatory variable) for observation i.
- β_0 is the intercept coefficient, representing the expected value of Y when $X = 0$.
- β_1 is the slope coefficient, representing the change in the expected value of Y for a one-unit change in X.
- ϵ_i is the error term (or disturbance), capturing random noise and variables omitted from the model.

The estimated regression line is:

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i \tag{10.2}$$

where $\hat{Y}_i$ is the predicted value of the dependent variable for observation i, and $\hat{\beta}_0$ and $\hat{\beta}_1$ are the sample estimates of β_0 and β_1.

10.1.2 Coefficient Estimation of Linear Regression

The Ordinary Least Squares (OLS) estimation criterion determines the coefficients $\hat{\beta}_0$ and $\hat{\beta}_1$ by minimizing the Sum of Squared Errors (SSE), which is the sum of the squared differences between the observed values Y_i and the predicted values $\hat{Y}_i$:

$$\text{SSE} = \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^{n} e_i^2 \tag{10.3}$$

The OLS estimators for the slope and intercept are calculated as:

$$\hat{\beta}_1 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} = \frac{\sum_{i=1}^{n} (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{n} (X_i - \bar{X})^2} \tag{10.4}$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} \tag{10.5}$$

where $\bar{X}$ and $\bar{Y}$ are the sample means of the independent and dependent variables, respectively.

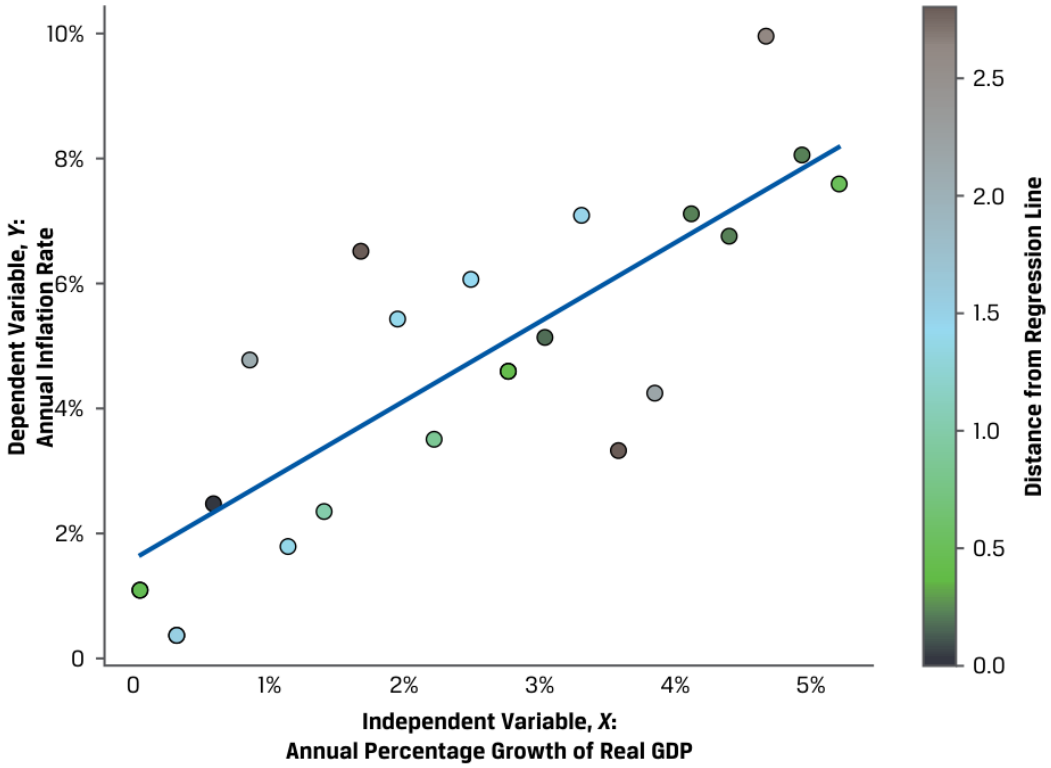


Figure 10.1: OLS Regression: Inflation versus Real GDP Growth

Tại sao OLS lại được ưa chuộng? Phương pháp này đảm bảo tổng các sai số bằng không ($\sum e_i = 0$) và đường hồi quy luôn đi qua điểm tọa độ trung bình mẫu ($\bar{X}, \bar{Y}$). Tuy nhiên, hãy nhớ rằng việc bình phương các phần sai số (e_i^2) sẽ phạt cực kỳ nặng các điểm dữ liệu dị biệt (outliers), từ đó làm lệch đường hồi quy nếu dữ liệu có nhiều biến động bất thường.

10.1.3 Estimation Residuals

The difference between the observed value Y_i and the estimated/predicted value $\hat{Y}_i$ is the residual (e_i):

$$e_i = Y_i - \hat{Y}_i \tag{10.6}$$

The residuals are the sample counterparts to the population error terms (ϵ_i). Figure 10.2 illustrates the residuals plotted around a horizontal line at $e_i = 0$, reflecting the vertical distance of each observation from the regression line.

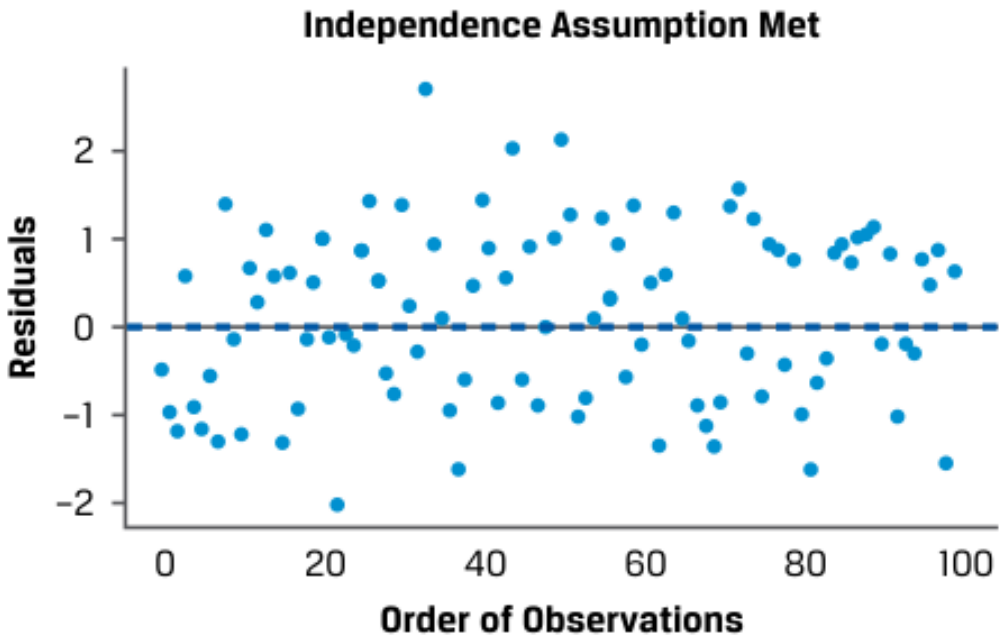


Figure 10.2: Residuals of the Estimated OLS Equation (Inflation vs Real GDP)

10.1.4 Data Types

Financial regression models typically utilize three types of data structures:

- **Cross-sectional data:** Observations on multiple entities (e.g., different companies, sectors, or countries) at a single point in time. It is indexed by $i = 1, \ldots, n$.
- **Time-series data:** Repeated observations of a single entity (e.g., a country's GDP growth or a company's stock return) across multiple time periods. It is indexed by $t = 1, \ldots, T$.
- **Panel data:** A combination of cross-sectional and time-series data, observing multiple entities repeatedly over time.

Example: Estimating Inflation using Real GDP Growth

An analyst estimates the linear relationship between annual inflation (Y, dependent variable) and annual real GDP growth (X, independent variable) for a country over 24 periods. The estimated OLS model yields the following equation:

$$\hat{Y}_i = 1.59\% + 1.27 X_i$$

Interpretation:

- **Intercept ($\hat{\beta}_0 = 1.59\%$):** If the real GDP growth rate is 0%, the expected annual inflation rate is 1.59%.
- **Slope ($\hat{\beta}_1 = 1.27$):** For every 1% increase in the annual real GDP growth rate, the expected annual inflation rate increases by 1.27%.

If the real GDP growth is projected to be 3%, the predicted inflation rate is:

$$\hat{Y} = 1.59\% + 1.27 \times 3\% = 5.40\%$$

10.2 Limitations of Linear Regression Models

b	describe and compare the assumptions of simple linear regression, identify violations through analyzing residuals, evaluate the estimated model's goodness-of-fit and regression coefficients, and results of ANOVA estimates

Một mô hình hồi quy tuyến tính OLS chỉ đáng tin cậy khi các giả định cơ bản được thỏa mãn. Nếu các giả định này bị vi phạm (như phương sai sai số thay đổi - Heteroskedasticity hoặc các sai số tự tương quan với nhau

- Autocorrelation), kết quả ước lượng và các khoảng tin cậy sẽ bị lệch lạc. Hãy học cách phân tích đồ thị phần dư (Residual Plots) - đây chính là công cụ trực quan mạnh mẽ nhất để phát hiện các lỗi hệ thống này.

10.2.1 Assumptions of the Simple Linear Regression Model

To ensure that the OLS estimators are Blue (Best Linear Unbiased Estimators) and that statistical inferences are valid, 5 key assumptions must hold:

1. **Linearity:** The relationship between the dependent variable Y and the independent variable X is linear in parameters.
2. **Homoskedasticity:** The variance of the error terms (ϵ_i) is constant across all observations: $\text{Var}(\epsilon_i) = \sigma^2$ for all i.
3. **Independence:** The error terms are independent of one another, implying they are uncorrelated across observations: $\text{Cov}(\epsilon_i, \epsilon_j) = 0$ for all $i \neq j$.
4. **Normality:** The error terms (ϵ_i) are normally distributed. This is crucial for small sample hypothesis testing.
5. **Exogeneity:** The expected value of the error term conditional on the independent variable is zero: $\text{E}(\epsilon_i | X_i) = 0$.

10.2.2 Residual Analysis

Residual plots (plotting residuals e_i against X_i or time t) are crucial diagnostics to identify violations of OLS assumptions:

- **Violation of Linearity:** Residuals show a distinct curved (e.g., U-shaped) pattern when plotted against X, suggesting that a linear form is inappropriate. Figure 10.3 illustrates this violation.

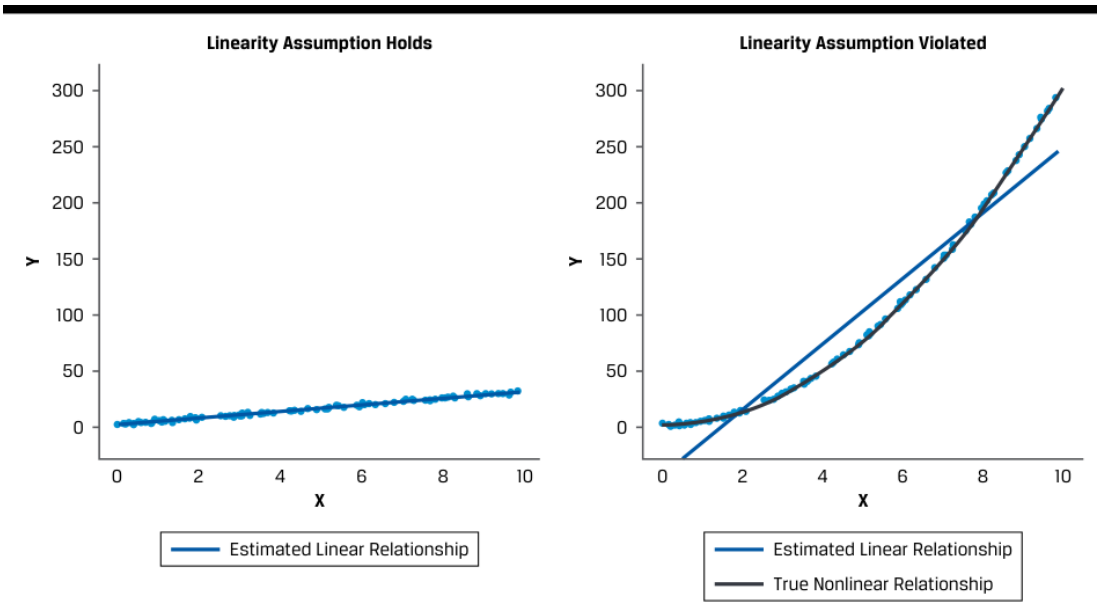


Figure 10.3: Fitting a Linear and a Non-Linear Model to Non-Linear Data

- **Heteroskedasticity:** The spread of the residuals increases or decreases systematically with X (creating a funnel shape). Non-constant variance invalidates standard error estimates, making t-statistics unreliable. Figure 10.4 shows a comparison of regressions on data with two variance regimes.
- **Autocorrelation (Serial Correlation):** Residuals exhibit a systematic trend or cyclical pattern over time (e.g., positive residuals followed by positive residuals). Common in time-series data, this violates the independence assumption. Figure 10.5 compares independent and autocorrelated residuals.

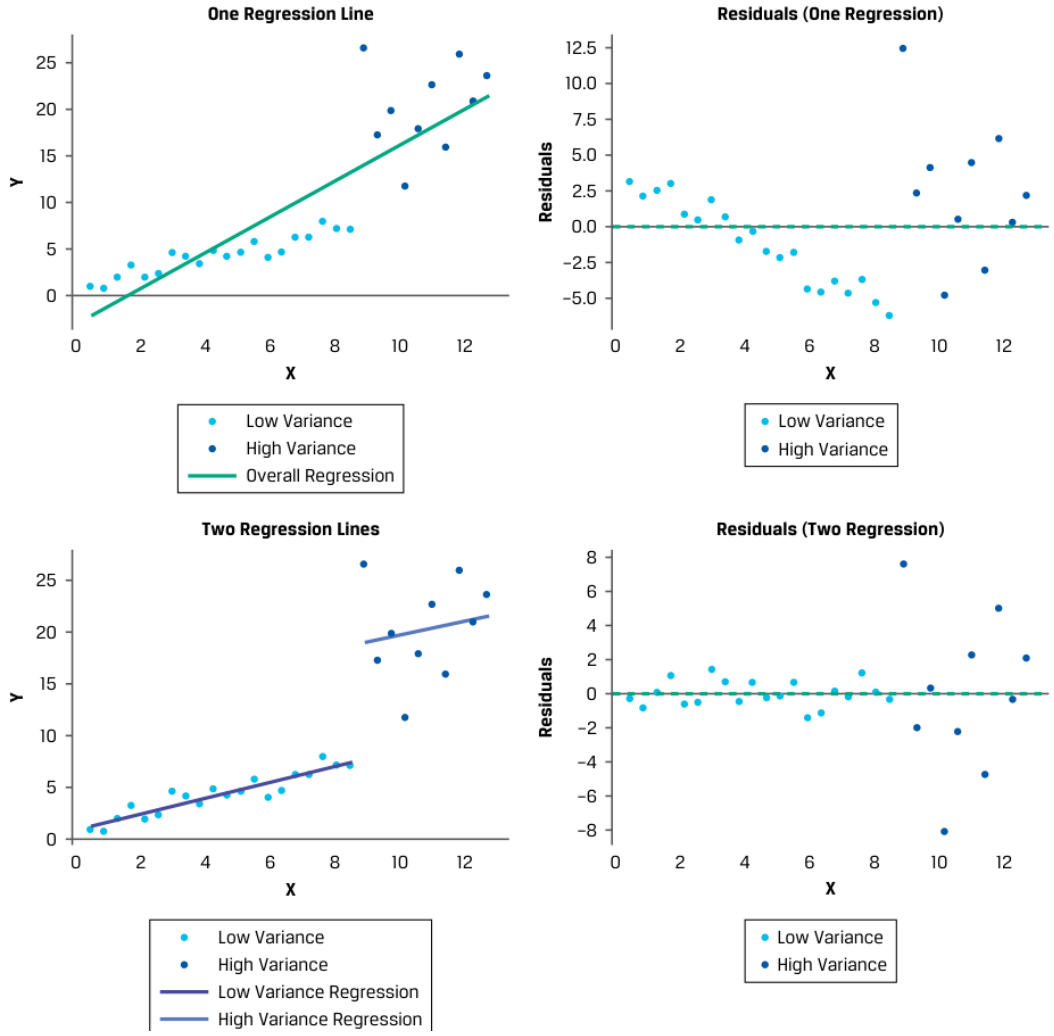


Figure 10.4: Regressions on Data with Two Different Regimes and Residuals

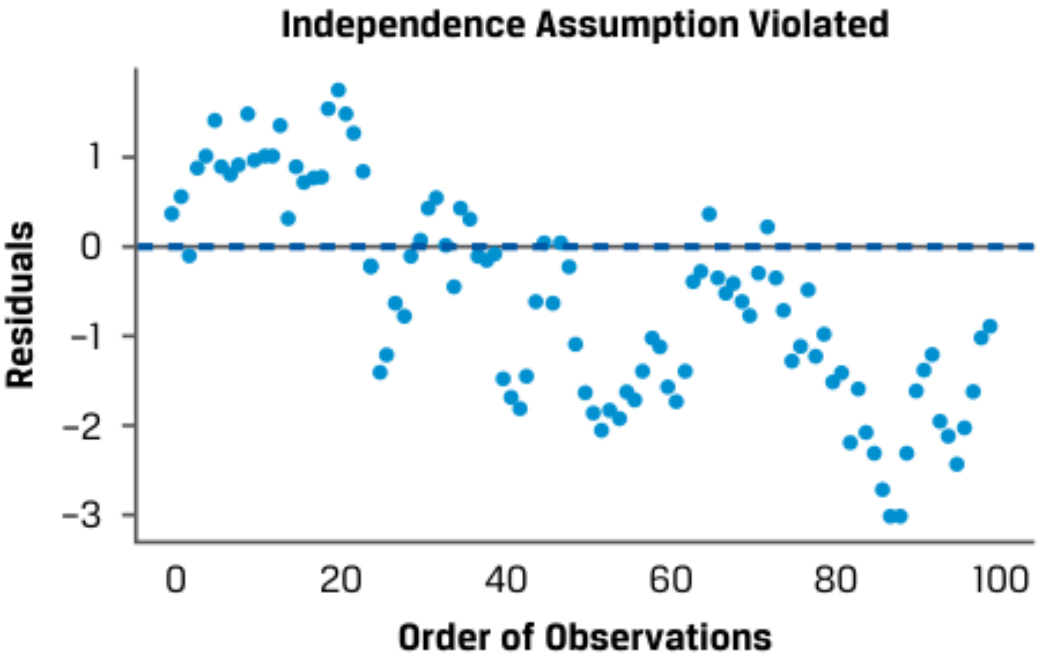


Figure 10.5: Comparison of Residual Plots: Independence (Left) and Autocorrelation (Right)

10.2.3 Evaluating Goodness-of-Fit and Regression Coefficients

Goodness-of-fit measures how well the independent variable explains variation in the dependent variable. Total variation is decomposed into:

$$\text{SST} = \text{SSR} + \text{SSE} \tag{10.7}$$

where:

- **Total Sum of Squares (SST):** Total variation of the dependent variable around its mean: $\text{SST} = \sum(Y_i - \bar{Y})^2$.
- **Regression Sum of Squares (SSR):** Variation explained by the regression line: $\text{SSR} = \sum(\hat{Y}_i - \bar{Y})^2$.
- **Sum of Squared Errors (SSE):** Unexplained variation (residuals): $\text{SSE} = \sum(Y_i - \hat{Y}_i)^2$.

The **Coefficient of Determination (R^2)** measures the proportion of total variation in Y explained by the regression model:

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}} \tag{10.8}$$

In simple linear regression, the correlation coefficient r is related to R^2 by:

$$r = \text{sign}(\hat{\beta}_1) \times \sqrt{R^2} \tag{10.9}$$

10.2.4 Hypothesis Testing and ANOVA Estimates

Analysis of Variance (ANOVA) summarizes the regression outputs to test the overall model significance. Table 10.1 shows the standard ANOVA layout.

Table 10.1: ANOVA Table for Simple Linear Regression

Source	df	Sum of Squares	Mean Square	F-statistic
Regression	1	SSR	$\text{MSR} = \text{SSR}/1$	$F = \text{MSR}/\text{MSE}$
Residual	$n-2$	SSE	$\text{MSE} = \text{SSE}/(n-2)$	–
Total	$n-1$	SST	–	–

The **Standard Error of the Estimate (SEE)** measures the dispersion of the observed values around the regression line:

$$s_e = \sqrt{\text{MSE}} = \sqrt{\frac{\text{SSE}}{n-2}} \tag{10.10}$$

To test whether the independent variable has a significant linear relationship with the dependent variable, we conduct a hypothesis test on the slope coefficient:

$$H_0 : \beta_1 = 0 \quad \text{vs} \quad H_a : \beta_1 \neq 0$$

The test statistic follows a t-distribution with $n-2$ degrees of freedom:

$$t = \frac{\hat{\beta}_1 - 0}{s_{\hat{\beta}_1}} \tag{10.11}$$

where the standard error of the slope estimator $s_{\hat{\beta}_1}$ is:

$$s_{\hat{\beta}_1} = \frac{s_e}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2}} \tag{10.12}$$

Example: Evaluating FIRN AG Stock Regression

An analyst regresses the excess returns of FIRN AG stock (Y) on the market risk premium (X) using 24 monthly observations. The regression output is shown below:

Source	df	SS	MS	F-statistic
Regression	1	0.029	0.029	63.81
Residual	22	0.010	0.000455	–
Total	23	0.039	–	–

Variable	Coefficient	Standard Error	t-statistic
Intercept	0.0077	0.0007	11.328
Market Premium	0.8260	0.1030	7.988

Calculations & Analysis:

1. **Goodness-of-Fit (R^2):**
$$R^2 = \frac{\text{SSR}}{\text{SST}} = \frac{0.029}{0.039} = 0.7436\,(74.36\%)$$
Interpretation: 74.36% of the monthly excess returns variation of FIRN AG is explained by the market risk premium.

2. **Correlation Coefficient (r):**
$$r = \sqrt{0.7436} = 0.8623$$
Since the slope coefficient is positive (0.8260), the correlation between FIRN AG and the market is +0.8623.

3. **Standard Error of the Estimate (SEE):**
$$s_e = \sqrt{\text{MSE}} = \sqrt{0.000455} = 0.0213$$

4. **Significance of the Slope:** At a 5% significance level, the critical t-value with 22 degrees of freedom is 2.074. Since the calculated t-statistic for the slope is $7.988 > 2.074$, we reject the null hypothesis $H_0 : \beta_1 = 0$. The slope is statistically significant.

10.3 Using Linear Regression Models for Prediction

c	calculate and interpret predicted values, the standard error of the estimate, and prediction intervals for the dependent variable in a simple linear regression model and describe different functional forms

Dự báo giá trị tương lai là mục đích tối thượng của phân tích hồi quy. Khi dự báo, chúng ta không chỉ tính ra một giá trị điểm đơn lẻ mà phải tính toán được khoảng dự báo (Prediction Interval) phản ánh sự không chắc chắn của dự báo đó. Ngoài ra, thế giới thực thường phi tuyến: hãy nắm vững các phương pháp biến đổi logarith (Log-Lin, Lin-Log, Log-Log) để làm mượt các mối quan hệ tăng trưởng mũ hay co giãn trong tài chính.

10.3.1 Prediction Using a Simple Regression

For a given value of the independent variable (X_p), the predicted value of the dependent variable ($\hat{Y}_p$) is:

$$\hat{Y}_p = \hat{\beta}_0 + \hat{\beta}_1 X_p \tag{10.13}$$

To capture the uncertainty of our prediction, we calculate the **Prediction Interval** for the dependent variable. The variance of the prediction error (s^2_{pred}) is:

$$s^2_{pred} = s^2_e \left[1 + \frac{1}{n} + \frac{(X_p - \bar{X})^2}{\sum_{i=1}^{n}(X_i - \bar{X})^2}\right] \tag{10.14}$$

The $1 - \alpha$ confidence level prediction interval for the estimated value is:

$$\hat{Y}_p \pm t_{crit} \times s_{pred} \tag{10.15}$$

where t_{crit} is the critical value of the t-distribution with $n - 2$ degrees of freedom at the significance level $\alpha/2$.

Exam Tip

Notice in Equation 10.14 that the variance of the prediction error increases as X_p moves further away from the sample mean $\bar{X}$. This means our predictions are much more precise when X_p is close to the average of our historical data, and become highly uncertain for extreme inputs!

10.3.2 Modeling Non-Linear Relationships

If the relationship between variables is non-linear, we can apply logarithmic transformations to achieve linearity. Table 10.2 summarizes the three primary logarithmic functional forms.

Table 10.2: Logarithmic Regression Functional Forms

Model	Regression Equation	Slope Coefficient Interpretation
Log-Lin	$\ln(Y_i) = \beta_0 + \beta_1 X_i + \epsilon_i$	A 1-unit change in X is associated with an approximate $\beta_1 \times 100\%$ relative change in Y.
Lin-Log	$Y_i = \beta_0 + \beta_1 \ln(X_i) + \epsilon_i$	A 1% relative change in X is associated with an approximate $\beta_1/100$ unit change in Y.
Log-Log	$\ln(Y_i) = \beta_0 + \beta_1 \ln(X_i) + \epsilon_i$	A 1% relative change in X is associated with an approximate $\beta_1\%$ relative change in Y (constant elasticity β_1).

*Làm thế nào để chọn đúng dạng hàm? 1. Sử dụng **Log-Lin** khi biến phụ thuộc tăng trưởng theo cấp số nhân (ví dụ: quy mô doanh thu của doanh nghiệp qua các năm). 2. Sử dụng **Lin-Log** khi tốc độ tăng trưởng giảm dần theo quy mô (hiệu ứng lợi ích cận biên giảm dần). 3. Sử dụng **Log-Log** để tìm độ co giãn của cầu theo giá hoặc tìm mức độ nhạy cảm không đổi giữa hai đại lượng tỷ lệ.*

Example: Log-Lin Regression of Swiss Company Cap on Revenue

An analyst collects a sample of 34 Swiss companies to evaluate the relationship between revenue per share (X, in CHF) and the natural logarithm of market capitalization ($Y = \ln(\text{Market Cap})$, in CHF billions). The OLS regression yields:

$$\ln(\text{Market Capitalization}_i) = 0.5987 + 0.2951 \times \text{Revenue per Share}_i$$

with $s_{\hat{\beta}_1} = 0.0077$.
Analysis:

1. **Interpretation of the Slope ($\hat{\beta}_1 = 0.2951$):** For every 1 CHF increase in revenue per share, the market capitalization increases by approximately 29.51%. Specifically, the exact relative change is:
$$e^{0.2951} - 1 \approx 0.3432 \, (34.32\%)$$
2. **Hypothesis Testing:** To test if the relationship is statistically significant at the 1% significance level ($t_{crit} \approx 2.75$ with 32 df):
$$t = \frac{0.2951}{0.0077} = 38.32$$
Since $38.32 > 2.75$, we reject H_0. The slope is highly statistically significant.

10.4 The Capital Asset Pricing Model (CAPM)

d | calculate and interpret the variable estimates of the capital asset pricing model (CAPM)

CAPM là ứng dụng kinh điển nhất của hồi quy tuyến tính đơn trong đầu tư tài chính. Phương trình hồi quy CAPM (còn gọi là Security Characteristic Line - SCL) biểu diễn phần bù rủi ro của tài sản ($R_i - R_f$) theo phần bù rủi ro của thị trường ($R_m - R_f$). Hệ số góc của phương trình này chính là hệ số Beta (β_i), đại diện cho mức độ rủi ro hệ thống của cổ phiếu. Nếu một cổ phiếu có Beta lớn hơn 1, nó sẽ biến động mạnh hơn thị trường và ngược lại.

10.4.1 CAPM Regression Framework

The CAPM asserts that the expected excess return on an asset is linearly related to the market risk premium. We express it as a simple linear regression:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i(R_{m,t} - R_{f,t}) + \epsilon_{i,t} \tag{10.16}$$

where:

- $R_{i,t} - R_{f,t}$ is the excess return of asset i over the risk-free rate at time t (dependent variable Y).
- $R_{m,t} - R_{f,t}$ is the market risk premium at time t (independent variable X).
- β_i is the systematic risk (or market beta) of the asset, reflecting its sensitivity to market movements.
- α_i is the intercept, representing the abnormal return (Jensen's alpha). Under market efficiency, the expected value of α_i is zero.

Case Study: Calculating LVMH's Beta

An analyst estimates the beta of LVMH Moët Hennessy Louis Vuitton (LVMH) using 104 weeks of historical data. The CAC 40 Index excess return is used as the proxy for the market risk premium (X), and LVMH's excess return is the dependent variable (Y). The OLS regression yields the following results:

Source	df	Sum of Squares	Mean Square
Regression	1	0.108622	0.108622
Residual	102	0.041994	0.000412
Total	103	0.150616	–

Parameter	Coefficient	Standard Error	t-statistic
Intercept (α)	0.0010	0.0020	0.5250
Market Slope (β)	1.3684	0.0840	16.2430

Interpretation & Key Estimates:

1. **Estimated Beta ($\hat{\beta}_{LVMH} = 1.3684$):** LVMH has a beta of approximately 1.37. Since $\beta > 1.0$, LVMH is more volatile than the market CAC 40. A 1% increase in the CAC 40 market risk premium is expected to lead to a 1.37% increase in the excess return of LVMH.
2. **Statistical Significance of Beta:** The calculated t-statistic for the slope coefficient is 16.243. Compared to the critical t-value at the 5% level of significance ($t_{crit} \approx 1.96$), $16.243 > 1.96$. Thus, LVMH's beta is statistically significant.
3. **Model R-squared (R^2):**

   $$R^2 = \frac{\text{SSR}}{\text{SST}} = \frac{0.108622}{0.150616} = 0.7212\,(72.1\%)$$

   This indicates that 72.1% of the variation in LVMH's weekly excess returns is explained by the market risk premium. The remaining 27.9% is due to firm-specific (unsystematic) factors.
4. **Intercept Alpha ($\hat{\alpha} = 0.0010$):** The intercept represents the abnormal excess return of LVMH. The calculated t-statistic is $0.525 < 1.96$, showing that the alpha is not statistically different from zero, which is consistent with the CAPM.

Module 11

Introduction to Financial Data Science

LEARNING OUTCOMES	
a	describe how big data, machine learning, and artificial intelligence are used in financial data science, fintech, and investment management

11.1 Financial Data and Big Data

a	describe how big data, machine learning, and artificial intelligence are used in financial data science, fintech, and investment management

Dữ liệu định lượng (quantitative) như giá cả, lợi suất đã rất quen thuộc. Nhưng cuộc cách mạng thực sự nằm ở dữ liệu định tính (qualitative) và dữ liệu thay thế (alternative data). Nhờ sự phát triển của công nghệ số, chúng ta có thể chuyển hóa các cuộc gọi thu nhập, tin tức hay bài đăng mạng xã hội thành số liệu để phân tích đồng thời. Hãy suy nghĩ: dữ liệu nào sẽ mang lại lợi thế cạnh tranh lớn hơn trong tương lai?

11.1.1 Quantitative vs. Qualitative Data

The investment decision-making process analyzes both quantitative and qualitative data:

- **Quantitative data** describes numeric information, such as prices, returns, interest rates, currency exchange rates, economic indicators, and accounting numbers, that can be analyzed using statistical methods.
- **Qualitative data** describes non-numeric information, such as market sentiment, management quality, industry trends, environmental factors (ESG), and geopolitical risks.

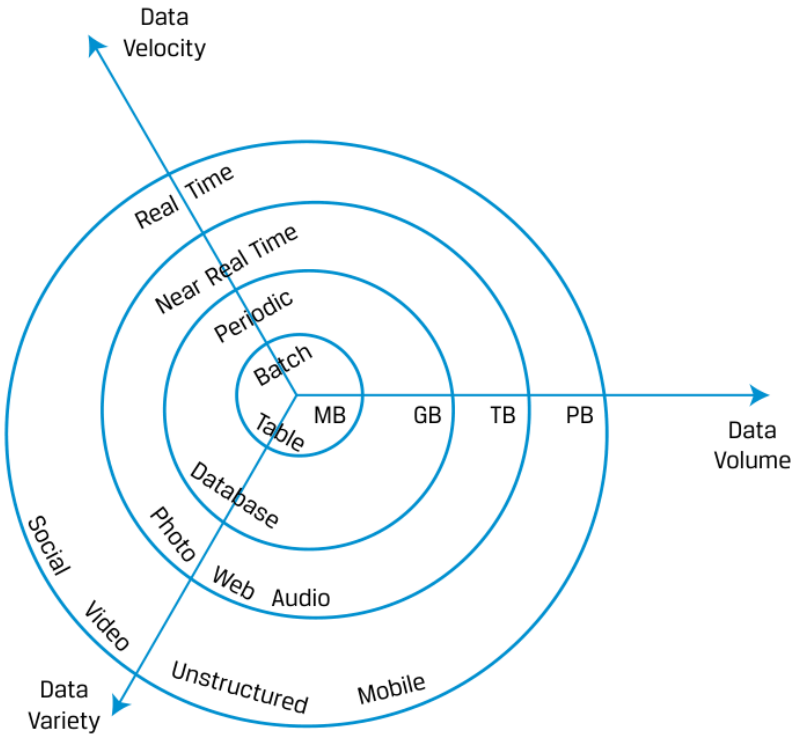
With the advance of digitization, qualitative data are increasingly captured digitally and processed algorithmically (e.g., using sentiment analysis), allowing for the integration of qualitative insights with quantitative analysis.

11.1.2 Characteristics of Big Data (The 4 Vs)

Datasets available for analysis are growing rapidly. Big data are characterized by four key dimensions:

- **Volume:** The vast amount of data collected (ranging from megabytes to gigabytes, terabytes, and petabytes), representing millions or billions of observations.
- **Velocity:** The speed at which data are communicated and updated. Real-time or low-latency data feeds have become the standard in financial applications.
- **Variety:** The diversity of data formats and sources, which can be structured, semi-structured, or unstructured.
- **Veracity:** The credibility, authenticity, and reliability of different data sources. Veracity is critical in big data because of the varying quality of large, non-traditional datasets.

Exhibit 1: Big Data Characteristics: Volume, Velocity, and Variety



Data	Volume Key	Bytes of Information
MB	Megabyte	One Million
GB	Gigabyte	One Billion
TB	Terabyte	One Trillion
PB	Petabyte	One Quadrillion

Figure 11.1: Characteristics of Big Data: Volume, Variety, and Velocity

11.1.3 Data Types and Sources

Data can be classified into three types based on structure:

- **Structured data:** Organized in tables (rows and columns) and commonly stored in databases where fields are defined (e.g., SQL tables, CSV files).
- **Unstructured data:** Disparate, unorganized data that do not fit into traditional tables. Examples include social media posts, emails, voice recordings, pictures, and video feeds.
- **Semi-structured data:** Blends structured and unstructured elements. The data do not fit into traditional tables but contain tags or markers (e.g., HTML/XML code, JSON files, tagged financial news).

Table 11.1 outlines common sources of big data in finance:

Table 11.1: Examples of Big Data Sources in Finance

Source	Examples
Financial markets	Equity, fixed income, futures, options, other derivatives, and commodities.
Businesses	Corporate financial statements, commercial transactions, credit card purchases, and supply chain data.
Governments	Trade, economic indicators, regulatory filings, taxation, and payroll data.
Individuals	Personal websites, online product reviews, internet search logs, email, and social media posts.
Sensors	Satellite imagery, aircraft tracking, shipping cargo data, and vehicle traffic patterns.
Internet of Things	Data from smart buildings (climate control, energy usage) and connected commercial devices.

11.1.4 Alternative Data and Risks

Alternative data refers to information collected from non-traditional sources, such as satellite imagery, web traffic, social media sentiment, mobile device geolocation, and retail point-of-sale scanners.

- **Data generated by individuals:** Personal activities like social media clicks, product reviews, and web searches. This data tends to be unstructured and qualitative.
- **Data generated by business processes:** Direct transactions like credit card purchases or point-of-sale scanner data. This structured data acts as a leading indicator of performance, whereas traditional corporate filings are lagging indicators.
- **Data generated by sensors:** Satellite imagery (e.g., counting cars in retail parking lots to predict earnings or monitoring flare gas at oil rigs) and Internet of Things (IoT) streams. It is largely unstructured and grows exponentially.

Lưu ý về đạo đức và pháp lý: Việc khai thác dữ liệu từ các trang web (Web scraping) có thể vi phạm các quy định bảo mật thông tin cá nhân (GDPR) hoặc bản quyền điều khoản dịch vụ của bên sở hữu dữ liệu. Khi các mô hình đầu tư phụ thuộc nhiều vào dữ liệu phi chính thống, tính tuân thủ pháp lý của nguồn dữ liệu trở thành yếu tố sống còn.

11.2 Financial Data Science and Fintech

11.2.1 Definitions

- **Data science** is an interdisciplinary field combining statistical methods, mathematical models, computational tools, and domain-specific knowledge to clean, process, and interpret big data.
- **Financial data science** applies data science principles specifically to financial datasets (prices, interest rates, returns, accounting statements, and macroeconomic variables) to support investment, portfolio management, trading, and risk decisions.

11.2.2 Unique Characteristics of Financial Data

Financial data exhibits unique challenges that require specialized analytical tools:

1. **High volume and speed:** Market transactions occur in microseconds, producing a massive continuous stream of tick data.
2. **Sequential and time-dependent structure:** Observations are chronologically ordered, requiring time-series analysis rather than standard independent assumptions.
3. **Noise and non-stationarity:** Financial data are highly noisy (containing random fluctuations) and non-stationary (statistical properties like mean and variance change over time).
4. **Interdependencies and non-linearity:** Financial variables have complex, non-linear relationships. Advanced tools like **Copulas** (mathematical functions joining multivariate distributions to capture dependency structures) and neural networks are used.
5. **Seasonal and cyclic patterns:** Recurring periods, such as quarterly corporate earnings releases or macro business cycles.
6. **Extreme events and fat tails:** Returns exhibit fat-tailed distributions, meaning extreme market crashes or jumps occur far more frequently than predicted by a standard Gaussian distribution.
7. **Data sparsity and missing values:** Emerging markets or private assets often lack complete historical data, requiring robust imputation methods to avoid estimation bias.

11.2.3 Fintech Evolution

Fintech (financial technology) represents technological innovation in the design and delivery of financial services and products. It has evolved through several stages:

- **Early stage:** Routine automation and simple data processing (e.g., calculating bookkeeping records).
- **Intermediate stage:** Rule-based decision execution systems ("if-then" systems).
- **Advanced stage:** Complex algorithmic and machine learning decision-making (e.g., robo-advisors that automatically construct, balance, and manage diversified portfolios based on client risk profiles and target return requirements).

11.3 Machine Learning and Artificial Intelligence

Trong học máy, triết lý cốt lõi là "find the pattern, apply the pattern" (tìm kiếm mẫu và áp dụng mẫu). Thay vì con người phải lập trình từng dòng quy tắc, thuật toán tự rút ra cấu trúc ẩn của dữ liệu từ quá trình huấn luyện (training). Hãy nắm chắc cách phân chia tập dữ liệu và sự khác biệt giữa các phương pháp học để trả lời chính xác câu hỏi trong kỳ thi.

11.3.1 Dataset Splitting and Time-Series Validation

In machine learning (ML), datasets are divided into three distinct subsets:

- **Training dataset:** The largest portion (typically 60%–80%) used by the algorithm to identify historical relationships between inputs and targets.
- **Validation dataset:** Used to validate and tune the model's parameters and refine its performance (10%–20%).
- **Testing dataset:** Used at the very end to evaluate the model's performance on unseen, out-of-sample data (10%–20%).

For cross-sectional data, data is split randomly. This randomization prevents bias and helps prevent **data leakage** (when information from outside the training set accidentally influences the model). For time-series data, simple randomization cannot be used because it disrupts the chronological sequence and introduces forward data leakage (predicting the past using future data). Instead, analysts use:

- **Time-based splits:** Earlier data is allocated to the training set and subsequent chronological blocks are used for validation and testing.
- **Rolling-window validation:** Shifts a fixed-size historical window forward in time, training on one window and validating on the next sequential period.

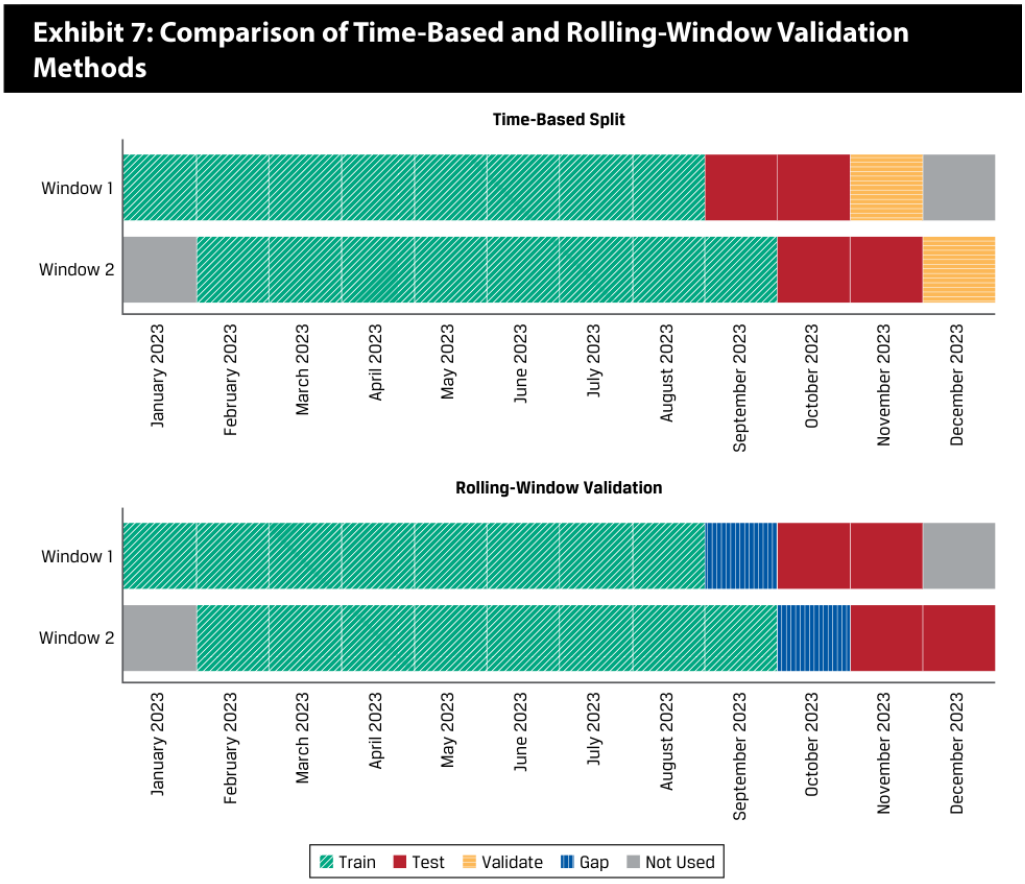


Figure 11.2: Time-Based Split vs. Rolling-Window Validation

11.3.2 Overfitting vs. Underfitting

- **Overfitting:** The model learns the training data too precisely, capturing random noise as if it were a true parameter. This leads to extremely low training error but high out-of-sample test error ($Error_{\text{train}} \ll Error_{\text{test}}$).
- **Underfitting:** The model is too simplistic and treats true relationships as noise, failing to learn the underlying structure in the training data.

11.3.3 Machine Learning Approaches

Machine learning is classified into four main approaches, summarized in Table 11.2:

11.3.4 Neural Networks and Generative Models

Neural networks are computational models inspired by biological brains. A neural network consists of layers of interconnected nodes (neurons):

- **Input layer:** Receives the raw input features (e.g., historical returns, financial ratios).
- **Hidden layers:** Intermediate layers where non-linear data transformations occur. Multiple hidden layers constitute a "deep" neural network.
- **Output layer:** Provides the final classification or predicted continuous value.

Table 11.2: Comparison of Machine Learning Approaches

Approach	Description	Applications	Advantages	Limitations
Supervised learning	Models learn from labeled historical training data where target output is known.	• Spam detection • Stock return prediction • Default classification	• Highly accurate • Well-understood methods	• Labeled data is costly • Fails to generalize to new scenarios
Unsupervised learning	Models analyze unlabeled data to discover hidden structures and groupings.	• Customer segmentation • Anomaly detection • Dimension reduction	• Discovers unexpected patterns • Reduces data dimensionality	• Subjective evaluation • Difficult to interpret results
Reinforcement learning	Models learn by trial-and-error interaction with an environment to maximize rewards.	• Portfolio optimization • Algorithmic execution	• Good for sequential decisions • Adapts to dynamics	• High training time • Sensitive to reward design
Deep learning	Models use multi-layered artificial neural networks to extract abstract patterns.	• Image recognition • NLP	• Automatically extracts features • Models non-linear structures	• "Black box" nature • Requires massive datasets

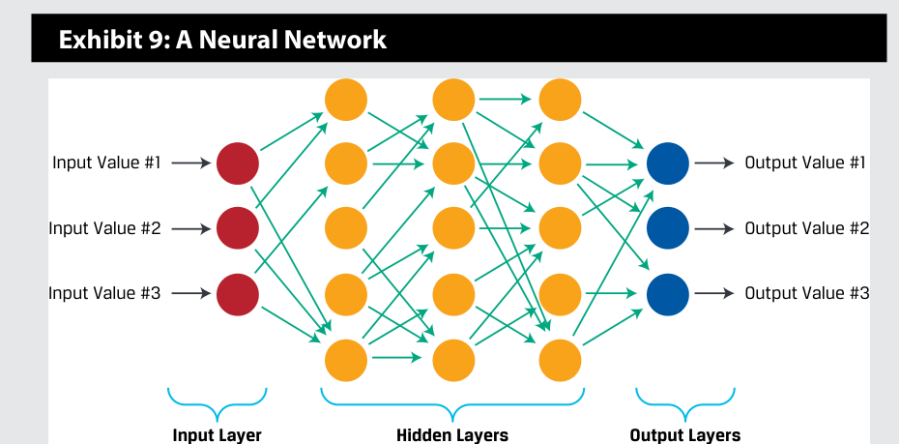


Figure 11.3: Structure of an Artificial Neural Network

Training is completed using **backpropagation**: 1. **Forward pass:** Data flows through the network to generate a prediction. 2. **Loss calculation:** Computes the error (loss) between the predicted and actual target. 3. **Backward pass (Backpropagation):** Calculates gradients of the loss function with respect to the network weights. 4. **Weight update:** Weights are updated slightly in the opposite direction of the gradient using **gradient descent**. This process is repeated over multiple cycles (**epochs**) until the network converges.

Two advanced generative neural network architectures are:

- **Generative Adversarial Networks (GANs):** Combines a generator network (creates synthetic data) and a discriminator network (evaluates authenticity) in an adversarial loop. Used to generate synthetic financial scenarios for stress-testing.
- **Variational Autoencoders (VAEs):** Compresses inputs into a latent, lower-dimensional space (encoder) and reconstructs them (decoder). Useful for dimensionality reduction and anomaly detection.

11.4 Natural Language Processing and Generative AI

11.4.1 Natural Language Processing (NLP)

- **Text analytics** extracts information and identifies patterns from unstructured text- or voice-based datasets (filings, transcripts, emails, news, etc.).
- **Natural language processing (NLP)** is a subfield at the intersection of computer science, AI, and linguistics focused on analyzing human language.

Common applications in finance include sentiment analysis (e.g., parsing earnings call transcripts to assign a positive/negative sentiment rating to detect subtle manager outlooks before analyst rating updates) and monitoring central bank communication (parsing Fed or ECB statements to extract tone shifts on interest rates).

11.4.2 Large Language Models (LLMs)

Large Language Models (LLMs), such as GPT models, represent a significant advancement in generative AI.

- **Architecture:** Rely on **transformers** to process and capture long-range contextual relationships within textual sequences.
- **Huấn luyện song miền (Dual-domain training):** Sourced via self-supervised learning on massive general corpora (to learn grammar, semantics) and specialized financial corpora (filings, earnings call transcripts to master domain jargon).
- **Tinh chỉnh (Fine-tuning):** Adapted via transfer learning for specialized applications (e.g., text generation, report summarization, financial sentiment analysis).
- **Rủi ro và hạn chế (Risks):** LLMs excel at language prediction but are prone to **hallucinations** (generating false information) and lack numerical calculation capabilities. Over-reliance can result in strategic and regulatory errors. A hybrid model combining automated outputs with human review is recommended.

11.4.3 Generative AI vs. Traditional Simulations

Generative AI creates new content or scenarios based on representations learned during training. Unlike Monte Carlo simulations (which assume predefined parametric distributions) or bootstrapping (which simply shuffles or resamples historical records), Generative AI learns complex underlying data structures to produce realistic, non-replication-constrained synthetic scenarios.

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